

15th

Interventional MRI Symposium



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**The Liberty Hotel
Boston, Massachusetts
October 8-9, 2026**

General Chairs:

Clare M. Tempany, MD
Thomas Kahn, MD
Jonathan S. Lewin, MD, FACR

Local Chairs:

Junichi Tokuda, PhD
Nobuhiko Hata, PhD

<https://imri2026.org/>

BOOK OF ABSTRACTS

2026



Brigham and Women's Hospital
Founding Member, Mass General Brigham



**Universitätsklinikum
Leipzig**

Medizin ist unsere Berufung.



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MEDICINE

Preface

Dear Colleagues and Friends,

Welcome to the 15th Interventional MRI Symposium (iMRI), October 8-9, 2026, at The Liberty Hotel in Boston, Massachusetts, United States. Similar to the last gathering in 2024 in Maryland, the symposium provides the opportunity for the interventional MRI community to interact with opinion leaders and innovators in the field, meet old friends and develop new collaborations to expand our field. This event is jointly organized by Brigham & Women's Hospital/Harvard Medical School, Johns Hopkins University and University of Leipzig. The iMRI Symposium provides an ideal platform for researchers, clinicians, healthcare professionals, and students alike to present their latest results. Sessions with both technical and clinical topics are composed of lectures by invited speakers and proffered scientific papers. We hope you enjoy Boston and everything the conference has to offer!

October 8-9, 2026
Boston, Massachusetts

Thomas Kahn
Clare M. Tempany
Jonathan S. Lewin

Table of Contents

Ferenc Jolesz Memorial Lecture	
From MRI-Guided Biopsy to Autonomous Intervention: A Decade Toward Self-Driving Surgery	1
<i>Axel Krieger</i>	
Session I – Body I	
A New Era of Body iMRI, Why Now?	2
<i>Frank Wacker</i>	
Beyond the Bore: Key Strategies for Building an iMRI Program	3
<i>Erica Knavel Koepsel</i>	
Precision Targeting of Small Liver Lesions Using MRI-Guided Freehand Biopsy: Outcomes and Safety	4
<i>Vanessa F. Schmidt, Olaf Dietrich, Sinan Deniz, Daniel Puhr-Westerheide, Osman Öcal, Moritz Schnitzer, Florian Obereisenbuchner, Mingming Wu, Florian Maier, Ricarda Seidensticker, Moritz Wildgruber, Jens Ricke, Matthias Philipp Fabritius and Max Seidensticker</i>	
MRI-Guided Hepatic Interventions Using Augmented Reality Optical Navigation: Initial Clinical Experience with the LUMENA System	5
<i>David Woodrum, Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Kenny Harlan, Purnima Rajan, Pezhman Foroughi, Alican Demir, Daniel Adamo and Aiming Lu</i>	
Toward Personalized Planning of Liver Tumor Microwave Ablation Using Patient-Specific Modeling	6
<i>Nino Avetikovi, Ida Burgers, Mariana De Melo Antunes, Manon Desclides, Olaf Dietrich, Max Seidensticker, Jean-Luc Battaglia and Valéry Ozenne</i>	
Assessing the Technical Feasibility of MR-Guided Irreversible Electroporation in a Porcine In Vivo Model in a 1.5 T Clinical MR Environment	7
<i>Othmar Belker, Simon Schröer, Thomas Gerlach, Max Joris Hubmann, Daniel Dux, Marcel Eisenmann, Christoph Fritsch, Julian Glandorf, Seben Sena Yerdelen, Inga Bruesch, Björn Hartleben, Sahar Arbabzadah, Jörg Mengwasser, Philipp Felgendreff, Georg Rose, Frank Wacker, Marcel Gutberlet and Bennet Hensen</i>	
Session II – Body II	
Proceed with Confidence: Overcoming MR Safety Barriers in MRI-Guided Interventions ..	8
<i>Aiming Lu</i>	
Preliminary Experience of MR-Guided Lateral Decubitus Breast Biopsy Using a Flexible Surface Coil	9
<i>Xin Weng, Pin Nie, Yanli Dang, Kaiguo Zhu, Jintao Hu and Baoying Chen</i>	
Safety and efficacy of MRI-guided cryoablation of renal cancer: a systematic review and meta-analysis	10
<i>Hans Jonas Meyer, Timm Denecke and Silke Zimmermann</i>	

Initial Evaluation of Interactive Imaging on a High Performance 0.55T System	11
<i>Nathan Ooms, Jason Scott, Jesse Roll, Axel Krafft and Florian Maier</i>	
Post-Ablation Margin Assessment using Free-Breathing Liver DCE MRI: An In-Silico Study	12
<i>Jayant D Sakhardande, Julia M Velikina, Alexey A Samsonov, Eric M Schrauben, Francisco Donato Jr, Silpa Babu, Desmond T Yeo, Shane A Wells and James H Holmes</i>	
Session IV – Technology I	
MRI-Guided Robotics to Enable Interventions	13
<i>Kevin Cleary</i>	
RF Receive Coil with Integrated MRI-Transparent Heater for Cryoablation	14
<i>Joseph Corea, Jayden Cox, Renato Hernandez and Pejman Ghanouni</i>	
MRI Hybrid Ablation: Using the MR Scanner Hardware as Power Source for RF Ablation and as receiving unit of thermal monitoring by MR thermometry	15
<i>Thomas Gerlach, Othmar Belker, Marcel Gutberlet, Bennet Hensen, Frank Wacker, Oliver Speck and Ralf Vick</i>	
Out-of-bore Respiratory Tracking for MRI-guided Percutaneous Interventions using a Noise Navigator	16
<i>Erik Arts, Witek Bosman, Evangelia Ilia and Wyger Brink</i>	
Flexible RF receive array coil optimized for MR-guided Interventions	17
<i>Jayant D Sakhardande, Jana Vincent, Victor Taracila, Fraser Robb, Francisco Donato Jr, Desmond T Yeo, Shane A Wells and James H Holmes</i>	
Session V - Cardiovascular	
Interventional Cardiac MR for Congenital Heart Disease: Is 0.55T the Answer?	18
<i>Aimee Armstrong</i>	
Initial In Vivo Validation of MRI-Guided and Actuated Robotic Ablation Catheter	19
<i>Ridaa Ali, Eser Tuna, Andrew Dupuis, Yen-Chun Chen, Ahmed Aydin, Tobias Cowles, Gary Pawlowski, Brendan Morgan, Michael Markley, Michael Kavran, Ajay Bhandari, Hiram Bezerra, Joseph Piktel, Mark Griswold, Daniel Herzka and Murat Cavusoglu</i>	
First in-vivo MR-Guided Venous Stenting with a Novel Non-Magnetic Delivery System ...	20
<i>Lisa Regler, Wibke Uller, Niklas Verloh, Alexander Maier, Simon Reiss, Klaus Düring, Areli Luttmann, Pooja Virkar and Michael Bock</i>	
Phantom-Based Evaluation of AI-Assisted MRI Guidance of a Marker-Enhanced Nonmetallic Microwire for Peripheral Endovascular Interventions	21
<i>Lena Marie Wilms, Paula Krüselmann, Darius Nuernberger, Jalil Ahmed, Benedikt Kamp, Samuel Mueller and Peter Minko</i>	
Development and characterization of a non-metallic aortic stent graft for Magnetic Resonance Imaging	22
<i>Franz Wegner, Marius R. Meyer, Valentine Gesché, Paul Borm, Samuel Müller, Franziska Seidensticker, Annika C. Dell, Georg Männel, Dennis Wendt, Thomas Friedrich, Thorsten M. Buzug, Jörg Barkhausen, Roman Kloeckner and Malte M. Sieren</i>	

Catheter Transmit Coils for Tracking and Intra-arterial Spin Labeling	23
<i>Felix Spreter, Alexander Maier, Michael Bock and Simon Reiss</i>	

Session VI – Pelvis (Prostate, Gyn & RadOnc)

Prostate Interventions: How I Do It	24
<i>Jurgen Fütterer</i>	
MRI-Guided In-Bore Percutaneous Transgluteal Prostate Biopsy at 0.55T	25
<i>Serdar Sagdic, Yun Jiang, Tejinder Kaur, Shane Wells, John Wei, Hero Hussain, Vikas Gulani and Elaine Caoili</i>	
MRI-Guided Cryoablation for Primary and Salvage Prostate Cancer Treatment: Initial Patient Experience	26
<i>Lauren Marlatt, Tom Lilieholm, Michael Risk, David Jarrard and Erica Knavel-Koepsel</i>	
Cost-Effective Needle Tract Verification Using Susceptibility-Weighted Imaging (SWI) in MR-Guided Prostate Biopsy	27
<i>Can Yüksel, Christiane K. Kuhl, Christoph Katemann, Oliver M. Weber and Narine Mesropyran</i>	
Accuracy and verification of an MR-compatible transperineal biopsy device	28
<i>J.G.R. Bomers, D.G.H. Bosboom, M. Willemen, M. Hoogenboom and J.J. Fütterer</i>	
Four-Year Experience with MRI-Guided Focal Prostate Cryoablation	29
<i>Vipul Sheth, Ryan Brunsing, Sharmila Sewell, Geoff Sonn, Richard Fan, Pejman Ghanouni, Negaur Iranpour and Rachelle Bitton</i>	
In-Bore MRI-Guided Trans-Perineal Partial Gland Cryoablation for Primary Prostate Cancer: A Prospective Study	30
<i>David Woodrum, Setayesh Sotoudehnia Korani, Ola Mohamed, Lance Mynderse, Scott Thompson, Daniel Adamo and Aiming Lu</i>	
MR guided brachytherapy of small liver lesions - optimizing dose delivery and local tumor control (Results of the MR BRIGHT trial)	31
<i>Matthias Philipp Fabritius, Alireza Haghpahan, Olaf Dietrich, Daniel Puhr-Westerheide, Vanessa F. Schmidt, Stefanie Corradini, Jens Ricke, Osman Öcal and Max Seidensticker</i>	
End-to-End MRI and MRT-guided cervical-cancer brachytherapy interventions	32
<i>Majd Antaki, Junghoon Lee, Michael Roumeliotis, Gayoung Kim, Khadija Sheikh, Bruce L Daniel, Daniel Song, Ravi Seethamraju, Yury Niatsetski, Junichi Tokuda, Robert A Cormack, Charles L Dumoulin, Wolfgang Loew, Ehud J Schmidt and Akila N Viswanathan</i>	
Automated detection of Syed-Neblett applicators and interstitial needles on PETRA MRI images	33
<i>Evangelia Kaza and Ivan Buzurovic</i>	

Session VII - Neuro

Radio-Pathomic Mapping of Brain Tumor Invasion	34
<i>Peter LaViolette</i>	

Brain Proton MRS Thermometry Enhanced by Fiber Orientation Susceptibility Correction	35
<i>Adisorn Yimsuk and Kagayaki Kuroda</i>	
Clinical Feasibility of an MRI-Compatible Motorized Alignment Device for MRI-Guided Deep Brain Stimulation: Initial Experience.....	36
<i>Pedro Moreira, Yang Wang, Zhanyue Zhao, Charles Bales, Shiyue Wang, Garth Cosgrove and Gregory Fischer</i>	
Investigating the thermal effect of interstitial photodynamic therapy for glioblastoma treatment: a combined experimental and numerical study	37
<i>Manon Desclides, Bastien Gouges, Anne-Sophie Dewalle, Gregory Baert, Benoit Derre, Quentin Vannod Michel, Jean-Luc Battaglia, Nadira Delhem, Nicolas Reyns and Valéry Ozenne</i>	
Trajectory Guide for Accelerated Minimally-Invasive Neurosurgery: Hardware Validation in Phantoms	38
<i>Thomas Lilieholm, Nirvaan N Jaitli, Terrence R Oakes, Azam S Ahmed, Andrew L Alexander and Walter F Block</i>	
An MRI-Guided Robot for Stereotactic Neurosurgery: An Initial Cadaver Study.....	39
<i>Gang Li, Atharva Paralikar, Ayush Nankani, Pavel Yarmolenko, Wolfgang Loew, Charles Dumoulin, Kevin Cleary, Chima Oluigbo and Reza Monfaredi</i>	

Special Invited Lecture VIII

From Intraoperative MRI to Intelligent Intervention: 30 Years of Image-Guided Therapy at Brigham and Women's Hospital	40
<i>Nobuhiko Hata</i>	

Session IX – Focused Ultrasound (FUS)

MR-Guided Focused Ultrasound Blood-Brain Barrier Opening for Next-Generation Gene Therapy Delivery Paradigms	41
<i>Nicholas Todd</i>	
MRI-Guided Transurethral Ultrasound Ablation (TULSA) vs Robotic Prostatectomy RCT: 6-Month Outcomes	42
<i>David Woodrum, Lance Mynderse, Pejman Ghanouni, Sandeep Arora, Steven Raman, Daniel Costa, Jovan Begovic, David Sella, Gregory Frey, Robert Princenthal, Martin Cohen, Katarzyna Macura, Temel Tirkes, Joseph Busch, Andrei Purysko, Pietari Makela, Gary Brahm, Chirag Patel, Mohammad Mahdi Bhurwani, Gina Clarke and Robert Staruch</i>	
Impact of Ablation Plan on PSA Response and Catheterization Duration After MRI-Guided TULSA in the CARE Registry.....	43
<i>David Sella, Gregory Frey, Chandler Dora, Ram Pathak, Andrea Acevedo Bado, Daniel Costa, Jovan Begovic, Xiaosong Meng, Naveen Kella, Edward Steiner, Adnaan Moin, Martin Cohen, Robert Princenthal, Mohammad Mahdi Bhurwani and Gina Clarke</i>	

Translational Assessment of a Rabbit-Model Trained MRgFUS Imaging Biomarker	44
<i>Amanpreet Singh, Samuel Adams, Sara Johnson, Ariana Page, Jill Shea, Henrik Odeen, Sarang Joshi and Allison Payne</i>	

Session X - Musculoskeletal

Feasibility of 3D visualization for intraprocedural coverage assessment in MR-guided spine cryoablation	45
<i>Mariana C. Bernardes, Vincent M Levesque, Junichi Tokuda and Jeffrey P. Guenette</i>	
MRI-Guided Multi-Regional Biopsy of Soft-Tissue Sarcomas	46
<i>Simon Reiss, Balazs Bogner, Alexander Runkel, Elena Fritsch, Matthias Jung, Jakob Weiss and Michael Bock</i>	
MRI-Guided Percutaneous Laser Ablation for Pediatric Vascular Anomalies: A Single-Institution Experience	47
<i>Setayesh Sotoudehnia Korani, Ola Mohamed, Erica Knavel Koespel, Scott Thompson, Aiming Lu, Daniel Adamo and David Woodrum</i>	
MRI-Safe Robot and Drill for Bone Biopsy: First Clinical Results	48
<i>Dan Stoianovici, Karun Sharma, Pavel Yarmolenko, Van Khanh Lam, Doru Petrisor, Jakub Piwowarczyk and Kevin Cleary</i>	

Session XI – Technology II

Deep Learning-Based Needle Tracking Technologies for MRI-Guided Interventions	49
<i>Holden Wu</i>	
Tissue and Needle Tracking on Real-Time Interventional Liver MRI Using the SAM2++ Foundation Model	50
<i>Christina Kerr, Wenqi Zhou and Holden Wu</i>	
Unsupervised Conditional INR for Referenceless PRF Thermometry	51
<i>Yueran Zhao, Shenyan Zong, Guofeng Shen and Chang-Sheng Mei</i>	
MR Thermometry-based validation of liver tumor microwave ablation computational modeling for lesion size prediction	52
<i>Nino Avetikovi, Ida Burgers, Mariana De Melo Antunes, Manon Desclides, Luigi Nardone, Pierre Bour, Thibaut Faller, Nestor Pallares-Lupon, Bruno Quesson, Olaf Dietrich, Max Seidensticker, Jean-Luc Battaglia and Valéry Ozenne</i>	
AI-Based Workflow for Automatic Needle Tracking and Dynamic Scan-Plane Control in Interventional MRI	53
<i>Mariana C. Bernardes, Pedro Moreira, Ravi Seethamraju, Lori Foley, Kemal Tuncali, Clare Tempny, Simon Leonard, Axel Krieger and Junichi Tokuda</i>	

Poster – Pelvic (Prostate and Gyn)

MRI-Guided Percutaneous Cryoablation for vaginal/paravaginal lesions - a case series	54
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Brian Welch, Scott Thompson, Daniel Adamo, Aiming Lu, Angela Bathke and David Woodrum</i>	

MR-Guided Percutaneous Cryoablation for Extrapelvic Endometriosis: Safety and Pain-Relief Outcomes in a Five-Patient Case Series.....	55
<i>Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Aiming Lu, Daniel Adamo and David Woodrum</i>	
Locoregional Tumor Control of Prostate Cancer Bone Metastasis using MRI-Guided Cryoablation: A Retrospective Cohort Study.....	56
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Lance Mynderse, Daniel Adamo, Aiming Lu, Angela Bathke, Scott Thompson and David Woodrum</i>	
MR-guided Cryoablation in the Treatment of Recurrent Prostate Cancer in the Prostate Bed: A 13-Year Prospective Study.....	57
<i>David Woodrum, Setayesh Sotoudehnia Korani, Ola Mohamed, Lance Mynderse, Scott Thompson, Angela Bathke, Derek Lomas, Daniel Adamo and Aiming Lu</i>	
AI-Assisted Quantification of Cryo-Injury in MRI-Guided Prostate Cryoablation Using Imaging-Derived Dose Metrics.....	58
<i>Pedro Moreira, Mijia Ma, Nicholas Fordham, Mark Vangel, Clare Tempany, Kemal Tuncali and Junichi Tokuda</i>	
Effect of Bead Presence, Composition, and Configuration on MRI and CT Artifact Characteristics.....	59
<i>Deborah Shinyanbola, Sharath Bhagavatula, Nobuhiko Hata and Clare Tempany</i>	
MRI-Guided Transperineal Prostate Biopsy in Patients Without Rectal Access: Technique and Feasibility.....	60
<i>Michael McKernan, Ola Mohamed, Setayesh Sotoudehnia Korani, Mrinal Singh, Derek Lomas, Scott Thompson, David Woodrum and Daniel Adamo</i>	
Imaging Sequences Used for Prostate Cryoablation and Biopsy Guidance.....	61
<i>Sarah Fasbender, Carly McKenzie, Angela Pooler, Heather Heimer, Connor Gleichner, Wayne Stenzel, Gehling Katie, Burke Courtney, Bathke Angela, Aiming Lu, Myungho In and David Woodrum</i>	
Image Guided Focal Therapy for Prostate Cancer: A Review of Patient Selection, Imaging, and Treatment Techniques.....	62
<i>Daniel Adamo, Setayesh Sotoudehnia Korani, Scott Thompson, Aiming Lu, Ola Mohamed, Lance Mynderse and David Woodrum</i>	
Image Quality Assessment of 0.6T and 1.5T MRI for MRI-Guided Prostate Interventions.	63
<i>Daniel Adamo, Setayesh Sotoudehnia Korani, Omer Burak Demirel, Zainudeen Kallumpurath, Spencer Waddle, Zhiqiang Li, Jacinta Browne, Tim Leiner, Aiming Lu, Scott Thompson and David Woodrum</i>	
MR-Guided Cryoablation of Penile Metastasis – a case series of three patients.....	64
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Scott Thompson, Daniel Adamo, Aiming Lu, Angela Bathke and David Woodrum</i>	

MRI-Guided Percutaneous Cryoablation of a Symptomatic Penile Neurofibroma in the Setting of Neurofibromatosis Type 1.....	65
<i>David Woodrum, Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Daniel Adamo and Aiming Lu</i>	
Poster - RadOnc	
MR guided catheter-based radiotherapy / brachytherapy of small liver lesions – advantages of a tip marked MRI safe wire to define exact catheter position for radiotherapy planning.....	66
<i>Matthias Philipp Fabritius, Vanessa F. Schmidt, Daniel Puhr-Westerheide, Jens Rieke, Olaf Dietrich, Osman Öcal and Max Seidensticker</i>	
Outcome after MR-Guided Ablation of Small Liver Lesions with Catheter-Based Brachytherapy	67
<i>Max Seidensticker, Alireza Haghpanah, Olaf Dietrich, Daniel Puhr-Westerheide, Vanessa F. Schmidt, Frederik Fuchs, Jens Rieke, Osman Öcal and Matthias Philipp Fabritius</i>	
Real-Time FBG Shape Feedback and Trajectory Prediction for Adaptive MRI-Guided Needle Interventions	68
<i>Yinsong Ma, Jacynthe Francoeur, Sichen Deng, Jin Seob Kim and Iulian Iordachita</i>	
Integration of Habitat Analysis in Novel Clinical Trial for Characterization of Ultra-early Response to Radiotherapy.....	69
<i>Adrian Breto, Veronica Wallaengen, Sandra Gaston, Matthew Abramowitz, Alan Pollack, Benjamin Spieler and Radka Stoyanova</i>	
Poster - Technology	
MR-compatible ultrasound tracker for fast alignment of US-MR images.....	70
<i>Chitresh Bhushan, Eric Fiveland, Afis Ajala, Mills David, Jhimli Mitra, Shane Wells, James Holmes and Desmond Yeo</i>	
Custom Hardware to Adapt Conventional Flexible MRI Coils for Interventional Applications	71
<i>Nirvaan N Jaitli, Walter F Block, Terrence R Oakes, Azam S Ahmed, Andrew L Alexander and Thomas Lilieholm</i>	
Design and Preliminary Evaluation of an MR-Compatible Microwave Applicator Holder with Motion-Adaptive Compliance for Interventional MRI Procedures	72
<i>Qiaozhi Chen, Othmar Belker, Marcel Gutberlet, Daniel Dux, Martin Volk, Anton Schlünz, Thomas Gerlach, Frank Wacker, Oliver Speck and Georg Rose</i>	
Dynamic Decoupling of RF Coil Arrays for Interventional MRI	73
<i>Isabella Duan and Feng Liu</i>	
Comparison of Three Real-Time Marker Detection Algorithms for Instrument Tracking of MRI-Guided Robotic Needle Interventions.....	74
<i>Gina Thieße, Marcel Gutberlet, Bennet Hensen, Frank Wacker, Anton Schlünz, Oliver Speck, Georg Rose and Ivan Vogt</i>	

Feasibility of Extended Reality Guidance for MRI-Guided interventions : A Phantom Evaluation in Cardiac and Prostate Applications	75
<i>Evangelia Ilia, Roy Damgrave and Wyger Brink</i>	
From "Stop-and-Go" to "Continuous" in MRI-guided Needle Steering: Addressing Delayed and Sparse Feedback	76
<i>Ryo Murakami, Yiwei Jiang, Satoshi Mori, Yang Wang, Charles Bales, Farid Tavakkolmoghaddam, Pedro Moreira, Junichi Tokuda, Gregory Fischer, Nobuhiko Hata and Haichong Zhang</i>	
Bidirectional Continuous Rotation and Variable Curvature Needle Steering (B-CURV) for MRI-Guided Robotic Prostate Biopsy	77
<i>Tae Wan Kim, Ryo Murakami, Kehan Yang, Eric Wang, Charles Bales, Nobuhiko Hata and Haichong Zhang</i>	
Framework for fast, inline 3D needle localization	78
<i>Dominique Franson, Jonas Faust, Gunter Ambruster and Florian Maier</i>	
Learning Magnetic Needle Models for Interventional Localization	79
<i>Sarah Garrow and William Grissom</i>	
Automated Ground-Truth Generation for Markerless Stereo RGB Needle Tracking in MRI-Guided Prostate Biopsy	80
<i>Nadeem Shah, Lucas Knull, Thomas Gerlach and Oliver Speck</i>	
Use of a robotic-assisted percutaneous ablation platform for MR-guided procedures: Proof of concept experiments	81
<i>Christopher Favazza, Eric Stinson, Myungho In, Aiming Lu, Christi Fernandes, Daniel Adamo and David Woodrum</i>	
Automatic Needle Tracking in 4D Ultrasound for MR-US Fusion-Guided Ablation	82
<i>Julia Chaffin, Kourosh Kalayeh, J. Brian Fowlkes, David Mills, Heather Chan, Desmond Yeo, Afis Ajala, Chitresh Bhushan, Jhimli Mitra, James H. Holmes, Christopher L. Brace and Shane Wells</i>	
Needle segmentation for patient-specific thermal model	83
<i>Hippolyte Salles, Nino Avetikovi, Manon Desclides, Jean-Luc Battaglia, Max Seidensticker, Olaf Dietrich and Valéry Ozenne</i>	
MRI-Guided Microwave Ablation: System Comparison of Shaft Cooling and Thermocouple Regulation on Thermal Dose Distributions and Artifacts	84
<i>Laura Maria Bauer, Luigi Nardone, Mingming Wu, Vanessa F Schmidt, Matthias P Fabritius, Jens Ricke, Max Seidensticker and Olaf Dietrich</i>	
Investigation of water-cooled MRI-guided microwave ablation system on electromagnetic interference and artifacts	85
<i>Mandy Yu, David Woodrum, Scott Thompson, Daniel Adamo and Aiming Lu</i>	
Projection-based Receiver Interference Separation Method for Electromagnetic Interference Suppression during MR-Guide Microwave Ablation	86
<i>Ning Jin, Jonas Frederik Faust, Hojjat Fallahi, Axel J. Krafft, Florian Maier and Jianing Pang</i>	

Poster - FUS

Technical Validation of Transurethral Ultrasound Ablation (TULSA) on Siemens 0.55T Free.Max MRI	87
<i>David Woodrum, Setayesh Sotoudehnia Korani, Lance Mynderse, Martin Requardt, Xingmin Guan, Joseph Busch, Axel Krafft and Ben Leung</i>	
Single-center experience with MRI-guided transurethral ultrasound ablation (TULSA): factors affecting treatment duration and outcome.....	88
<i>Andrea Acevedo, Robert Staruch, David Sella, Gregory Frey, Chandler Dora and Ram Pathak</i>	
MRI-Guided Transurethral Ultrasound Ablation (TULSA) for Treating Native Prostate Cancer, a prospective single center study	89
<i>Setayesh Sotoudehnia Korani, Ben Leung, Ola Mohamed, Scott Thompson, Aiming Lu, Daniel Adamo and David Woodrum</i>	
A Simulation-to-Robot Workflow for Directional Needle-Based Therapeutic Ultrasound Ablation	90
<i>Kehan Yang, Yiwei Jiang, Yang Wang, Andrew Daudelin and Gregory S. Fischer</i>	
Characterization of MRgFUS Ablation Effects through Radiology- Pathology Data Integration.....	91
<i>Justina Bonaventura, Amanpreet Singh, Sophie Zhao, Samuel Adams, Sarang Joshi, Beatrice Knudsen and Allison Payne</i>	
Poster – Cardiovascular	
SPION-polymer balloon catheters in 3T MRI: a phantom study	92
<i>Paul E Hermann</i>	
Evaluation of an Interventional MRI User Interface Extension for Real-Time Vascular and Cardiac Navigation	93
<i>Nathan Ooms, Jason Scott, Günter Armbrüster, Fang Dong, Jesse Roll and Florian Maier</i>	
Plane-Specific Anomaly Detection for Passive Marker Identification and Localization in iCMR.....	94
<i>Yixuan Liu, Ateeb Butt, Jianli Wei, Samuel Ting, Aimee Armstrong and Orlando Simonetti</i>	
Dexterous Intramyocardial Needle Ablation (d-INA): Initial Design, Fabrication, and In-Vivo Validation	95
<i>Chang Zhou, Charles Hong, Yifan Wang, William Stevenson, Ehud Schmidt, Junichi Tokuda, Aravindan Kolandaivelu and Yue Chen</i>	
Poster - MSK	
MR-guided cryoablation of gluteal metastases	96
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Scott Thompson, Daniel Adamo, Aiming Lu, Angela Bathke and David Woodrum</i>	

MR-guided laser ablation of a painful glomus tumor of the left dorsal forefoot and right knee in a 14-year-old male patient after failed surgical resection – a case report	97
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Scott Thompson, Haraldur Bjarnason, Megha Tollefson, Emily Bendel, Daniel Adamo, Angela Bathke, Aiming Lu and David Woodrum</i>	
MR guided bone biopsy	98
<i>David Woodrum, Ola Mohamed, Setayesh Sotoudehnia Korani, Scott Thompson, Daniel Adamo and Aiming Lu</i>	
Combined CT- and MR- Guided Ablation of a Sacral Tumor: A Case Report	99
<i>Heather Heimer, Setayesh Sotoudehnia Korani, Ola Mohamed, Angela Pooler, Carly McKenzie, Sarah Fasbender, Wayne Stenzel and David Woodrum</i>	
MSK MRI-Based Personalized Osteochondral Arthroplasty with Tissue-Specific Robotic Resection: A Feasibility Framework	100
<i>Sangeun Song, Ulas Bagci, Ramsey Kinney and Jutta Ellermann</i>	
Hybrid CT/MRI-Guided Cryoablation of Osteoblastic Prostate Cancer Bone Metastases . .	101
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Scott Thompson, Daniel Adamo, Aiming Lu, Angela Bathke and David Woodrum</i>	
MRI-Guided Percutaneous Cryoablation for Pediatric Vascular Anomalies: Safety and Technical Outcomes	102
<i>Setayesh Sotoudehnia Korani, Ola Mohamed, Erica Knavel Koespel, Scott Thompson, Aiming Lu, Daniel Adamo and David Woodrum</i>	
MR-Guided Cryoablation for Desmoid Fibromatosis: Technical Feasibility, Safety, and Symptomatic Outcomes	103
<i>Setayesh Sotoudehnia Korani, Brian Welch, Daniel Adamo, Ola Mohamed, Aiming Lu and David Woodrum</i>	
MR-Guided Ablation in Kaposiform Hemangioendothelioma	104
<i>Hasan Usmani, Nessa Aghazadeh Mohandesi, Megha Tollefson, Scott Thompson, Katelyn Anderson and David Woodrum</i>	
MR guided cryoablation for treatment of painful large to medium slow-flow venous malformations in the plantar foot – a retrospective study	105
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Haraldur Bjarnason, Juna Musa, Scott Thompson, Megha Tollefson, Emily Bendel, Daniel Adamo, Angela Bathke, Aiming Lu and David Woodrum</i>	
Sequential MR-guided laser and cryoablation of a recurrent painful slow-flow venous malformation in a 6-year-old	106
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Scott Thompson, Haraldur Bjarnason, Megha Tollefson, Emily Bendel, Daniel Adamo, Angela Bathke, Aiming Lu and David Woodrum</i>	
MRI-Guided Cryoablation for Recurrent Pediatric Hand Desmoid Tumor: A Longitudinal Case Report	107
<i>Laurel Woodrum, Setayesh Sotoudehnia Korani, Scott Thompson, Aiming Lu and David Woodrum</i>	

MRI-Guided Cryoablation of a Bilateral Plantar Fibromatosis	108
<i>Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Aiming Lu, Daniel Adamo and David Woodrum</i>	
Poster - Body	
MRI-Guided Microwave Ablation of Hepatic Metastases Using a High-Energy MR-Compatible Microwave System: Initial Clinical Series	109
<i>Aiming Lu, Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Daniel Adamo and David Woodrum</i>	
MR-guided cryoablation for local control of adrenal gland metastases – a case series	110
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Scott Thompson, Daniel Adamo, Aiming Lu and David Woodrum</i>	
MRI-guided Microwave Ablation for Primary and Metastatic Liver Tumors: a Comprehensive Single-center Experience Including Dual-antenna and Combined Cryoablation Techniques	111
<i>Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Aiming Lu, Daniel Adamo and David Woodrum</i>	
MR-Guided Renal Cryoablation: Local Control and Technical Parameters in a Single-Center Cohort	112
<i>Daniel Adamo, Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Aiming Lu and David Woodrum</i>	
MRI-Guided Percutaneous Liver Biopsy: 15-Year Single-Center Experience at a Tertiary Referral Institution	113
<i>Daniel Adamo, Setayesh Sotoudehnia Korani, Andrew Mason, Matthew Augustine, Brian Welch, Ola Mohamed, Aiming Lu and David Woodrum</i>	
An End-to-End Framework for Real-Time 2D Multislice MRI-Guided Liver Interventions .	114
<i>Wenqi Zhou, Qing Dai, Shu-Fu Shi, Timoteo Delgado-Esbenshade, David Lu, Jason Chiang and Holden Wu</i>	
MR-Guided Microwave Ablation of Liver Tumors: Outcomes in Local Tumor Control and Determinants of Treatment Success	115
<i>Vanessa F. Schmidt, Philipp Linden, Olaf Dietrich, Sinan Deniz, Daniel Puhr-Westerheide, Osman Öcal, Mingming Wu, Luigi Nardone, Florian Maier, Ricarda Seidensticker, Moritz Wildgruber, Jens Ricke, Matthias Philipp Fabritius and Max Seidensticker</i>	
An Adaptable, Patient-Conforming Robotic Template for MRI-Guided Abdominal Needle Interventions	116
<i>Samuel Wilcox, Aiming Lu, Junichi Tokuda, David Woodrum and Yue Chen</i>	
MRI-Guided Percutaneous Cryoablation of Hepatic Tumors in Pericardiac Locations: A Case Series	117
<i>Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Aiming Lu, Daniel Adamo and David Woodrum</i>	

Optimization of Flip Angle for Dynamic Contrast-Enhanced MR Lymphangiography with Gadopichlenol.....	118
<i>Eric Stinson, David Woodrum, Jeremy Collins and Scott Thompson</i>	
First-in-Human Safety and Feasibility of Nodal MR Lymphangiography with Gadopichlenol.....	119
<i>Wayne Stenzel, Eric Stinson, David Woodrum, Jeremy Collins, Emily Bendel, Robert McDonald and Scott Thompson</i>	
Feasibility of MR guided lung biopsy at 0.55T using 3D stack-of-spirals UTE.....	120
<i>Pan Su, Nathan Ooms, Florian Maier, Joshua Krieger, Bradley Bolster, Jesse Roll and Jianing Pang</i>	
Combined MRI-Guided Microwave Ablation and Cryoablation for Hepatic Tumors in a Single Session: A Single-Center Feasibility and Safety Report.....	121
<i>David Woodrum, Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Daniel Adamo and Aiming Lu</i>	
MRI-Guided Cryoablation of Primary and metastatic Liver Tumors: A Five-Year Single-Center Experience.....	122
<i>David Woodrum, Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Daniel Adamo and Aiming Lu</i>	
Poster - Neuro	
MRI-Guided Cryoablation for Residual Limb Neuroma Pain After Transtibial Amputation and Targeted Muscle Reinnervation: A Case Report.....	123
<i>Setayesh Sotoudehnia Korani, Ola Mohamed, Scott Thompson, Aiming Lu, Daniel Adamo and David Woodrum</i>	
MR-guided cryoneurolysis for painful neuropathy – a case series.....	124
<i>Ola Mohamed, Setayesh Sotoudehnia Korani, Anil Kurup, Scott Thompson, Daniel Adamo, Aiming Lu and David Woodrum</i>	
Methods for simulating advanced MR guided cortical brain gene therapy.....	125
<i>Dalton Bermudez, Aidan Burich and Walter Block</i>	
Brain Laser Interstitial Thermal Therapy (LITT) in Drug-Resistant Epilepsy: Case Examples.....	126
<i>Mrinal Singh, Kai Miller, Timothy Kaufmann and Vance Lehman</i>	
Poster – MR Thermometry	
Microwave ablation of liver tumor assisted by magnetic resonance thermometry based on proton resonance frequency displacement method.....	127
<i>Zhengyu Lin, Rui Guo, Bingyu Huang and Jin Chen</i>	
Respiratory-Triggered Real-Time PRF Thermometry in the Liver at 0.55 T.....	128
<i>Jonas Frederik Faust, Axel J. Krafft, Hojjat Fallahi, Bradley D. Bolster and Florian Maier</i>	
Volumetric 3D Stack-of-Spirals PRFS Thermometry at 0.55 T for MR-Guided Hepatic Microwave Ablation: Initial Feasibility.....	129
<i>Dominik Horstmann, Othmar Belker, Daniel D��x, Hojjat Fallahi, Jonas Faust, Moritz Gutt, Florian Maier, Frank Wacker, Bennet Hensen and Marcel Gutberlet</i>	

A subvoxel correction of boiling-induced susceptibility artifacts in MR thermometry and dosimetry for monitoring MW thermoablation.....	130
<i>Eber Dantas, Rebecca Lafont, Pierre Bour, Thibaut Faller, Bruno Quesson, Helcio Orlande and Valéry Ozenne</i>	
MR-Compatible Testbed for MR Thermometry in Liver Microwave Ablation with Heat-Sink Effect	131
<i>Yuqing Fan, Suelia Fleury Rosa, Junichi Tokuda and Mariana Bernardes</i>	
Using Fiber Bragg Grating Sensor Arrays to validate MR Thermometry	132
<i>Ana Sofia Garcia, Thomas Gerlach, Anbarasi Pugazandi, Marcel Gutberlet, Othmar Belker, Bennet Hensen, Frank Wacker, Georg Rose and Oliver Speck</i>	
Poster – Safety	
Safety and Efficacy of Live Intraprocedural Neuromonitoring During Interventional MRI Guided Ablation: A Retrospective Analysis	133
<i>Setayesh Sotoudehnia Korani, Matt Hoffman, Tatsuya Oishi, Ola Mohamed, Scott Thompson, Daniel Adamo, Aiming Lu and David Woodrum</i>	
Technical Performance and Quality Assessment of Clinical Interventional MR Microwave Applicators for Microwave Ablation Procedures	134
<i>Laura Maria Bauer, Luigi Nardone, Mingming Wu, Vanessa F Schmidt, Matthias P Fabritius, Jens Ricke, Max Seidensticker and Olaf Dietrich</i>	
Effective Wire Heating Risk Mitigation for Intraoperative Neurophysiological Monitoring during MRI-Guided Ablations.....	135
<i>Aiming Lu, Christopher Favazza, Matt Hoffman, Myungho In, Eric Stinson, Tatsuya Oishi, Scott Thompson, Daniel Adamo and David Woodrum</i>	
Reliability of vendor provided gradient data for interventional off-center MR Imaging	136
<i>Janis Seibt, Thomas Gerlach, Marcel Gutberlet and Oliver Speck</i>	
Mitigation of RF Heating and Needle Artifacts in MRI-Guided Interventions Using RF Chokes.....	137
<i>Aiming Lu, Phillip Rossman, Daniel Adamo, Scott Thompson, Christopher Favazza, Myungho In, Eric Stinson and David Woodrum</i>	
Quantitative Investigation of Off-Centre Positioning Effects on MR Image Quality in an Interventional Setting based on Phantom study.....	138
<i>Li Xiao, Philip Lee, O L Chan, Oscar Sin, H F Tse, Jimmy Siu, Allen Li, Juan Wei and K Y Kwok</i>	
Large Language Model-Supported Multi-Agent Pipeline for Speech-Based Workflow Analysis in Interventional Radiology: A Proof-of-Concept Study.....	139
<i>Hui Chen, Enrico Pannicke, Frank Wacker, Marcel Gutberlet and Bennet Hensen</i>	
High impedance dielectric cable combination to reduce burn risk from Intraoperative Neuromonitoring during MRI	140
<i>Daniel Hernandez, Suin Choi, Taewoo Nam, Eunwoo Lee, Aiming Lu, Christopher P. Favazza, Eric G. Stinson, David A. Woodrum, Kyoung-Nam Kim and Myung-Ho In</i>	

Electromagnetic-Thermal Analysis of RF-Induced IONM Lead Heating at 1.5T MRI: Quantitative Effects of Geometric Parameters	141
<i>Suin Choi, Daniel Hernandez, Taewoo Nam, Eunwoo Lee, Aiming Lu, Christopher P. Favazza, Eric G. Stinson, David A. Woodrum, Kyoung-Nam Kim and Myung-Ho In</i>	
Safety in Zone 4 of the Interventional MRI Suite during MR Guided Cryoablation – From an MR Technologist Perspective.....	142
<i>Sarah Fasbender, Heather Heimer, Katie Gehling, Courtney Burke, Ang Bathke, Christopher Favazza, Aiming Lu, Myungho In, Eric Stinson and David Woodrum</i>	
Ergonomic Comparison of CT- and MR-Guided Needle Interventions Using Digital Human Model Simulation	143
<i>Steven Fischer, Justin Davidson, Jan Brandt, Joshua Krieger, Sean Chambers and Eric Brandner</i>	
MR-guided Ablation in Patients with Active Implanted Medical Devices.....	144
<i>Angela Pooler, Eric Stinson, Scott Thompson, Heather Heimer and David Woodrum</i>	
Safety in Zone 4 of the Interventional MRI Suite during MR Guided Cryoablation – A Nursing Perspective.....	145
<i>Katie Gehling, Courtney Burke, Ang Bathke, Christopher Favazza, Aiming Lu, Eric Stinson, Myungho In, Sarah Fasbender, Wayne Stenzel, Setayesh Sotoudehnia and David Woodrum</i>	
CPT Code Based Reimbursement Disparity Between Prostate Cryoablation (55873) and Radical Prostatectomy (55866): Implications for Interventional Radiology Practice and Value Based Care	146
<i>Moises Cruz, Setayesh Sotoudehnia Korani, Lauren Lewey, Joshua Krieger, Scott Thompson, Daniel Adamo and David Woodrum</i>	
Poster – Policy	
Cost-Effectiveness of MR-Guided versus CT-Guided Biopsy of Small Focal Liver Lesions: A Decision-Analytic Comparison across the US and German Health Systems.....	147
<i>Moritz Ludwig Schnitzer, Vanessa F. Schmidt, Olaf Dietrich, Daniel Pühr-Westerheide, Sinan Deniz, Moritz Wildgruber, Jens Rieke, Matthias Philipp Fabritius and Max Seidensticker</i>	
Reimbursement Disparity Between CT Guided and MR Guided Percutaneous Ablation Procedures Under the CY2026 Medicare Physician Fee Schedule.....	148
<i>Moises Cruz, Setayesh Sotoudehnia Korani, Lauren Lewey, Joshua Krieger, Scott Thompson, Daniel Adamo and David Woodrum</i>	

From MRI-Guided Biopsy to Autonomous Intervention: A Decade Toward Self-Driving Surgery

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Robot-assisted image-guided intervention has transformed procedural precision, but nearly every motion still requires direct human control; a bottleneck that limits speed, consistency, and access to advanced procedures. Our goal is not to replace the physician, but to increase their productivity and extend the reach of advanced procedures by closing that gap with autonomy. This talk traces a path from image-guided teleoperation toward autonomous, learning-based surgical systems.

I will begin with early work on an MRI-guided prostate biopsy robot, which demonstrated how image guidance can offload targeting precision from the surgeon's hand and the limits of that approach when the task still depends entirely on human motion. I will then turn to my lab's core focus: closing that gap with autonomy. This includes the Smart Tissue Autonomous Robot (STAR), which achieved supervised-autonomous laparoscopic anastomosis, and the hierarchical Surgical Robot Transformer (SRT-H), an imitation-learning framework that enables robots to execute complex, multi-phase surgical procedures from demonstration, recently published in *Science Robotics*.

Building on these foundations, I will describe how we are extending autonomy back into image-guided intervention itself, including new work with Philips Healthcare on autonomous robotics and intelligent imaging for endovascular care, and close with a brief outlook on where autonomous, image-guided intervention is headed.

A New Era of Body iMRI, Why Now?

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Interventional MRI (iMRI) has long been recognized for its superior soft-tissue contrast, multiplanar and real-time imaging capabilities, and access to unique functional information. However, widespread clinical adoption has remained limited. Recent advances in MRI-compatible devices, treatment monitoring, and workflow integration are changing this landscape. At the same time, growing demand for image-guided local therapies in interventional oncology, supported by emerging evidence demonstrating particular benefit in selected patient populations and smaller lesions, has created a renewed opportunity for MRI-guided interventions.

Key drivers of this evolution include improved lesion conspicuity, access to MRI-only targets, functional and molecular imaging, radiation-free procedures, and real-time treatment monitoring including MR thermometry. In parallel, recent advances in MRI-compatible instruments, together with growing commercial commitment from major industry leaders, have significantly improved the feasibility and scalability of MRI-guided procedures in routine clinical practice.

Rather than serving solely as an alternative guidance modality, iMRI is evolving into a comprehensive therapeutic platform that enables radiologists to find, target, treat, and verify interventions within a single environment. Current technological developments and clinical evidence suggest that interventional oncology represents the most immediate opportunity for broader iMRI adoption, while future applications may extend to increasingly complex interventions.

Beyond the Bore: Key Strategies for Building an iMRI Program

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MRI-guided interventions, including focal prostate ablation, represent a rapidly expanding field of minimally invasive therapy. As interest in these procedures grows, successful program development requires careful planning across clinical, technical, operational, and administrative domains. This presentation will review key considerations for developing a prostate ablation practice, including patient selection, multidisciplinary collaboration, MRI safety, equipment and workflow requirements, billing and coding, program sustainability, and research. Drawing on experiences from program development at academic institutions, practical lessons learned and common implementation challenges will be discussed. The goal of the presentation is to build a template for starting and growing a safe, effective, and sustainable iMRI service.

Precision Targeting of Small Liver Lesions Using MRI-Guided Freehand Biopsy: Outcomes and Safety

Vanessa F. Schmidt (MD)¹, Olaf Dietrich (PhD)¹, Sinan Deniz (MD)¹, Daniel Puhr-Westerheide (MD)¹, Osman Öcal (MD)², Moritz L. Schnitzer (MD)¹, Florian Obereisenbuchner (MD)¹, Mingming Wu (MD)¹, Florian Maier (PhD)³, Ricarda Seidensticker (MD)¹, Moritz Wildgruber (MD, PhD)¹, Jens Ricke (MD)¹, Matthias P. Fabritius (MD)¹#, Max Seidensticker (MD)¹#

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Purpose

To evaluate clinical success, procedure times, and complication rates in MRI-guided real-time biopsies of small focal liver lesions (FLL) <20 mm in a routine clinical setting using a closed-bore scanner.

Material and Methods

This retrospective study included patients who underwent MRI-guided real-time biopsies of small FLLs (<20 mm) using a freehand technique. All procedures were performed under conscious sedation and local anesthesia on a closed-bore 1.5T MRI system. Gadoteric acid was administered for contrast enhancement, followed by insertion of a coaxial needle and lesion targeting using real-time gradient-echo fluoroscopic sequences. Clinical and periprocedural data—including clinical success, procedure times, and complication rates classified according to CIRSE guidelines (1)—were analyzed.

Results

MR-guided biopsies were technically successful in 61/61 patients (100%; mean age, 61.6±12.7 years; mean BMI, 24.8±4.8 kg/m²). The median FLL diameter was 11 mm (IQR, 9-15mm). The median lesion-to-target distance was 86 mm (IQR, 69.5-103mm), and a median of 3 samples (IQR, 2.5-4) were obtained. Clinical success of biopsies was achieved in 58/61 cases (95.1%). Mean total room time was 36.9±11.0 minutes, with a mean intervention time of 19.8±7.9 minutes. One major complication occurred: a postprocedural intercostal hemorrhage managed by transarterial embolization, without subsequent sequelae (1.6%).

Conclusion

The superior soft tissue contrast of MRI facilitates freehand biopsies with high clinical success rates in routine practice, enabling reliable tissue acquisition even in very small FLLs.

Acknowledgements

Siemens Healthineers (Erlangen, Germany) provided access to the real-time sequences and associated software.

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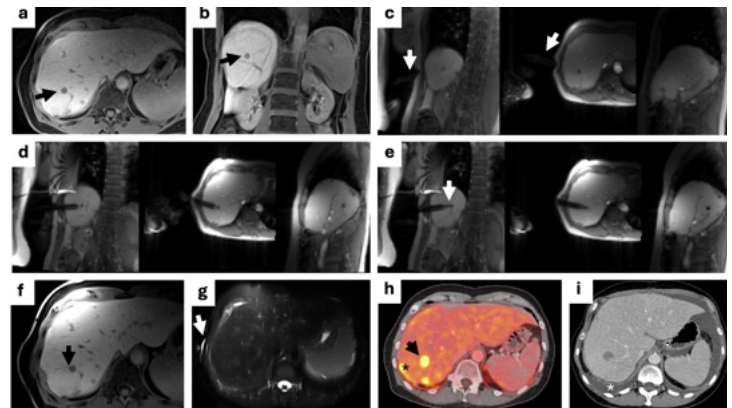


Figure 1. 58-year-old female patient with known ovarian carcinoma and new suspected FLL located in liver segment 7 undergoing MRI-guided liver biopsy with an intercostal approach.

a-b. Preprocedural axial and coronal T1-weighted (T1w) gradient echo (GRE) sequence images after liver-specific contrast agent application revealed a small target lesion with a diameter of 8 mm (arrows). c. Defining of entry point and target lesion as well as finger-pointing (arrows) for procedure planning in use of three-dimensional real-time T1w GRE sequences. d. Three-dimensional real-time T1w GRE sequences confirmed guidance and angulation of 16-gauge coaxial needle while interactive graphical modification of the slice geometry as well as revealed the exact position of the needle tip at the lesion border. e. Firing of the biopsy system under real-time monitoring of three-dimensional T1w GRE sequences (arrow). f. Postprocedural axial T1w GRE sequences showed two thin hypodense lines before and after the small FLL, compatible with the corresponding puncture tract (arrow). g. Postprocedural axial T2-weighted (T2w) fat-saturated turbo spin-echo (TSE) sequences showed a thin peripheral fluid rim along the puncture tract, without progression on subsequent ultrasound follow-up. h-i. Consistent with the histological diagnosis of ovarian carcinoma, imaging at 6-month follow-up shows progressive enlargement of the lesion in segment 7, along with newly identified peritoneal lesions (h, asterisk) and ascites (i, asterisk).

MRI-Guided Hepatic Interventions Using Augmented Reality Optical Navigation: Initial Clinical Experience with the LUMENA System

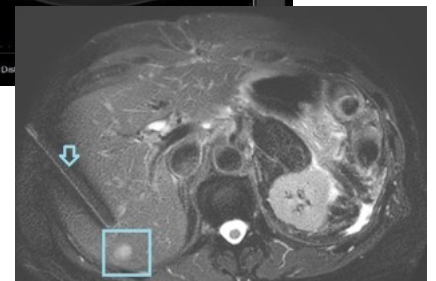
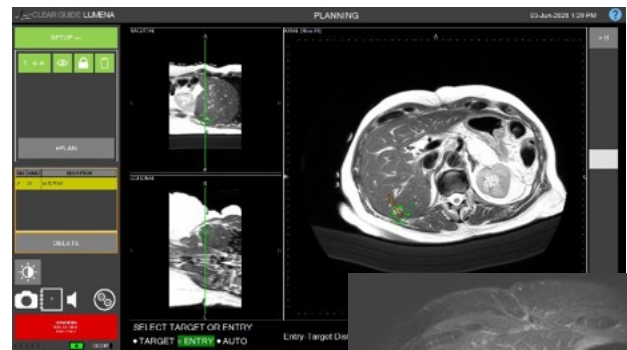
David A. Woodrum Setayesh Sotoudehnia Korani, Ola A.E. Mohamed, Scott M. Thompson, Kenny Harlan, Purnima Rajan, Pezhman Foroughi, Alican Demir, Daniel A. Adamo, Aiming Lu

Purpose: Conventional MRI-guided hepatic interventions rely on an iterative advance-and-check workflow requiring multiple confirmation scans. The LUMENA system (Clear Guide Medical Inc., Baltimore, MD) is an augmented reality optical navigation platform that projects real-time needle guidance cues directly onto the patient's skin surface within the interventional MRI suite, with AI-based needle tip tracking to reduce intermediate confirmatory scans. The purpose of this study is to report our initial clinical experience using LUMENA for MRI-guided hepatic ablation and biopsy.

Materials and Methods: This is a single-institution retrospective case series of patients who underwent MRI-guided hepatic intervention with LUMENA-assisted navigation from September 2025 through June 2026. The system uses optically tracked fiducial markers and a camera-projector unit mounted on an MR-safe articulating arm. Entry point and trajectory guidance are projected onto the patient, with real-time display of needle tip-to-target distance and needle pseudo-shadow. Procedures were performed under general anesthesia with combined ultrasound and MRI guidance. Ablation modalities included microwave ablation (MWA), cryoablation, and combined MWA plus cryoablation. Position confirmation and treatment monitoring were performed with intraoperative MRI. Technical success was defined as complete lesion coverage on post-procedure contrast-enhanced MRI.

Results: Ten patients (6 male, 4 female) underwent LUMENA-assisted MRI-guided hepatic intervention. Primary malignancies included neuroendocrine tumor (n=2), colorectal cancer (n=2), prostate cancer (n=3), renal cell carcinoma (n=1), and pancreatic cancer (n=1), with one patient undergoing biopsy for a suspected focal nodular hyperplasia. Ablation modalities were MWA alone (n=6), cryoablation alone (n=2), combined MWA and cryoablation (n=1), and percutaneous biopsy (n=1). A total of 19 lesions were treated across 9 ablation procedures. MWA parameters ranged from 35 to 75 W with durations of 1 to 6 minutes per position. Cryoablation was performed using 3 freeze-thaw cycles with continuous MRI iceball monitoring. Technical success with complete lesion coverage was achieved in all cases. No post-procedure complications were observed. The LUMENA optical navigation system facilitated accurate needle placement while reducing the number of intermediate confirmatory MRI acquisitions required for trajectory optimization.

Conclusion: LUMENA-assisted augmented reality optical navigation is technically feasible and clinically applicable across a range of MRI-guided hepatic interventions including MWA, cryoablation, combined ablation, and biopsy. Initial experience demonstrates good needle placement with favorable procedural efficiency.



Toward Personalized Planning of Liver Tumor Microwave Ablation Using Patient-Specific Modeling

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Purpose

Microwave tumor ablation is a minimally invasive technique used to destroy tumor tissue through localized heating [1]. However, no reliable tool currently exists to accurately plan the intervention and predict the extent of ablation in clinical practice. Prediction remains particularly challenging due to the heat sink effect, in which large blood vessels locally cool the ablation zone, increasing the risk of incomplete treatment. Personalized numerical modeling offers a promising solution for improving treatment planning. In this study, a 3D patient-specific MWA modeling framework was developed and applied to three ablation cases, incorporating patient-specific liver anatomy and vasculature to improve prediction of the ablation zone extent.

Material and Methods

Three-dimensional patient-specific simulations were developed using a retrospective clinical dataset of three microwave ablation procedures performed under MR guidance. T1-weighted images acquired after needle insertion were used to segment the patient's anatomy, including the liver and blood vessels (Fig. 1), and to localize the ablation probe. The probe was automatically detected from the images, and the coordinates of its tip and entrypoint were extracted. Comsol Multiphysics was used to simulate microwave heating by modeling the coupling between electromagnetic field propagation and bioheat transfer [2]. The ablation probe geometry and heating characteristics were investigated in a previous study. Simulated temperature distributions were used to estimate tissue damage using the CEM43 thermal dose model for cellular necrosis. Predicted ablation zones were then compared with ground-truth ablation lesions segmented from post-ablation contrast-enhanced T1-weighted images after image registration.

Results

The Dice similarity coefficients between simulated and observed ablation volumes were 0.81, 0.83, and 0.87 across the three cases (Fig.2) .

Conclusion

This study demonstrates the feasibility of patient-specific 3D numerical modeling for microwave ablation planning. The strong agreement between simulated and observed ablation zones highlights its potential for improving treatment planning. Future work will focus on validation in a larger cohort.

Acknowledgements

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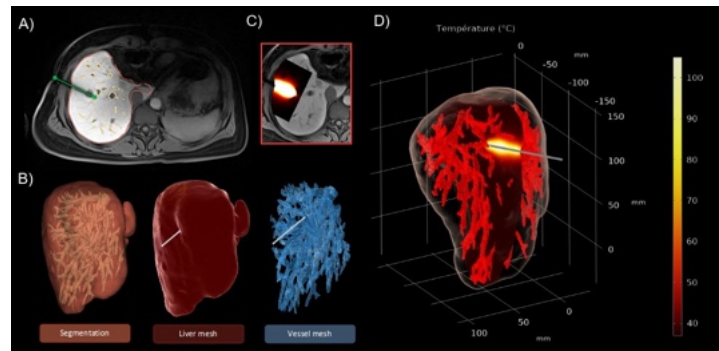


Fig 1: MWA model personalization workflow

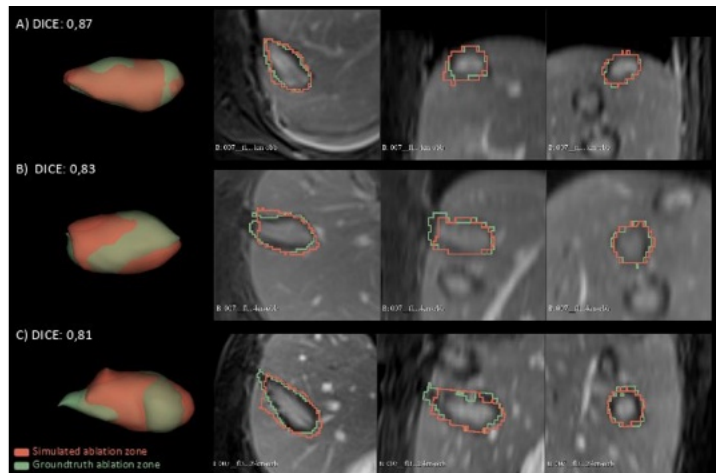


Fig 2: Comparison between simulated ablation zones and ground-truth segmentations.

Assessing the Technical Feasibility of MR-Guided Irreversible Electroporation in a Porcine In Vivo Model in a 1.5 T Clinical MR Environment

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Purpose

Irreversible electroporation (IRE) is a non-thermal ablation technique. Cell death is induced by applying high-voltage electrical pulses, thereby avoiding heat-sink effects, and preserving vascular structures [1,2], but requiring precise alignment of the electrodes [3]. Despite these beneficial attributes and the advantages of MRI for precise needle guidance, IRE is currently not performed under MR-guidance, primarily due to technical challenges such as severe artifacts due to the absence of MR-conditional electrodes and electromagnetic noise induced by the pulse delivery system. This study investigates the technical feasibility of in-vivo ablation in porcine kidneys using real-time MR-guided IRE with prototype equipment [4] in a clinical 1.5 T environment.

Material and Methods

Three swine underwent MR-guided IRE in healthy renal tissue under general anesthesia in compliance with German animal welfare laws and European Directive 2010/63/EU (approved by the Lower Saxony State Office for Consumer Protection & Food Safety, AZ: 33.19-42502-04-23-00414). Experiments were performed on a 1.5 T clinical MRI (MAGNETOM Aera, Siemens Healthineers AG, Germany) with a real-time balanced steady state free precession sequence (bSSFP, two orthogonal slices, TE: 1.55 ms; TR: 605.48 ms; FoV: (450 x 450) mm², resolution: (2.4 x 2.4 x 4.5) mm³; slice thickness: 4.5 mm; flip angle: 70°) for electrode guidance and a 3D-radial gradient echo sequence (3D-GRE, TE: 1.55 ms; TR: 4 ms; FoV: (450 x 450 x 80) mm³; resolution: (1.4 x 1.4 x 2) mm³; flip angle: 12°) for position verification. A prototype IRE generator configured for the use in an MR environment [4] and custom MR-conditional electrodes from 18 G coaxial cannulas (Innovative Tomography Products, Germany) were utilized to deliver 100 pulses (100 µs, 1 to 1.5 kV) per ablation with a pre-treatment evaluation of radiofrequency noise (vendor sequence, bandwidth: 1 MHz). Delayed contrast-enhanced T1-weighted MRI (3D-GRE) was acquired approximately 4 hours post ablation to assess the extent of the treatment area.

Results

The electrodes were successfully placed in the kidney parenchyma using the real-time bSSFP. Delayed contrast enhanced MRI revealed T1-hypointense areas in the kidney parenchyma 4 hours post treatment, correlating with the electrode position. The T1-hypointense area overextended the active zone into the intrinsic back musculature. Pre-treatment evaluation of radiofrequency noise showed a minor increase of electromagnetic interference with the electrodes connected and the generator in standby compared to the baseline.

Conclusion

Real-time MR-guided renal IRE in a porcine model can be performed in a clinical setting with minimal impact on the image quality. Delayed contrast-enhanced MRI suggests electroporation of the renal parenchyma. While the custom electrodes provide a well-balanced MR artifact, the T1-hypointense areas extend over the active zone of the electrodes indicating insufficient insulation.

Acknowledgements

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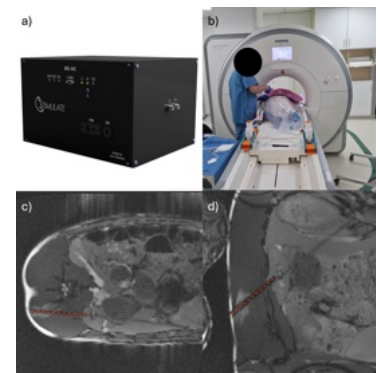


Fig 1: a) IRE generator configured for the use in an MR environment b) Procedure setup showing the physician preparing the insertion of the second electrode. c) & d) One time frame of the bSSFP sequence (cropped) illustrating the electrode (dashed red line) and target (renal parenchyma) visibility.

Proceed with Confidence: Overcoming MR Safety Barriers in MRI-Guided Interventions

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MRI-guided interventions offer powerful imaging advantages but require careful management of safety and operational challenges associated with the MR environment. Overcoming these barriers requires translating MR physics and safety principles into reliable workflows and effective safeguards. This presentation will address key safety considerations in interventional MRI practice, including personnel safety, device safety and compatibility, imaging optimization, patient monitoring, and emergency preparedness. Examples will illustrate practical, generalizable strategies that can be adopted across institutions to establish safe and efficient workflows and enable MRI-guided interventions to be performed with greater consistency and confidence.

Preliminary Experience of MR-Guided Lateral Decubitus Breast Biopsy

Using a Flexible Surface Coil

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Purpose

For MRI-only suspicious lesions, MR-guided breast biopsy is the diagnostic gold standard. Conventional prone positioning, however, is limited in Asian women with small breasts, lesions in special sites (e.g., nipple-areolar complex, axillary tail), potentially causing inadequate immobilization, restricted puncture, and chest wall injury. Furthermore, some patients with prior breast-conserving surgery, radiotherapy, or physical comorbidities have difficulty tolerating prolonged prone positioning. To resolve these issues, we propose a novel MR-guided lateral decubitus breast biopsy technique with a 3D-printed MRI-compatible support device and flexible surface coil, and preliminarily assess its feasibility, safety, and diagnostic efficacy.

Methods and Materials

Fifty consecutive patients (mean age 45 years; 27-72 years) who underwent MR-guided lateral decubitus breast biopsy between July 2024 and November 2025 were retrospectively enrolled. All procedures were performed on a Philips 3.0T Ingenia DNA MRI scanner with a 3D-printed support device and a flexible surface coil, and patients were positioned in the lateral decubitus orientation tailored to the lesion site. Using histopathological results or at least 5 months of imaging follow-up as the reference standard, we evaluated diagnostic accuracy, high-risk lesion underestimation rate, and false-negative rate.

Results

Fifty patients underwent 63 biopsies with a 100% technical success rate and no conversion to open surgery. Histopathology revealed 79.4% benign, 12.7% high-risk, 3.2% DCIS, and 4.8% invasive carcinoma. With pathology or ≥ 5 -month imaging follow-up as the reference standard, diagnostic accuracy, high-risk lesion underestimation rate, and false-negative rate were 100%, 0%, and 0%, respectively. Adequate specimens were obtained in small breasts and lesions at special sites. MRI image quality and needle tip visualization were satisfactory. In addition, patients reported good acceptance and comfort with the lateral decubitus position, and no procedure was interrupted intraoperatively due to intolerance to the position.

Conclusions

MR-guided lateral decubitus breast biopsy with 3D-printed support and flexible surface coil shows considerable feasibility, high patient comfort, and excellent diagnostic accuracy. It is particularly advantageous for Asian women with small breasts or lesions in challenging locations, providing an effective alternative to conventional prone biopsy.

Acknowledgements

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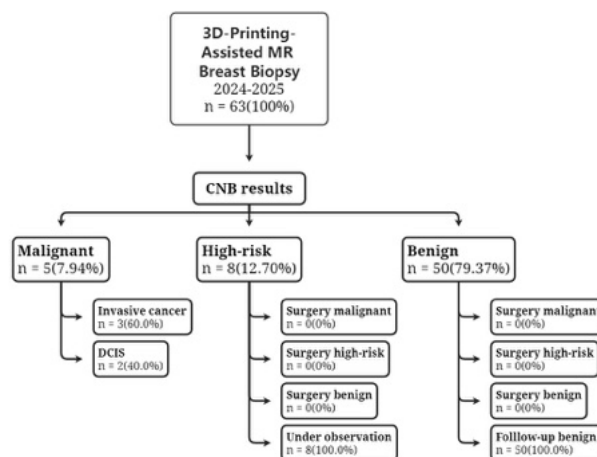


Fig 1: Biopsy Pathology and Patient Follow-Up

Safety and efficacy of MRI-guided cryoablation of renal cancer: a systematic review and meta-analysis.

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Purpose Local-ablation therapy is a recommended therapy for localized renal cell carcinoma (RCC). Due to its excellent tissue contrast, some studies highlighted the superiority of magnetic-resonance imaging (MRI)-guided ablation over standard computed-tomography (CT)-guidance. The present meta analysis aims to provide comprehensive data of the MRI-guided cryoablation therapy in RCC.

Materials and methods MEDLINE library, EMBASE and SCOPUS databases were screened for papers analyzing the outcome for MRI-guided cryoablation in RCC up to February 2026. The primary endpoint of the systematic review was the efficacy and oncological outcome after cryoablation. In total, four studies were suitable for the analysis and included into the present study.

Results The included studies comprised 244 patients undergoing MRI-guided cryoablation. The overall technical success was 224/244 (91.58%). The pooled local-recurrence rate was 9.1%. The pooled 3-years overall survival was 68.4% and the pooled 5-years overall survival was 63.7%. The pooled complication rate was 13.6% with no major complication.

Conclusion MRI-guided cryoablation has a high technical success rate with a low complication rate. However, the results are similar to previous reported studies using CT-guidance.

Initial Evaluation of Interactive Imaging on a High Performance 0.55 T System

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Purpose

Real-time MR-guided interventions require a balance between spatial resolution, temporal resolution, and image quality. Advances in gradient performance, scanner design, and receiver coil technology have the potential to improve procedural visualization while maintaining interactive workflow [1]. The purpose of this comparison was to evaluate the performance gains during interactive imaging achieved with the transition from a standard diagnostic 0.55 T system to the high performance 0.55 T system.

Materials and Methods

A comparative evaluation was performed using optimized *BEAT interactive* interventional sequence protocols routinely performed at Cook Advanced Technology on a 0.55 T MAGNETOM Free.Max (S1) (Siemens Healthineers, Erlangen, Germany). These protocols were imported into the 0.55 T MAGNETOM Free.XL (S2) (510k pending, Siemens Healthineers, Erlangen, Germany). Combined, the optimized spine coil and a high-performance gradient system provide improved SNR performance and increased temporal resolution - capabilities critical to supporting real-time interventional workflows.

Results

At matched spatial resolution ($2.5 \times 2.5 \times 5.0 \text{ mm}^3$), S2 achieved 3.2 fps compared with 2.3 fps on S1, representing a 39.1% increase in imaging speed while simultaneously providing improved signal-to-noise (SNR) ratio. Real-time adjustment of slice thickness from 5 mm to 10 mm increased frame rate to 3.5 fps and produced an improvement in SNR, providing additional flexibility for procedural guidance.

When prioritizing spatial resolution, S2 achieved $2.08 \times 2.08 \times 3.5 \text{ mm}^3$ resolution at 1.7 fps (Fig. 1). This represented a 51.6% reduction in voxel volume and a twofold increase in spatial resolution compared with the standard interventional acquisition.

Marked improvements were also observed in single plane high frame-rate imaging. S1 achieved 6.8 fps at $2.55 \times 2.55 \times 8.0 \text{ mm}^3$ resolution, whereas S2 achieved 10.9 fps at $2.0 \times 2.0 \times 10.0 \text{ mm}^3$ resolution using an optimized protocol for S2 (Fig. 2). This corresponded to a 60.3% increase in frame rate while simultaneously improving in-plane spatial resolution by 21.6%.

Conclusion

Preliminary comparison testing demonstrated substantial improvements in interventional imaging performance with the XL platform. Enhanced gradient capability and upgraded coil technology enabled higher frame rates, improved SNR, and greater flexibility in the tradeoff between temporal and spatial resolution. These improvements expand the operating range available during MRI-guided procedures and may facilitate more efficient targeting, device visualization, and procedural confidence. The results suggest that the XL system represents a meaningful advancement toward a dedicated interventional MRI platform capable of supporting increasingly complex MRI-guided interventions.

Acknowledgements

We would like to acknowledge Siemens Healthineers, Cook Advanced Technology, and the iMRI Division of Cook Medical for our continued partnership and mutual support in advancing interventional MRI.

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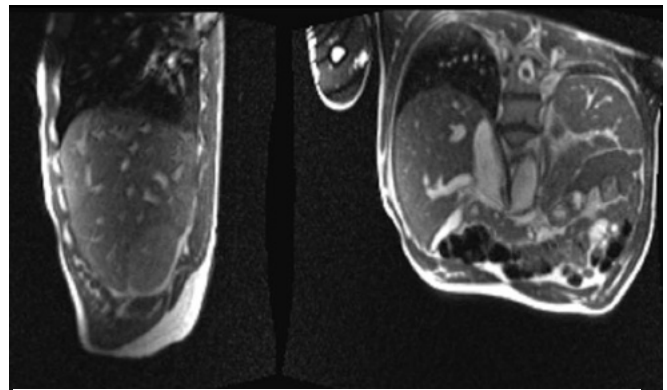


Fig 1: High resolution interactive real time. 1.7 frames per second at $2.08 \times 2.08 \times 3.5 \text{ mm}^3$ voxels.

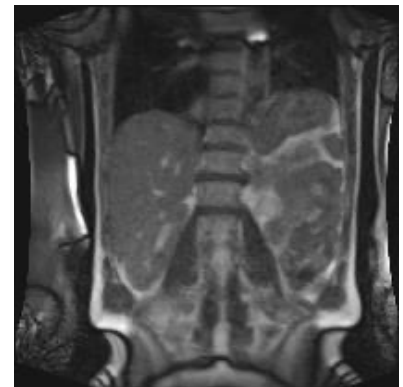


Fig 2: High frame rate still example, 10.9 fps. $2.0 \times 2.0 \times 10 \text{ mm}^3$

Post-Ablation Margin Assessment using Free-Breathing Liver DCE MRI: An In-Silico Study

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Purpose

Directed therapies including thermal ablation are commonly used for primary and metastatic liver cancer. The efficacy of ablation relies on complete tumor destruction, making accurate ablation margin assessment critical to minimize local tumor recurrence. Current clinical practice relies on guidance using CT or Ultrasound followed by assessment of treatment response using dynamic contrast-enhanced (DCE) MRI several weeks post-procedure. While MR-guided interventions may enable immediate on-table ablation margin verification, current DCE-MRI reconstructions may introduce temporal blurring and artificial rim enhancement artifacts that compromise ablation characterization and can mimic residual tumor¹. This study aims to develop and validate a free-breathing liver DCE MRI reconstruction framework to improve ablation margin assessment using a digital simulation.

Material and Methods

The MRXCAT digital reference Object (DRO³) was used to simulate a time-resolved liver DCE stack-of-stars acquisition including golden angle multicoil complex k-space data. Three clinically relevant ablation scenarios were simulated: 1) treatment planning with a hepatocellular carcinoma (HCC) tumor, 2) a successful ablation demonstrating characteristic peripheral hyperemia surrounding necrosis, and 3) an incomplete ablation with both residual viable tumor and necrosis. Adaptively self-gated data were reconstructed with MOCCO and compared against compressed sensing with temporal differences (CS-TD⁴).

Results

For the pre-treatment scenarios, MOCCO demonstrated superior temporal fidelity and more closely matched the reference signal compared to CS-TD (Fig 1). The ablation simulations confirmed that residual reconstruction errors were relatively small and did not manifest as artifacts that could obscure assessment of the relevant enhancing features including visualization of the hyperintense hyperemia regions or the incomplete ablation margins (Fig 2).

Conclusion

This work demonstrates the feasibility of combining radial-sampling, adaptive self-gating, and MOCCO reconstruction in the setting of a digital simulation of free-breathing liver DCE-MRI in ablation assessment. MOCCO achieved a more optimal balance between spatial quality and temporal accuracy. Ablation simulations showed that temporal signal errors at the edges of enhancing structures remained low and did not impact visual inspection.

Acknowledgements

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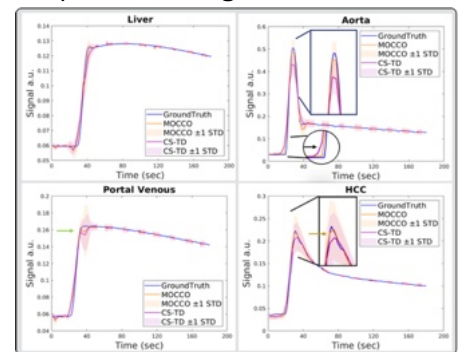


Fig 1: Enhancement curves for relevant tissues. Shaded regions show ± 1 SD for MOCCO and CS-TD. MOCCO curves align more closely with ground truth, during arterial-phase enhancement (black arrow), portal venous phase (green arrow) and HCC (gold arrow).

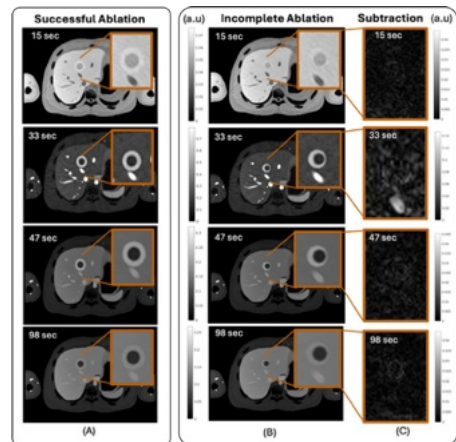


Fig 2: (A) Successful ablation: non-enhancing ablation zone with peripheral hyperemia. (B) Incomplete ablation: ablation zone with residual eccentric enhancing tumor (individually windowed level). (C) Subtracted image (MOCCO - Ground truth) in the incomplete ablation case, shows very minimal error around the edges of the ablation zone.

MRI-Guided Robotics to Enable Interventions

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Abstract

Our research team has developed several low profile MRI-compatible robots for minimally invasive interventions. The first robot is a four degree of freedom (DOF) system called ArthroBot for shoulder arthrography in pediatric patients. The second robot is a six DOF system called PainReliefBot for perineural injections to treat pain in adult and pediatric patients. The third robot is a two DOF system for use with an MRI compatible drill for long bone biopsy. The fourth robot is a concentric tube system for intracerebral hemorrhage evacuation under MRI. In this talk I will describe these systems and the issues involved in developing these robots and moving them to clinical practice.

Acknowledgements

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RF Receive Coil with Integrated MRI-Transparent Heater for Cryoablation

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Purpose

The precision of MR-guided cryotherapy demands ultra-high-resolution imaging, which is currently crippled when standard receive coils are displaced by sterile cryoprobe hardware and bulky water based skin heating methods, potentially blinding clinicians to critical structures (**Fig 1**). While thin and flexible electric heaters could mitigate skin freezing while bringing receive arrays closer, current heat pads introduce ferromagnetic artifacts and RF safety hazards by negatively interacting with the static (B0) and RF transmit (B1) fields of the MRI. This work presents a novel, high signal-to-noise ratio (SNR) two-channel MR receive coil featuring an integrated, MRI-transparent heater designed to optimize image quality and patient safety while allowing placement directly within the surgical field. This prototype device allows for needles to be inserted through the device to allow the coil to be within the surgical field and maintain high image quality.

Material and Methods : In this work a hardware prototype was experimentally measured for imaging and heating performance.

The prototype features a two-channel shared-rung receive coil for GE 3.0T scanners integrated with an MRI-transparent heater, minimizing B0 and B1 distortions (**Fig 2**). A grid pattern of openings allows for cryoprobe insertion. The device draws power directly from the scanner eliminating additional cables. The test array was compared to a commercial 16-channel GE GEM Anterior Array on a 3.0T GE MR750W scanner using an ellipsoidal, tissue-mimicking 9.4L water phantom (21.6x15.6x41.1cm; 5g/L NaCl, 3.75 g/L NiSO₄*6H₂O). The commercial array was tested flush (ideal) and offset by 12 cm to simulate cryotherapy hardware and heating packs. SNR [1], B0 homogeneity (GRE), and B1 homogeneity (flip-angle mapping [2]) were assessed. Surface temperature was measured via a FLIR C2 thermal camera.

Results: **Fig 3A** and **B** shows that the SNR of the 2-channel prototype array is equal to the 16-channel commercial coil in the region of interest. When the 16-channel coil is lifted off the phantom, as would be the case in cryotherapy, the 2-channel coil has more than double the SNR of the commercially available device. **Fig 3C** shows that there is negligible impact to the scanner's B0 and B1 fields even while the heater was able to far surpass the 41°C heat supplied by a chemical heat pack (**Fig 3D**).

Conclusion : By unifying skin-protection, probe access, and coil proximity, the prototype device is able to dramatically improve image quality, ease of use, and ultimately improve the standard of care patients receive.

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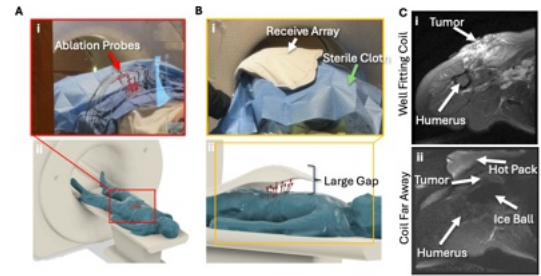


Fig 1: Cryotherapy hardware increases array distance, degrading image quality. A) Photo (i) and rendering (ii) showing cryoprobes present during cryotherapy. B) Photo (i) and rendering (ii) of receive array positioned over cryoprobes and hot packs. C) MRI scans taken with array at (i) ideal distance without probes and (ii) offset to make room for cryoprobes and hot packs.

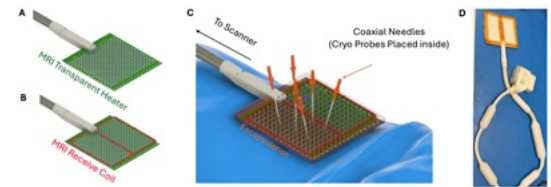


Fig 2: Renderings of coil stack highlighting A) the MRI transparent heater array, B) the 2-channel shared rung receive array, C) proposed device encapsulated with openings in a grid pattern for needle access. D) Photo of prototype device with receive coil and MRI-transparent heater. Encapsulation was not used for the prototype.

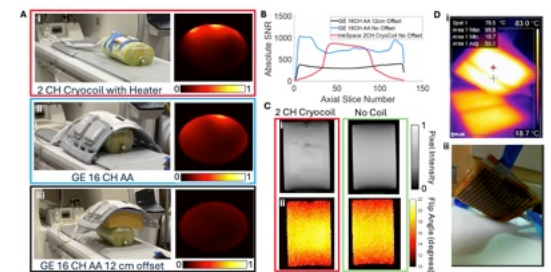


Fig 3: SNR evaluation and imaging compatibility of a 2-channel cryocoil during heat application. A) Photos of experimental setup with an axial SNR map at the center of the phantom. B) SNR between 3 experimental setups in SNR units moving through the phantom superior to inferior extent. C) i. GRE images and ii. flip angle maps with and without the coil present showing no significant difference. D) Thermal image and photo of device showing that significant heat can be supplied to the patient using the scanner power.

MRI Hybrid Ablation: Using the MR Scanner Hardware as Power Source for RF Ablation and as receiving unit of thermal monitoring by MR thermometry

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Introduction

MRI-compatible radiofrequency ablation (RFA) electrodes are available, but clinically applicable RF generator systems that do not interfere with MR thermometry remain limited [1]. The concept of the MRI hybrid ablation system aims to drive the ablation electrode through the RF power amplifier of the MRI system and use the scanner itself as RF power source [2]. Simultaneously, the electrode can be utilized as receive coil for MR-thermometry. This ex vivo study in chicken breast evaluates the feasibility of RF ablation and its thermal monitoring by combining an RF electrode with the Transmit/ Receive (TX/RX) unit of the MRI system.

Material and Methods

A bipolar ablation electrode was connected to the MRI RF infrastructure via the coil interface of a 3T MR scanner. Using Transmit-Receive switches, the electrode was utilized for RF heating, MR excitation/reception, and thermometry monitoring with either the electrode itself or an external imaging coil. System control and temperature-dependent power regulation were implemented via an interface for scanner remote control. RF heating and MR thermometry were performed alternately using a 30 s RF pulse train (Gaussian pulse, 200 V reference amplitude, TR 9.2 ms, pulse length 3 ms) followed by PRFS-based MR-thermometry imaging (FoV 200 mm, TR 50ms, TE 15 ms, flip angle 10°, 96*96 pixel, BW 260Hz/Px).

Results

Using the electrode for both heating and MR thermometry resulted in a characteristic sawtooth-like temperature profile due to alternating heating and imaging phases. An external imaging coil enabled more homogeneous thermometry but reduced heating efficiency due to RF losses. Temperature-controlled power regulation successfully limited tissue temperatures to 85 °C to avoid tissue carbonisations. Thermometry maps demonstrated localized heating near the electrode tip. However, electrode-based imaging showed increased noise and susceptibility artifacts caused by tissue dehydration and air cavity formation during ablation.

Conclusion

The study demonstrates the feasibility of using an MRI system as RF power source for ablation while simultaneously enabling MR thermometry. An interface for scanner remote control further enables sensor-based monitoring and temperature-controlled power regulation. Although SAR limitations currently restrict the usable RF power to approximately 30 W, potential applications may exist in cardiology and pain therapy.

Acknowledgements

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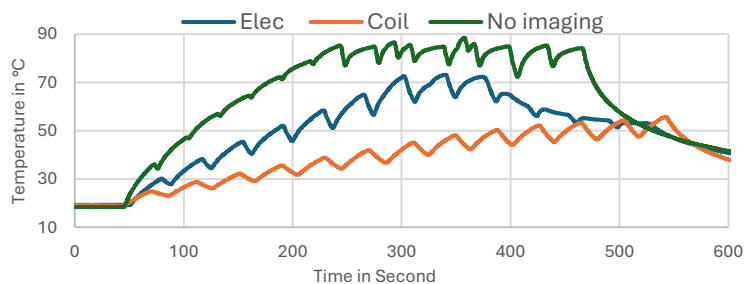


Fig 1: Temperature development using the MRI Hybrid Ablation concept: Blue: Using the RF electrode as receive unit for thermometry. Red: External receive coil for thermometry. Green: No MR-Thermometry.

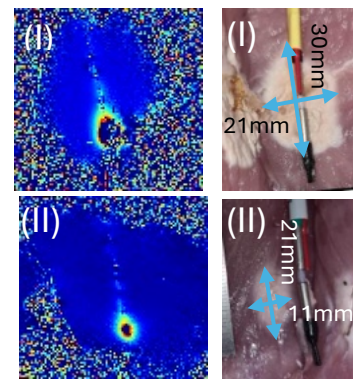


Fig 2: Comparing MR Thermometry and Ex-Vivo results. (I) Using the electrode as receive unit thermometry. (II) Using the external receive coil for thermometry.

Out-of-bore Respiratory Tracking in MRI-guided Percutaneous Interventions using a Noise Navigator

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Purpose

MR-guided percutaneous interventions in the abdomen are challenged by respiratory motion and operator access when the patient is inside the bore. Passive thermal-noise navigation allows respiratory tracking outside the bore, and may provide subject-specific guidance for initial needle placement.

Material and Methods

Three healthy volunteers were scanned on a 1.5T MRI system (Magnetom Aera, Siemens Healthineers, Forchheim, Germany) using both anterior and posterior RF coils. High temporal resolution sagittal 2D bSSFP data (TR/TE = 2.16/1.08 ms, FA = 42°, FOV = 500 mm, voxel size = $3.1 \times 3.1 \times 10 \text{ mm}^3$) were acquired for 3 minutes both in-bore as well as out-of-bore (e.g. noise-only). Subjects were instructed to perform three breath-holds of 10 s. A moveable phantom container with contrasting targets were first moved up and down in-bore and then left stationary for out-of-bore needle placement and 3D bSSFP confirmation imaging (TR/TE = 3.12/1.56 ms, FA = 63°, FOV = 384 mm, voxel size = $2 \times 2 \times 2 \text{ mm}^3$).

Image data were used to determine the liver dome displacement as respiratory reference. Noise data were processed via a principal component and gaussian process analyses to produce a respiratory trace [1]. To assess the potential interference created from the needle intervention, a biopsy needle was manually positioned on top of the anterior coil. For validation of needle positioning accuracy, a 3D rendering of the phantom as well as the needle was visualized on the in-room screen using optical tracking.

Results

Noise navigator traces reproduced breathing cycles with similar amplitude and timing, indicating reliable discrimination of the respiratory phase (Fig. 1). Deep irregular breaths degraded agreement. Vicinity of the operator and needle did not perturb noise-derived motion traces (Fig. 2). Out-of-bore needle placement using on-screen guidance demonstrated an average needle tip error of $9.3 \pm 4.9 \text{ mm}$ (Fig. 3).

Conclusion

A thermal noise-based motion navigator can predict the respiratory state of the subject outside the bore, enabling interventional guidance without compromising operator access. This method can establish a practical workflow during initial needle placement, followed by in-bore finalization under real-time image guidance.

Acknowledgements

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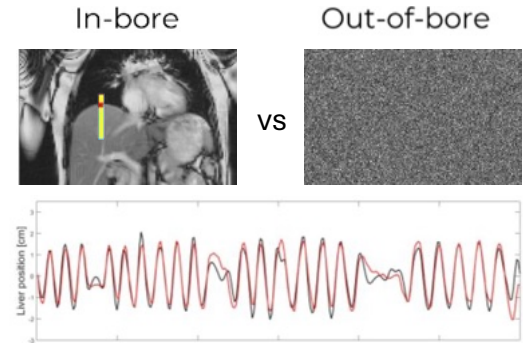


Fig 1: Concept of out-of-bore respiratory tracking based the noise navigator. Based on the noise statistics, a personalized motion state can be predicted, enabling interventional guidance without compromising operator access.

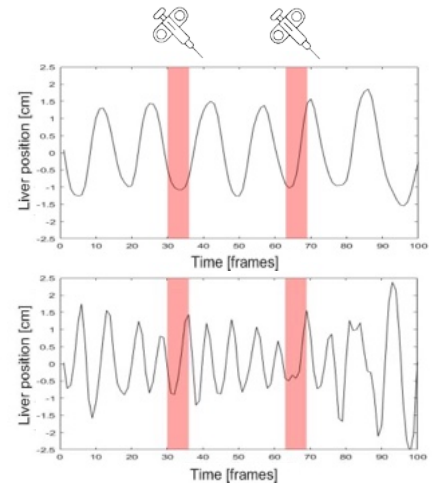


Fig 2: Out-of-bore respiratory signal with simulated needle positioning (red). The respiratory traces do not show any irregularities produced by operator interaction or needle vicinity.

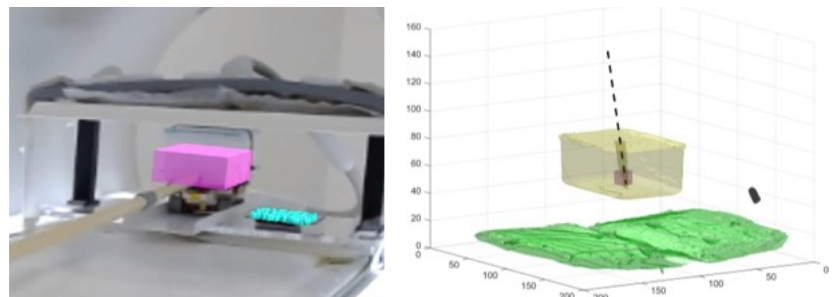


Fig 3: Phantom validation study involving needle placement. A 3D rendering of the targets within the phantom was shown on-screen (left), enabling operator guidance during needle placement (right).

Flexible RF receive array coil optimized for MR-guided Interventions

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Purpose

Real-time magnetic resonance (MR) interventions offer substantial promise for precise procedure guidance, yet technical limitations in coil design remain a barrier. Conventional rigid copper-loop arrays impose constraints on patient positioning and lack the flexibility for optimal anatomic conformance¹. With advances in coil array technology, the recently introduced Adaptive Image Receive (AIR) coil architecture, employs flexible conductor elements which offers high anatomic conformity and exhibit reduced electromagnetic coupling between elements compared to conventional designs². Prior work has established that coil overlap geometry influences both SNR and receive signal uniformity, particularly at extended imaging depths³. In this proof-of-concept study, we demonstrate a receive array with an aperture optimized for MR-guided procedures including liver ablation. We characterized the prototype coil performance both independently and in combination with the integrated MRI table receive array through in-vivo experiments.

Material and Methods

An intervention-compatible prototype RF receive array was designed using selective removal of 8 elements from a commercial 30-element array, yielding an 11 x 21 cm aperture (Fig 1A). The locations for specific coil loops to be removed were determined through geometric modeling that mapped coil element distribution and connecting circuitry relative to the anticipated oblique device insertion trajectory. Imaging was performed on a clinical 3.0T MRI (Premier, GE Healthcare, WI) in this IRB approved study. Coil performance was assessed in healthy volunteers under two configurations: ablation array only (signal received exclusively from the prototype coil) and ablation array + table array (signal received from both the prototype coil and the MRI table's built-in receivers). Relative SNR maps were computed for each configuration. Additionally, in-vivo acquisitions were qualitatively assessed by an abdominal radiologist with 15 years of experience in liver MRI.

Results

Figure 1B demonstrates the interventional setup and technical feasibility of the prototype array to provide access for IR procedures. In-vivo imaging confirmed expected modest reduction in SNR in the center of the access portal of the ablation array only configuration due to coil element removal (orange arrows). Further, integration with the table receivers helped to increase SNR across the full anatomic field of view (Fig 2). Qualitative radiologist assessment indicated the ablation array only provided adequate SNR but that the preferred configuration was when the ablation array was used in combination with the table array to improve posterior signal (Tab. 1).

Conclusion

This proof-of-concept study represents the first iteration of a prototype flexible RF receive array specifically designed for MR-guided interventions. The ablation array when used in combination with the table array offers a practical solution for intervention-compatible imaging, allowing access for placement of IR devices while maintaining sufficient coil coverage over the full liver.

Acknowledgements

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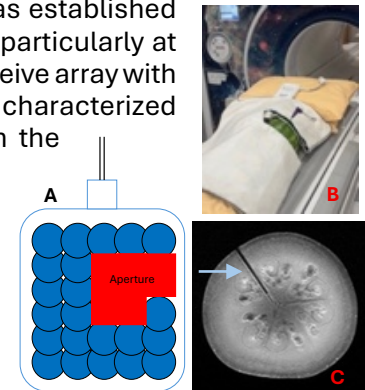


Fig 1: (A) Schematic of coil element placement (Blue circles) and optimal element-removal sites to minimize needle obstruction (red region) while maintaining electrical connections. (B) Mockup interventional setup with ablation needle inserted into a watermelon (C) T2w FSE image of the watermelon with inserted microwave ablation needle (blue arrow).

Criteria	Image A (Ablation Array)	Image B (Ablation + table array)
Liver SNR	2	4
Liver to portal vein CNR	3	4
IQ and Diagnostic confidence	3	5

Table 1: Qualitative evaluation of 2D T2w SSFSE by an expert abdominal radiologist. Scoring used a 5-point Likert scale (5 = excellent 1 = non-diagnostic)

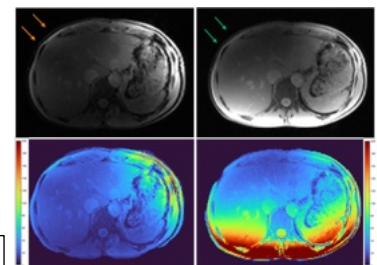


Fig 2: Central-slice of a 2D abdominal image stack (SPGR 2-point Dixon) and corresponding relative SNR maps of a human volunteer, comparing the standalone Ablation array (left) to the Ablation + table array setup (right)

Interventional Cardiac MR for Congenital Heart Disease: Is 0.55T the Answer?

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Cardiac catheterization for congenital heart disease and associated lymphatic disease leads to a high radiation burden on vulnerable pediatric patients who often require multiple lifetime catheterizations. MR-guided cardiac catheterization and lymphatic intervention could eliminate radiation from these procedures. The field has been slow to take off, due to lack of purpose-built catheterization equipment usable in MR. Mid-field (0.55T) MRI reduces radiofrequency-induced device heating and may enable the safe use of standard catheters, guidewires, embolization coils, and stents with just minor modifications. Mid-field scanners may be the catalyst needed for broader clinical acceptance of interventional cardiac MR (iCMR), but close collaboration among industry, academic researchers, and clinicians will be essential for successful translation into clinical practice.

Initial In Vivo Validation of MRI-Guided and Actuated Robotic Ablation Catheter

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Purpose

More than 100,000 catheter ablation procedures are performed annually to treat refractory arrhythmia¹. Our group has proposed a MRI guided and magnetically actuated robotic catheter system that offers precise tip navigation as well as shorter and more effective procedures through dexterous and stable catheter control². MRI offers reduced radiation and enhanced visualization compared to traditional imaging³. This abstract presents the first study exploring the *in vivo* feasibility of the system.

Material and Methods

The catheter system consists of microcoil actuators wound around the lumen of a microcatheter (outer diameter: 2.667 mm, inner diameter: 1.88 mm, material: NEUSOFT UR842A distal section, PeBax 35D, mid section, PeBax 72D proximal section, length: distal 80mm, mid 35mm, proximal 1420mm). The prototype had one actuation coilset (outer diameter: 5mm, 15Fr), consisting of two pairs of Helmholtz side coils wound over an axial solenoid coil. Actuation is achieved through Lorentz forces generated in response to controlled currents passed through these coils in the presence of the MR scanner's static (B_0) field⁴. Actuator currents were controlled using a linear current amplifier. The system was tested in a clinical 3T MRI scanner (Vida, Siemens, Erlangen, Germany). Through femoral vein access, the catheter was inserted into the right atrium of a 70 kg Yorkshire swine under x-ray guidance, followed by trans-septal puncture into the left atrium and transfer to the MRI suite. MR image guidance was used to localize and steer the catheter system. Because the current-based actuation method creates artifacts in MR images around the actuators that can obstruct view of the catheter and anatomy, device actuation and image acquisition were interleaved at a high switching frequency⁵. Multiple imaging sequences were used. The evaluated outcomes for this pilot study were: 1) Can the device be inserted to a live animal, 2) Can the device actuate inside a live animal, and 3) Can MRI guidance be achieved.

Results

The device was successfully inserted into the left atrium of the swine. As shown in Fig 1, MRI-based visualization of the anatomy and device after insertion was achieved. The device actuated inside the swine heart with and without the artifact-reducing interleaving scheme. When interleaving was done, MRI guidance was achieved without obstruction of the device or anatomy by artifacts associated with catheter actuation.

Conclusion

The proposed robotic catheter system can operate inside a live animal under MR guidance. This is a promising step towards clinical adoption of the technology, enabling improved device operability and visualization during procedures.

Acknowledgements

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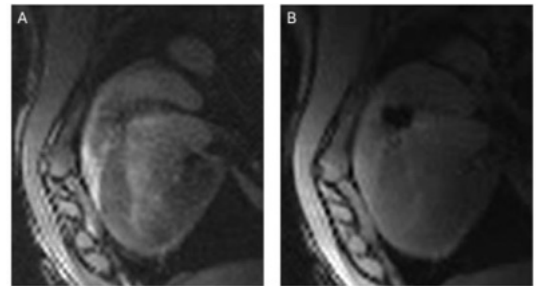


Fig 1: MR images of the proposed robotic catheter system actuating inside a live swine heart with (A) and without (B) artifact-reducing imaging-actuation interleaving. It can be seen that the proposed system can operate inside a live animal with MRI guidance.

First in-vivo MR-Guided Venous Stenting with a Novel Non-Magnetic Delivery System

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Purpose

Iliac vein compression syndromes are preferably treated by venous stenting¹. For this procedure, X-ray fluoroscopy is the established method for image guidance even though MR-guidance could provide additional functional information (e.g., venous flow increase). Unfortunately, commercial delivery systems create strong artifacts and may pose safety hazards. Recently, a novel, entirely polymeric delivery system in combination with a non-magnetic braided NiTi venous stent, that allows repositioning up to three times, was introduced and tested in-vitro^{2,3}. In this study, the first in-vivo MR-guided stenting procedure using this system is presented.

Material and Methods

The venous stenting procedure was performed in a German landrace pig (60 kg) at a Siemens PrismaFit 3 T MRI system. Access to both femoral veins was established using a 14 F and a 16 F sheath (Abbott Medical, Germany). At each side, the prototype, non-magnetic delivery system (Fig. 1) was advanced over an MR-safe guidewire (0.035" stiff, MaRVis, Germany) to the caudal v. cava. Real-time imaging was performed using a bSSFP sequence (Fig. 2): 2 frames/s, 2 orthogonal slices, FOV: 320 x 320 mm², res.: 0.9 x 0.9 mm², SL = 10 mm, TR/TE = 2.1/1.05 ms. As the delivery system allows for repositioning of the stent before the final release, each stent was relocated three times and finally deployed in a ballerina configuration (Fig. 3A). Finally, high resolution T1w images were obtained to verify the stent placement (VIBE DIXON, FOV: 309 x 380 mm², pix = 0.85 x 0.85 mm², SL = 2 mm, TR/TE = 5.4/3.7 ms).

Results

The MR-visible marker at the tip of the delivery system (Fig. 2) was clearly visible during real-time imaging, and the stent could be effectively guided and placed at the target location. The left/right stent had a deployed length of 140/135 mm (Fig. 3A). The delivery system enabled successful implantation of the stents with appropriate wall apposition (Fig. 3B).

Conclusion

This study demonstrates the feasibility of MR-guided venous stenting using a novel, non-metallic delivery system and a braided stent. This system may fill a critical gap in the availability of MR-compatible delivery systems.

Acknowledgements

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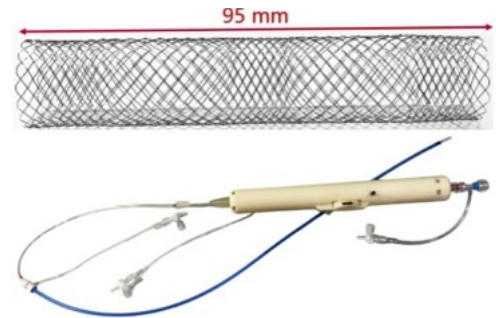


Fig 1: The prototype, non-magnetic braided venous stent used for the animal trial with an expanded length of 95 mm and the polymeric delivery system which allows repositioning up to three times.

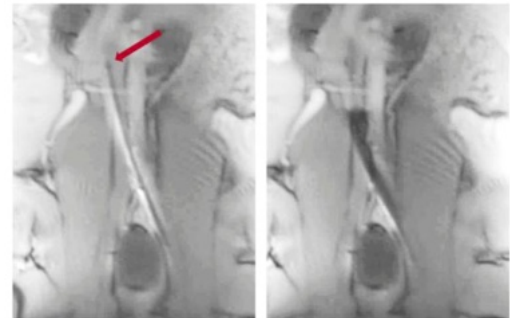


Fig 2: Real-time tracking during deployment of the left stent. Left: the tip of the system's marker is clearly visible. Right: partial deployment of left stent before repositioning.

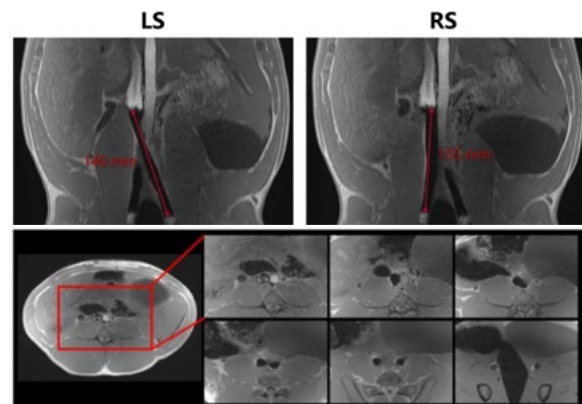


Fig 3: Both deployed braided venous stents. The left stent (LS) and right stent (RS) extended from the vena cava into the left iliac vein and right iliac vein, respectively. Top: Orthogonal slices along the stents' long axis were extracted and displayed as coronal projections. Deployed length of LS and RS were 140 mm and 135 mm, respectively. Bottom: Transversal slices showing the deployed stents.

Phantom-Based Evaluation of AI-Assisted MRI Guidance of a Marker-Enhanced Nonmetallic Microwire for Peripheral Endovascular Interventions

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Purpose

Limited MRI visibility of nonmetallic microwires remains a major challenge for endovascular interventions. While MRI-compatible devices are available for selected cardiovascular interventions [1], peripheral interventions are constrained by the lack of MRI-visible microdevices such as guidewires. This phantom study assessed the MRI visualization of a novel marker-enhanced nonmetallic microwire and the influence of different MRI sequences on detectability by human readers and artificial intelligence (AI).

Material and Methods

A custom-made knee phantom consisting of a 3D-printed knee model and a silicone vessel phantom (0.8 mm lumen, Y-shaped bifurcation) was imaged in a water-filled acrylic container. The investigated nonmetallic microwire comprised a glass-fiber composite encapsulated in polytetrafluoroethylene (PTFE) tubing (0.022") with three ring-shaped iron oxide nanoparticle markers spaced 2 cm apart at the distal end.

Real-time MRI (clinical 1.5T MRI, MAGNETOM Sola, Siemens Healthineers) was performed during microwire advancement using three sequences: GRE (TR/TE 5.25/2.76 ms, FA 4°, resolution $2.3 \times 2.3 \text{ mm}^2$), standard bSSFP (bSSFP-1: TR/TE 2.82/1.41 ms, FA 70°, resolution $2.3 \times 2.3 \text{ mm}^2$), and high-resolution bSSFP (bSSFP-2: TR/TE 3.21/1.61 ms, FA 45°, resolution $1.6 \times 1.6 \text{ mm}^2$). Device visualization was assessed independently by two blinded readers using a 5-point Likert scale for overall visibility, tip/marker visibility, background contrast, and reader confidence. In parallel, a fine-tuned U-Net model was used for automated marker detection and evaluated against manual annotations.

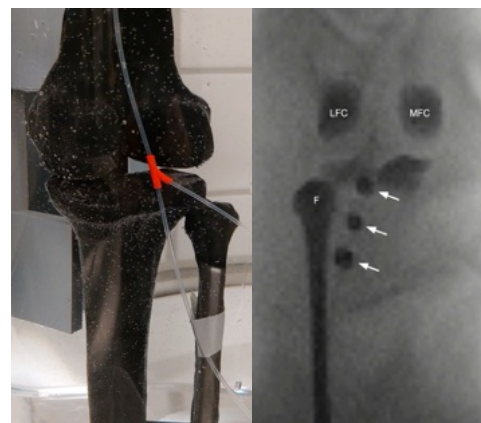


Fig 1: Left: Phantom used in the study. A 3D printed knee (black) placed in a acrylic tube filled with water. The vessel phantom consists of a silicon tube with a y-shaped bifurcation (red). Right: Representative bSSFP-2 image frame acquired after insertion of the

Results

The marker-enhanced nonmetallic microwire was reliably identified on MR images in the knee phantom, with mean scores of 4.3 ± 0.5 for overall device visibility, 4.2 ± 0.8 for tip visibility, and 4.3 ± 1.0 for background contrast/delineation. Reader confidence was high for both tip/marker (4.3 ± 0.5) and overall device identification (5.0 ± 0.0). All three markers were visible in the evaluated image sets, enabling reliable device localization. Sequence-specific human reader assessment showed the highest marker visibility for GRE (5/5), followed by high-resolution bSSFP-2 (4/5) and bSSFP-1 (3/5). AI-assisted detection performance, expressed as the percentage of frames with correctly detected markers, was also highest for GRE (80%), followed by high-resolution bSSFP-2 (60%) and bSSFP-1 (23%).

Conclusion

MRI-based visualization of a marker-enhanced nonmetallic microwire is technically feasible in a peripheral vascular phantom. Both human and AI-based assessments identified GRE as the most effective sequence for marker detection, outperforming high-resolution bSSFP imaging.

Acknowledgements

Supported by the German Federal Ministry for Economic Affairs and Energy (BMWE) (grant numbers: KK6015201AJ5 and KK6020401AJ5, PERIPHER).

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Development and characterization of a non-metallic aortic stent graft for Magnetic Resonance Imaging

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Purpose

Endovascular stents are typically made of metals whose ferromagnetic properties cause significant artefacts in MRI [1]. This complicates follow-up examinations and potential MRI-guided EVAR (endovascular aortic repair) procedures [2]. The aim of this study was to develop a non-metallic aortic stent graft and to characterize its mechanical and MR-imaging properties.

Material and Methods

The stent struts made from biocompatible PEEK (polyetheretherketone) polymer filaments [3] were printed in a Z-shaped configuration onto a tubular graft textile using a custom-built rotating 3D printer. The stent graft was deployed via an introducer tube (diameter: 20 mm) into a 3D-printed aortic phantom (Vero/Agilus, Shore 70A), which was filled with a gadolinium-based contrast agent (1:200 dilution) and examined using a clinical MRI scanner. Stent expansion and structural integrity were assessed macroscopically. Strut thickness was measured using a digital caliper. The radial force of the stent graft and a commercially available reference product was determined using a flat plate test in a universal testing machine. To evaluate potential imaging artefacts, signal-to-noise ratios (SNR) were calculated inside and outside the stent. The apparent strut thickness in MRI was measured at six locations and averaged.

Results

The printing process enabled a stable bond between the struts and the textile. The stent graft (diameter: 30 mm, length: 100 mm, strut thickness: 1.6 mm) fully expanded without loss of structural integrity. The radial force was 0.06 N/mm (approximately 67% of the reference product). In MRI, the struts were clearly delineable (diameter: 1.7 ± 0.1 mm). The SNR measured 86.4 inside the stent and 88.1 outside.

Conclusion

A non-metallic stent graft can be successfully manufactured using 3D printing with PEEK. The material provides adequate mechanical properties and artefact-free MR-visualization, demonstrating strong potential for future MRI-guided endovascular applications.

Acknowledgements

This project (MEVAR) was funded by Federal Ministry of Research, Technology and Space (BMFTR) under grant number 13GW0608F and coordinated by Nano4Imaging.

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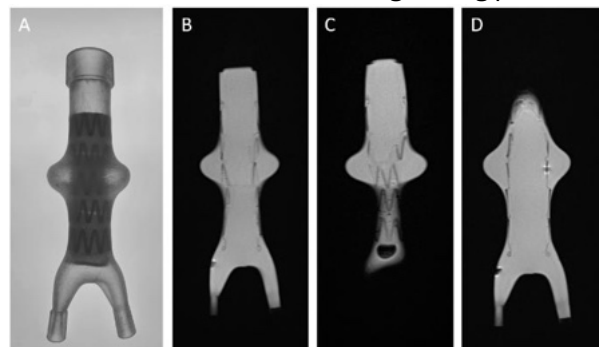


Fig 1: (A) shows the MEVAR stent graft implanted in an idealistic, 3D-printed aortic phantom. In (B)-(D) the stent graft is visualized by T1w-imaging without any stent related artefacts in a 1.5T MR-scanner (Magnetom Sola, Siemens, Erlangen, Germany).

Catheter Transmit Coils for Tracking and Intra-arterial Spin Labeling

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Purpose

Catheter-mounted RF coils have been used primarily in receive (Rx) mode for visualization and tracking during MR-guided interventions. Here, the catheter coils are used also in transmit mode to enable catheter-tip visualization outside of the image plane and to generate a vessel-selective contrast using intra-arterial spin labeling^{1,2} (iASL). Thus, both targeted angiography and quantitative perfusion measurements without contrast agents become possible.

Material and Methods

A rectangular RF coil was attached to a 6F catheter (Tiger II, Cordis) and connected as a Tx/Rx coil to an external RF-generator. For tracking, a 250 μ s-long hard pulse was added to a real-time FLASH sequence that was transmitted solely through the catheter coil, resulting in an increased flip-angle at the coil position. In Rx mode this creates a hyperintense signal in the vicinity of the catheter tip regardless of the catheter position to the imaging slice, so that the catheter signal is projected onto the image plane.

For vessel-selective angiography and myocardial perfusion measurements, the Tx coil is used for labeling of the blood using iASL. An additional RF pulse is transmitted through the coil that saturates the magnetization of the blood flowing around the catheter tip. Images with and without saturation are subtracted after labeling to visualize the vasculature (selective angiography). To measure myocardial perfusion, the catheter tip is placed in a coronary artery, and ECG-triggered FLASH images are acquired in a single breath-hold in a single-shot short axis view. The first half of the data are acquired as control without labeling, and the second half with the labeling pulse. An adjusted flow model is used to quantify myocardial perfusion from the iASL images. The methods were tested at a 3T system (Siemens, PrismaFit) in both flow phantoms and in a pig (German landrace, 60kg).

Results

The position of the Tx/Rx coil is clearly projected onto the real-time images (Fig. 1) allowing to detect the position even outside of the current imaging slice. Vessel-selective angiography can visualize the intra-renal arterial vasculature (Fig. 3) with iASL, and in the myocardium the catheter could be used for quantitative perfusion mapping (Fig. 4). The strong signal inside the LAD indicates a successful application of the label.

Conclusion

A catheter-mounted Tx/Rx coil provides a versatile platform for MR-guided interventions. In addition to improved device visualization that is independent on the position of the imaging slice, it enables vessel-selective iASL angiography and quantitative perfusion imaging without contrast agents.

Acknowledgements

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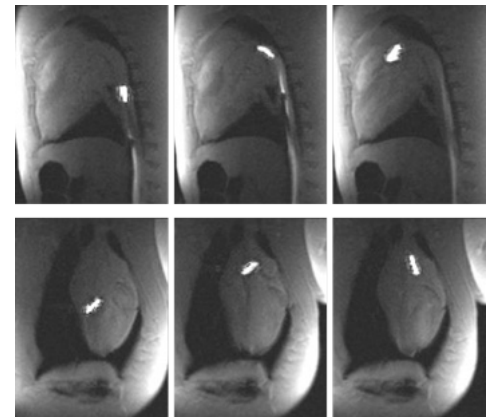


Fig 1: Real-time images of the catheter with the Tx/Rx coil in the aorta. Top: coil in slice. Bottom: coil position projected onto slice.

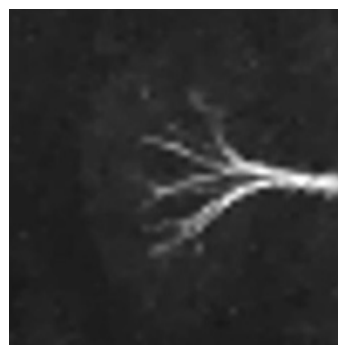


Fig 2: Maximum intensity projection of renal artery angiography in pig, measured with intra-arterial spin labeling.

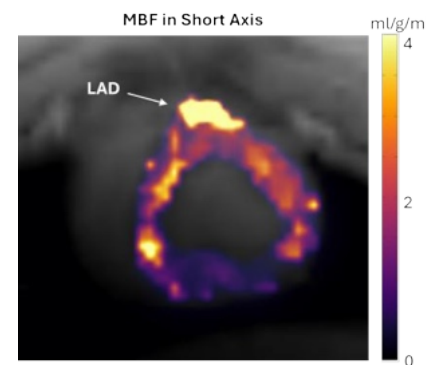


Fig 3: Myocardial blood flow map with the coil inside the left coronary artery, measured with iASL and quantified with a flow model.

Prostate Interventions: How I Do It

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Background

Multiparametric MRI enables accurate localization, targeting, and real time monitoring during prostate interventions. This presentation provides a practical approach to MRI guided biopsy and focal treatment of prostate cancer.

Approach

A practical workflow is presented through clinical cases, covering patient selection, preparation, lesion identification, device placement, biopsy confirmation, treatment planning, and procedural monitoring. Techniques include in bore biopsy, transurethral ultrasound ablation, cryoablation, and focal laser ablation. Their advantages, limitations, and technical requirements are discussed.

Conclusion

Successful prostate intervention depends on high quality imaging, accurate tissue diagnosis, appropriate treatment margins, and protection of critical anatomy. Careful surveillance with multiparametric MRI and targeted and systematic biopsies remains essential. Integrating imaging, biopsy, and minimally invasive therapy may provide personalized cancer control while preserving urinary, sexual, and bowel function.

MRI-Guided In-Bore Percutaneous Transgluteal Prostate Biopsy at 0.55T

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Purpose

To assess the performance of MRI-guided in-bore percutaneous transgluteal prostate biopsy using a wide-bore 0.55T MRI scanner.

Material and Methods

In this prospective single-center study, thirty-seven men (mean age 70.6±7.9 years; prostate-specific antigen (PSA) 8.8±6.4 ng/mL; prostate volume 54±45 mL) underwent targeted in-bore transgluteal prostate biopsy between August 2023 and June 2026 at 0.55T (Free.Max, Siemens Healthineers). All biopsies were performed prone under moderate sedation without antibiotic prophylaxis. A 16-G (ITP, Bochum, Germany) or 17-G (Cook Medical, Bloomington, IN) coaxial introducer was advanced to the target under serial T2W TSE guidance (**Figure 1**; scan time 45 s). An 18-G automatic biopsy device (BioPince, Argon Medical Devices, Plano, TX or ITP, Bochum, Germany) was then used to obtain cores. Total procedure time and time-to-first core were recorded. Histopathologic results were analyzed at the lesion and patient levels.

Results

A total of 42 lesions were biopsied, comprising 11 PI-RADS 3, 13 PI-RADS 4, and 18 PI-RADS 5 lesions (**Figure 2**). PSA and PSA density increased progressively with higher PI-RADS categories. Most lesions (35/42, 83%) were in the peripheral zone, with a mean size of 1.63±0.98 cm. Two to five cores per lesion were obtained. Overall, 29/42 (69%) lesions were malignant, with the malignancy detection rate increasing by lesion size (50% for ≤1 cm, 69% for 1.1–1.5 cm, 82% for >1.5 cm; **Table 1**). Histopathologic grade groups (GG) were as follows: benign or high-grade prostatic intraepithelial neoplasia (13/42), GG1 (12/42), and clinically significant cancer (≥GG2, 17/42). Imaging and pathological outcomes are summarized in **Table 2**. Clinical management changed in 25 of 37 patients (68%), comprising 20 new prostate cancer diagnoses, 1 grade upgrade, and 4 new benign diagnoses. Compared with literature, positivity for PI-RADS 3 lesions was similar (18% vs ~20%), whereas PI-RADS 4 lesions showed considerably higher positivity (85% vs ~50%), and PI-RADS 5 lesions showed slightly higher positivity (89% vs ~80%). This may reflect improved targeting of small lesions, given that PI-RADS 4 and 5 differ primarily in lesion size. Most patients (73%) had no rectal access due to prior surgery or rectal disease. Mean procedure time was 72±25 min, with a mean time to first core of 50±13 min. All procedures were completed without complications, and patients were discharged the same day.

Conclusion

MRI-guided in-bore percutaneous transgluteal prostate biopsy at 0.55T is a clinically feasible and accurate procedure with diagnostic yield increasing with lesion size. It is particularly suited for patients with limited rectal access, providing a safe targeted biopsy option when conventional approaches are not available.

Acknowledgements

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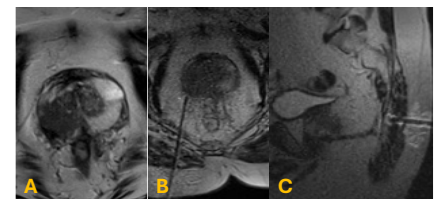


Figure 1: A 62-year-old patient with abdomino-perineal resection history for ulcerative colitis who had undergone two prior negative transperineal biopsies. Axial T2 image shows PI-RADS 5 lesion in the right peripheral zone with extracapsular extension (**A**). Intraprocedural T2 axial (**B**) and sagittal (**C**) images show the needle directed to the target.

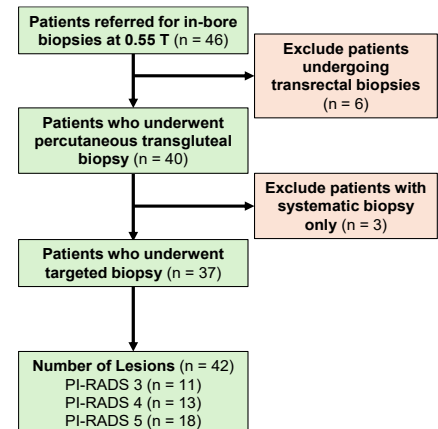


Figure 2: Patient selection flowchart.

Lesion Group	Mean Size (cm)	# Lesions	# Positive	% Positive
≤1.0 cm	0.82±0.18	12	6	50%
1.1–1.5 cm	1.28±0.15	13	9	69%
>1.5 cm	2.48±1.02	17	14	82%
Total	1.63±0.98	42	29	69%

Table 1: Summary of cancer detection rates stratified by lesion size, with mean size and the number of lesions per group.

Grade Group	# Lesions	PIRADS 3	PIRADS 4	PIRADS 5	Lesion size (cm)
Benign or HGPIN	13	9	2	2	1.2 ± 0.5
GG1	12	1	5	6	1.5 ± 0.7
GG2	12	1	5	6	1.7 ± 0.8
GG3	2	0	1	1	2.6 ± 1.9
GG4	1	0	0	1	1.8
GG5	2	0	0	2	3.9 ± 2.3
Total	42	11	13	18	
Total Positive	29	2	11	16	
Percent Positive	69%	18%	85%	89%	

Table 2: Summary of imaging and pathological outcomes of all biopsied lesions. HGPIN: High-grade prostatic intraepithelial neoplasia.

MRI-Guided Cryoablation for Primary and Salvage Prostate Cancer Treatment: Initial Patient Experience

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Purpose

To evaluate the efficacy of cryoablation for primary and recurrent prostate cancer (PCa), tracking prostate-specific antigen (PSA) levels and morbidities.

Material and Methods

At the University of Wisconsin-Madison, 73 patients underwent MRI-guided prostate cryoablation between 2022 and 2026 using previously reported techniques (Figure 1) {2}. Cryoablation is a minimally invasive treatment for both primary and recurrent prostate cancer due to its low risk of morbidity, and high potential in salvage therapy {3}. Patients with biopsy-proven, focal or nondiffuse multifocal PCa were cryoablation candidates.

Mean patient age was 72.4 ± 6.6 years, and mean BMI was 30.2 ± 5.5 . By race, 68 (93.2%) patients were white, 4 (5.5%) were black, and 1 (1.4%) was Asian. There were 42 (57.5%) patients with primary PCa and 31 (42.5%) with recurrent PCa. By Gleason score (GS), 6 cases were GS6, 55 GS7, 5 GS8, and 6 GS9. Recurrence was monitored via serial PSA measurements and MRI or PET/CT imaging.

Post-cryoablation biochemical recurrence (BCR) was determined via Phoenix Criteria: PSA levels rising 2 ng/mL from the post-treatment nadir, as done in other work {1,3}.

Results

BCR was observed in 11 (15.1%) cases. In the 62 (84.9%) non-BCR cases, post-treatment PSA dropped by a significant ($p < 0.0001$) 80.1% on average, with follow up monitoring showing no significant PSA increases (Fig. 2).

Among the 11 BCR cases, 5 showed recurrence outside of the treatment zone, suggesting microscopic, diffuse disease spread pre-treatment. Of the remaining 6, imaging showed local tumor progression in 5 cases, while the remaining 1 showed no clear evidence of disease on follow up imaging.

Long-term complications were assessed. Many patients had baseline erectile dysfunction (ED) and urinary symptoms pre-procedure. Notably, 2 patients experienced improved urinary symptoms post-cryoablation. Seven (9.5%) patients reported a decrease in erectile function and six (8.2%) separate patients reported stress incontinence post-procedure. There has been only 1 (1.4%) case of post-procedure meatal stricture.

Conclusion

MRI-guided prostate cryoablation offers safe, targeted, minimally invasive cancer control while sparing healthy structures, making it attractive for recurrent cancers, where treatment options are limited. It is also an appealing option for patients who choose to prioritize urinary and sexual function or are otherwise poor surgical candidates due to its low reported rates of morbidity with regards to ED and urinary symptoms.

Acknowledgements

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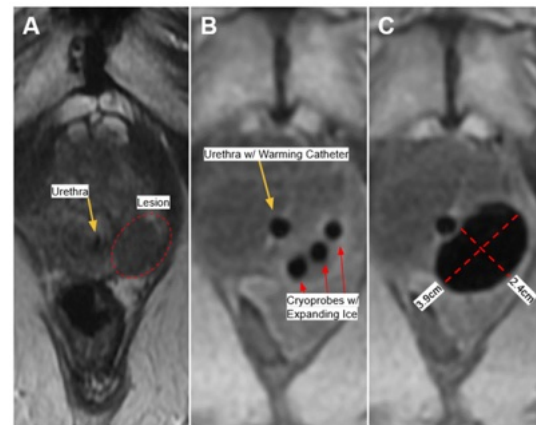


Fig 1: Images taken intraprocedurally show how, in the planning phase (A), the location and extent of a targeted lesion can be visualized and used to plan the positioning and count of several cryoprobes to ensure therapeutic coverage. These probes are then driven into the tissue using the prior-described workstation (Fig 2). Once in place, the probes are turned on (B), and ice (signal void) expands about them, monitored under real time imaging, until the extent of the ice covers the lesion (C).

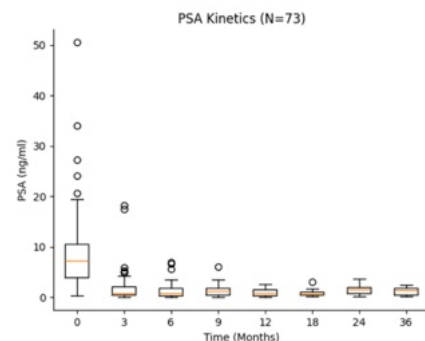


Fig 2: Median prostate-specific antigen levels at each timepoint shows a statistically-significant drop (median 80.1%, $p < 0.0001$) in the biomarker from pre- to post-treatment, with no other significant changes in PSA thereafter, for the dataset in aggregate.

Cost-Effective Needle Tract Verification Using Susceptibility-Weighted Imaging (SWI) in MR-Guided Prostate Biopsy

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Purpose

MR-compatible biopsy needles for in-bore MR-guided prostate biopsy are costly and not universally available. In contrast, bare-metal needles are inexpensive but cannot be reliably visualized during real-time MR guidance. This study evaluated a susceptibility-weighted imaging (SWI)-based workflow for immediate post-biopsy needle tract verification, aiming to enable cost-effective needle use while preserving targeting quality assurance.

Material and Methods

In this prospective single-center quality-improvement cohort, consecutive patients with clinical indications for in-bore MR-guided prostate biopsy were included. All procedures were performed at 3T using axial and sagittal T2-weighted imaging for lesion identification, procedural planning, and Soteria-based robotic trajectory guidance. A newly implemented near-isovolumetric, AI-reconstructed compressed-sensing 3D SWI sequence was acquired immediately before and after biopsy. Pre-biopsy SWI was used to exclude baseline susceptibility artifacts, whereas post-biopsy SWI was used to visualize the biopsy-related needle tract after needle withdrawal. SWI findings were compared with Soteria trajectory overlays on T2-weighted images. Technical success was defined as diagnostic SWI acquisition with visualization of a biopsy-related susceptibility tract. Two radiologists independently assessed tract visibility and trajectory concordance using 5-point Likert scales. Inter-reader agreement was assessed using intraclass correlation coefficients.

Results

Twenty patients with 20 biopsy targets were included (mean age, 74 ± 8 years). SWI workflow implementation was technically successful in all procedures (20/20, 100%), and post-biopsy SWI demonstrated a biopsy-related susceptibility tract in all targets (20/20, 100%). For both readers, median tract-visibility score was 4 (IQR, 3–5), and median trajectory-concordance score was 5 (IQR, 4–5). Inter-reader agreement was good for both tract visibility and trajectory concordance, with ICC values ranging from 0.78 to 0.86. Diagnostic SWI image quality was achieved in all cases, with an additional acquisition time of approximately 3 minutes per SWI scan. All biopsy cores were diagnostically adequate. No SWI-related workflow failures or adverse events were observed. Use of bare-metal needles reduced estimated material costs by approximately €135 per target compared with MR-compatible needles, based on local German institutional procurement costs.

Conclusion

Post-biopsy SWI enabled reliable needle tract verification after in-bore MR-guided prostate biopsy. This workflow may support cost-effective use of bare-metal biopsy needles while maintaining immediate targeting confirmation, diagnostic adequacy, and procedural safety.

Acknowledgements

None

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N/A.

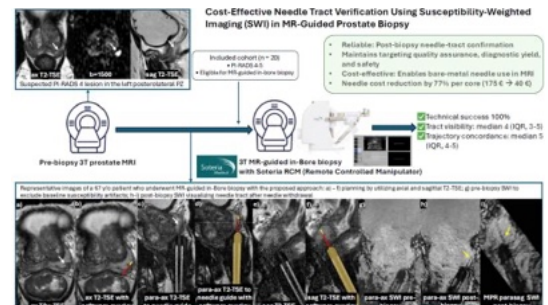


Fig 1: Representative in-bore MR-guided prostate biopsy workflow using SWI-based needle tract verification. T2-weighted images are used for lesion identification and Soteria-based trajectory planning. Pre-biopsy SWI excludes relevant baseline susceptibility artifacts, while post-biopsy SWI demonstrates the biopsy-related susceptibility tract and confirms concordance with the planned trajectory.

while maintaining immediate targeting

Accuracy and verification of an MR-compatible transperineal biopsy device

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Purpose In-bore MR-guided transperineal prostate biopsies are gaining interest because they may provide improved targeting, better sampling for certain lesions (for instance, anterior lesions), and reduce infection risk compared with conventional transrectal approaches. Existing freehand and fixed-grid techniques limit or prevent needle angulation, restrict workflow flexibility, and access to challenging targets. This study aimed to verify the targeting accuracy, maximum target range, and time to reach a target of a novel MR-compatible transperineal biopsy device that enables real-time in-bore MR-guided needle insertion under variable angles.

Material and Methods This experimental phantom study was performed on a 3T MRI system (MAGNETOM Skyra, Siemens Healthineers, Erlangen, Germany). A newly developed MR-compatible transperineal biopsy device (Figure 1a, Soteria Medical, Arnhem, The Netherlands), equipped with a tube-shaped MR-visible marker at the front, was used to hold and guide an MR-compatible coaxial needle. An agar phantom, containing three berries (4–10 mm) and three wax spheres (4–7 mm) as targets, was used to mimic a patient. A high-resolution 3D trueFisp (Trufi) sequence was acquired from the phantom to localize the targets. Oblique coronal and sagittal Trufi sequences were planned along the selected entry point aligned with the target. Under real-time MR imaging the marker was steered with a joystick driven mechanism into the planned trajectory and the needle was advanced through the marker by the use of a slider after achieving the correct angle and height. After the coaxial needle was visually aligned with the target, the titanium stylet was replaced by a plastic rod to reduce artefacts and a new high-resolution 3D Trufi sequence was acquired to localize the needle tip relative to the target. This workflow was repeated for all targets. Primary outcome measures were: (1) the shortest 3D distance from target to needle tip and (2) the time from trajectory planning to the final needle position.

Results All predefined targets in the agar phantom were successfully reached with the device. The 3D distance from the target center to the needle tip ranged from 1.1 to 2.6 mm for the berries and from 3.2 to 4.2 mm for the wax spheres. Mean time to reach a target was 3:49 minutes (range 2:13–5:20 minutes), with one outlier procedure lasting 14:09 minutes due to difficulty achieving the proper angulation at the limits of the system's range. The mechanical range of motion allowed access to all targets in the phantom from a single perineal configuration, illustrating that the device can reach a broad transperineal target range (Figure 1c).

Discussion and conclusion The difference in accuracy between the two types of targets can be explained by image artefacts of the wax spheres. Steering the device with the joystick required some experience. It is expected that these times could decrease once the learning curve is completed.

The MR-compatible transperineal biopsy device enabled accurate and fast needle placement to small targets under real-time MR guidance in a phantom. The observed accuracy and procedure times support the feasibility of integrating this system into in-bore MR-guided transperineal prostate biopsy workflows and warrant further evaluation in clinical studies.

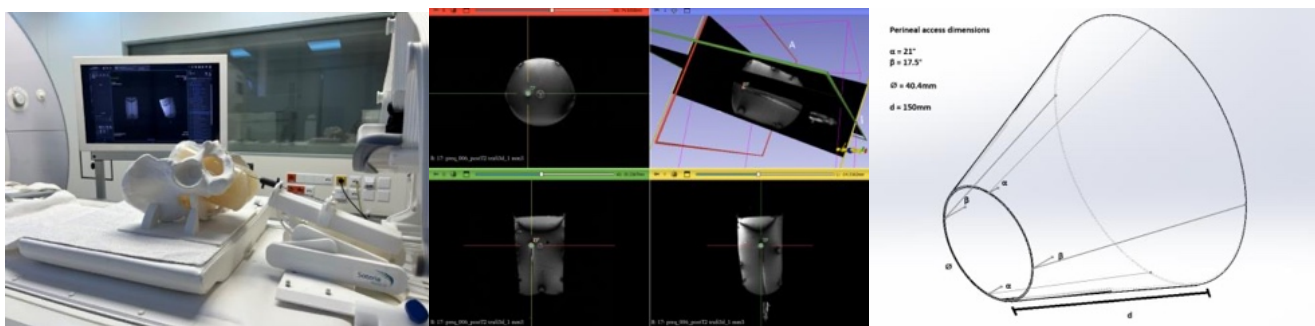


Figure 1: Left: setup of an MR compatible transperineal biopsy device with an agar phantom, and real time -in room- monitoring on the screen next to the MR-bore. **Middle:** The MRI scan images below show the axial, coronal and sagittal slices as well as the 3D render of the phantom with the target position in green, the reached position in red, and the needle with its marker. **Right:** determined target range of the device.

Four-Year Experience with MRI-Guided Focal Prostate Cryoablation

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Purpose

We evaluated oncologic and functional outcomes of MRI-guided focal cryoablation used as primary therapy for localized prostate cancer over four years, with emphasis on in-field control, frequency of recurrence outside the treated field, and changes in urinary and sexual function. Focal therapy for prostate cancer has shown favorable functional outcomes, but potentially lower long-term oncologic control, and remains classified as an experimental technology by the NCCN and AUA/ASTRO guidelines [1]. Most of the studies published to date are based on cryoablation with ultrasound (with and without MRI Fusion) or CT guidance. Results from MRI guided cryoablation have been most studied in the context of salvage therapy for recurrence after radiation or surgery [2,3].

Material and Methods

This retrospective study included 33 men with localized prostate cancer treated with MRI-guided focal cryoablation using a 3T MRI system (GE Architect) and the Visual ICE cryoablation system. Gleason Grade Groups (GG1–GG4) included 5, 14, 13, and 1, patient(s) respectively. A perineal grid was utilized for guiding cryoprobe placement. A perineal grid guided probe placement. Target volume included the lesion plus a 1 cm margin. Treatments consisted of 2–3 freeze–thaw cycles. A urethral warming catheter protected the urethra during freezing. Follow-up included PSA every 3 months in year 1 and annually thereafter; MRI at 6, 12, and annually. Biopsy was performed if PSA rose after nadir or MRI was suspicious. SHIM and IPSS were recorded; changes were analyzed with paired tests. The study was conducted in accordance with institutional review board guidelines.

Results

Median follow-up was 6–46.3 months (IQR). Clinically significant intra-field recurrence (GG ≥ 2) occurred in 3/33 (9%). No biopsy-proven clinically significant disease outside the treated field was identified; however, 22/33 patients were not biopsied during follow-up due to stable PSA and absence of suspicious MRI findings. SHIM scores declined from baseline (median 19) to 3 months (median 14; $p < 0.01$); declines persisted at later timepoints (exact values and timepoints provided in table/figure). IPSS remained stable with no significant changes at any timepoint ($p > 0.19$).

Conclusion

MRI-guided focal prostate cryoablation can achieve in-field control with low observed intra-field recurrence and preservation of urinary continence, but was associated with reduced sexual function. Given the retrospective design and the absence of a comparator group, longer follow-up and prospective trials are required to define oncologic efficacy and functional outcomes.

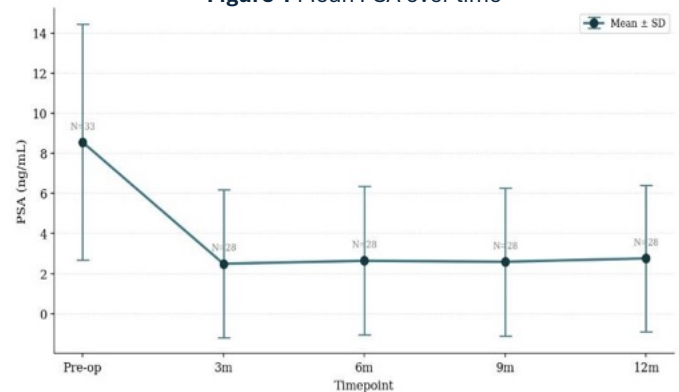
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Table 1: Patient and Treatment Characteristics:

Variable	Median (IQR)
Age (years)	71 (66–75)
PSA (ng/mL)	5.9 (4.3–10.9)
PSA density	0.2 (0.1–0.3)
Gland vol (cc)	42.5 (33–62)
Ablation vol (cc)	21.9 (14.5–32)
Fraction ablated	47% (40–59%)
Follow-up (months)	16.1 (6.0–46.3)

Figure 1 Mean PSA over time



Category	N	%
In-field failure (GG ≥ 2)	3	9%
Out-of-field cancer (all GG1)	4	12%
Biopsy negative	4	12%
Not biopsied	22	67%

Table 2: Biopsy outcomes (median time of followup 16.1 months, IQR 6.0–46.3 m). 11/33 patients were biopsied in the follow up period. All out of field cancers were GG1. 1 patient was biopsied for out of field target on day of treatment due to suspicious lesion (excluded from failure analysis).

In-Bore MRI-Guided Trans-Perineal Partial Gland Cryoablation for Primary Prostate Cancer: A Prospective Study

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Purpose

To assess biopsy-confirmed recurrence and adverse events in 100 patients with primary prostate cancer (Grade Group 2–3) treated by in-bore, MRI-guided trans-perineal focal cryoablation.

Material and Methods

This IRB-approved prospective observational registry (protocol 20-003822) enrolled 133 patients with biopsy-proven primary prostate cancer who underwent MRI-guided cryoablation between December 2020 and November 2025; 100 patients completing ≥ 12 months of follow-up (mean 46.6 months) are presented. All patients had Gleason 7 disease (Grade Group 2–3). Focal cryoablation was performed using a 1.5T wide-bore MRI suite (Philips Ingenio) with the Visual ICE™ MRI-compatible cryoablation system (Boston Scientific) under general anesthesia via a trans-perineal grid template. Four cryoneedles (range 3–6) and three freeze–thaw cycles to -40°C were employed; a urethral warming catheter (41°C) was used in all cases. Post-ablation dynamic contrast-enhanced MRI confirmed ablation zone adequacy. Follow-up included PSA every 3 months, multiparametric MRI every 6 months, and protocol biopsy at 1–2 years. Patient-reported outcomes (AUASS, EPIC, SF-12) were collected at baseline, 6, and 12 months. Statistical analysis used paired t-tests, Kaplan–Meier survival analysis, and mixed-effects models.

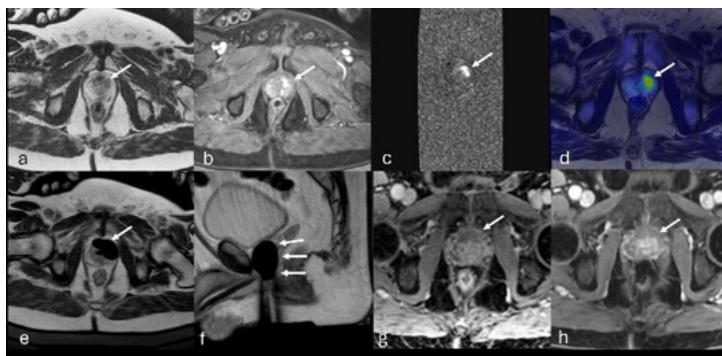


Fig 1: MRI and PET imaging of a 76-year-old male with a Gleason 3+4, PIRADS 5 lesion who experienced a significant PSA reduction from 23 ng/mL to 0.36 ng/mL after cryoablation treatment. (a) T2-weighted MRI: 1.5 cm lesion in the left anterolateral prostate. (b) Gadolinium-enhanced imaging. (c) Diffusion-weighted imaging. (d) PSMA PET imaging: increased metabolic activity within the lesion. (e) Treatment: T2-TSE MRI with iceball around the lesion. (f) Treatment sagittal T2-TSE MRI of the iceball. (g) Post-treatment gadolinium-enhanced images: lack of lesion enhancement at 6 months and (h) 2 years post-ablation.

Results

Median PSA declined from 5.95 [IQR 5.18–9.18] ng/mL at baseline to 1.90 [1.35–2.85] ng/mL at 3 months, 2.25 [1.45–2.88] ng/mL at 6 months, and 2.35 [1.50–3.10] ng/mL at 12 months (median reductions: –68.0%, –68.7%, –67.4%; all $p < 0.0001$). Kaplan–Meier recurrence-free survival was 100% at 1 year, 94.9% (95% CI 90.5–99.3%) at 2 years, and 93.4% (95% CI 88.2–98.6%) at 3 years. Six biopsy-proven recurrences occurred at 13–29 months (5 in-field/marginal, 1 out-of-field). Urinary symptoms improved significantly (AUASS 7.65 ± 6.5 at baseline vs. 6.02 ± 5.4 at 12 months; $p = 0.007$). Sexual function (EPIC) and quality-of-life (SF-12) remained stable ($p > 0.05$). Mild adverse events occurred in 9/100 patients (9%); no major complications (fistula, stricture, rectal injury) were observed. representative imaging in Fig 1.

Conclusion

In-bore MRI-guided focal cryoablation demonstrates promising mid-term cancer control with a favorable functional outcome profile for carefully selected patients with favorable and unfavorable intermediate-risk (Grade Group 2–3) primary prostate cancer. The procedure achieves durable PSA reduction, 93.4% 3-year recurrence-free survival, urinary symptom improvement, and preservation of sexual function and quality of life, with no major complications. These findings support its role as a viable focal therapy option; larger comparative studies with longer follow-up are needed to confirm long-term durability.

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MR guided brachytherapy of small liver lesions - optimizing dose delivery and local tumor control (Results of the MR BRIGHT trial)

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Aim: To evaluate the feasibility and safety of MR-guided catheter-based radiotherapy/brachytherapy for primary and secondary liver tumors, to compare non-target healthy liver radiation exposure with that of a propensity score-matched CT-guided cohort, and to assess 12-month local tumor control.

Materials and Methods: Between June 2023 and April 2024, 27 patients with 54 liver lesions were treated within a prospective single-center trial investigating MR-guided catheter-based radiotherapy (MR BRIGHT trial). Procedures were performed under conscious sedation and local anesthesia using a 1.5T MRI system (Magnetom SolaFit, Siemens Healthineers) equipped with a 15-cm loop coil. For contrast enhancement, gadoteric acid (0.1 mmol/kg body weight) was administered in each procedure. Lesions were accessed using an 18G coaxial needle (ITP) under real-time gradient-echo MR fluoroscopy (TE 3.4 ms, TR 7.3 ms, FA 30°) with the iMRI UI Extension (Research Software Package). After lesion access, the needle was exchanged over a nitinol guidewire for a 6F hydrophilic angiographic sheath accommodating a 6F brachytherapy catheter. Subsequently, 3D T1-weighted MRI datasets were transferred to the radiation oncology treatment planning system for high-dose-rate (HDR) brachytherapy using an Ir-192 afterloading unit. Prescribed target doses ranged from 15–25 Gy depending on tumour entity, including hepatocellular carcinoma (33.3%), colorectal cancer metastases (22.2%), gastrointestinal stromal tumours (11.1%), neuroendocrine tumours (11.1%), and metastases from other primaries (22.2%). After treatment delivery, the catheters were removed and the access tract was sealed with gelatin sponge. Outcomes were compared with a propensity score-matched CT-guided cohort derived from a control population of more than 1000 CT-guided brachytherapy procedures.

Results: In the MR-guided cohort, the median lesion diameter was 11 mm (IQR 9–17), with a median clinical target volume (CTV) of 1.8 cm³ (IQR 0.25–11.9). MR-guided procedures demonstrated significantly lower non-target liver irradiation compared with CT guidance (V5Gy: 15% vs. 43%). Procedural complications were rare; one patient (3.7%) experienced minor bleeding that did not require transfusion or intervention. At 12 months, local tumour control was higher in the MR-guided cohort compared with the matched CT-guided cohort (95.1% vs. 79.9%).

Conclusions: MR-guided catheter-based radiotherapy for liver tumours is feasible and safe. Improved soft-tissue visualization enables more precise tumour targeting and better sparing of healthy liver tissue, resulting in reduced non-target radiation exposure and improved local control rates. With low complication rates and precise dosimetry facilitated by advanced imaging, MR-guided brachytherapy represents a promising approach for the treatment of primary and secondary liver malignancies.

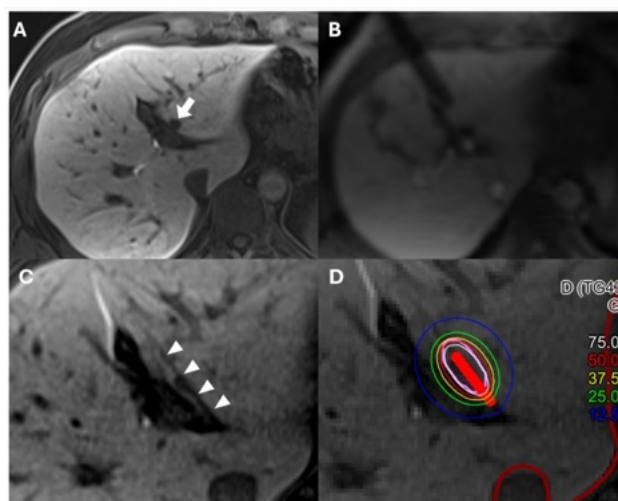


Figure 1. Case example

45-year-old patient with solitary colorectal liver metastasis. (A) Diagnostic hepatobiliary-phase T1 VIBE MRI shows a 8 mm liver lesion in segment 2, directly adjacent to the left portal vein (arrow). (B) Intraprocedural fluoroscopy image shows needle placement. (C) After brachytherapy catheter placement, planning 3D T1-weighted gradient-echo MRI confirms catheter position within the target lesions (arrow heads). (D) Dosimetry map demonstrates conformal target coverage with a prescribed dose of 25 Gy.

End-to-End MRI&MRT-guided cervical-cancer brachytherapy interventions

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Purpose: MR-Tracked (MRT) insertion of location-tracked metallic brachytherapy needles into cervical cancer was developed in 2010 (1), performed clinically at MGB 2012-2016 and ported to JHU in 2017. More than 70 interventions were performed between 2017-2026, with outcomes relative to non-MRT controls (2) demonstrating improved local control and disease-free survival 80 months post-procedure, with the greatest benefits being for large tumors.

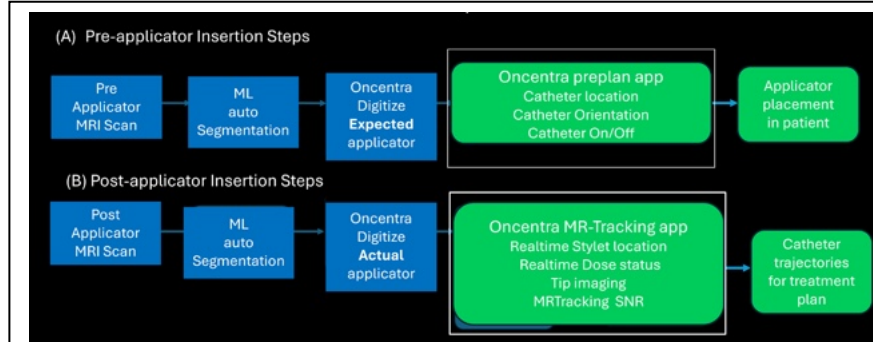


Figure 1: End to End MRI/MRT procedure workflow. (A) Before applicator placement and (B) after applicator placement. (A) is used to optimize Venezia applicator configuration and location and the needles to be inserted through applicator. It includes imaging and tissue auto-segmentation. (B) Since anatomy is perturbed after applicator inserting, imaging, segmentation and optimization are repeated, followed by MR-Tracked needle insertion and construct of needle trajectories, which can be used for the final radiation treatment plan.

Materials and methods: Starting in 2019, an end-to-end MRI/MRT workflow was developed that includes T2-weighted Machine-Learning based automated segmentation of extended tumors (HR-CTVs) and organs at risk (OARs) (3), Applicator and needle optimization pre-plans, followed by accurate 0.5mm³ Variable Flip Angle MRT needle insertion at 12-20 frames-per-second (fps) (4), with real-time on-line dosimetry at 4 fps informing clinician intra-procedurally of improvement in HR-CTV dose and reduction in OAR dose over time, ending in digital needle trajectories that can be used in an MRI-based radiation treatment plan (WIP).

Results (Fig.2): 2A; Preplan UI for needle and applicator optimization. 2B; User Interface during MRT needle insertion shows clinicians how close needles are to targets. 2C; There are multiple MRT stylets for multiple uses and two KMRI fields. 2D; plot that integrates real-time dosimetry from all times during the procedure shows how the tumor dose improves over time. Equivalent plots show how OARD dosimetry drops over time;

Conclusions: Procedure delivers accurate and complete dose at ~1min/needle speed.

Acknowledgements:

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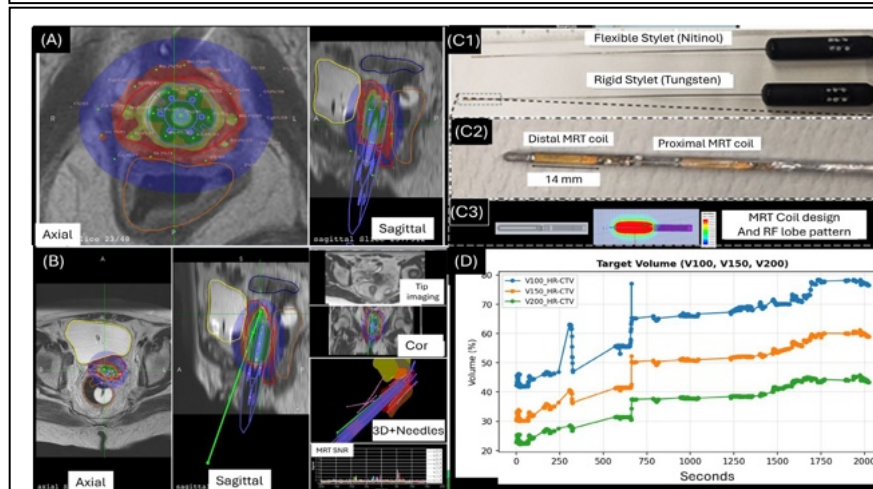


Figure 2: Key workflow components (A) **Prescan** optimizes needle locations. Thin red and orange lines show segmented extended tumor (HR-CTV) and rectum boundaries. Thick red and blue colors denote 200 and 50% dose contours. Numbers about needles denote each needle's dose contribution. (B) **Real-Time user interface** clinician sees during MRT needle insertion. Images shown are at the current MRT needle location (Green line), with previously inserted needles shown as circles. Red and Blue wash show current 200 and 50% dose contours, which change at 4fps to inform the clinician how needle location-changes improve dose. Lower right shows current MRT SNR, providing QA during use. (C) **MR-Tracked stylets:** Flexible and Rigid stylets used for navigation in soft and hard tissues. **Variable Flip Angle MRT** improves speed from 12 to 20 fps (D) **Real-time dosimetry plot over complete procedure** (33 minutes, 25 needles) as a function of time shows how HR-CTV dose improves over the procedure. Green, Orange, and Blue curves denote 100%, 150% and 200% volume HR-CTV dose contours.

Automated detection of Syed-Neblett applicators and interstitial needles on PETRA MRI images

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Purpose

PETRA MR sequences are increasingly used for MR-only treatment planning in gynecological interstitial high-dose rate (HDR) brachytherapy. Manual tracking of gynecological applicators and interstitial needles is subjective and tedious. We aimed to develop an algorithm for automatic applicator and interstitial needle tracking on PETRA images.

Material and Methods

Four patients with endometrial adenocarcinoma were treated with HDR brachytherapy utilizing Syed-Neblett template and ProGuide Sharp Needles 6Fx294mm. Patients underwent 3D PETRA MRI on a 3T Siemens Vida after marker removal (TR/TE 3.32/0.07 ms, voxel size 0.9 mm³ isotropic, 416 slices). PETRA series were analyzed with MATLAB R2024b. Images were cropped in each dimension to reduce computational burden. Detecting circles and most prominent signal minima on axial images aimed to identify the obturator and probable needle locations in tissue, respectively. A linear fit of obturator centers provided the obturator angle in the sagittal and coronal direction. Edge lines perpendicular to the obturator at its inferior range were considered as template borders. Prominent signal minima which lay within Euclidean distance of 2 pixels between subsequent axial slices, appeared on at least 5 continuous axial slices and did not deviate more than 5 degrees from the obturator angle on sagittal and coronal images were attributed to interstitial needles. Obtained results were visually assessed for accuracy.

Results

The obturator was detected. Analysis on multiple coronal and sagittal slices confirmed the template edge location as perpendicular to the obturator. The applied continuity and angle criteria allowed tracking of all interstitial needles along their whole length (fig1) while avoiding spurious signal minima from other structures (ex. bones).

Conclusion

Obturator, templates and needles can be detected using signal intensity characteristics and knowledge of applicator geometry. Algorithm implementation may expedite MR-only treatment planning in interstitial HDR brachytherapy.

Acknowledgements

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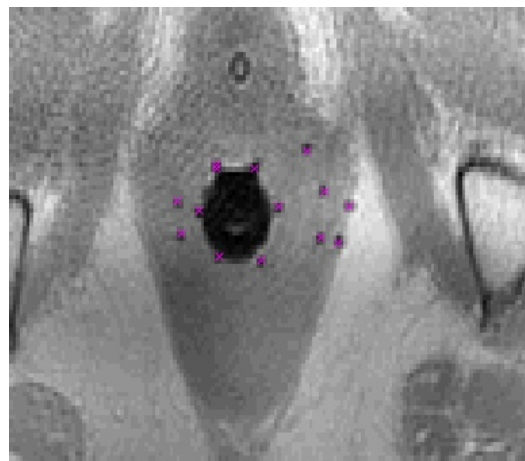


Fig 1: Zoomed axial PETRA image. Magenta crosses indicate detected needle positions

Radio-Pathomic Mapping of Brain Tumor Invasion

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In glioblastoma, tumor cells routinely infiltrate beyond the contrast-enhancing and FLAIR-hyperintense margins defined by conventional MRI. Radio-pathomic mapping addresses this blind spot by training machine learning models on spatially co-registered pairs of clinical MRI and digitized whole-mount autopsy histology. The resulting voxel-wise maps estimate histological features such as cell density and extracellular fluid density across the entire brain. This talk will review how these models were built from our neuro-oncology focussed brain bank, and present external validation studies in independent multi-institutional cohorts highlighting generalization across scanners and sites. Clinical applications, such as differentiating true tumor progression from treatment-related pseudoprogression, and the inclusion of our maps to inform surgical planning, including extent-of-resection targets and intraoperative or interventional MRI guidance, will also be discussed.

Brain Proton MRS Thermometry Enhanced by Fiber Orientation Susceptibility Correction

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Purpose

Proton magnetic resonance spectroscopy (MRS) thermometry estimates brain temperature from the frequency difference between water and metabolite resonances, such as N-acetlaspartate (NAA) [1]. However, because NAA resides predominantly in highly oriented axons in white matter, its resonance is expected to be more sensitive to fiber-orientation-dependent susceptibility perturbations than metabolites with more isotropic distributions. Water resonance frequencies may also be affected. These effects potentially introduce systematic bias into MRS temperature estimates [2]. Thus, this study was performed to investigate whether an orientation-dependent correction model can reduce these effects in the mouse brain.

Material and Methods

To measure the fiber-orientation dependence of chemical shift, a fiber phantom made of a hemostatic pad (Hakujuji) absorbed with 100-mM NAA and 20-mM TSP solution (Fig. 1) was used. Proton spectra were obtained by PRESS with the fiber angle θ set at 0°, 30°, 45°, 60°, and 90° relative to B_0 in a 9.4-T vertical system. The orientation dependencies of NAA-TSP and water-TSP chemical shifts were characterized as $\sin^2(\theta)$ [3]. To examine the effect of fiber orientation dependence in vivo, PRESS and DTI were acquired from bilateral brain regions, including the motor cortex (MO), corpus callosum (CC), anterior corpus callosum (CCA), cerebral peduncle (CPD), and hypothalamus (HY) in five mice at 7 T. Fiber angles relative to B_0 were estimated from DTI and used to apply phantom-derived orientation-dependent corrections to the water and NAA chemical shifts independently. Since phantom measurements used TSP as an external reference whereas in vivo measurements relied on the water-NAA frequency difference directly, corrections were applied to each resonance separately before computing the difference.



Fig. 1. Hemostatic pad phantom soaked with NAA and TSP solution

Results

The fiber-orientation dependencies of NAA and water chemical shifts measured in phantom are shown in Fig. 2. The PRESS voxels placed bilaterally in the five regions are shown in Fig. 3(a). A total of 22 valid bilateral region pairs were analyzed. The left-right asymmetries of the water-NAA chemical shift before and after orientation-dependent correction is shown in Fig. 3(b). Regional reductions were 0.0026, 0.0020, 0.0032, 0.0002, and 0.0029 ppm in MO, CC, CCA, CPD, and HY, respectively, and 21 of 22 region pairs (95.5%) showing reduction after correction. These results indicate that the orientation-dependent correction, based on the phantom-derived model and DTI-estimated fiber orientation, effectively reduced bilateral asymmetry and improved the water-NAA chemical shift consistency measurement.

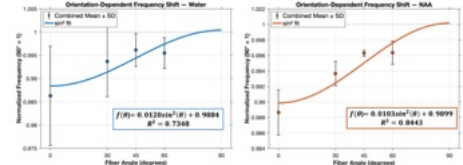


Fig. 2. Fiber-orientation dependencies of water-TSP (left) and NAA-TSP (right) chemical shifts measured in the phantom as a function of $\sin^2(\theta)$.

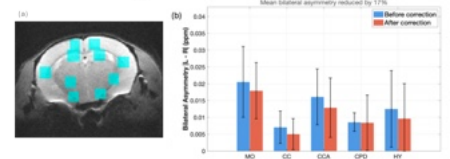


Fig. 3. Validation of the orientation-dependent correction. (a) Bilateral PRESS voxel placement in the mouse brain. (b) Comparison of left-right asymmetry in water-NAA chemical shift before and after correction

Conclusion

The proposed orientation-dependency correction technique is expected to enhance the accuracy and reliability of noninvasive brain thermometry based on the water-NAA chemical shift.

Acknowledgements

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Clinical Feasibility of an MRI-Compatible Motorized Alignment Device for MRI-Guided Deep Brain Stimulation: Initial Experience

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Purpose

Deep brain stimulation (DBS) is a neurosurgical procedure involving inserting electrodes (also referred to as leads) in specific brain areas to deliver electrical impulses. Lately, intraoperative magnetic resonance Imaging (iMRI) guidance has emerged as a cutting-edge technique for precise lead placement in the brain that has the ability to account for intraoperative brain shift. Utilizing iMRI enhances the accuracy of targeting specific brain regions, reducing risks, and improving treatment outcomes. This first-in-human feasibility study evaluated the safety and clinical workflow integration of an MRI-compatible motorized alignment device during MRI-guided DBS procedures.

Material and Methods

A prospective first-in-human feasibility study was conducted to evaluate the safety and workflow integration of a novel MRI-compatible motorized alignment device for MRI-guided DBS lead placement (Fig. 1). Prior to patient enrollment, multiple workflow dry-runs were performed in the Advanced Multimodality Image Guided Operating (AMIGO) suite using phantom models, and the study protocol was reviewed by the institutional AMIGO Safety Committee. The study was approved by the institutional review board (IRB# 2025P000304) and classified as a non-significant risk device study. Five adult patients undergoing standard-of-care MRI-guided DBS for Parkinson's disease and essential tremor at Brigham and Women's Hospital were enrolled after informed consent. Before initiation of the clinical procedure, the investigational alignment device was mounted to the custom-made head fixation system. After the patient's head was fixed, the conventional clinical MRI images were acquired to target and entry point selection (T1 weighted SPGR, T2 weighted MRI, SPGR White matter nulled and SWI sequence). The neurosurgeon defined target and entry point that were used to automatically align the motorized guiding device to the planned trajectory. MRI scans were acquired of a fiducial-integrated instrument guide to assess alignment relative to the planned trajectory. No surgical instruments were introduced through the device in this initial feasibility study, and no changes were made to the standard clinical DBS workflow. After image acquisition, the device was removed and the procedure continued according to routine clinical practice. The primary endpoint was safety, assessed through the incidence of adverse events and adverse device effects related to device operation. The study was designed to evaluate the feasibility of device deployment in the clinical MRI environment and establish a foundation for future trials involving motorized alignment device-guided electrode placement.

Results

Five patients undergoing MRI-guided DBS successfully completed all study-related procedures. Device setup, trajectory alignment, and MRI-based assessment were completed in all cases, after which the standard DBS procedure proceeded according to routine clinical practice. No device-related adverse events, adverse device effects, or serious adverse events were observed. The device was used safely within the AMIGO MRI environment, supporting the feasibility of integrating the system into the clinical workflow.

Conclusion

The primary safety endpoint was achieved, with no device-related adverse events observed across all participants. Successful execution of the workflow in a clinical MRI environment establishes a foundation for subsequent studies assessing the accuracy and clinical utility of MRI-guided robotic assistance for DBS lead placement.

Acknowledgements

This study was supported by AiM Medical Robotics, the manufacturer of the investigational device, under Research Agreement 2025A012246.

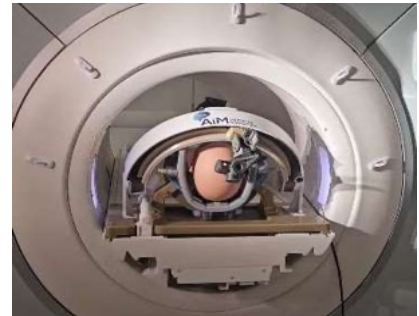


Figure 1 – The actuated stereotactic frame shown in the BWH AMIGO 3T Siemens MRI scanner.

Investigating the thermal effect of interstitial photodynamic therapy for glioblastoma treatment: a combined experimental and numerical study

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Purpose

Interstitial photodynamic therapy (iPDT) is an emerging loco-regional treatment for glioblastoma based on intracerebral illumination delivered through stereotactically implanted optical fibers. Despite encouraging oncological results, treatment-related morbidity remains significant [1], and the contribution of unintended light-induced thermal effects has never been experimentally investigated [2].

This study aimed to quantify ex vivo thermal effects of iPDT using MR-derived temperature map and numerical simulation to predict iPDT-induced in vivo thermal effects.

Material and Methods

Illumination (635 nm, 57 min) was performed in ex vivo calf brain (n=11) using three 3-cm diffusing fibers at three inter-probe distances (2, 1, 0.6 cm). Experiments were performed in a 1.5 T MRI scanner. Probe localization was confirmed on a pre-illumination T1w anatomical sequence, and temperature was monitored throughout using PRFS-based GRE-EPI thermometry [3,4] (3×20 min, 1.875×1.875×4 mm, 128×128×35 slices).

Numerical simulations were performed using COMSOL Multiphysics® based on a 3D anatomically-derived mesh and coupling the Bioheat Transfer Equation with the Photon Diffusion Equation. The model

was first validated ex vivo across the three inter-probe distance conditions by comparison with MRI thermometry, then extended to prospective in vivo conditions by incorporating blood perfusion effects.

Results

- MRI thermometry enabled precise temperature monitoring during iPDT (accuracy ~0.5°C).
- Cumulative heating was inter-probe distance-dependent, with peak temperatures of 43.7±0.7°C, 46.6±5.9°C, and 53.5±10.2°C at 2.0, 1.0, and 0.6 cm spacing, respectively.
- The numerical model showed good agreement with MRI thermometry across all three conditions (RMSE: 1.34, 1.52, and 2.34°C at 2.0, 1.0, and 0.6 cm). (**Fig. 1**)
- Prospective in vivo simulations incorporating blood perfusion predicted a maximum temperature rise of +10°C above baseline, suggesting a risk of thermal damage over a 1-hour illumination.

Conclusion

MRI thermometry proved feasible and precise for iPDT temperature monitoring. Ex vivo heating was non-negligible, and the ex vivo-validated numerical model predicted clinically relevant temperature rises in vivo, supporting its use as a digital twin for future patient-specific iPDT planning.

Acknowledgements

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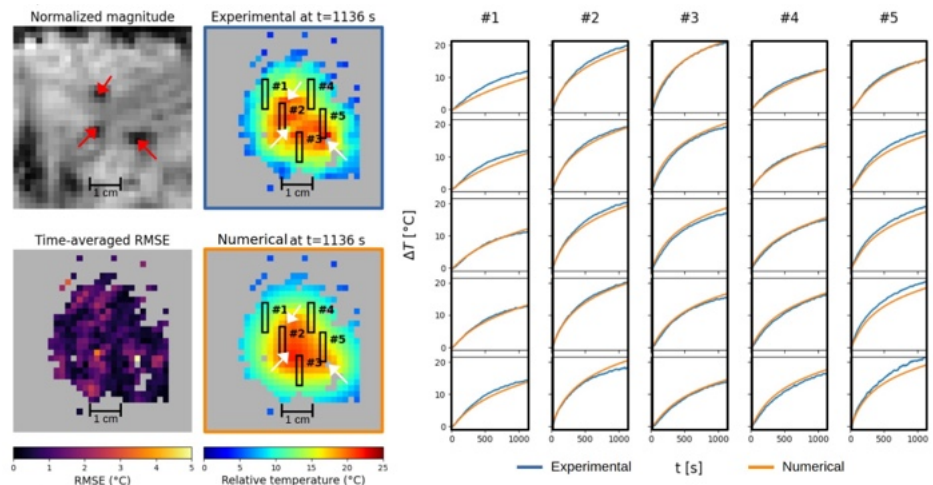


Fig 1: Ex vivo comparison between experimental and numerical thermal fields for d=1 cm. Images : (left up) normalized magnitude of MR-thermometry map at t=1136 s showing probe positions (red arrows) and (right up) its corresponding temperature map ; (right down) numerical temperature map (same time, same position) ; (left down) voxel-wise RMSE map (averaged over time) Graphs : temporal evolution of (blue) experimental and (orange) numerical temperature of voxels in corresponding columns #N in images

Trajectory Guide for Accelerated Minimally-Invasive Neurosurgery: Hardware Validation in Phantoms

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Purpose

In the past 18 months, MR-guided intraparenchymal gene delivery has been FDA-approved (Kebiliidi, PTC Therapeutics) for one pediatric disease while the AMT-130 Trial for Huntington's Disease appears very promising.^{1,2} Currently ≥ 12 hours are needed to complete these surgeries, much of which is spent iteratively imaging and manipulating trajectory guides to aim convection-enhanced delivery (CED) catheters into the brain. Faster trajectory alignment would enable wider clinical adoption. We present here the AccuGyd, a quantized neurosurgical trajectory guide, which only requires one imaging and manipulation step to reach alignment.³

Materials and Methods

AccuGyd is a small-profile rigid trajectory guide (Fig.1A, purple box) with MRI-detectable fiducials inset. Associated software uses 1) a scan of the fiducials adjacent to anatomy and 2) a target point on the same scans to calculate three device configuration settings which will align insertable devices with deep tissue targets based on only a single MRI acquisition.

A water tank (Fig.1A) with spike-lined bottom (Fig.1B) was fabricated to measure accuracy under repeat target trials. Aiming at the tips of the spikes (Fig.1B), 19 borosilicate glass rods were aimed and inserted in series (Fig.1C). Radial error logged (Fig.1D) using a 3T GE Premier scanner and a standard near-isotropic 3D T2w FSE sequence. This was repeated using a fruit-based phantom (Fig.2).

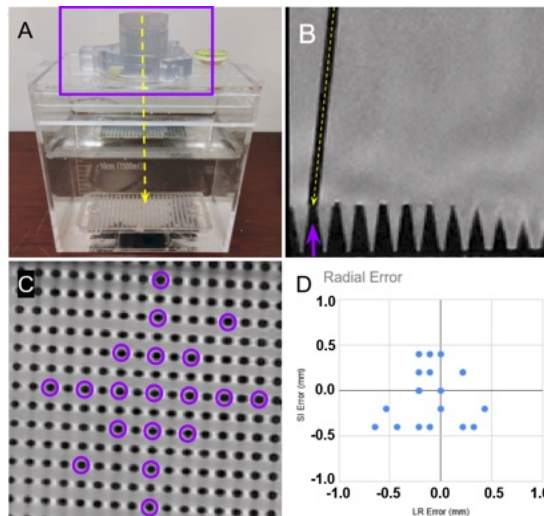


Figure 1: A water tank (A) was used to assess the accuracy of inserted rods (B) aimed at spikes (C). Errors were logged (D).

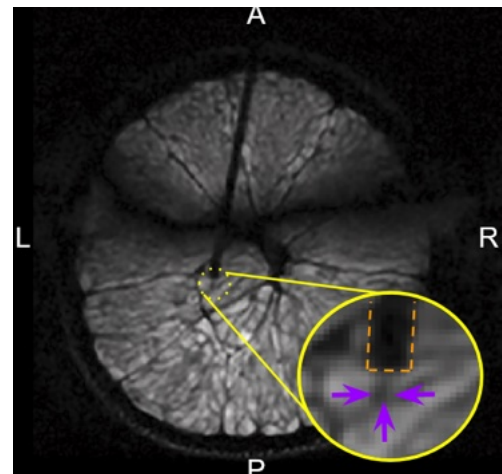


Figure 2: Similar insertions targeted points in a fruit. Here, the needle (orange) was inserted 1mm short of target (purple)

Results

Using the water tank phantom, mean radial error was 0.38 ± 0.17 mm at a depth of 150 mm across 19 trials. In the fruit phantom, radial error was 0.76 mm at 69.6 mm depth.

Conclusion

AccuGyd has demonstrated excellent device placement accuracy in *ex vivo* models. Future work will progress to preclinical animal trials in swine and nonhuman primates.

Acknowledgements

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An MRI-Guided Robot for Stereotactic Neurosurgery: An Initial Cadaver Study

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Purpose

The goal of this project is to develop a novel MRI-guided robotic system with an integrated imaging coil for precise single- and multi-target stereotactic neurosurgical procedures performed within the MRI bore. The system will enable real-time MRI visualization of the brain during instrument placement for a variety of neurosurgical interventions, including epilepsy treatment, tumor biopsy, laser interstitial thermal therapy (LITT), and deep brain stimulation (DBS).

Material and Methods

We developed a 5 degrees-of-freedom (DOF) MR-Conditional robot [1] with custom-designed pneumatic motors [2] to position and orient the instrument. The robot consists of a 4-DOF stereotactic frame [3] and a 1-DOF needle driver [4]. The stereotactic frame is designed with a parallel mechanism, which consists of two arcs. Each arc can provide 2-DOF rotational motion, mimicking the motion of the Leksell stereotactic frame.

The needle driver can provide 1-DOF needle insertion with coupled rotation. Experiments were performed to validate the MRI compatibility of the robotic system in a 1.5 T Siemens MRI scanner. The joint level positioning accuracy study was conducted using an optical tracker (Certus, NDI, Canada). An initial cadaver study was conducted to validate the clinical workflow and system feasibility.

Results

The mean SNR variation was less than 12.26%, and the maximum 2D geometric distortion was below 0.20%. The rotational accuracy of the robot was $0.09 \pm 0.11^\circ$ for the inner slide, $0.29 \pm 0.37^\circ$ for the inner rotation, $0.10 \pm 0.11^\circ$ for the outer slide, and $0.12 \pm 0.12^\circ$ for the outer rotation. In a composite tissue phantom, the automatic needle driver achieved a mean positioning accuracy of 2.38 ± 1.00 mm. An initial cadaver study validated the clinical workflow and demonstrated the feasibility of the system, achieving a mean targeting error of approximately 2.7 mm across two trials.

Conclusion

The proposed pneumatically actuated robot provides the motion capability and accuracy required for stereotactic neurosurgical procedures. Negligible MRI image degradation was observed from the custom-designed pneumatic motors, enabling simultaneous robot motion and real-time visualization of both the surgical instrument and patient anatomy. An initial cadaver study validated the proposed clinical workflow and demonstrated the feasibility of the system.

Acknowledgements

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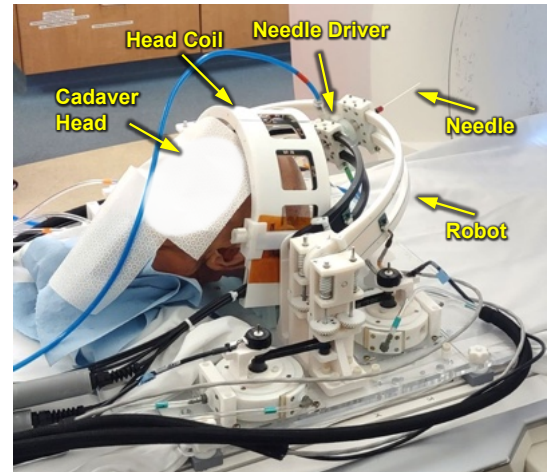


Fig 1: Experiment setup of the cadaver study inside an MRI scanner.

From Intraoperative MRI to Intelligent Intervention: 30 Years of Image-Guided Therapy at Brigham and Women's Hospital

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Background

Brigham and Women's Hospital (BWH) installed the GE Signa SP open 0.5 T intraoperative MRI in 1994, launching a program in MRI-guided therapy. In 2008, therapy moved to closed-bore 3T scanners on the L1 floor of Radiology: Siemens MAGNETOM Verio scanners for needle-based procedures and a GE scanner with the InSightec ExAblate for focused ultrasound. In 2011, the AMIGO suite added a ceiling-mounted Verio. We reviewed the program's 30-year output across three periods, marked by the Interventional MRI Symposia hosted in Boston in 1994, 2012, and 2018.

Methods

We searched PubMed from 1994 to September 2026 for articles with a BWH-affiliated author and the terms "MRI-guided," "MR-guided," or "intraoperative MRI." We supplemented this with investigator-name and affiliation-variant searches, and, from 2012, with NCIGT P41 grant (EB015898, EB028741) searches. Articles on MR-guided focused ultrasound and MR-Linac radiotherapy, which run as separate programs, were counted and set aside. The remaining articles were classified by procedure, and neurosurgical articles by technique, across three periods: 1994–August 2012, September 2012–October 2018, and October 2018–September 2026. We also searched ClinicalTrials.gov for BWH-sponsored interventional trials.

Results

From 1994 to 2012, 123 articles were identified, and 38 of them addressed focused ultrasound. Of the remaining 85, prostate procedures accounted for 31% and neurosurgery for 24%. From 2012, 238 articles were identified. Of these, 45 addressed focused ultrasound, which has now exceeded 500 thalamotomies [1], and 27 addressed MR-Linac radiotherapy. The remaining 168 numbered 100 in 2012–2018 and 68 since 2018. Across the three periods, neurosurgery accounted for 24%, 19%, and 37% of articles, and prostate procedures for 31%, 25%, and 28%. Gynecologic brachytherapy fell from 17% to 6% after 2018. The 44 neurosurgical articles since 2012 covered intraoperative MRI-guided resection, navigation, and brain-shift correction ($n = 29$), laser interstitial thermal therapy (LITT; $n = 4$), deep brain stimulation (DBS; $n = 4$), intraoperative mass spectrometry ($n = 4$), and ablation or biopsy ($n = 3$). Three of four DBS articles and three of four LITT articles appeared after 2018. Mass spectrometry in AMIGO identified IDH-mutant gliomas within minutes [2]. Asleep DBS placed 98% of 205 leads within 1 mm of the target [3], and non-cooled LITT matched direct temperature measurements ($r = 0.96$) [4]. In the prostate, robotic guidance reached about 3 mm accuracy [5], and 53 multidrug microdevices tested drug response in 14 patients [6]. Collaborative oncolytic virotherapy prolonged survival in recurrent glioblastoma [7]. Four of six BWH-sponsored trials started after 2018, and four remain active [8–10].

Conclusion

Over 30 years, prostate and neurosurgical procedures have remained the core of MRI-guided intervention at BWH. Since 2018, neurosurgery has grown to span resection, brain-shift correction, LITT, DBS, and intraoperative mass spectrometry, alongside large focused ultrasound and MR-Linac programs. These findings set the requirements for AMIGO 2.0, now in planning with industry partners.

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MR-Guided Focused Ultrasound Blood-Brain Barrier Opening for Next-Generation Gene Therapy Delivery Paradigms

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The blood-brain barrier (BBB) remains a major obstacle to the development of gene therapies for neurological disease, limiting the efficient delivery of systemically administered therapeutics to the central nervous system. MR-guided focused ultrasound (MRgFUS), combined with intravenously administered microbubbles, provides a noninvasive and spatially targeted approach for transiently increasing BBB permeability. This technology offers the potential to address several key challenges in CNS gene therapy, including achieving efficient delivery to specific brain regions while reducing systemic exposure and avoiding invasive intracranial administration. This presentation will provide an overview of our work developing MRgFUS-mediated BBB opening as a platform for targeted delivery of next-generation genetic therapeutics. Studies spanning small- and large-animal models will highlight applications to viral vectors and emerging non-viral delivery platforms, as well as ongoing efforts to improve delivery efficiency, spatial coverage, and clinical translatability.

MRI-Guided Transurethral Ultrasound Ablation (TULSA) vs Robotic Prostatectomy RCT: 6-Month Outcomes

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Purpose

Ablative therapies aim to reduce the significant risk of urinary incontinence and erectile dysfunction associated with radical prostatectomy, while preserving oncologic control. CAPTAIN (NCT05027477) is the first, fully enrolled, multicenter randomized controlled trial in this space, performing head-to-head comparison of MRI-guided transurethral ablation (TULSA) vs. robotic prostatectomy (RP) in men with localized, intermediate-risk prostate cancer (PCa). Here we report one-month perioperative outcomes, three-month recovery, and the six-month primary functional endpoint.

Material and Methods

211 men with localized, GG2-3, PSA $\leq$ 20 ng/mL PCa were treated at 23 high-RP volume centers across USA, Canada, and Europe. Allocation to TULSA:RP was 2:1 balancing for key prognostic factors; crossover between arms was not allowed. Ablation ranged from hemi- to whole-gland ablation and one repeat TULSA was allowed; RP was performed per local standard of care. The following measures of perioperative morbidity and return to baseline activity were captured: estimated blood loss and length of stay post-treatment; patient diary to 1 month (EQ-5D-5L overall health) and penile shortening; return to work and hospitalization for complications by 3 months; and primary functional endpoint at 6 months, defined as preservation of both IIEFQ2 $\geq$ 2 erections sufficient for penetration and EPICQ5 pad-free continence.

Results

At baseline, median (IQR) age, PSA and ISUP 2/3 proportions were similar for TULSA, RP: 63 (58-68) y vs 65 (60-69) y; 6.5 (4.9-9.6) ng/mL vs. 7.2 (5.6-9.7) ng/mL; 76%/24% vs 77%/23% for RP (all $p>0.05$). For TULSA, 68% had whole-gland ablation, 32% subtotal. For RP, 95% had nerve-sparing, 89% bilaterally. After TULSA, there was zero blood loss vs. median 100 (IQR 100-200) mL for RP. Length of stay after TULSA (none overnight) was significantly shorter: 0.28 (0.28-0.34) d vs. 1.16 (1.06-1.31) d. Overall health (EQ-5D-5L) declined less and recovered to baseline faster for TULSA vs RP, and TULSA patients had 4 mm less penile shortening. TULSA patients returned to work within 10 (4-15) d vs. 19 (10-41) d after RP. TULSA patients had fewer complications requiring hospitalization (0.7% vs. 6.3%). At 6 months, TULSA achieved primary superiority on preservation of erectile function and urinary continence (risk ratio 2.1, $p=0.004$).

Conclusion

TULSA demonstrated favorable perioperative safety, more rapid return to baseline activities including paid employment, and fewer complications requiring hospitalization compared with RP. TULSA was also superior to RP on the primary functional endpoint at 6 months, comprising preservation of pad-free continence and erections sufficient for penetration. Oncologic outcomes will be reported with additional follow-up.

Acknowledgements

This study was sponsored by Profound Medical.

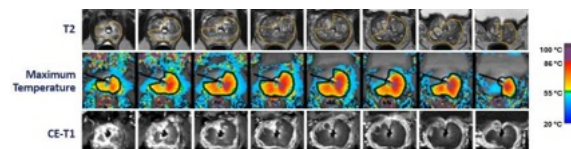


Fig 1: Treatment images from the TULSA procedure: T2-weighted images (top row) on which the physician-prescribed treatment target is contoured; maximum temperature maps, from MRI thermography, used to monitor heating to ablative temperature (55°C) in real-time within automatic, closed-loop feedback (middle row); contrast-enhanced T1-weighted images indicating the perfusion defect, acquired immediately post-treatment (bottom row). Columns represent the 8 independently-controlled transducer elements, from apex-base, left-right.

Impact of Ablation Plan on PSA Response and Catheterization Duration After MRI-Guided TULSA in the CARE Registry

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Purpose

MRI-guided transurethral ultrasound ablation (TULSA) delivers directional ultrasound with real-time MR thermometry-based feedback to thermally coagulate a physician-prescribed prostate volume, enabling individualized prostate cancer treatment from focal to whole-gland ablation. This study evaluated associations between ablation extent, PSA response and catheterization duration in the multicenter CARE Registry (NCT05001477).

Material and Methods

CARE Registry participants treated for prostate cancer with documented ablation extent were included. Ablation extent was categorized as whole-gland, subtotal >50%, or focal ≤50%. Outcomes included PSA nadir, percent reduction from baseline PSA to PSA nadir, biochemical recurrence (BCR) defined by Phoenix criteria (PSA nadir + 2 ng/mL), and catheterization duration. Baseline Grade Group distribution was compared across ablation categories.

Results

A total of 197 patients from five sites were included: 79 whole-gland (40.1%), 62 subtotal >50% (31.5%), and 56 focal ≤50% (28.4%). Median PSA follow-up was 24 months (IQR 12–36). Baseline median PSA was 6.42 ng/mL (IQR 4.86–9.50), prostate volume was 40.0 cc (IQR 30.0–53.0), and number of lesions was 1 (IQR 1–2). Baseline Grade Group was GG1 in 20/197 (10.2%), GG2 in 125/197 (63.5%), GG3 in 45/197 (22.8%), and GG4 in 7/197 (3.6%). GG3+ disease was present in 35.4% of whole-gland, 24.2% of subtotal >50%, and 16.1% of focal ≤50% cases (p=0.038).

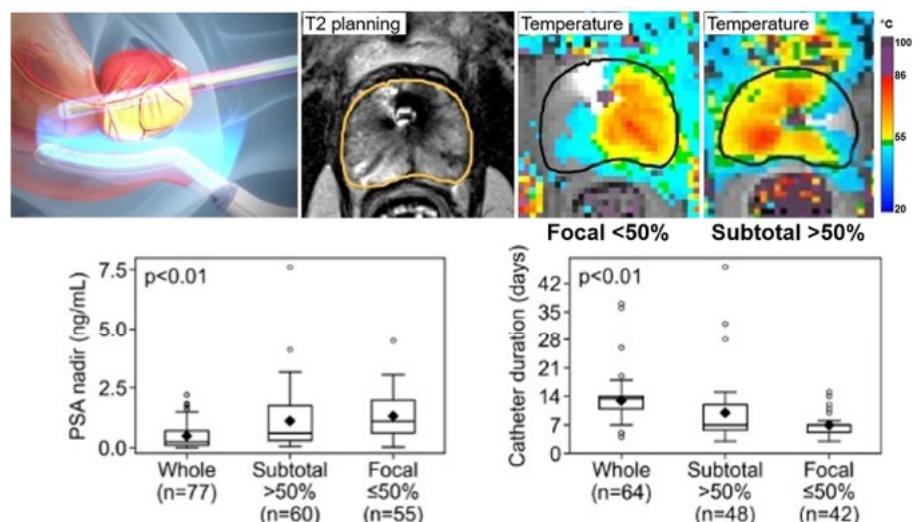
PSA nadir differed significantly by ablation extent: 0.23 ng/mL after whole-gland ablation, 0.60 after subtotal >50%, and 1.10 after focal ≤50% (p<0.001). Median PSA nadir reduction was 96.8%, 92.2%, and 79.3%, respectively (p<0.001). BCR occurred in 11/79 patients (13.9%) after whole-gland ablation, 14/62 (22.6%) after subtotal >50%, and 5/56 (8.9%) after focal ≤50% ablation, without a significant difference by ablation extent (p=0.11). Median catheterization duration was shorter with less extensive ablation: 13.5 days after whole-gland ablation versus 7.0 days after subtotal >50%, and focal ≤50% ablation (p<0.001).

Conclusion

Greater ablation extent was associated with deeper PSA reduction after MRI-guided TULSA ablation. Higher Grade Group disease was more frequently treated with greater ablation extent, suggesting risk-adapted treatment selection in real-world practice. Phoenix-defined BCR did not significantly differ by ablation extent. Less extensive ablation was associated with shorter catheterization duration, supporting MRI-guided tailoring of treatment extent to balance PSA response with early post-procedural recovery.

Acknowledgements

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TULSA-Pro treatment concept, MRI planning, and MR thermometry examples are shown above. Bottom panels show PSA nadir and catheterization duration by ablation extent. Boxes represent median and interquartile range, diamonds indicate means, and circles indicate outliers. PSA nadir decreased with greater ablation extent, whereas catheterization duration was shorter with less extensive ablation; both p<0.01.

Translational Assessment of a Rabbit-Model Trained MRgFUS Imaging Biomarker

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Purpose

Accurate assessment of magnetic resonance-guided focused ultrasound (MRgFUS) treatment efficacy immediately following ablation remains challenging because conventional biomarkers such as thermal dose and non-perfused volume (NPV) often incompletely reflect the final extent of tissue necrosis. We previously developed a deep learning-based multiparametric MRI biomarker [1] capable of predicting delayed NPV following MRgFUS treatment in a VX2 rabbit tumor model. While promising results were obtained within the training species, the ability of such models to generalize across different biological systems and imaging environments remains unknown. The objective of this study was to evaluate the translational robustness of a rabbit-trained MRgFUS imaging biomarker on independent goat and human datasets without retraining.

Material and Methods

Two deep neural network models previously trained using data acquired from rabbit (N=18) VX2 tumors undergoing MRgFUS ablation were evaluated as fixed inference models. The first model utilized the complete multi-parametric MRI (mpMRI) feature set developed during the original study [2], including pre- and post-treatment T1-w, T2-w MRI, MR-thermometry and quantitative T1 and T2 maps. The second model was trained using only a Multi Pathway Multi Echo (MPME) [3] sequence and MR thermometry maps. Independent datasets from goat (N=1) and human breast cancer (N=3) MRgFUS procedures were processed using the same registration, normalization, and pre-processing pipeline employed during rabbit model development. No transfer learning, or species-specific adaptation was performed. Predicted treatment volumes from both models were compared against MRI-derived NPV measurements serving as the reference standard. Performance was evaluated using the Dice similarity coefficient.

Results

Quantitative evaluation demonstrated that the current clinical standard achieved a Dice similarity coefficient of 0.68 in the goat cohort. The rabbit-trained deep learning model achieved a Dice score of 0.61 when utilizing the full imaging feature set and 0.58 when using only MPME sequences and temperature-derived data. While both models exhibited a degree of generalization across species, performance remained below that of the clinical standard in goats. In contrast, both rabbit-trained models failed to provide reliable predictions in the human cohort. This performance degradation is likely attributable to the substantial domain shift between rabbit and human data, including differences in tissue architecture and resulting imaging characteristics, which were not represented during model training.

Conclusion

These preliminary results suggest that deep learning models trained on preclinical MRgFUS data can capture imaging features that generalize to other preclinical models. While performance in the goat cohort is promising, limited success in the human breast cancer cohort highlights the challenges of cross-species translation. Additional data from larger and more diverse cohorts will be required to validate these findings and assess the clinical utility of the proposed approach.

Acknowledgements

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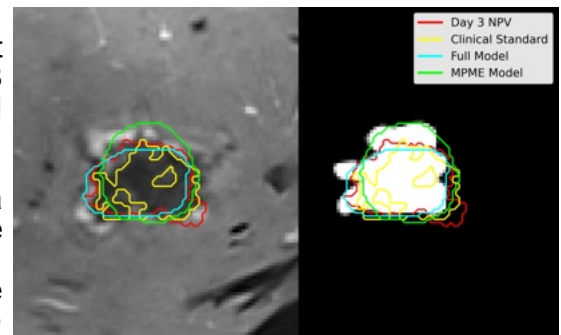


Fig 1: Representative goat coronal MRI slice with Day 3 NPV (red), clinical-standard (yellow), full model predictions (cyan), and MPME model predictions (green).

Feasibility of 3D visualization for intraoperative coverage assessment in MR-guided spine cryoablation

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Purpose: Percutaneous MR-guided cryoablation is a minimally invasive option for treating painful or locally aggressive spine lesions. MRI provides excellent visualization of the cryoablation ice ball and adjacent soft-tissue anatomy near critical structures such as the spinal cord, nerve roots, and blood vessels [1]. However, coverage assessment can be challenging because tumor boundaries, evolving ice-ball extent, and nearby anatomy must be interpreted across multiplanar images. We evaluated the technical feasibility of a 3D Slicer-based visualization workflow using existing AI-assisted segmentation tools to generate 3D models for coverage assessment during MR-guided spine cryoablation.

Material and Methods: Six MR-guided cryoablation cases were included. Two early cases were used for retrospective workflow development, followed by application during four subsequent procedures. Two complementary nnUNet-based segmentation approaches were used [2]: TotalSegmentator [3] for automatic spine/vertebral labeling from preprocedural CT/MRI, and nnInteractive [4] for prompt-based segmentation or refinement of tumor, critical structures, or ablation ice-ball on intraoperative MRI. Clinically acquired T2-weighted TSE, STIR, and/or multiplanar T2-weighted images were used for ice-ball monitoring. Preprocedural images and corresponding segmentations were manually registered to intraoperative MRI when feasible. 3D models were displayed with intraoperative MR images in 3D Slicer. Technical feasibility was assessed qualitatively based on interpretable visualizations and retrospective review with the treating interventional neuroradiologist.

Results: 3D visualizations were generated in all six cases, including lumbar, sacral, and cervical targets. In three of the four intraoperative cases, preprocedural spine segmentations derived from CT/MRI were registered to intraoperative MRI. In one case, the preprocedural CT did not fully cover the target region, and differences between diagnostic and procedural MRI neck positioning caused deformation that limited transfer of preprocedural segmentations. Nevertheless, tumor and nerve segmentations were generated directly on intraoperative MRI using nnInteractive during intraoperative planning. The models displayed the target lesion, vertebral anatomy, ablation ice-ball, and nearby structures at risk in a common 3D view (Fig. 1). In a left S2 osteoid osteoma/osteoblastoma case, the anterior nerve branch was partially within the segmented ice-ball volume periphery; the patient subsequently experienced transient numbness that resolved over weeks to months without intervention, consistent with stunning of the nerve without cryoneurolysis. In a brachial plexus case requiring five freeze cycles, sequential ice-ball segmentations enabled 3D displaying of overlapping freeze zones to estimate cumulative ablation extent. Retrospective review supported the visualization utility for coverage assessment.

Conclusion: This feasibility study shows that 3D segmentation-based ice-ball coverage of tumor and adjacent nerves can be depicted in near-real time from preprocedural CT/MRI and intraoperative MRI during MR-guided spine cryoablation. Combining automatic spine segmentation with prompt-based intraoperative segmentation supported visualization across anatomically variable cases, including situations with limited preprocedural registration. This approach may be useful for complex ablations requiring assessment of tumor coverage, overlapping ice-ball volumes, cumulative ablation extent, and proximity to neural or vascular structures. Future work will focus on quantifying segmentation and ablation-volume accuracy.

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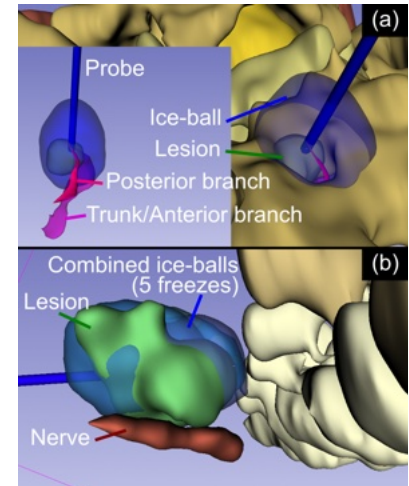


Fig 1: 3D visualization in MR-guided spine cryoablation. (a) Left S2 osteoid osteoma with lesion, probe, ice ball, and sacral nerve branches. (b) C4-C7 brachial plexus lesion, nerve, and combined ice-ball volumes.

MRI-Guided Multi-Regional Biopsy of Soft-Tissue Sarcomas

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Purpose

Soft-tissue sarcomas (STS) are heterogeneous malignancies¹ and CT- or US-guided core needle biopsies are typically performed to assess subtype and grade, often from a single non-targeted core². Multiparametric pre-biopsy MRI (mpMRI) depicts regional variation in cellularity and perfusion, but it is not routinely integrated for target selection³. MR-guided biopsies have been described, but without systematic multi-regional sampling. Building on *ex vivo* MRI to histology correlations^{4,5}, this study tests *in vivo* multi-regional MR-guided biopsies with mpMRI-informed target selection for phenotype-guided sampling.

Material and Methods

Ten patients with suspected STS underwent pre-interventional 3T mpMRI (Siemens Vida) for tumor characterization and biopsy planning, including T2w, pre-/post-contrast T1w, DWI ($b = 40/500/1000 \text{ s/mm}^2$), and DCE with Tofts-based K^{trans} estimation. For each lesion, 1-3 intralesional targets were selected based on distinct mpMRI patterns, typically low ADC/high K^{trans} and high ADC/low K^{trans} regions, while avoiding macroscopic necrosis. MR-guided biopsies were subsequently performed at 1.5T (Siemens Aera). T2w TSE images were acquired for target and entry-site localization, with patients positioned 5-8 cm off-isocenter toward the bore opening. Needle placement was guided by interactive real-time bSSFP imaging (2 views, 2.3 frames/s). A coaxial MR-compatible 16 G cannula and biopsy punch were used, with cannula-tip position confirmed by repeated T2w TSE before sampling.

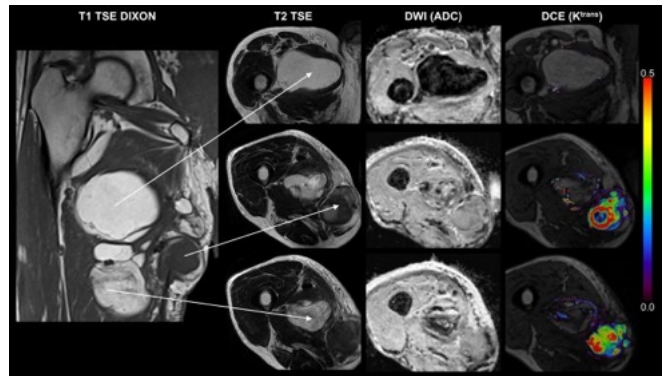


Fig 1: Multiparametric imaging of an STS. Three target sites were selected for MR-guided biopsy showing distinct patterns of T2-weighted signal intensity, ADC and K^{trans} .

Results

Multi-regional biopsy was feasible in all 10 patients with no major complications. An example with three targets is shown in Fig. 1. All mpMRI-defined targets were clearly visible during intervention on real-time imaging. Few-slice T2w TSE enabled fast cannula-position verification (Fig. 2). Average navigation time was $(5.9 \pm 6.2) \text{ min}$ per target (21 targets in total). Extracted tissue was adequate for histopathological analysis enabling successful tumor classification in all cases (60% intermediate-to-high grade).

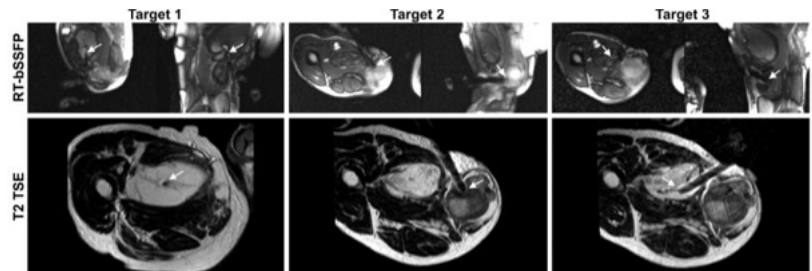


Fig 2: Real-time bSSFP images in two views of the cannula tip at the target locations (upper row). Bottom row shows corresponding T2-weighted TSE images in transverse for position verification.

Conclusion

This study demonstrates the feasibility of *in vivo* multi-regional, mpMRI-informed sampling before neoadjuvant radiotherapy. The approach may improve diagnostics by reducing under-grading from single, non-targeted cores and provides a tool to investigate correlations between mpMRI parameters and tumor biology for treatment stratification.

Acknowledgements

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MRI-Guided Percutaneous Laser Ablation for Pediatric Vascular Anomalies: A Single-Institution Experience

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PURPOSE

Pediatric vascular anomalies including venous malformations (VM), arteriovenous malformations (AVM), and kaposiform hemangioendothelioma (KHE) present significant treatment challenges due to their heterogeneous morphology, anatomic complexity, and risk of incomplete response to conventional sclerotherapy. MRI-guided percutaneous laser ablation offers precise, real-time thermal monitoring with simultaneous soft-tissue characterization, enabling safe treatment of anatomically challenging lesions in a growing pediatric population. The purpose of this study was to evaluate the technical feasibility, safety, and outcomes of in-bore MRI-guided percutaneous laser ablation across a spectrum of pediatric vascular anomalies.

MATERIALS AND METHODS

This retrospective single-institution study included all pediatric patients (age 0 to 17 years) who underwent in-bore MRI-guided percutaneous laser ablation for vascular anomalies between June 2014 and May 2025. Procedures were performed under general anesthesia using combined ultrasound and MRI guidance on a 1.5 MRI system. Laser fibers used included the Visualase system (Medtronic, Minneapolis MN), ECHO laser (Elesta, Florence Italy), and CLS fiber systems (Rare Medical, New York NY). Continuous MRI thermometry was employed for real-time temperature monitoring. Patient demographics, lesion type, anatomic location, number of treatment sessions, technical success (defined as ablation zone encompassing the target lesion), and procedural complications were recorded and analyzed. Lesion types were classified as VM, AVM, or KHE based on radiology reports and clinical records.

RESULTS

A total of 76 laser ablation procedures were performed in 46 pediatric patients (mean age 11.2 years, SD 4.9, range 1 to 17 years). Lesion types included VM in 44 procedures (57.9%), AVM in 24 procedures (31.6%), and KHE in 8 procedures (10.5%) across 3 patients. Anatomic sites included the thigh and femur (n = 21, 27.6%), lower leg and calf (n = 14, 18.4%), chest wall (n = 10, 13.2%), foot and ankle (n = 9, 11.8%), trunk and retroperitoneum (n = 7, 9.2%), upper extremity (n = 6, 7.9%), knee region (n = 5, 6.6%), and shoulder (n = 1, 1.3%). The mean number of procedures per patient was 1.7 (range 1 to 8). Minor complications occurred in 4 procedures (5.3%): 2 small skin blisters (less than 1 cm) at the ablation site, 1 self-limited peri-malformation hematoma, and 1 minor superficial heat injury at the insertion site requiring primary closure. No major complications, no nerve injuries, and no procedure-related hospitalizations were recorded.

CONCLUSION

MRI-guided percutaneous laser ablation is technically feasible and safe for the treatment of pediatric vascular anomalies across a broad spectrum of lesion types and anatomic locations. Real-time MRI thermometry enables precise thermal control in anatomically critical regions, including periarticular and intramuscular sites not amenable to conventional sclerotherapy. Repeat sessions may be necessary for complex or multifocal anomalies such as KHE, and a structured treatment algorithm is warranted in this subgroup.

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MRI-Safe Robot and Drill for Bone Biopsy: First Clinical Results

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Purpose: Our team has developed an MRI-safe robot and drill to enable bone biopsy under MRI guidance. The workflow allows the patient to be moved in and out of the scanner for MRI-guided needle-guide targeting and confirmation while the biopsy is done through the needle-guide locked on target with the patient out of the scanner. We describe the system and our first clinical case.

Materials and Methods: The MRI-safe robot uses pneumatic actuation and optical encoding to ensure no interference with the MRI field. The robot has 4 passive degrees of freedom (DoF) for positioning, two Remote Center of Motion actuated axes to orient the needle-guide towards the target, and one actuated DoF to set the depth of needle insertion. The MRI-safe drill is pneumatically actuated by a high power motor that we built for drilling through the bone cortex (15 W at 0.275 MPa).

Results: FDA IDE and IRB approvals were granted for a pilot clinical trial in pediatric patients. Informed consent was obtained for an MRI-guided biopsy of a marrow-replacing lesion in the right femoral diaphysis of an 8-year-old male. As there was clinical concern for malignancy, a bone specimen was required to make diagnosis. The patient was placed under general anesthesia, brought to the MRI room (1.5T Philips Achieva) and positioned on the scanner table. Bottom elements of a SENSE Body MRI coil were placed under the legs and the top coil elements were secured above the area of interest, with the opening centered on the lesion. The skin overlying the target was prepped and draped in sterile fashion and the robot was centered over the target and covered with a sterile bag. T2-weighted axial MRI images were obtained (2D turbo spin echo sequence (factor = 25); TE/TR = 120/4451 ms; in-plane resolution=1.38 mm (RL) x1.5 mm (AP); slice thickness/gap 1.4/0.14 mm; matrix 116x187; 100 slices; Scan duration: 5 min 43 s) and used to identify the target and skin entry point. The robot was registered with the images using integrated fiducials and moved autonomously to align the needle-guide towards the target. The pneumatic drill was used with a commercial bone biopsy kit (ARROW® OnControl® Bone Lesion Biopsy System Tray). The outer 11-gauge 10 cm access needle was advanced to the target. The inner stylet was removed and three needle core specimens were obtained with a 13-gauge 14 cm bone biopsy needle. After the biopsy, a 22-gauge MR compatible needle was inserted through the access needle, which was then removed, and repeat MRI was obtained to confirm the needle path and the location from which the sample was removed. The case was completed successfully in less than 2 hours on the MRI scanner, without any complications. The biopsy results were analyzed by pathology, resulting in a definitive diagnosis of a rare condition known as Juvenile Paget's disease.

Conclusions: This case showed feasibility of performing an MRI-guided bone biopsy using an MR safe robot and bone drill. MRI ensured an accurate biopsy of the target area avoiding the ionizing radiation exposure.

Acknowledgements and References

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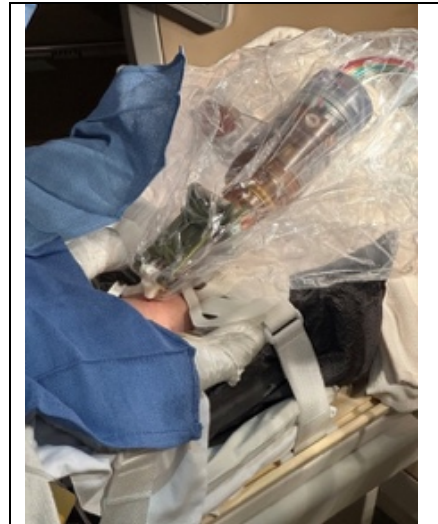


Figure 1: Patient and robot on table with needle-guide oriented on target, before needle drilling

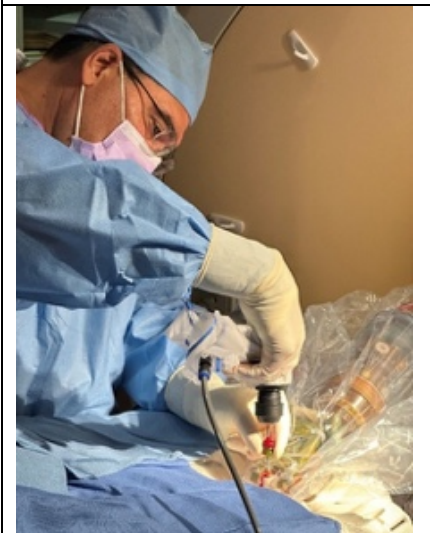


Figure 2: Dr. Sharma using the pneumatic drill through the needle-guide to obtain biopsy sample

Deep Learning-Based Needle Tracking Technologies for MRI-Guided Interventions

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Needle localization and tracking on intra-procedural MRI provides crucial information for the guidance, navigation, and confirmation of minimally invasive needle-based interventions. Interventionalists with expert knowledge and experience can interpret and adjust intra-procedural MRI to identify anatomical landmarks, target tissues, and needle position throughout an intervention. However, such manual determination of the needle position adds cognitive burden over the multiple steps in the interventional workflow and can compromise the time efficiency and procedural success. This talk will present recent developments of deep learning-based technologies for automated, rapid, and accurate needle tracking on MRI for guiding interventions. These new needle tracking technologies hold promise to assist and augment the capabilities of interventionalists, ensure consistent procedural success, and enable new real-time workflows for MRI-guided interventions.

Tissue and Needle Tracking on Real-Time Interventional Liver MRI Using the SAM2++ Foundation Model

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Purpose

Magnetic Resonance Imaging (MRI) enables guidance of minimally invasive procedures such as microwave ablation of liver cancers [1, 2]. However, accurate, low-latency tracking of tissue and interventional devices on real-time MRI remains challenging [3], and limited real-time MRI interventional datasets hampers the development of task-specific deep learning models. This study evaluates the SAM2++ foundation model [4], trained only on natural images and videos, without fine-tuning, to determine its ability to generalize to track rigid liver motion, non-rigid tissue targets, and needles on 2D real-time MRI.

Material and Methods

SAM 2++ Model: The SAM2++ model is a unified model built to perform video tracking across multiple granularities including key points and bounding boxes (Fig 1a). It uses an adaptive memory mechanism based on the specified tracking granularity to reduce computational time [5].

Dataset: The dataset consisted of 23 real-time sagittal liver MRI series acquired from 5 pig subjects using a 2D radial sequence with sliding-window reconstruction at 100 ms temporal resolution [6]. Each series contained approximately 100 frames spanning 1–2 respiratory cycles, acquired with TR/TE = 6.31/2.85ms, field-of-view: 300x300 mm², in-plane resolution: 1.56x1.56 mm², flip angle: 12, slice thickness: 5 mm.

Real-time Processing: A streaming pipeline was developed for online implementation of SAM2++ (Fig 1c).

Evaluation Metrics: Liver bounding boxes, tissue keypoints, and interventional needle tip locations were manually annotated as reference trajectories. Tracking success was defined as localization error <5mm relative to the reference annotations [8]. Motion curves were compared to assess rigid (bounding box) versus non-rigid (keypoint) motion. Real-time applicability was evaluated by measuring the inference time per frame.

Results

SAM2++ achieved an RMS error of 4.7mm for keypoints and bounding box tracking across all data sets. On an NVIDIA RTX A100 GPU, inference time was 247 ms per frame. Processing rates could be reduced by exploiting faster hardware or using multiple GPUs in parallel to limit increasing latency.

Conclusion

SAM2++ enables assessment of rigid and non-rigid liver motion through bounding box and key point tracking, respectively. The proposed real-time pipeline demonstrates the feasibility of low-latency MRI feature tracking using a pre-trained foundation model without fine-tuning.

Acknowledgements

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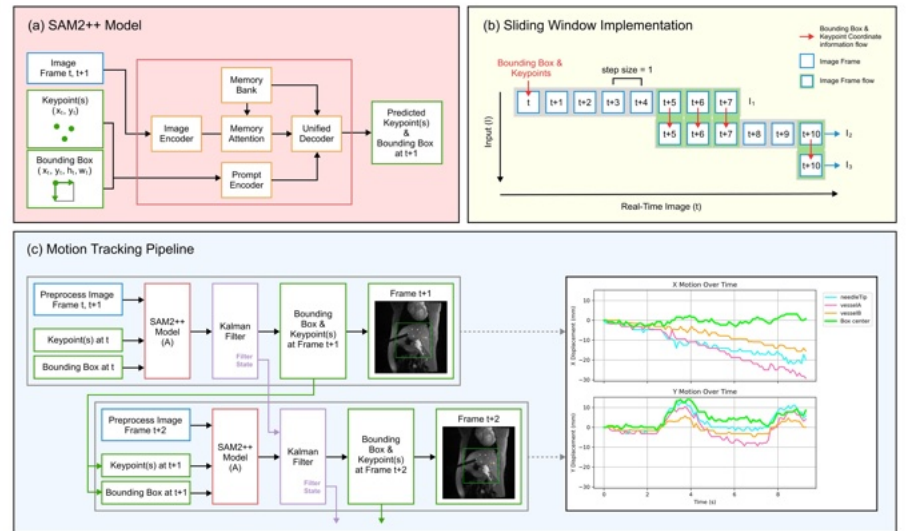


Fig 1: Real-time SAM2++ motion tracking pipeline. (a) SAM2++ framework. (b) Sliding-window implementation with 8-frame windows and step size of 5 frames. The first frame in the image series used manual keypoints and bounding box for initialization. Following windows used propagated coordinates to initialize tracking. (c) Complete real-time tracking pipeline integrating SAM2++ predictions, Kalman filtering, and motion visualization [7].

Unsupervised Conditional INR for Referenceless PRF Thermometry

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Purpose

To develop an unsupervised conditional implicit neural representation (INR) framework for referenceless PRF thermometry that estimates the background complex image inside the heated ROI without requiring pseudo background supervision.

Material and Methods

The study used the same transcranial MRgFUS dataset as our previous work [1], comprising 32 patients (24 training, 4 validation, and 4 test patients) and 1416 images (1110 training, 160 validation, and 146 test images). For referenceless PRF thermometry on this dataset, temperature estimation requires recovering the background complex signal inside the heated ROI from surrounding non-heated tissue, while the true background signal within the ROI is not directly observable [2]. To address this inverse problem without pseudo background supervision, the framework uses Conditional Implicit Neural Representation (Conditional INR) to model the background complex image as a continuous coordinate-dependent field [3], which supports smoothness and boundary-continuity constraints. A context encoder extracts conditioning features from masked real/imaginary inputs and the ROI mask, and Fourier-embedded spatial coordinates are passed to a conditional MLP to reconstruct the complex background signal. Training uses outside-ROI data consistency, boundary-ring continuity, and Laplacian-based spatial smoothness constraints.

Results

Preliminary experiments demonstrated that the framework can recover background phase and temperature estimates. Representative center-temperature curves, Bland-Altman analysis, and regression results are shown in Fig. 2, achieving a center MAE/RMSE of 0.60/0.74°C. Compared with our previous supervised framework, the proposed method reduced the center MAE and RMSE by approximately 6% and 10%, respectively. These results support the feasibility of unsupervised INR-based background reconstruction for referenceless temperature estimation. In addition, the estimated temperature curves demonstrated good temporal agreement with the reference measurements throughout the heating process.

Conclusion

Conditional INR provides a feasible continuous-field representation for referenceless thermometry, while unsupervised physical constraints enable background recovery without baseline-derived pseudo supervision. Preliminary results demonstrate that accurate referenceless temperature estimation can be achieved without direct background supervision.

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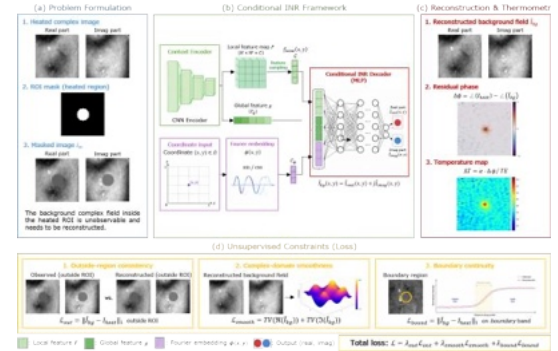


Fig 1: Conditional INR framework for unsupervised referenceless thermometry.

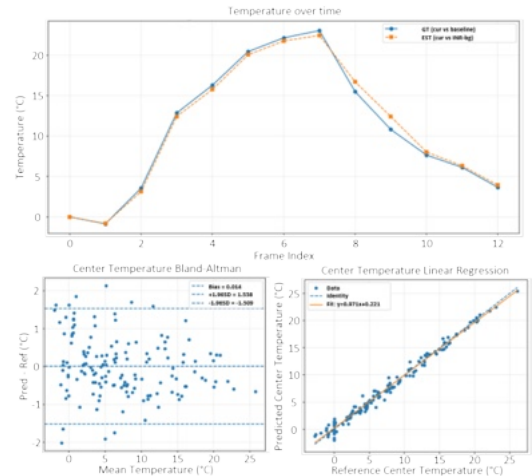


Fig 2: Representative center-temperature curves, Bland-Altman analysis, and regression analysis for temperature estimation.

MR Thermometry-based validation of liver tumor microwave ablation computational modeling for lesion size prediction

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Purpose

Microwave ablation computational (MWA) modeling may support personalized treatment planning and lesion-size prediction, but clinical translation requires validation against experimental data. This study validated a simplified 2D axisymmetric finite-element MWA model using volumetric magnetic resonance (MR) thermometry.

Material and Methods

Ex vivo bovine (N=24) and *in vivo* porcine (N=30) livers were ablated under MR guidance using an MR-compatible microwave system. Temperature monitoring was performed with phase mapping based on the proton-resonance frequency-shift method. Finite element simulations were conducted using COMSOL Multiphysics to model the coupling between the electromagnetic field and bioheat transfer phenomena. To reduce computational cost, a 2D axisymmetric model was employed, assuming a homogeneous and symmetric tissue environment [1].

An automated pipeline was developed to map simulated temperature fields onto the experimental MR-thermometry grid for voxel-wise comparison. Micro-CT imaging was used to characterize the geometry of the microwave antenna. Model parameters (delivered power and blood perfusion rate) were calibrated using an optimization framework, and generalizability was assessed with a hold-out validation strategy. For *in vivo* cases, MR thermometry-derived and simulation-derived ablation zones were compared with post procedural T1-weighted lesion segmentations using the Dice similarity coefficient and sensitivity.

Results

In *ex vivo* experiments, root-mean-squared (RMSE) error between experimental and simulated temperatures ranged between 3.1 and 12.1 °C for target temperatures between 60–100 °C. *In vivo* model achieved an average RMSE of 6.0±2.26 °C across the cohort, with good qualitative spatiotemporal agreement. Lesion prediction improved when using simulation-derived dosimetry compared with MR thermometry-based dosimetry, with Dice and sensitivity of 0.72±0.10 and 0.83±0.09 versus 0.61±0.10 and 0.72±0.14, respectively.

Conclusion

3D validation and calibration of MWA thermal modelling based on volumetric MR thermometry has been demonstrated using a total of 15 million informative measurement points. The proposed retrospective framework improved lesion prediction relative to MR thermometry-based dosimetry alone.

Acknowledgements

This work received financial support from the CNRS through the MITI interdisciplinary programs (CHAUPATAT) through its exploratory research program and from the French National Investments for the research programs ANR-DFG (SMART-HEAT)

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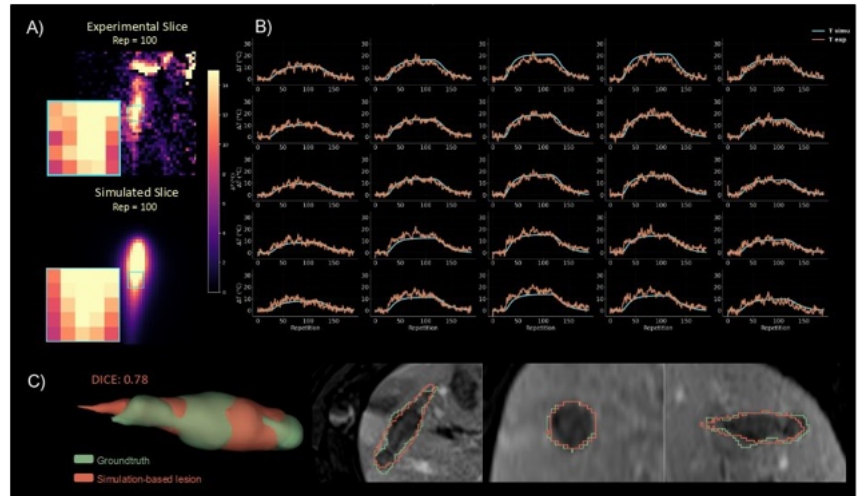


Fig 1: Comparison between measured and simulated temperatures and lesion sizes for in vivo microwave ablation.

A. MR thermometry slice at a given time point (top) with the corresponding simulated temperature map (bottom). **B.** Temperature time plots within a 5×5 region of interest. Experimental measurements are shown in orange and simulations in blue. **C.** Ground truth (green) and simulation-based (red) lesions overlaid on post ablation gadolinium-enhanced T1w images. A 3D rendering of the lesions is displayed on the left, followed by axial, coronal and sagittal views.

AI-Based Workflow for Automatic Needle Tracking and Dynamic Scan-Plane Control in Interventional MRI

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Purpose: Continuous needle tip localization during MRI-guided interventions could enable earlier detection of targeting error and provide feedback for trajectory adjustment during needle advancement. However, fast MRI tradeoffs can limit imaging volume and risk loss of tip visibility, while needle deflection and variable artifact appearance complicate automatic localization. Building on prior work in MRI needle localization, real-time scan-plane control, and our previous synthetic-data training framework [1-6], we developed an AI-enabled real-time MRI workflow that localizes the needle tip from real-time MR images and uses the estimated tip position to update orthogonal scan planes during needle insertion to maintain tip visibility.

Material and Methods: We used an image-based tracking strategy based on fast GRE imaging, orthogonal slice groups, and AI-based tip localization. GRE was selected because it supports rapid acquisition and is sensitive to needle-induced susceptibility effects, enabling passive visualization of the needle without additional tracking hardware. Alternating sagittal and coronal GRE slice groups were used to provide complementary high-resolution views while limiting acquisition time. Incoming magnitude and phase images were converted to two-channel complex-valued inputs. A 2D U-Net segmented the needle shaft and tip, and postprocessing extracted the three-dimensional needle tip position. Estimated tip coordinates were sent back to the scanner to update scan-plane placement during ongoing acquisition. The U-Net was trained using synthetically generated complex-valued GRE images with known needle tip locations, avoiding the need to acquire and annotate a large real image dataset. The workflow was evaluated during *in vivo* MRI-guided needle insertion experiments performed with a robotic needle guide in two porcine models on a clinical 3T MRI system. Final tracked tip positions were compared with manually annotated confirmation MRI.

Results: The workflow was evaluated across eight *in vivo* needle insertions. Dynamic orthogonal scan-plane updated maintained needle tip visualization throughout all insertions, with successful in-plane tracking in every acquisition cycle and full 3D localization in most cycles. Final 3D localization error was 1.0 ± 0.4 mm relative to confirmation MR images. Processing time remained below the image acquisition interval, enabling online scan-plane updates during insertion.

Conclusion: This work demonstrates a real-time MRI workflow with AI-based needle tracking and dynamic scan-plane updates for maintaining needle-tip visibility during *in vivo* MRI-guided needle insertion. By combining fast GRE imaging, complex-valued AI-based tip localization, and automatic scan-plane updates, the workflow enables continuous localization within limited imaging volumes and provides intraprocedural feedback for trajectory adjustment during needle advancement.

Acknowledgements: This work was supported by NIH (R01EB020667, R01CA235134, R01EB030539, R01EB036015, and P41EB028741) and the Brigham and Women's Hospital Department of Radiology (GR0131914).

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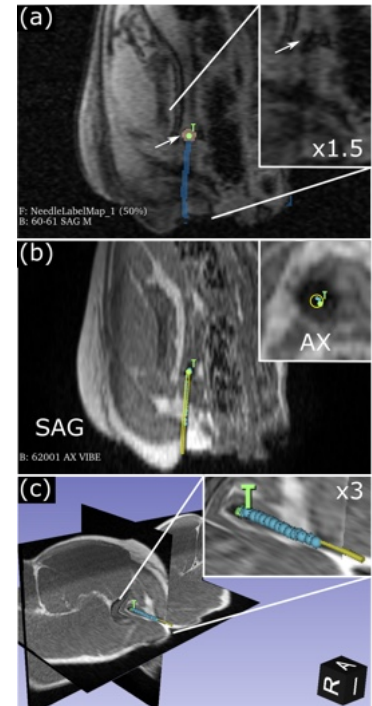


Fig 1: Representative *in vivo* AI-enabled needle tracking result. (a) Sagittal real-time GRE image with U-Net segmentation of the needle shaft (blue) and tip (yellow), tracked tip position (green), and magnified tip artifact (arrow). (b) Post-insertion confirmation VIBE with manually annotated needle position (yellow), final tracked tip, and axial tip view. (c) 3D visualization of the tracked tip trajectory (blue) relative to the confirmation image, with magnified final trajectory and tip position.

MRI-Guided Percutaneous Cryoablation for vaginal/paravaginal lesions - a case series

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Purpose: To evaluate the feasibility and technical success of MRI-guided cryoablation for vaginal/paravaginal lesions in previously irradiated or surgically altered anatomy.

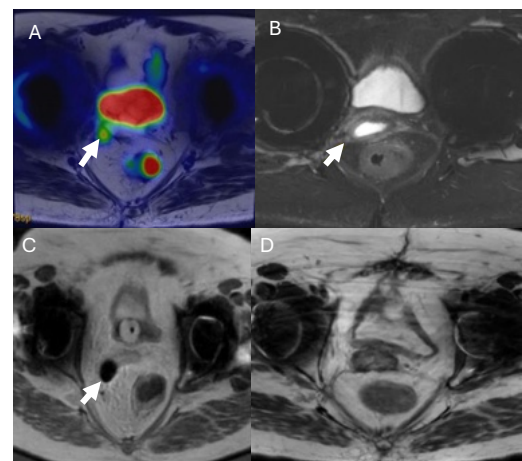
Materials and Methods: This retrospective single-center study (2013–2026) included 16 procedures (1–4 procedures per patient) in 12 patients (mean age, 64 years; range, 27–88) with metastatic gynecologic malignancies or desmoid tumor. Lesions involved the vaginal cuff, paravaginal space, periurethral region, and uterus. Procedures were performed under general anesthesia using a transperineal guidance grid, with adjuncts (rectal saline displacement and urethral warming) as needed. Outcomes included technical success (complete nonenhancement on immediate post-ablation MRI), complications (Society of Interventional Radiology classification), and local control rate.

Results Seven of 12 patients (58%) had prior failed treatments, including radiation, chemotherapy, or surgery. All patients underwent MR-guided cryoablation; one also received subsequent laser ablation. Procedures utilized 2–8 cryoprobes with a mean of three freeze-thaw cycles. Technical success was achieved in all cases (100%) with complete ablation coverage on post-procedure MRI. At 6 months, local control was achieved in 9 of 12 patients (75%), demonstrating near-complete or complete resolution on post-gadolinium MRI after a single session. Residual disease occurred in 2 of 12 patients (17%): one resolved after brachytherapy, and one underwent repeat cryoablation with follow-up pending. One patient developed recurrence outside the ablation zone at 6 months and achieved complete resolution after repeat cryoablation followed by laser ablation for lesions too tiny for iceball ablation. Patients with initial resolution maintained disease control over roughly 6 years of follow-up. Minor adverse events occurred in 2 of 12 patients (17%), including symphysisitis (managed with antibiotic) and transient urinary retention; no major complications were observed. Increased vaginal drainage occurred in four patients and was considered expected post-procedural change.

Conclusion: MRI-guided cryoablation is a feasible and safe modality for the treatment of recurrent vaginal/paravaginal lesions in anatomically complex pelvic regions.

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Paravaginal metastasis from endometrial cancer in a 65-year-old with prior failed radiation therapy. (a) PET/MRI: diffuse vaginal thickening with FDG-avid nodularity in the right fornix, adherent to the lateral pelvic wall (white arrow). (b) T2-intense focal nodularity in the right fornix (white arrow). (c) Intraprocedural T2: ice ball fully encompassing the lesion. (d) Twelve-month post-ablation MRI: expected post-cryoablation changes without recurrence.

Title: MR-Guided Percutaneous Cryoablation for Extrapelvic Endometriosis: Safety and Pain-Relief Outcomes in a Five-Patient Case Series

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Purpose: To evaluate the safety and efficacy of MRI-guided cryoablation for symptomatic extrapelvic endometriosis, with focus on pain relief, residual disease, and complications across varied anatomical locations.

Materials and Methods: Retrospective review of five women (mean age 35 years; range 26 to 45) treated between 2018 and 2025. Lesion locations: anterior abdominal wall (n=4) and sciatic notch (n=1). Lesion size ranged from 1.8 to 5.0 cm. All procedures were performed under general anesthesia using combined MRI and ultrasound guidance, with 2 to 6 cryoprobes and 1 to 3 freeze-thaw cycles. Hydrodissection and intraoperative neural monitoring were used when lesions were adjacent to critical structures. Technical success, pain scores, residual disease, and complications were recorded [1,2].

Results: All five procedures were technically successful. Mean follow-up was 58 months (range 5 to 88). All patients (100%) reported substantial to complete resolution of targeted pain. One patient with sciatic notch endometriosis (20%) developed a permanent peroneal nerve deficit with motor weakness and sensory loss despite achieving complete pain relief with adjunctive hormonal therapy. One abdominal wall patient (20%) had pain recurrence successfully treated with subsequent MRI-guided laser ablation. Imaging demonstrated decreased lesion size in all cases with variable residual tissue.

Conclusion: MRI-guided cryoablation provides durable pain relief for extrapelvic endometriosis [1,2]. Abdominal wall lesions can be treated safely with low complication rates. Lesions adjacent to major nerves carry a substantial risk of permanent neurologic injury, underscoring the critical importance of patient selection and pre-procedural counseling.

Acknowledgements:

The authors thank the interventional radiology nursing and anesthesia teams at Mayo Clinic for their contributions to procedural care.

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Locoregional Tumor Control of Prostate Cancer Bone Metastasis using MRI-Guided Cryoablation: A Retrospective Cohort Study

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Purpose

To evaluate the efficacy of MRI-guided cryoablation for achieving locoregional tumor control of prostate cancer osseous metastases.

Materials and Methods

This single-institution retrospective study (October 2017–July 2025) included 38 patients (mean age 70.6 ± 7.4 years; range, 57–89) who underwent 46 MRI-guided cryoablation procedures for metastatic bone lesions (most commonly in pelvis [n = 14]) with a mean largest diameter of 15.5 mm. Patients had received on average 6 (maximum 14) prior treatments including prostatectomy, chemo-hormonal therapy, radiation, and radioactive therapy. Twelve out of the 46 (26.1%) targeted lesions had failed prior procedures. Follow-up with 11C-choline/PSMA PET-CT was performed every 3–6 months with a median follow-up of 3 years and a range of 0.5–5 years. Outcomes included locoregional control, disease-free status, recurrence at the treated site, and PSA change at 3 months. Complications were graded per Society of Interventional Radiology (SIR) criteria.

Results

Locoregional tumor control was achieved in 30/38 patients (78.9%) at the first follow-up (mean 3 months). Of the 30, 5 patients (16.7%) were disease-free for the entire extent of follow-up for 2–4 years. Mean PSA decreased from 3.33 ng/mL pre-ablation to 2.71 ng/mL post-ablation (19.2% decrease) at 3 months. Minor complications occurred in 7 patients (18.4%), including paresthesia (n = 2), minor post-ablation non-displaced fractures (n = 4), and bruising (n = 1); no major complications were observed.

Conclusion

MRI-guided percutaneous cryoablation is a safe and effective treatment for treating osseous metastases from prostate cancer providing good locoregional control rates.

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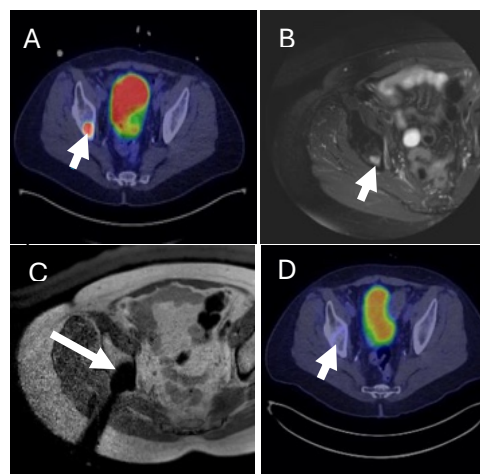


Figure 1. A 57-year-old man with prostate cancer metastasis to the right iliac bone just above the acetabulum. (a) PET/CT demonstrated a PSMA-avid lesion at the margin of the previous ablation prior to treatment (white arrow). (b) The corresponding iliac lesion was visible on T2-weighted MRI pre-ablation (white arrow). (c) Iceball formation during MR-guided cryoablation was observed on MRI (white arrow). (d) Follow-up PET/CT at 102 days post-ablation demonstrated no PSMA uptake at the ablation zone (white arrow).

MRI-Guided Salvage Cryoablation of Localized, Recurrent Prostate Cancer: 13-Year Prospective Study

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Purpose

To evaluate the long-term oncological, functional, and safety outcomes of in-bore MRI-guided salvage cryoablation for biopsy-proven locally recurrent prostate cancer following prior definitive therapy.

Material and Methods

This prospective, IRB-approved single-center study enrolled 90 men with histologically confirmed recurrent prostate adenocarcinoma in the prostate or prostatic fossa who underwent in-bore MRI-guided cryoablation between September 2011 and March 2024. Prior treatments included radical prostatectomy (RP) only (n=17, 18.9%), radiation therapy (RT) only (n=37, 41.1%), or combined RP and RT (n=36, 40.0%). All procedures were performed within a 1.5T wide-bore MRI suite using an MRI-compatible cryoablation system. Cryoprobes were placed transperineally under real-time MRI guidance; a urethral warming catheter and hydrodissection were employed as protective measures. Primary outcomes were PSA reduction at 6 and 12 months and biopsy-proven recurrence rate. Secondary outcomes included procedural safety and patient-reported functional outcomes (AUASS, SHIM, EPIC). Statistical analysis included paired t-tests, Kaplan–Meier survival analysis, chi-squared tests, and one-way ANOVA.

Results

Median follow-up was 109.5 months (range 29.5–161.5). Mean PSA decreased from 3.57 ± 4.38 ng/mL at baseline to 1.41 ± 3.70 ng/mL at 6 months (60.5% reduction; $p < 0.001$) and 1.52 ± 4.88 ng/mL at 12 months (57.5% reduction; $p < 0.001$). Local recurrence-free survival was 95.4%, 90.3%, and 86.2% at 1, 3, and 5 years, respectively. Overall recurrence-free survival was 91.1%, 71.9%, and 62.6% at 1, 3, and 5 years. In-field recurrence rates were statistically similar across all prior treatment subgroups (11.8–13.9%; $p = 0.98$). Higher Gleason grade correlated with increased out-of-field recurrence (40.0% for Gleason 8 vs. 0% for Gleason 6). Of 34 patients with recurrence, 33 (97.1%) underwent repeat cryoablation with favorable outcomes. Urinary function remained stable post-procedure (AUASS 9.9 vs. 9.7; $p = 0.84$). A statistically significant decline in sexual function was observed (SHIM 8.1 to 4.5; $p = 0.002$). One major complication occurred: a vesico-rectal fistula (1.1%; Clavien–Dindo ≥ 3). Minor complications included urinary retention (11.1%), hematuria (8.9%), and dysuria (5.6%).

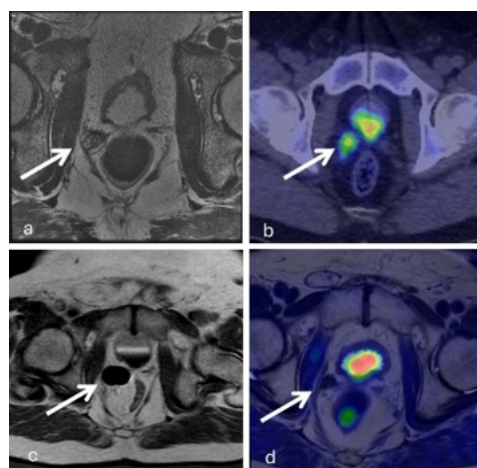


Fig 1: 68-year-old male with local recurrence after RP, demonstrating a durable response to MRI-guided cryoablation. PSA level decreased from 3.6 ng/mL to undetectable post-treatment. (a) Pre-ablation axial T2-weighted MRI - recurrent lesion in the right prostate bed (arrow). (b) Corresponding PET imaging avid uptake within lesion. (c) Intraprocedural T2-weighted MRI captures iceball formation fully encompassing the tumor. (d) PSMA PET imaging 5 years post-ablation-complete resolution of metabolic activity.

Conclusion

In-bore MRI-guided salvage cryoablation achieves durable local tumor control with a favorable safety profile across patients with prior RP, RT, or combined therapy, with median follow-up exceeding 9 years. The procedure preserves urinary function while sexual function decline reflects the heavily pre-treated nature of this cohort. Recurrence, particularly out-of-field, is influenced by tumor grade rather than prior treatment modality. These data support MRI-guided cryoablation as a viable and repeatable salvage option for locally

recurrent prostate cancer.

Acknowledgements

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AI-Assisted Quantification of Cryo-Injury in MRI-Guided Prostate Cryoablation Using Imaging-Derived Dose Metrics

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Purpose

Focal therapy is a partial-gland treatment that offers a therapeutic middle ground between active surveillance and definitive whole-gland interventions. Its primary goal is to kill focal regions of cancerous tissue while simultaneously maximizing the preservation of healthy prostate parenchyma and adjacent critical structures. The goal of this study is to develop an AI-assisted workflow to parameterize damage to EUS and NVB quantitatively during the procedure, and test if those parameters can predict postoperative urinary and erectile dysfunction.

Material and Methods

A retrospective analysis approved by the Institutional Review Board (IRB) was conducted on MRI data from 42 MRI-guided prostate cryoablation procedures, comprising 402 multi-slice image sets (patient age range: 57–84 years). Cryoablation was performed using an MRI-compatible cryoablation system (SeedNet MRI, Boston Scientific, Natick, MA) following the standard protocol of 2 cycles of 15 minutes. A T2-weighted Turbo Spin Echo (TSE) MRI was acquired for treatment planning (TR/TE = 4800/96 ms, flip angle = 150°, matrix = 256×197, in-plane resolution = 0.625 mm, slice thickness = 3 mm, 40 slices). During freezing, the iceball was monitored every 3 minutes using a separate T2-weighted TSE sequence (TR/TE = 2430/104 ms, flip angle = 120°, matrix = 192×134, in-plane resolution = 0.94 mm, slice thickness = 3 mm, 18 slices). Planning and monitoring images were exported in DICOM format and de-identified prior to analysis. The final dataset included 42 planning scans and 402 monitoring scans.

Deep learning models based on 3D U-Net architectures were developed to automatically segment anatomical structures (EUS, NVB, and prostate gland) and the evolving iceball. Using these segmentations, three “quasi cryo-dose” metrics were defined to estimate tissue injury: (i) maximum spatial overlap between the iceball and target structures, representing extent of coverage; (ii) a time-based metric capturing cumulative exposure to freezing; and (iii) a temperature-proxy metric based on distance from the iceball boundary, approximating thermal gradients.

These quantitative metrics were correlated with postoperative urinary and erectile dysfunction identified through chart review and analyzed using logistic regression models. We fitted separate logistic regression models for postoperative incontinence and erectile dysfunction using the proposed Quasi Cryo-Dose estimation approaches. We performed likelihood ratio tests to compare each fitted model against a null model at a significance level of $p=0.05$. Wald tests were further performed for each model to confirm whether the variables are collectively significant or not.

Results

Metrics incorporating temporal and spatial characteristics of cryo-injury demonstrated significant associations with postoperative outcomes. EUS injury was significantly associated with urinary dysfunction in both time-based and temperature-proxy models ($p < 0.05$), while NVB injury was significantly associated with erectile dysfunction in the time-based model ($p < 0.05$). In contrast, the maximum overlap (volume-based) metric did not significantly predict either outcome. Effect size analysis indicated medium-to-large associations for the significant models, supporting their clinical relevance.

Conclusion

Our findings demonstrate that EUS and NVB injuries are predictors of postoperative morbidity. The proposed AI-assisted framework enables objective cryo-dose assessment and provides a foundation for quantitative injury metrics aimed at reducing side effects and improving the safety of focal cryoablation.

Acknowledgements

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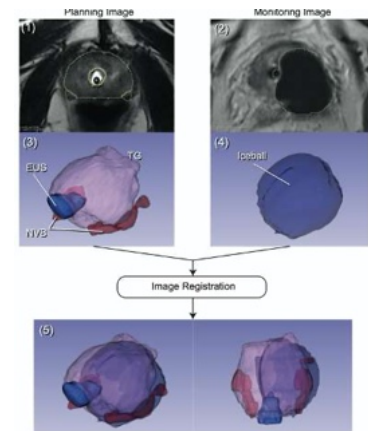


Fig 1: The total gland (TG), external urethral sphincter (EUS), and neurovascular bundles (NVB) were segmented on the planning image.

Effect of Bead Presence, Composition, and Configuration on MRI and CT Artifact Characteristics

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Abstract:

Implantable microdevices (IMDs) have emerged as a promising tool for in situ assessment of therapeutic susceptibility within intact tumors. However, conventional IMDs lack sufficient MRI conspicuity, leading to placement uncertainty and limiting correlation between imaging and therapeutic response. In this study, we systematically evaluated how bead presence, composition, and configuration influence MRI and CT artifact characteristics using phantom, ex vivo, and clinical deployments, with the goal of optimizing IMD design for reliable image-guided implantation and retrieval. Devices were imaged using standardized MRI sequences (Axial TSE T2, Axial 3D VIBE, Coronal HASTE) and CT, with CT serving as the reference modality for localization. Among device configurations, ferromagnetic and metal-anchored devices yielded the largest mean artifactual volumes, frequently exceeding 350 mm³, followed by AxTSE CorHaste. Several low-artifact devices were not detectable on selected sequences in ex-vivo tissue, highlighting tradeoffs between visibility and artifactual size. Standard CT demonstrated lower variability than MRI across all conditions. This study provides a foundation for the design and implementation of a reliable IMD for prostate cancer treatment

MRI-Guided Transperineal Prostate Biopsy in Patients Without Rectal Access: Technique and Feasibility

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Purpose

To evaluate the safety and feasibility of MRI-guided transperineal prostate biopsy in patients without a rectum for standard transrectal ultrasound access.

Materials and Methods

This single-institution retrospective study included 132 patients without a rectum for standard transrectal ultrasound access who underwent MRI-guided transperineal prostate biopsy. Patient characteristics collected included age, PSA level, and PSA density. Imaging findings included prostate volume and PI-RADS score. Procedures were performed under general anesthesia. MRI T2 weighted, diffusion, and post-Gadolinium images were obtained for accurate targeting with fusion of a transperineal guidance grid to the T2 weighted imaging. Percutaneous access to the prostate was performed using an introducer catheter (14Ga Angiocatheter) followed by tissue sampling using an 18-gauge biopsy needle (ITP MR biopsy device). Biopsies were acquired performing both targeted and systematic biopsies. Biopsy-specific variables collected included biopsy type (targeted, systematic, or combined), number of targeted lesions, and lesion location. Pathologic outcomes from targeted and systematic biopsies were analyzed separately with location of positive biopsy findings, and Grade Group. Concordance between MRI findings and biopsy pathology was assessed.

Results

Data analysis is ongoing. Outcomes of interest include overall cancer detection rate, clinically significant prostate cancer detection rate (Grade Group ≥ 2), concordance between MRI findings and pathology, location of positive biopsy findings, procedure duration, technical success, and complication rates.

Conclusion

MRI-guided transperineal prostate biopsy represents an important alternative for prostate biopsy in patients where the rectum has been removed.

Title: Imaging Sequences Used for Prostate Cryoablation and Biopsy Guidance

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Purpose:

To describe the role of multiparametric MRI sequences in differentiating prostate tissue characteristics and guiding targeted biopsy and cryoablation procedures.

Methods:

Multiparametric MRI of the prostate was reviewed, including T2-weighted imaging, diffusion-weighted imaging (DWI) with multiple b-values, and dynamic contrast-enhanced (DCE) imaging. The contribution of each sequence to lesion detection and procedural planning was evaluated from both the technologist and physician perspectives.

Results:

T2-weighted imaging provides detailed anatomical visualization of the prostate and assists in identifying areas of abnormal tissue signal. Malignant lesions frequently demonstrate low T2 signal intensity, although inflammation and fibrosis may produce similar appearances. DWI evaluates tissue cellularity and highlights areas of restricted diffusion, which are commonly associated with prostate cancer. DCE imaging assesses tissue perfusion and vascularity, identifying regions of abnormal enhancement that may correspond to tumor angiogenesis and clinically significant disease.

Conclusion:

Multiparametric MRI combines anatomical and functional imaging to improve lesion characterization and localization. The integration of T2-weighted, DWI, and DCE sequences provides valuable information for accurately targeting prostate lesions during biopsy and cryoablation procedures.

Image Guided Focal Therapy for Prostate Cancer: A Review of Patient Selection, Imaging, and Treatment Techniques

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Purpose

Whole gland therapy with prostatectomy or radiation has historically been considered the standard of care for localized prostate cancer, but carries a substantial side effect profile including urinary incontinence and erectile dysfunction. Focal therapy targets the index tumor focus responsible for the majority of total tumor volume, recurrence, and Gleason grade, offering a favorable side effect profile without precluding subsequent therapeutic options. This educational exhibit reviews patient selection, imaging requirements, and image guided treatment modalities for primary, locally recurrent, and metastatic prostate cancer.

Materials and Methods

This exhibit synthesizes institutional experience and published literature on focal therapy for prostate cancer. Patient selection criteria are reviewed, emphasizing high quality multiparametric MRI and targeted biopsy for accurate localization and grading of low to intermediate risk disease, with multidisciplinary review prior to treatment. Technical specifications for diffusion weighted, T2 weighted, and dynamic contrast enhanced sequences are detailed, alongside the complementary roles of PET and MRI in local staging and detection of extraprostatic extension or metastatic disease. Representative cases illustrate the major focal therapy modalities, including MRI guided and ultrasound guided cryoablation, high intensity focused ultrasound, laser ablation, and irreversible electroporation, with attention to technique and monitoring. Application of these modalities in the locally recurrent and metastatic settings is also reviewed, along with institutional follow up protocols using serial PSA, repeat mpMRI, and biopsy of the ablation margin.

Results

Focal therapy provides a targeted alternative to whole gland treatment, with favorable rates of urinary continence and erectile function preservation. High quality mpMRI is essential for lesion detection and accurate disease localization, while PET imaging, particularly with PSMA based tracers, improves assessment of recurrent and metastatic disease. Reported short term outcomes for MRI guided cryoablation demonstrate local control rates of approximately 90% and complication rates under 5%, with potency and continence maintained in the majority of treated patients. Biochemically recurrent disease following prostatectomy or radiation occurs in approximately 20 to 40% of patients, and targeted focal therapy, most often MRI guided cryoablation or laser ablation, can be applied effectively as salvage treatment in this setting. Focal therapy can additionally be used to treat select osseous, nodal, or solid organ metastases, particularly when other therapies such as radiation are contraindicated.

Conclusion

Image guided focal therapy is a maturing, organ sparing alternative to whole gland treatment for appropriately selected patients with primary, recurrent, or metastatic prostate cancer. High quality multiparametric MRI, complementary PET imaging, and multidisciplinary patient selection are central to safe and effective application of these techniques. This exhibit provides a practical framework for modality selection, imaging technique, and follow up in image guided focal therapy of prostate cancer.

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Image Quality Assessment of 0.6T and 1.5T MRI for MRI-Guided Prostate Interventions

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Purpose

High-quality visualization of prostate anatomy is essential for MRI-guided prostate interventions. Recent advances in low-field MRI systems may improve accessibility and reduce system costs, but their suitability for prostate interventions remain unclear¹. The purpose of this study was to compare prostate image quality between 0.6T and 1.5T MRI systems using matched imaging protocols acquired in the same subjects.

Material and Methods

Six healthy volunteers (age 38.8±10.8 years, weight 86.7±22.3 kg) underwent prostate MRI examinations on a 1.5T Ingenia system and a 0.6T Ambition X system (Philips Healthcare, Best, The Netherlands). Comparable imaging protocols included axial T2-weighted fiducial localization imaging for procedural planning, diffusion-weighted imaging (DWI), T2-weighted imaging for assessment of cryoablation freeze-thaw cycles, and dynamic T1-weighted gradient-echo imaging (eTHRIVE). Images were anonymized and independently reviewed by two experienced readers. For each patient, four corresponding sequences were compared in a side by side and field strength blinded format: axial T2 weighted imaging, high b value (b1600) diffusion weighted imaging, T1 weighted fat suppressed three dimensional gradient echo imaging (THRIVE), and an intraprocedural thermal monitoring sequence (FREEZE). Two independent readers graded each sequence pair using a five point comparative scale ranging from 1 (0.6T superior to 1.5T) to 5 (1.5T much better than 0.6T), with 3 representing equivalent quality. Per patient scores were summed across the four sequences, yielding a possible range of 4 to 20, with 12 representing overall equivalence.

Results

Total protocol acquisition times were 14 min 20 s and 10 min 19 s for the 0.6T and 1.5T systems, respectively. All volunteers successfully completed imaging on both systems. Representative images demonstrated preservation of key prostate anatomical structures at both field strengths (**Fig. 1**). Across 48 total paired comparisons (24 sequence comparisons per reader), both field strengths were rated equivalent in 46% of comparisons, 1.5T was favored in 33%, and 0.6T was favored in 21%. Mean per patient total score was 12.4 of 20 (range 10.5 to 13.5), indicating overall near equivalence with a modest tilt toward 1.5T. By sequence, 1.5T was preferred most consistently for THRIVE (mean 3.50) and FREEZE (mean 3.42), while axial T2 weighted imaging slightly favored 0.6T (mean 2.67). High b value diffusion weighted imaging showed mixed preference between readers (combined mean 2.83).

Conclusion

Low-field 0.6T MRI enabled visualization of prostate anatomy relevant to MRI-guided interventions using clinically practical acquisition times. In this blinded paired comparison, 0.6T and 1.5T MRI demonstrated comparable diagnostic image quality for prostate interventional planning, with 1.5T showing a modest advantage on fat suppressed T1 weighted and intraprocedural thermal monitoring sequences. These findings support the feasibility of 0.6T MRI as an alternative platform for prostate interventional guidance.

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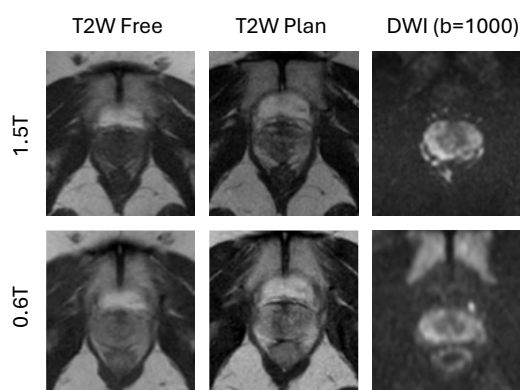


Figure 1. Representative prostate MRI examinations acquired in the same volunteer at 0.6T and 1.5T.

Sequence	0.6T (acq time, res, slice)	1.5T (acq time, res, slice)
DWI	6:09, 2.5 × 3.5 mm, 22 slices	3:17, 2.31 × 2.40 mm, 15 slices
Sagittal T2W	2:20, 0.86 × 0.89 mm, 16 slices	2:03, 0.85 × 1.00 mm, 16 slices
Axial T2W	1:30, 1.50 × 1.65 mm, 12 slices	1:14, 1.50 × 1.78 mm, 18 slices
T2W Fiducial Loc	3:40, 1.70 × 1.20 mm, 60 slices	3:13, 1.20 × 1.34 mm, 60 slices
Dynamic T1W	0:41, 2.0 × 2.0 mm, 60 slices	0:32, 2.0 × 2.0 mm, 60 slices

Table 1. Summary of matched prostate MRI protocols acquired on 0.6T and 1.5T systems, including acquisition time, spatial resolution, matrix size, and sequence parameters used for image quality comparison. (acquisition time, resolution, slices)

MR-Guided Cryoablation of Penile Metastasis – a case series of three patients

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Purpose:

To assess the feasibility and safety of MR-guided cryoablation of penile metastases as an organ-sparing approach to avoid penectomy.

Materials and methods:

Three male patients (mean age 68, two alive and 1 deceased) underwent six MR-guided cryoablation procedures (1 patient received 4 sessions) for penile metastases. The primary malignancy was prostate cancer in two patients and chordoma in one patient. Angiocath guide needles (BD Angiocath™ Special IV Catheter, Becton, Dickinson and Company, Franklin Lakes, NJ, USA) were placed under ultrasound guidance, which were then exchanged for cryoprobes (average 5 cryoprobes). Position was confirmed with MRI. Saline hydrodissection was used in one of three patients to displace and protect the penile skin. A urethral warming catheter was utilized in one of three patients to protect adjacent critical structures. On average, three freeze–thaw cycles were performed per procedure. Gelfoam tract embolization was performed along the needle track in one of three patients to reduce the risk of bleeding after probe removal.

Results:

The first patient underwent a total of four MR-guided cryoablation procedures for recurrent penile metastatic disease. Following each procedure, locoregional tumor control was maintained for approximately 12 months. The second patient demonstrated a favorable treatment response, with decreased lesion size and reduced contrast enhancement on 3-month follow-up imaging. These findings remained stable through 12 months of follow-up. The third patient recently underwent MR-guided cryoablation and is currently awaiting 3-month follow-up imaging to assess treatment response.

Conclusion:

MR-guided cryoablation of penile metastases is a feasible and safe method of as a potential organ-spacing method for locoregional tumor control.

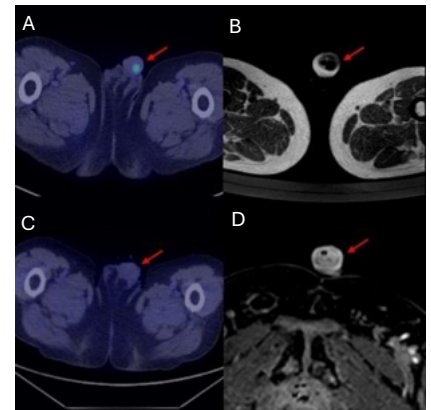


Figure 1. 66-year-old male with solitary penile metastasis from prostate adenocarcinoma. (A) Pre-ablation axial C-Choline PET/CT fusion shows a hypermetabolic metastatic lesion in the glans penis (arrow). (B) Intraprocedural T2-weighted MRI displays the signal void of the iceball (red arrow). (C) Three-month post-ablation axial PET/CT demonstrates complete resolution of metabolic activity in the glans (arrow). (D) T2-weighted MRI showing complete resolution of the lesion.

MRI-Guided Percutaneous Cryoablation of a Symptomatic Penile Neurofibroma in the Setting of Neurofibromatosis Type 1: A Case Report

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Purpose

Neurofibromatosis type 1 (NF1) is an autosomal dominant neurocutaneous disorder characterized by widespread neurofibromas arising along peripheral nerve sheaths. Pelvic and perineal involvement, including along the pudendal nerve and its terminal branches, is well recognized but rare and may produce debilitating pain and sexual dysfunction. We describe the first case of MRI-guided percutaneous cryoablation of a symptomatic neurofibroma at the base of the penis in a patient with NF1 presenting with pelvic pain and erectile dysfunction.

Materials and Methods

A 28-year-old male with established NF1 presented with penile pain and erectile dysfunction of approximately one year duration. Pre-procedural pelvic MRI demonstrated extensive plexiform neurofibromatosis involving the lumbar and sacral plexus and the entire course of the right pudendal nerve, as well as a 2.2 x 1.5 cm solitary neurofibroma located at the left lateral base of the penis immediately anterior to the bulbous spongiosum and adjacent to the dorsal nerve of the penis. After multidisciplinary discussion and exhaustion of conservative options, the patient was referred for MRI-guided cryoablation of the penile neurofibroma. The procedure was performed under monitored anesthesia care in a 1.5T intra-operative MRI suite. The patient was positioned supine with legs in semi-frog-leg position. A transperineal guidance grid was placed against the perineum and fused with localizing MRI to plan a safe needle trajectory to the target neurofibroma. Four IceSeed cryoneedles were placed transperineally into the neurofibroma under real-time MRI guidance with a urethral warmer employed throughout to protect the urethra. Three freeze-thaw cycles were performed with continuous intra-bore MRI monitoring of iceball propagation. Dynamic contrast-enhanced MRI was obtained at procedure completion to confirm treatment coverage.

Results

The procedure was technically successful with no immediate or delayed complications. Post-ablation dynamic contrast-enhanced MRI confirmed good iceball coverage of the target neurofibroma with no identifiable residual enhancement. Follow-up MRI at approximately three months demonstrated interval size reduction of the ablated neurofibroma from 2.2 cm to 1.9 cm in maximum diameter with new homogeneous T2 hypointensity and near-complete absence of contrast enhancement, consistent with successful ablation. Clinically, the patient reported partial improvement in pelvic pain. Erectile function did not improve, attributed to ongoing contributions from the extensive untreated plexiform neurofibromatosis involving the right pudendal nerve and sacral plexus.

Conclusion

MRI-guided percutaneous cryoablation of a penile neurofibroma is technically feasible and safe, achieving radiographic ablation success with partial symptomatic benefit. Real-time intra-bore MRI guidance with a transperineal grid approach and urethral warming permits precise targeting in this anatomically critical region while protecting adjacent neurovascular structures. This case establishes proof of concept for MRI-guided cryoablation as a minimally invasive treatment option for symptomatic penile and perineal neurofibromas in NF1 patients who are poor surgical candidates.

Acknowledgements

The authors have no conflicts of interest to disclose.

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MR guided catheter-based radiotherapy / brachytherapy of small liver lesions – advantages of a tip marked MRI safe wire to define exact catheter position for radiotherapy planning

Matthias P. Fabritius, Vanessa F. Schmidt, Daniel Puhr-Westerheide, Jens Ricke, Olaf Dietrich, Osman Öcal, Max Seidensticker

Introduction: Local ablation using catheter-based radiotherapy/brachytherapy is gaining increasing clinical relevance. Due to the MR compatibility of the material, procedures can be readily performed in the interventional MRI setting. In particular, small tumors are amenable to treatment via MR-guided brachytherapy. However, accurate visualization of the implanted brachytherapy catheter, especially precise localization of the tip, remains challenging, as it appears primarily a signal void on MR imaging. Differentiation from adjacent vessels or extrahepatic structures exhibiting similarly low signal intensity is difficult. The use of an MR-safe wire with a permanent tip marker (0.035“ Amber Wire, stiff 60cm, Marvis) might enable precise spatial identification of the brachytherapy catheter tip, thereby facilitating accurate MR-based dosimetry planning. Thus, an additional CT for radiation treatment planning might be avoided.

Methods: In 15 patients with 23 liver lesions, tip localization was assessed on T1-weighted fluoroscopic MR sequences after MR-guided placement of the brachytherapy catheter, on both with and without an MR-safe wire with tip marker. Subsequently, identification of the catheter course and catheter tip was evaluated using a 5-point Likert scale regarding confidence of catheter/catheter tip localization (1 = not at all confident, 2 = slightly confident, 3 = moderately confident, 4 = very confident, 5 = extremely confident).

Results: Treatment of 23 lesions with a median diameter of 11mm (range, 4-25mm) was analyzed. With the MR marker wire in place, visualization of the catheter course was superior compared to the planning sequence acquired without the wire. Wire tip visibility was rated significantly higher than catheter visibility without wire (median 5 vs. 2; $p < 0.001$; Wilcoxon signed-rank test, $n = 23$). In 17 out of 23 cases, improved catheter tip visualization on fluoroscopic sequences frequently or consistently affected catheter tip definition for radiotherapy planning. In none of the cases an additional CT was needed for radiation planning purposes.

Conclusion: The use of an MR-compatible wire with tip marker improves identification of the brachytherapy catheter course and enables precise tip localization, facilitating fully MR-based radiation treatment planning. This approach reduces the need for post-implantation CT acquisitions for radiation treatment planning and thereby improves the workflows of this iMRI technique.

Outcome after MR-Guided Ablation of Small Liver Lesions with Catheter-Based Brachytherapy

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Aim:

To evaluate the technical performance, safety, and early local control of MR-guided catheter-based high-dose-rate brachytherapy for small liver lesions.

Materials and Methods:

Between November 2024 and April 2026, 62 patients with 130 small liver lesions, defined as lesions of primary or secondary liver cancer with a maximum diameter of 30 mm, underwent MR-guided catheter-based radiotherapy and were retrospectively analyzed. All procedures were performed under conscious sedation and local anesthesia using a 1.5T MRI system (MAGNETOM Sola Fit, Siemens Healthineers) with a 15-cm loop coil.

For lesion visualization, gadoxetic acid was administered at a dose of 0.1 mmol/kg body weight. An 18G coaxial needle was inserted and advanced toward the target lesion under real-time gradient-echo fluoroscopy sequences (TE 3.4 ms, TR 7.3 ms, flip angle 30°) using the iMRI UI Extension research software package. The needle was subsequently exchanged over a wire for a 6F hydrophilic angiography sheath harbouring a 6F brachytherapy catheter.

3D T1-weighted MRI datasets were transferred to the radiation oncology treatment planning system for high-dose-rate brachytherapy using an Iridium-192 afterloading unit. Prescribed target doses ranged from 15 to 25 Gy depending on tumor entity. Treated lesions included hepatocellular carcinoma (24%), neuroendocrine tumor metastases (18%), colorectal cancer metastases (13%), and metastases from other primary tumors (35%). After brachytherapy, catheters were removed while sealing the catheter tract with gelatin sponge.

Clinical and periprocedural data were collected, including dosimetric and volumetric parameters, procedure duration, complication rates according to CIRSE guidelines, and local control. Clinical and imaging follow-up was conducted longitudinally, with follow-up duration varying between patients. Lesion-based progression-free interval was defined as the time from intervention to radiological evidence of local progression or censoring at last follow-up.

Results:

A total of 130 lesions were treated. Median lesion diameter was 9 mm (range of 3–29). Median overall treatment duration in iMRI was 64 minutes (IQR 47–74). The mean delivered dose per lesion, expressed as D100, was 19.5 ± 3 Gy.

Periprocedural and postprocedural complications were uncommon. Minor bleeding occurred in 2 patients (3%) and did not require blood transfusion or additional intervention. No major complications were observed.

After a median follow-up duration of 6 months, local recurrence was observed in 2 lesions (2%). The estimated local control rate was 99% at 12 months, with a 95% confidence interval of 96–100%. Median local control was not reached.

Conclusions:

MR-guided catheter-based high-dose-rate brachytherapy of small liver lesions was feasible and safe in this single-center retrospective cohort. The technique enabled treatment of very small lesions with low complication rates and promising early local control. Longer follow-up and larger prospective studies are warranted to confirm durability of local tumor control.

Real-Time FBG Shape Feedback and Trajectory Prediction for Adaptive MRI-Guided Needle Interventions

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Purpose

MRI-guided needle interventions, including targeted prostate biopsy, focal therapy, and image-guided brachytherapy workflows, require accurate placement of long, flexible instruments into small targets while avoiding nearby critical structures. MRI provides excellent soft-tissue contrast for target definition, treatment planning, and post-placement verification, but it does not continuously measure the full needle shape during insertion. Needle deflection caused by tissue interaction, layered anatomy, and sliding interfaces can therefore lead to targeting error, repeated imaging, and prolonged plan-insert-check cycles. Building on prior FBG sensorized-needle work, this study adds real-time shape reconstruction during continuous robotic insertion and short-horizon trajectory prediction to support adaptive MRI-guided robotic needle placement.

Material and Methods

We evaluated the framework using dynamic robotic needle insertion experiments and a preliminary prediction study. A 20 cm 18G bevel-tip needle instrumented with a four-core multicore fiber containing quasi-distributed FBGs was inserted into homogeneous and layered gel phantoms using a 4-DOF robotic linear stage at 5, 10, and 20 mm/s. FBG-based 3D shape reconstructions were acquired at approximately 100 Hz and compared with synchronized stereo-vision ground truth to quantify time-resolved reconstruction accuracy and needle deflection. To explore predictive feedback, we implemented a configurable prediction framework and evaluated an initial linear-regression baseline. In this feasibility study, the prediction horizon was set to 30 mm, with training data comprising 50,000 synthetic samples from a Lie-group-based model and 1,000 real phantom insertion samples.

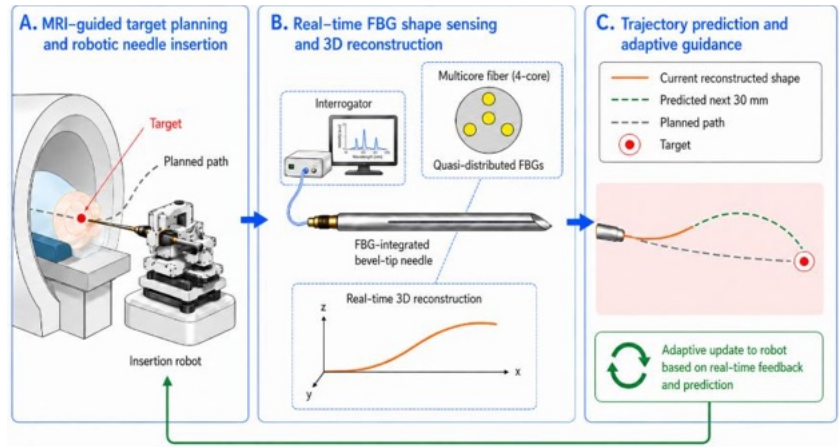


Fig 1: Overview of the proposed workflow for adaptive MRI-guided robotic needle placement. MRI defines the target and planned path; an FBG-integrated needle, connected to an optical interrogator, provides real-time 3D shape feedback; and short-horizon trajectory prediction estimates the next 30 mm of needle trajectory to support adaptive guidance.

Results

During continuous robotic insertion, the FBG system achieved a mean shape RMSE of 0.70 ± 0.31 mm and a mean tip error of 1.28 ± 0.69 mm without temporal smoothing. Insertion velocity significantly affected tip error, whereas needle deflection was primarily driven by phantom configuration over the tested velocity range. In the preliminary 30 mm prediction task, under a single-layer, constant-curvature setting, a linear-regression baseline trained with the hybrid dataset achieved a validation mean tip prediction error of 0.239 ± 0.011 mm, a submillimeter result that is promising for millimeter-scale MRI-guided needle placement.

Conclusion

These results support high-rate FBG sensing as a feasible real-time feedback modality for MRI-compatible robotic needle insertion. By combining dynamic shape reconstruction with early trajectory prediction, the proposed framework may help clinicians monitor needle deviation, anticipate future trajectory, and adapt placement before clinically meaningful targeting errors accumulate. This novel capability is intended to reduce reliance on repeated intraoperative imaging and to support more efficient adaptive MRI-guided needle placement workflows.

Acknowledgements

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Integration of Habitat Analysis in Novel Clinical Trial for Characterization of Ultra-early Response to Radiotherapy

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Purpose: To integrate and assess the performance of the Habitat Risk Score (HRS)¹, an approach for prostate mpMRI analysis, in a novel clinical trial, the Miami UAdapt trial (NCT06111313; PI: Spieler). UAdapt's primary objective is to evaluate the efficacy and modulation of response to Lattice Extreme Ablative Dose (LEAD) radiotherapy (RT)². HRS is utilized at several critical clinical points in UAdapt: **(i)** HRS-Ultrasound fused (HRS-US) guided biopsy (Bx) of the prostate; **(ii)** HRS is used as an initial estimate of Gross Tumor Volume (GTV) that is considered by the physician for defining of radiation boost delivery; and **(iii)** mapping the tumor habitats and radiation dose distribution following delivery of LEAD to guide targeted biopsies. HRS is also utilized for longitudinal analysis of the post-RT mpMRI, acquired at 3, 9 and 12 months.

Materials and Methods: UAdapt is an innovative clinical study designed to enable deep, multiscale characterization of prostate cancer (PCa) before RT, immediately after RT, and throughout longitudinal follow-up. To investigate the biological mechanisms of response, tissue and liquid biopsies are collected at multiple timepoints in order to assess potential markers of progression. HRS was developed to map tumor heterogeneity by assigning a score from 1 to 10 to individual tumor habitats in order of increasing aggressiveness. HRS is implemented at three key timepoints of trial (**Figure**): **(i)** At baseline, patients on UAdapt undergo MRI-US fused Bx during fiducial marker placement. HRS is generated and Bx targets are assigned to sample different habitats. HRS6 (volume defined by all pixels with HRS=6) is assumed to delineate the tumor volume as HRS6 showed strong linear concordance with radical prostatectomy volumes. Three types of targets are assigned: Aggressive habitats (> HRS6); HRS6; and Adjacent Normal (outside of HRS6). **(ii)** HRS6 is transferred to the planning MRI and used for delineation of the GTV. **(iii)** Three weeks after delivery of LEAD RT, a new mpMRI is acquired and HRS is again generated. The patient undergoes an early post-treatment biopsy, where the original habitats are overlaid with the radiation dose distribution to study the interplay between tumor aggressiveness and RT dose. HRS volumes are evaluated at each timepoint post-RT to assess early treatment effects.

Results: Since the inception of UAdapt (October 2024) 18 patients have undergone HRS-US biopsies. In 4 patients the sample from the highest habitat resulted in upgrade: in 2 cases from non-significant PCa, Grade Group (GG) 1 to significant, GG2. In 8 patients the highest GG matched the diagnostic. In 6 patients the HRS-biopsy results in lower grade. It should be noted that fewer Bx cores were acquired relative to the diagnostic Bx. No cancer was found outside and adjacent to HRS6 volumes in 12/18. Following HRS-US biopsy at 3 weeks post-LEAD, cancer cells were detected in 8/18 patients.

Consistently, the HRS Bx cores had a high yield (60 to 100%) tumor, which supports further genomic and spatial transcriptomics studies. The post-tx HRS showed differential patterns; in some patients, volumes remained similar, while in others they were significantly reduced. The associations of these patterns with trial endpoints may result in discovery of an early biomarker for treatment response. To date HRS is generated for 52 exams (baseline 18; and post-RT: 3 weeks: 17; 3 mo – 13 and 9 mo – 4).

Conclusion: HRS has been fully integrated into RT trial. UAdapt was designed to generate unique and novel insights into the early biological response to treatment. HRS supports the identification and assignment of biopsy targets. HRS provides initial estimate of GTV, and volumes that will be queried for quantitative changes over time. Early results suggest that post-RT alterations in HRS may provide valuable, hypothesis-generating information about treatment response.

Acknowledgements: NCI/NIH: P30CA240139, U01CA239141, U01CA271400, R01CA294824.

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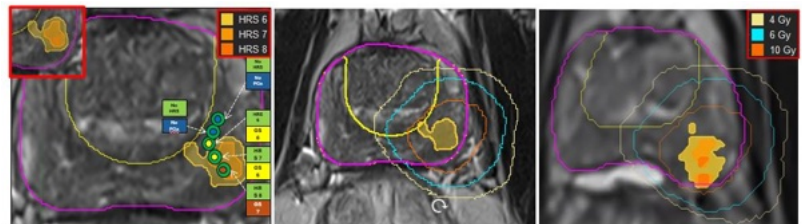


Figure: HRS Implementation at Key at UAdapt Trial Time Points: **Left panel**, HRS for targeted biopsy: Axial T2w with prostate and HRS6 to HRS8 maps delineated. Five biopsy cores (Bx) were acquired: 2 outside of HRS and 3 - HRS6 to HRS8 areas (insert); Both Bx outside of HRS6 were negative for cancer. The biopsy within HRS6-7 resulted in GS6 and in HRS8 – GS7. **Central panel**, HRS for GTV: The volume of HRS6 is used to delineate the GTV and generate radiation plan. **Right Panel**, HRS for post-RT biopsy: Baseline HRS and radiation iso-dose contours are translated to post-RT imaging. Note the presence of hydrogel, implanted on the baseline procedure.

MR-compatible ultrasound tracker for fast alignment of US-MR images

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Purpose: It is widely recognized that fusion of ultrasound (US) and MR images can improve effectiveness of image-guided intervention by leveraging excellent soft tissue contrast from MR [1]. Although several deformable fusion approaches have been proposed [1-3], most of these rely on manual, time-consuming pre-alignment or use trackers that are incompatible with MRI. Electromagnetic (EM) trackers are widely used but their efficacy degrades in strong magnetic fields, making them unsuitable for use inside MRI scanners [1,3]. Optical tracking is compatible with MRI but require continuous direct line of sight, which makes it cumbersome in practical clinical use. We present a novel probe tracker approach for rigid alignment of US and MRI images that is accurate, fast and compatible with MRI environment.

Material and Methods: MRI and US images are acquired simultaneously using an MR-compatible 3D US probe [4], which can be strapped to the body. To enable a simplified practical workflow, we designed a custom 3D-printed US probe adapter that incorporates dedicated slots for 13 fiducials (Fig-1). This design ensures that the fiducials maintain a fixed spatial relationship with the US probe (and in US coordinates) regardless of probe placement. The designed probe adapter does not impede acoustic coupling and uses MR-compatible materials for 3D printing. We use small ~7mm spherical shaped oil-based fiducial markers that produce high-contrast MRI signals, which enables reliable and efficient detection in MRI. We use Otsu thresholding [6] to obtain the MR-coordinates of these fiducials in the pre-operative MRI images (optional operator verification). We parameterize transformation by unit quaternion to avoid gimbal lock. The fiducial locations are strategically chosen to maximize 3D spatial coverage and to ensure that there is plane of symmetry, which is critical to achieve a unique solution [5]. We use closed form solution for landmark-registration [5], to obtain the alignment of fiducial locations in MR and US coordinates in least-squares sense.

Our approach is evaluated in a study with an abdomen phantom and a human subject (under IRB approval). Quantitative error in alignment is estimated by computing distance between homologous locations marked manually by an expert annotator on US and MR images

Results: Fig-2 demonstrate quality of alignment achieved with our approach. The mean distance errors were 1.31mm for phantom and 4.74mm for human subject (supine: 3.37mm; Left Decubitus: 3.36mm; Right decubitus: 7.47mm). Mean processing time was 0.25 sec. on a 16-core 1.4 GHz Intel® processor.

Conclusion: Use of our novel 3D printed US probe adaptor may improve liver intervention workflow with MRI on the scanner table itself. Results demonstrate effective rigid alignment regardless of subject positioning and with sub-second processing time. This would improve workflows for hybrid MRI+US guidance while reducing the burden of more direct image-only registration methods. Overall, hybrid MR+US brings rich soft-tissue MR contrast for needle guidance and enables practical workflow with MRI scanners for liver ablation procedures. Although no actual intervention was performed in this study as the accuracy of image alignment can be evaluated without it, future work aims to integrate this into image guided procedures.

Acknowledgements

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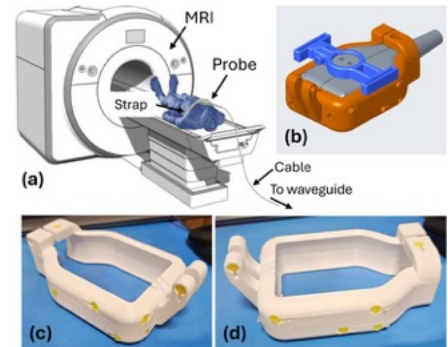


Fig 1: (a) Experiment setup on the MRI table. (b) 3D model showing US probe, strap-swivel, and designed probe-adaptor (orange) with built-in space for MRI visible fiducial marker. (c-d) 3D printed probe adaptor with oil-based fiducial markers (yellow) that is used in this study.

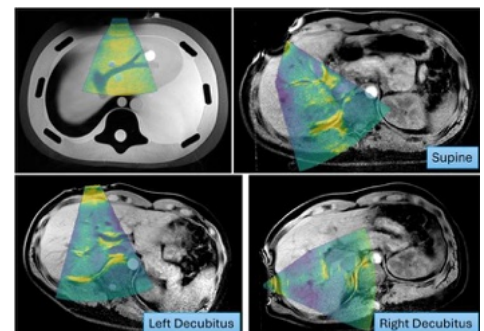


Fig 2: Illustration of rigid alignment achieved using our approach. Transformed US image is overlaid on MR image for abdomen phantom (top left) and on human volunteer for different subject positioning. Note that most anatomical features align across MR and US.

Custom Hardware to Adapt Conventional Flexible MRI Coils for Interventional Applications

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Purpose

Capital costs for dedicated interventional MRI coils are hard to justify given the number of procedures and limited hardware lifespans. We present a simple, rigid, support frame that holds a flexible, clinical GE Air Coil over a body site while maintaining free space for clinicians to interact with the subject. The coil can be lifted off the frame for better patient access, and replaced as needed for imaging. Accommodations for interventional MRI often bear some SNR loss; this is measured via a standard GE phantom.

Materials and Methods

An arch-shaped frame was modeled in CAD software, 3D printed in PLA plastic, and assembled using epoxy resin adhesives (Fig. 1A). SNR analysis was performed using a standard T1w BRAVO GE sequence. Two images of a uniform, spherical GE phantom were acquired using a 36 channel head coil (Fig. 1B and E), as is typical in diagnostic clinical imaging. This was repeated using the arch frame with a flexible GE Aircoil draped over top (Fig. 1C and F) and the table's posterior coils for consistent signal along the A/P axis. SNR was calculated via a difference image. A human volunteer was imaged to assess image quality in tissue.

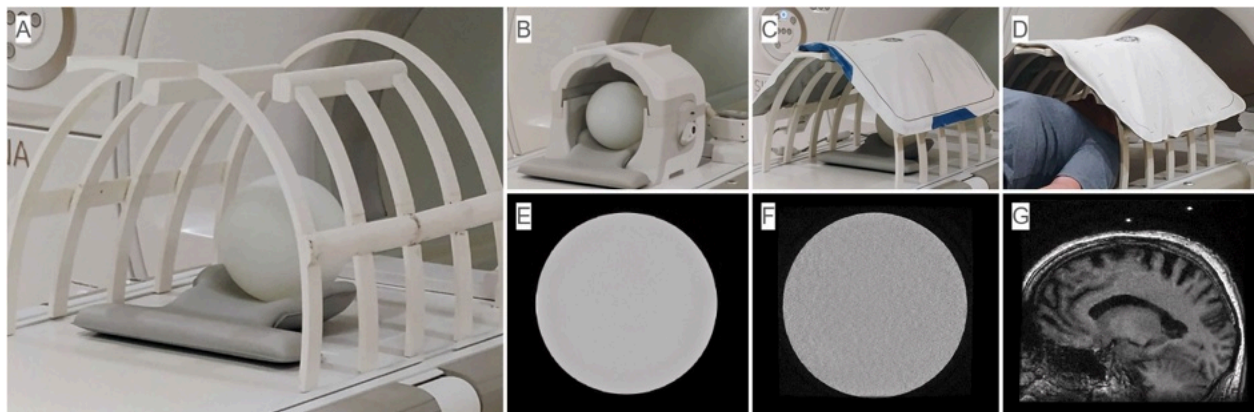


Figure 1: The arch support (A) was used to acquire images in a pseudo-interventional setup (C and D) and compared against a standard diagnostic receive coil paradigm (B and D). As the arch holds the flexible receive coil further from the scan subject, some loss in SNR is expected. Conversely, the typical diagnostic set up shown in B does not provide space for interventional work. In image G, two points of signal superior to the head show the fiducials of a neurointerventional trajectory guide.¹

Results

In the standard diagnostic imaging setup, SNR was found to be 136.08, while under the arch-enabled pseudo-interventional conditions, SNR was 21.09. Anatomical images still bear discernable device fiducials and anatomical structures, indicating preserved interventional utility.

Conclusion

The developed hardware is able to support a flexible coil over patient anatomy while providing clinicians with sufficient space to perform their interventions. Future work will trade off some vertical height for higher SNR by allowing coil closer to the body.

Acknowledgements

We acknowledge GE Healthcare for research support.

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Design and Preliminary Evaluation of an MR-Compatible Microwave Applicator Holder with Motion-Adaptive Compliance for Interventional MRI Procedures

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Purpose

MRI-guided hepatic microwave ablation requires prolonged manual stabilization of instruments, with reported durations of 11.7 ± 5.6 minutes per tumor¹. Limited access within the MRI bore forces the surgeon to maintain an ergonomically unfavorable posture during therapy, motivating the development of MR-compatible holding arms². However, these approaches primarily address static stabilization, whereas respiratory-induced organ displacement introduces dynamic disturbances that challenge accurate applicator placement and require continuous compensation³. This study aims to develop and preliminarily evaluate an MR-compatible microwave applicator holder that provides stable macro-positioning while enabling controlled micro-scale compliance to accommodate respiratory motion.

Material and Methods

An MR-compatible holder consisting of a patient-table docking base, a reconfigurable articulated arm, and a compliant applicator adapter was designed and evaluated in a phantom-based setup. Stable macro-positioning was achieved through a form-fit docking interface and articulated support arm, while controlled micro-scale compliance was provided by passive elastic elements within the applicator adapter. A polyvinyl alcohol cryogel phantom containing different embedded target structures was mounted on an MR-compatible mechanical actuation unit to simulate respiratory-induced organ displacement⁴. MRI data were acquired on a 1.5 T MRI system using T1-weighted BEAT and VIBE sequences. The primary outcome was holder stability during simulated breathing cycles using dynamic 2D BEAT image series (voxel size: $2.34 \times 2.34 \times 4.5$ mm³), evaluating whether the articulated arm remained fixed under respiratory loading. Gadolinium-filled markers (2.2 mM) were integrated into the arm to facilitate MRI visualization. Secondary outcomes were based on analysis of 3D VIBE images (voxel size: $1.5 \times 1.5 \times 1.5$ mm³). Needle-tip landmarks were manually identified in 3D Slicer. Embedded phantom targets served as internal references to determine needle-tip displacement with and without the holder in the superior-inferior, anterior-posterior, and left-right directions.

Results

Real-time MRI assessment demonstrated stable fixation of the holder throughout repeated breathing cycles, with no observable displacement of the articulated arm. MRI-based landmark analysis showed a superior-inferior needle-tip displacement of approximately 6 mm without the holder, which was reduced to 1.2 mm with the holder. In contrast, both anterior-posterior (1.5 mm without vs. 1.9 mm with) and left-right (3.0 mm without vs. 3.4 mm with) displacements remained essentially unchanged.

Conclusion

This proof-of-concept study demonstrates the feasibility of integrating an MR-compatible microwave applicator holder into an MRI-guided phantom environment while maintaining stable macro-positioning under simulated respiratory loading and cable forces. MRI-based evaluation indicated effective suppression of superior-inferior needle-tip displacement, whereas no meaningful effects were observed in the anterior-posterior and left-right directions within the limits of the imaging resolution. These findings highlight the need for further optimization of the compliant stabilization concept and systematic investigation of physiological loading conditions, insertion geometries, and phantom configurations for future studies.

Acknowledgements

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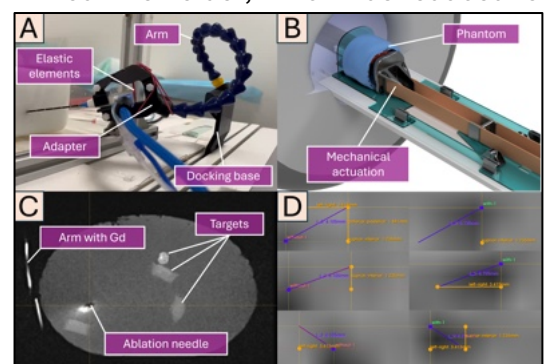


Fig 1: MR-compatible microwave applicator holder and evaluation workflow. (A) Holder assembly. (B) Phantom and actuation unit setup. (C) MR image of arm, needle and target structures. (D) MRI-based displacement analysis in 3D Slicer.

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Dynamic Decoupling of RF Coil Arrays for Interventional MRI

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Purpose

Interventional MRI demands specialized radiofrequency (RF) coil arrays capable of adapting to dynamic surgical environments. These coil arrays must be highly flexible, with individual coil elements independently positionable to conform closely to patient anatomy and accommodate interventional devices [1,2]. However, independent positioning dynamically changes the relative orientation of the coil elements, severely destabilizing decoupling networks, degrading image quality, and potentially causing localized RF heating [3]. In this work, we present a novel dynamic decoupling strategy.

Material and Methods

We decoupled the coil array using floating loops placed above the RF coil elements. Normally, the coupling between coil loops can be canceled by overlapping the loops by a critical distance (~23% of the diameter). We therefore used this as a reference to evaluate image quality under two conditions: 50%-overlapped and zero-overlapped, with and without floating loops. The coil elements were circular loops with a diameter of 85 mm and were constructed from soft, insulated copper wire (Fig. 1, black) segmented by capacitors. Each coil element was connected to a low-input-impedance preamplifier for preliminary decoupling and was actively detuned during RF transmission. The floating loops were an oval loop for the 50%-overlapped array, and a figure-8 loop for the zero-overlapped array (Fig. 1, red). Images were acquired from a phantom on a 3T GE Signa Premier scanner using a 3D FSPGR sequence (TR = 30 ms, TE = 3.2 ms, flip angle = 20°, FOV = 256 × 256 mm², matrix = 256 × 256, slice thickness = 1 mm, NEX = 1, and BW = 31.25 kHz). The coil elements were mounted on a 3D-printed cylindrical former with the phantom positioned inside (Fig. 1c).

Results

Compared with the well-decoupled reference case (Fig. 2a, 2d, and Fig. 3a, 3d, green lines), the 50%-overlapped array without the floating loop produced severe image distortion, and the SNR decreased to only 29.6% of the reference value (Fig. 2b, 2d, red lines). In contrast, with the floating loop, the image remained smooth, and the SNR increased to 83.7% of the reference value (Fig. 2c, 2d, blue lines), representing a 2.8-fold improvement over the configuration without the floating loop. Similarly, in the zero-overlapped array without the floating loop, the SNR decreased to 27.3% (Fig. 3b, 3d, red lines). With the floating loop, the SNR increased to 80.2% (Fig. 3c, 3d, blue lines), representing a 2.9-fold improvement.

Conclusion

Floating loops can significantly improve decoupling between RF coil elements across a wide range of coil configurations, providing a practical and effective decoupling strategy for RF coil arrays that require dynamic positioning in interventional MRI.

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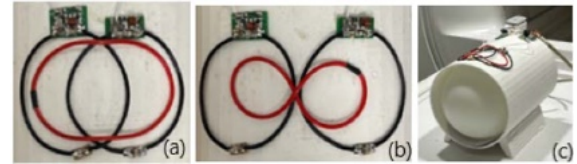


Fig 1. Constructed two-element coil arrays (in black) with floating decoupling loops (in red). (a) Coil loops overlapped by 50%. (b) Coil loops with zero overlap. (c) The coil array with a phantom on the MRI scanner.

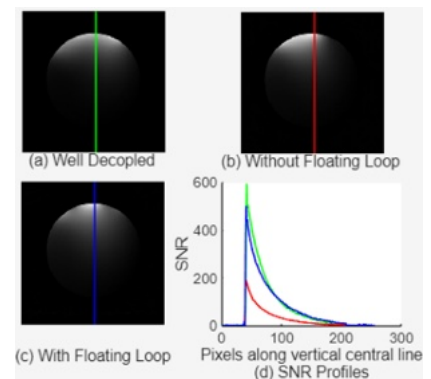
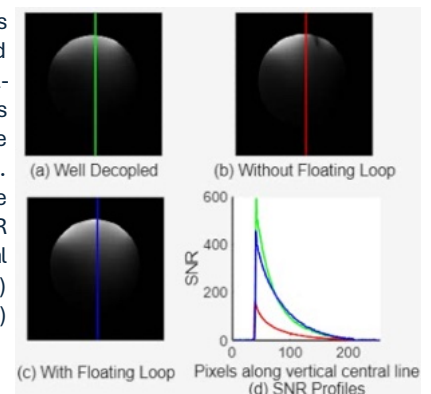


Fig 2. Acquired images from the 50%-overlapped coil array. (a) The well-decoupled image as reference. (b) The image without the floating loop. (c) The image with the floating loop. (d) The SNR profiles along the vertical center line of image (a) (green), image (b) (red) and image (c) (blue).

Fig 3. Acquired images from the zero-overlapped coil array. (a) The well-decoupled image as reference. (b) The image without the floating loop. (c) The image with the floating loop. (d) The SNR profiles along the vertical center line of image (a) (green), image (b) (red) and image (c) (blue).



Comparison of Three Real-Time Marker Detection Algorithms for Instrument Tracking of MRI-Guided Robotic Needle Interventions

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Purpose

Reliable real-time marker localization is essential for instrument tracking of MRI-guided robotic needle interventions. However, tracking information must be extracted from imaging with low spatial resolution and low signal-to-noise ratio due to the rapid acquisition. To identify the most suitable method for MR-based marker tracking under these conditions, this study extends a comparison of phase-only cross-correlation (POCC) and Hough Circle Transform (HCT) for interventional MRI marker detection [1] by including Blob detection and evaluating all three methods with respect to detection accuracy and processing time.

Material and Methods

POCC [2], HCT [3] and Blob detection [4] were evaluated in a phantom-based setup using a custom-made robotic assistance system [μ RIGS, 5] for MRI guided punctures at 30 randomized configurations. Based on a marker concept adapted from Reichert et al. [6], two cylindrical markers encoded rotational and tilting motion, while one spherical marker was linked to the needle feed (Fig. 1(a)). Imaging was performed on a 3T Siemens Skyra using two T1-weighted gradient-echo (GRE) slices capturing cross-sections of the cylindrical markers and one true fast imaging with steady-state precession (TRUFI) slice parallel to the needle axis and intersecting the spherical marker (Fig. 1(a)). Four spatial resolutions were investigated per sequence type (GRE: 1.1–2.5 mm; TRUFI: 1.0–3.0 mm). Localization accuracy was quantified as Euclidean distance between detected and manually annotated reference marker centers. Runtime was measured per detection step, and statistical comparisons used mixed-effects models with corrected post-hoc testing.

Results

Blob achieved the lowest localization errors, outperforming POCC and HCT for the cylindrical markers by approximately 1.5-fold and 1.7-fold in the combined analysis, respectively (Fig. 1(b)), and for the spherical marker by approximately 2.4-fold and 2.2-fold across all investigated resolutions. HCT showed greater variability and false detections in some cases (e.g., noise, phantom structures). POCC required the longest runtimes, whereas Blob showed approximately 10-fold lower median runtime than POCC in both tracking scenarios. Compared with HCT, Blob showed comparable runtime for GRE and approximately 2.7-fold lower runtime for the TRUFI slice, leaving the largest temporal margin for downstream processing.

Conclusion

Compared to POCC and HCT, Blob detection provided the best balance between localization accuracy, robustness, and computational efficiency for the specific μ RIGS marker geometry investigated in this tracking scenario. Future work will integrate Blob-based marker detection into a real-time tracking and control workflow and validate its performance under more realistic dynamic conditions to assess its potential for more adaptive and reliable robotic assistance in MRI-guided needle interventions.

Acknowledgements

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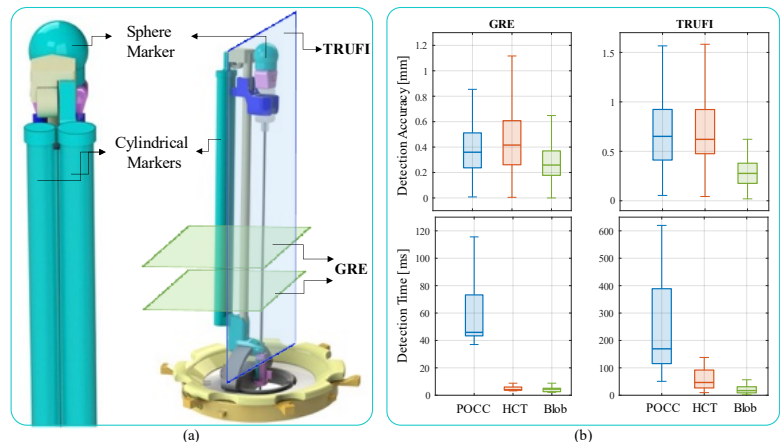


Fig 1: (a) μ RIGS – marker and tracking slice positioning. **(b)** Boxplots – detection accuracy (upper row) and detection time (lower row) for POCC, HCT, and Blob detection for GRE and TRUFI, averaged over all spatial resolutions.

Feasibility of Extended Reality Guidance for MRI-Guided interventions : A Phantom Evaluation in Cardiac and Prostate Applications

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Purpose

MRI-guided interventions rely on 2D image planes for real-time guidance. Although 3D models may also be used, they are confined on conventional 2D displays. This may limit depth perception and understanding of spatial relationships and could increase the cognitive load^{1,2}. This work investigates the potential benefit of using an Extended Reality app for 3D navigation during MRI-guided interventions.

Material and Methods

An Extended Reality(XR) app was created with Unity version 2019.4f and installed on a Microsoft Hololens v1 headset. Bidirectional communication between headset and MRI scanner(Siemens Magnetom Aera 1.5T) was achieved using a previously described architecture³ and utilized to receive tracking data in the app. A reference segmentation of each phantom was created with a coronal 3D bSSFP sequence.

Two interventional scenarios were tested, a cardiac intervention and a prostate biopsy (fig. 1, 2). For the cardiac one, an anthropomorphic cardiac phantom was used along with a commercially available active tracking ablation catheter. A real-time (RT) tracking sequence was used. 10 users, including 3 physicians, completed 2 tasks: reaching the isthmus (flutter ablation) and passing through the septum. Task completion time was recorded and compared to using a conventional screen. For the prostate biopsy, a prostate phantom was made to mimic the pelvic anatomy. A robotic needle manipulator was used to perform an in-bore biopsy and 3 custom active tracking coils provide orientation and needle position information. A RT imaging and tracking sequence (2D bSSFP) was used and 9 users, including 1 physician, were given the task to reach 3 targets inside the prostate. Task completion time and accuracy were recorded and compared to using a conventional screen with a RT imaging sequence (2D bSSFP).

Results

In the cardiac study, user tests showed a 25% reduction in task completion time for reaching the isthmus and 39% reduction for passing through the septum(fig. 3). In the prostate study, user tests showed a 52% reduction in completion time and a 36% offset reduction(fig. 4). Repeat experiments showed similar improvements (52%,32% respectively; n=5).

Conclusion

Preliminary results suggest that Extended Reality guidance could potentially improve navigation and targeting performance during MRI-guided interventions by providing intuitive 3D visualizations of the instrument and the target anatomy. Larger-scale evaluations are needed to validate these observations.

Acknowledgements

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Figure 1: Cardiac test setup and XR navigation including the catheter.



Figure 2: Prostate test setup and XR navigation including the robotic needle manipulator.

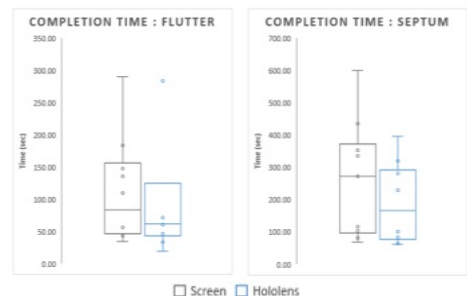


Figure 3: Cardiac User Tests (n=10): Task Completion comparison between Screen and XR guidance.

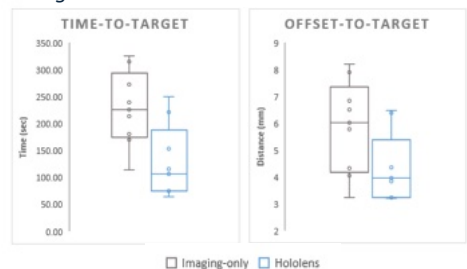


Figure 4: Prostate User Tests (n=9) : Task Completion and Offset comparison between Imaging-only and XR guidance.

From “Stop-and-Go” to “Continuous” in MRI-guided Needle Steering: Addressing Delayed and Sparse Feedback

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Purpose

Image-guided percutaneous procedure, such as MRI and CT, inherently face temporal sensing constraints, including multi-second latencies (e.g., 1.97-5.56 s) and long sensing periods (e.g., 1.4-3.3 s) [1]. These limitations severely degrade the accuracy of continuous closed-loop steering, forcing a time-consuming “stop-and-go” workflow. The objective of this study is to quantify the control error degradation caused by sensing latency and sensing period in continuous closed-loop needle steering. Furthermore, it tests the hypothesis that a delay-aware predictive estimation mechanism lowers control errors under conditions of long sensing periods and latencies.

Material and Methods

This study utilized a simulation and experimental setup mimicking MRI-guided percutaneous interventions. The asymmetric-tip needle was molded using a unicycle kinematic model. The baseline algorithm for continuous closed-loop control was B-CURV [2]. To mitigate sensing limitations, the authors proposed eB-CURV, an enhanced control architecture integrating a delay-aware predictive state estimator based on an Extended Kalman Filter (EKF). Validations were conducted across three environments: A MATLAB simulation environment, a homogeneous gelatin tissue-mimicking phantom, an ex vivo beef block. The hardware consisted of a 2-DoF needle insertion robot and an electromagnetic (EM) tracker, which was intentionally down sampled (1 to 5 s period) and delayed (0 to 5 s) via software to replicate imaging constraints. The primary evaluation metric was the root mean square error (RMSE) from the target trajectory. Performance was evaluated using a one-sided t-test under the most severe constraints (5 s latency and 5 s sensing period).

Results

Simulation and phantom studies demonstrated that increases in both sensing period and latency exacerbate control errors. Notably, latency exhibited a steeper error degradation gradient compared to the sensing period, indicating a more critical impact on control accuracy. Under the severe temporal constraints of a 5 s sensing period and 5 s latency, the proposed 3B-CURV algorithm significantly reduced tracking errors compared to the baseline B-CURV algorithm. In the ex vivo experiment, eB-CURV achieved a 34% reduction in mean error compared to B-CURV, demonstrating statistical significance ($p = 0.046$).

Conclusion

The integration of a delay-aware predictive estimation laser (eB-CURV) successfully mitigates the severe degradation of control accuracy caused by sparse and delayed feedback. These findings indicate the feasibility of transitioning from traditional “step-and-go” workflows to continuous needle steering interventions. This shift could significantly reduce procedure times, decrease the likelihood of prologed tissue deformation, and enable continuous closed-loop control on widely accessible imaging systems with lower temporal specification.

Acknowledgements

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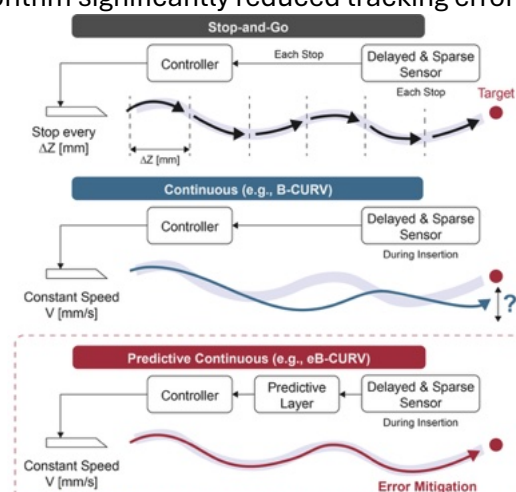


Fig 1: Concept illustration of the proposed approach. MRI sensing constraints conventionally force “stop-and-go” needle steering. While transitioning to continuous insertion is demanded, naïve implementation suffer from delayed and sparse feedback. The proposed predictive approach (eB-CURV) mitigates these limitations.

Bidirectional Continuous Rotation and Variable Curvature Needle Steering (B-CURV) for MRI-Guided Robotic Prostate Biopsy

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Purpose

Prostate cancer is the second-leading cause of cancer-related deaths in the US, and accurate needle placement in trans-perineal biopsy is critical for reliable diagnosis. MRI offers superior tissue visualization, but manual insertions remain vulnerable to tissue heterogeneity and needle-tissue forces that deflect the needle, often requiring repeated insertions with risks of infection and hematoma. Robot-assisted biopsy has been developed to address this limitation. A medical robot employing the Bidirectional Continuous Rotation and Variable Curvature (B-CURV) algorithm in open-loop control has shown potential to reduce tissue trauma and improve needle placement accuracy in homogeneous phantoms while MRI-compatible. However, its accuracy is unknown when it inserts a needle in a homogeneous or non-homogeneous tissue-like environment with or without MRI-guidance. This work investigates the open-loop and closed-loop MRI-guided B-CURV control in said environments.

Material and Methods

A 2-DOF MRI-compatible robot (PLA/brass, ultrasonic motors) was integrated with a GE SIGNA Premier 3.0T scanner via OpenIGTLink, and controlled through 3D Slicer, MATLAB, Arduino, and Galil (Fig. 1). Three experiments were performed: (1) open-loop B-CURV vs. no control insertion in a homogeneous gelatin phantom, (2) open-loop B-CURV in homogeneous vs. three-layer non-homogeneous gelatin phantom (40/100/200 g per 1500 mL), (3) open-loop vs. closed-loop B-CURV in the non-homogeneous phantom with MRI-guidance. Five insertions were completed for each case, however for closed-loop only four insertions were utilized for analysis. Independent t-tests were computed for comparative analysis.

Results

In the homogeneous phantom, B-CURV achieved 2.49 mm root mean square error (RMSE) vs. 8.84 mm for no control-loop steering insertion ($p = 0.02$). In the non-homogeneous phantom, open-loop control accuracy had an error of 8.07 mm RMSE ($p < 0.01$ vs. homogeneous open-loop control). Closed-loop MRI-guided feedback then had a RMSE of 4.44 mm ($p = 0.03$).

Conclusion

B-CURV improves steering in homogeneous tissue compared to no control, but open-loop accuracy degrades predictably in non-homogeneous environments. Without proper feedback on the position of the needle tip during insertion, any discrepancies between internal model predictions and reality can cause compounding error in needle biopsy. Closed-loop MRI feedback recovered targeting accuracy across phantoms and showed improved targeting accuracy in more clinically relevant situations. The data supports that real-time image guidance can improve accuracy under non-homogeneous tissue conditions. This work represents the first demonstration of open- and closed-loop B-CURV inside an MRI bore with a fully compatible robot, establishing a foundation for future clinical translation.

Acknowledgements

The authors thank the WPI Robotics Engineering program and the AiM lab, FUSION lab, and SNR lab for their support.

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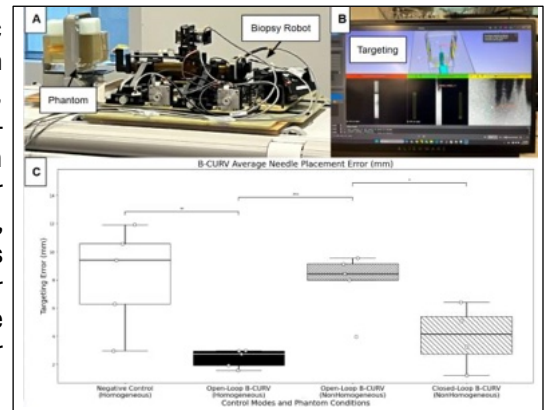


Fig 1: A). Phantom experiment setup. B). MRI guidance and targeting. C). Needle placement error results.

Framework for fast, inline 3D needle localization

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Purpose

MR-guided interventions require continuous device visibility. If a device leaves the imaging slice, the operator must manually realign it, requiring expertise and time. The objective of this project was to implement a framework for automatic needle localization and slice alignment integrated inline on the scanner.

Material and Methods

The proposed research framework integrates a 3D white marker (WM) acquisition, a non-uniform fast Fourier transform (NUFFT) image reconstruction, a convolutional neural network (CNN)-based needle localization algorithm, and a “snap-to-device” user interface button (Fig. 1). The acquisition and localization have been described [1], though with offline processing and without automatic slice alignment. The acquisition uses undersampled radial sampling with an extra gradient moment for WM contrast. The volume of reconstructed magnitude images is passed to the localization algorithm – trained on images from porcine tissue phantoms with a needle in varying poses – which outputs device position and orientation. The “snap-to-device” button triggers automatic alignment of the imaging slices to the needle.

Testing was performed at 0.55T (MAGNETOM Free.XL (510K pending), Siemens Healthineers, Erlangen, Germany). An aspiration needle (Cook Medical, Bloomington, Indiana, USA) was inserted into a porcine phantom $\approx 90^\circ$ to B_0 . Real-time imaging used a 2D balanced steady-state free precession

sequence, with slices initially misaligned from the device. The 3D WM sequence was run with the following parameters: spoiled gradient echo contrast, 256 radial lines, FOV = (256mm)³, matrix = (64px)³, acquisition time ≈ 3.8 s, flip angle = 10°, BW = 160Hz/px. The “snap-to-device” button was selected to align the slices, and real-time scanning resumed. A screen capture recorded the operator’s view of the integrated workflow.

Results

The proof-of-concept phantom experiment showed that all framework components integrated together on the scanner. The framework, including the NUFFT and localization algorithm, runs on standard scanner hardware with no additional computing power. Fig. 2 shows images before and after slice alignment. Before, the device artifact is not visible in either slice. After, the slices orthogonally intersect on the visible artifact.

This framework enables needle localization within the prescribed volume, but outside of the imaging slices, without prior knowledge of the device position. Currently, the operator must stop the real-time sequence to run the localization module separately; future work will merge the two for more streamlined scanning. Localization accuracy was previously reported [1]. This framework could be modified to include other device imaging sequences and localization algorithms, such as for needles with markers.

Conclusion

This work builds on a previously published needle localization by integrating it into an inline framework with a “snap-to-device” feature for slice positioning. Next steps will include testing in an animal model.

Acknowledgements

Cook Medical and Cook Advanced Technologies for supplying the needles and procedural input.

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Slice becomes misaligned from device

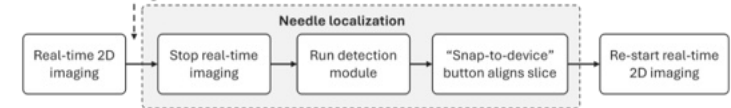


Fig 1: Proposed workflow for automatic needle localization. The localization module includes a 3D white marker acquisition and a CNN-based localization algorithm. A “snap-to-device” button in the UI allows the user to automatically align the imaging slices to the device for subsequent real-time imaging.

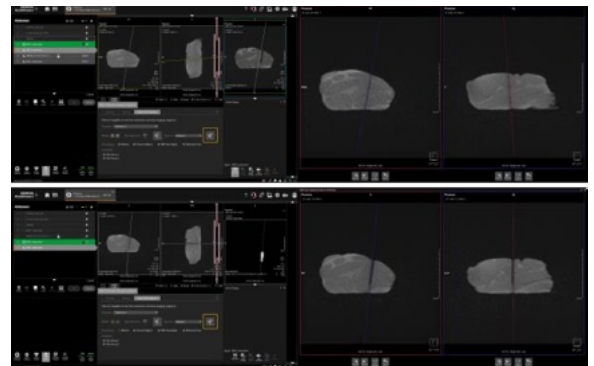


Fig 2: Real-time images of a porcine phantom with a needle inserted. On the top, the two imaging slices are misaligned from the needle, which is not visible. The bottom shows the slices after selecting the “snap-to-device” button. The slice position reference lines now overlap with the hypointense needle artifact. Images are displayed on the right screens using a research software interface for needle interventions. The yellow box indicates a new button to automatically align slices with a saved device position and orientation.

Learning Magnetic Needle Models for Interventional Localization

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Purpose: Metallic instruments used in MR-guided interventions can cause distortions that obscure their location and introduce treatment or biopsy error. This research is aimed toward providing more accurate and precise position information of these metallic instruments.

Methods: We use a differentiable needle modelling approach to localization, in which a magnetic model of a needle is trained using images acquired with the needle in a range of orientations with respect to B_0 , in concert with a forward model of those images that relates them to the field distortion produced by the needle's magnetic basis functions. The model is then deployed to recover the needle's position from novel interventional images by updating its position and orientation to minimize the difference between the forward model-simulated images and the acquired image.

The needle model was defined as the superposition of weighted dipole basis functions, the coefficients of which were trained using phantom images with the needle embedded in a known position, taken in different orientations. Each dipole was simulated individually as $H_z \propto (x^2 + y^2 - 2z^2)/(\sqrt{x^2 + y^2 + z^2})^5$. Amplitudes were tuned separately, then evaluated over a spatial grid and summed to generate an off-resonance map. A nitinol needle's (OD = 1.1 mm) dipole coefficients were trained by matching simulated images to measured images using gradient descent with mean-squared error (MSE) between acquired and simulated images as a loss function. The needle's true position was calculated using fiducial markers embedded in the phantom holder, and dipole positions were distributed in a ring around the known location.

Experiments were performed on a 3T MRI scanner. The nitinol needle was scanned in a jig 3D-printed with clear resin that held the needle in a known fixed location submerged in water, acquired at 5 angles with respect to B_0 . GRE images were acquired with TE = 1.62 ms, readout BW = 1502 Hz/pixel, resolution = $0.8 \times 0.8 \times 2 \text{ mm}^3$. Simulation imaging parameters were matched to experimental parameters. To validate the calculation, a straight needle's modeled map was compared with the analytic solution of the field disturbance from an infinite cylinder (**Figure 1**)¹. Directly modeling the field obviates the need for a convolution to convert susceptibility changes into magnetic field changes⁴. Distorted images were calculated from the simulated field map $\Delta\phi$ ([rad/s]) according to: $img = \mathcal{F}(p e^{-i\Delta\phi TE} e^{-ik_x x_i \Delta\phi T_{readout}})^2$, where p is initial magnetization. Simulated images were 2x oversampled, then downsampled to target resolution to capture intravoxel effects. Simulated and acquired images were normalized and high pass filtered before loss calculations.

For needle location recovery, an initial estimate was calculated from the center of the void caused by the needle. The model was aligned to this void and initialized the gradient descent calculations, that optimized 3D translations and rotations to find the position that minimized the loss between the simulated image and the target measured image.

Results: **Figure 1** shows the dipole summation model matches the analytic solution of an infinite cylinder in a magnetic field, with a Pearson correlation of **0.9821**. **Figure 2** shows the acquired images of the needle in two representative orientations, and the output of the optimized simulated image data. For each image, the needle was jointly optimized across the remaining 4 data sets. Across all 5 volumes tested in a leave-one-out training scheme, the tip of the needle was found with an average accuracy of **1.55 mm RMSE**.

Conclusion: Here we showed a novel method of modelling a needle in a magnetic field that can be applied flexibly to different needle types and different pulse sequences. We demonstrated the ability to train the dipole representation and invert that forward model for the purposes of localization.

Acknowledgements: This work was supported by NIH grant R01 NS120518 and Siemens Healthineers.

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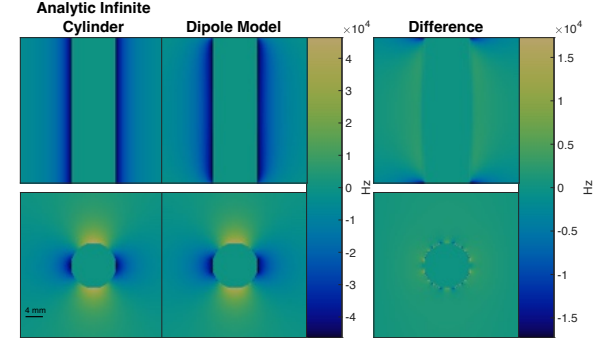


Fig 1: Shown on the left is the analytic solution of a magnetic field in the presence of an infinite cylinder. Shown center is the field in the presence of a cylinder but calculated using dipole methods. On the right is the difference between the two methods. The volumes have a Pearson correlation of 0.9821

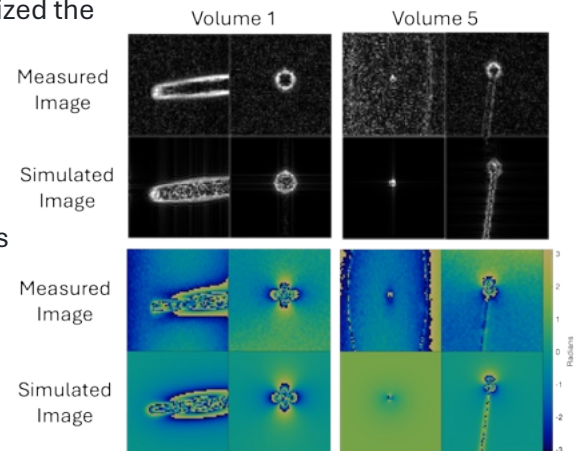


Fig 2: Comparison of measured images and simulated images generated from the trained needle models for 2 representative angles. For both volume a view along the length of the needle and a cross section is shown. Top: High-pass filtered magnitude images for 5 volumes. Bottom: Phase images for 5 volumes.

Automated Ground-Truth Generation for Markerless Stereo RGB Needle Tracking in MRI-Guided Prostate Biopsy

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Purpose

Magnetic Resonance Imaging (MRI)-guided prostate biopsy requires accurate needle localization to support safe and precise targeting. Markerless needle tracking using computer vision and deep learning has the potential to simplify workflow compared with conventional marker-based approaches, but development of such methods is limited by the lack of large, accurately annotated datasets and reliable ground truth required for training and evaluation of learning-based tracking systems [1]. Manual annotation is labor-intensive and difficult to scale. This work presents an automated stereo framework in which marker-based pose estimation is used solely for ground-truth generation, providing a scalable foundation for developing and evaluating future markerless stereo needle tracking approaches in MRI-guided interventions.

Material and Methods

Two cameras (Logitech MX Brio 705) were used in a stereo configuration. Intrinsic and stereo calibration were performed using a ChArUco calibration board. An 18G biopsy needle was instrumented with a rigid cube carrying ArUco markers [2]. For each synchronized stereo image pair, marker detections were matched across views and triangulated to estimate the 3D pose of the cube. Needle tip position relative to the cube was determined through a pivot calibration procedure in which the needle tip was held fixed while the cube was rotated through 30 distinct poses. Using the recovered cube pose and calibrated needle geometry, needle base, shaft, and tip keypoints were automatically transformed into a common 3D coordinate frame, projected into both image views, and converted into heatmap-based training annotations. Automated quality checks, including reprojection and geometric consistency measures, were applied before samples were accepted into the dataset. An overview of the complete pipeline is shown in Fig. 1. The framework generates synchronized stereo image pairs, per-view heatmaps, corresponding 2D image-space keypoints and 3D needle keypoint annotations without manual labeling.

Results

Initial experiments demonstrated successful end-to-end generation of needle annotations from synchronized stereo image pairs. Single-camera calibration achieved mean reprojection errors of 0.26 and 0.24 pixels for the left and right cameras, respectively, while stereo calibration yielded a Root Mean Square (RMS) reprojection error of 0.42 pixels. Pivot calibration using 30 needle poses resulted in a residual fitting error of 1.29 mm. The framework successfully generated stereo heatmaps and corresponding 2D/3D annotations for needle base, shaft, and tip locations.

Conclusion

The proposed framework enables automated generation of high-quality training annotations for markerless stereo RGB needle tracking. Motivated by the requirements of MRI-guided prostate biopsy, it eliminates manual labeling while generating synchronized stereo images with heatmaps, 2D keypoints, and 3D ground-truth annotations. This provides a scalable pathway for creating datasets and evaluating heatmap-based detection, direct keypoint regression, and stereo reconstruction approaches. Future work will focus on validating the final annotation accuracy and investigating marker-independent model training.

Acknowledgements

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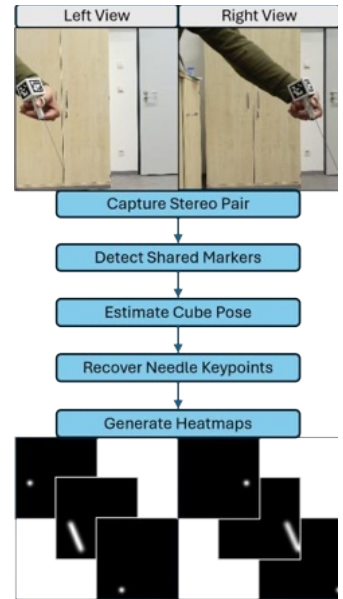


Fig. 1: Automated ground-truth generation pipeline for markerless stereo RGB needle tracking. ArUco detections are used to generate heatmaps and 2D/3D needle annotations.

Use of a robotic-assisted percutaneous ablation platform for MR-guided procedures: Proof of concept experiments

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Purpose

To determine the feasibility of using a robotic-assisted platform for targeting and needle insertion during MR-guided procedures.

Material and Methods

A robotic-assisted intervention system (Epione, Quantum Surgical), hereafter referred to as the Robot, was evaluated for compatibility and targeting performance within an interventional MRI (Ingenia, Philips Healthcare) environment. In the first experiment, commercially available multimodality skin fiducials (IZI Medical) were attached to the Robot patient reference device and positioned directly over ceramic markers used for automatic device registration in CT images (Figure 1). The modified reference device was imaged with MRI to assess susceptibility artifacts. Subsequently, the reference device was attached to a multimodality phantom and MR images were acquired. The images were imported into the Robot, the ceramic marker locations were manually identified, and the system's internal registration accuracy assessment was performed.



Figure 1. Modified Robot patient reference device with skin markers placed over embedded ceramic markers.

In a second experiment, three carbon fiducial markers were implanted into a porcine tissue phantom. The modified reference device was secured to the phantom and MR images were acquired. The images were imported into the Robot, and the ceramic markers were manually labeled to establish image-to-phantom registration. Multiple insertion trajectories targeting the carbon fiducials were planned within the Robot software. Subsequently, the patient table was removed from the scanner room, and the Robot was used to position a cryoprobe along each planned trajectory (Figure 2). Post-procedural CT imaging was performed, and cryoprobe placement accuracy was quantified by measuring the deviation between the planned target locations and the achieved probe positions.



Figure 2. Porcine phantom with modified reference device removed from MRI scan room and in position for Robot assisted needle insertions.

Results

The ceramic markers produced limited susceptibility artifact. The multimodality skin fiducials exhibited image distortion but remained visible, enabling manual localization of the underlying ceramic markers. Manually labeled ceramic marker positions were accepted by the Robot software and demonstrated a registration accuracy within the system tolerance threshold of 0.5 mm (Figure 3).

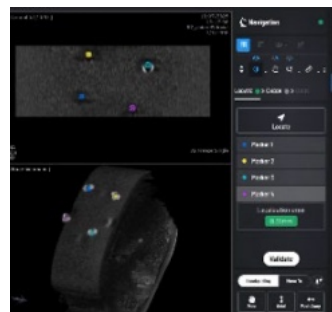


Figure 3. Robot software interface showing modified reference device and manually labeled ceramic marker locations within software tolerance threshold.

For the porcine phantom experiment, post-procedural CT imaging measurements yielded an average cryoprobe to target separation of 9.6 mm (range, 7.3–11.3 mm). Sources of targeting error included uncertainty in identifying the carbon fiducial targets within the MR images in the Robot's viewer and potential probe displacement during transfer from the MRI table to the CT scanner.

Conclusion

These proof-of-concept experiments demonstrate the feasibility of integrating a commercially available robotic-assisted ablation platform into an MR-guided intervention workflow using multimodality fiducial markers for image registration. The proposed approach enabled successful targeting and cryoprobe placement with a minimum targeting error of 7.3 mm. Further optimization of MR imaging to reduce artifact and improve target visualization, may improve accuracy and support future clinical translation.



Figure 4. 3D rendering post-needle insertion CT images showing (A) needle trajectories and (B) needle tip positions relative to carbon fiducials implanted in porcine tissue.

Automatic Needle Tracking in 4D Ultrasound for MR-US Fusion-Guided Ablation

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Purpose

Accurate needle placement is critical to interventions such as thermal ablation but is limited by intraprocedural imaging. MR-ultrasound (US) fusion-guided interventions can improve target visualization. However, needle detection and tracking often require manual, real-time manipulation of imaging planes, which is challenging and highly skill-dependent. Automated needle tracking using 3D US can enable continuous volumetric detection of the needle and reduce operator-dependent manual adjustments. Prior work has demonstrated the feasibility of an MRI-compatible 4D US system [1]. In this abstract, we introduce a straightforward yet effective framework for automated needle detection and tracking in an MRI-compatible 4D US (3D + time) that could support integration into MR-US fusion-guided intervention systems.

Methods

Our proposed algorithm integrates ROI-based foreground detection, 3D clustering, and Kalman temporal tracking. User-provided insertion angle and entry point define a cylindrical region of interest that constrains processing, after which adaptive background subtraction (MOG2) identifies candidate foreground. The Kalman filter leverages prior velocity and insertion direction to predict the needle tip location, update estimates, and provide fallback predictions during temporary loss of detection, thereby improving robustness to shadowing or low contrast.

Numerical simulations: Simulated 4D US data were generated with a homogeneous background containing Rayleigh-distributed speckle noise to mimic tissue appearance, and a Gaussian point spread function to model acoustic blurring. Needle insertion was modeled as a cylinder that progressively extended from an entry face along a configurable trajectory.

Phantom studies: To further validate the algorithm, we tested it on a 3D Multi-Modality Abdominal Biopsy Phantom (Sun Nuclear, Melbourne, FL) using a GE Vivid E95 with a custom MR-compatible 3D matrix array ultrasound probe.

Results

Detection performance was evaluated by comparing the algorithm's predicted binary masks with ground-truth needle masks from the 4D US simulator. All 19 frames containing a simulated needle were correctly identified, with no false positives. At the voxel level, the mean Dice coefficient was 0.70 ± 0.02 , indicating substantial overlap between predicted and ground-truth needle masks. High sensitivity (0.92 ± 0.06) confirms that most needle voxels were captured, while lower precision (0.54 ± 0.02) indicates some over-segmentation near needle boundaries. Preliminary testing on an abdominal phantom also demonstrated successful needle tracking (Figure 1), although quantitative analysis has not yet been performed.

Conclusion

The proposed method enables fully automated needle detection and tracking in the context of MRI-compatible 4D US, providing localization and trajectory information during insertion. These results demonstrate the potential to reduce operator dependence and enable improved needle guidance for MR-US fusion-guided interventions.

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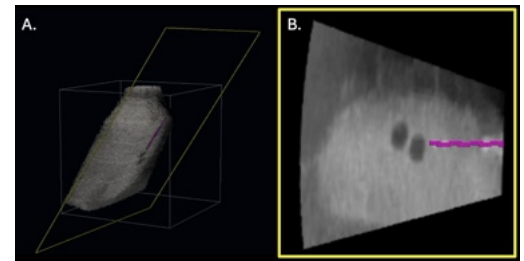


Figure 1: Visualization output of the needle-tracking algorithm in abdominal phantom. (A) 3D rendering of ultrasound volume with the needle track in purple and best plane of visualization outlined yellow. (B) 2D cross-section of the best plane for visualizing the needle.

Needle segmentation for patient-specific thermal models

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Purpose

Microwave ablation (MWA) is a minimally invasive technique used to destroy tumor tissue through controlled heating [1]. However, the presence of blood vessels complicates the procedure, as they act as heat sinks, deforming the ablation zone and increasing the risk of incomplete treatment. Currently, no reliable clinical tool exists for accurate procedure planning. Numerical modeling offers a promising solution through the development of patient-specific thermoablation models. Accurate MWA modeling requires precise needle localization within patient anatomy, as vascular structures strongly influence local heat distribution. This study evaluates the use of a convolutional neural network (CNN) for automatic needle segmentation in this context.

Material and Methods

Fifty-six patients underwent microwave ablation under MR guidance. Post-navigation T1-weighted images were acquired to verify correct needle positioning within the tumor. A 3D U-Net [3] was trained on 139 of those images to automatically identify the needle tip and entry point locations. The deep learning task was performed using the nnLandmark [2] self-configuring framework for landmark detection. Default preprocessing and training parameters were used and the dataset was randomly split into a training set of 111 images and a test set of 28 images. The network was trained for 400 epochs using 5-fold cross-validation, requiring approximately 12 hours in total on a single NVIDIA H100 GPU (with 40 GB VRAM usage). A Binary Cross-Entropy TopK20 loss was employed, focusing learning on the 20% of voxels with the highest loss values. Model performance was evaluated using the Mean Radial Error (MRE), defined as the average Euclidean distance between predicted and reference landmark coordinates.

Results

A MRE of 4.77 ± 3.36 mm for the entry point and 3.87 ± 2.75 mm for the tip was found. Figure 1 shows the error as a function of the likelihood of prediction. Three cases (circled) were excluded from the results due to degraded image contrast, resulting in errors greater than 80 mm. The best and worst cases are shown in Figure 2.

Conclusion

Precise needle localization in T1W MRI images can be achieved using a deep learning-based approach, enabling automated generation of patient-specific models. When appropriately deployed, the framework may also serve as an adjunctive tool to support clinicians during needle placement in clinical workflows.

Acknowledgements

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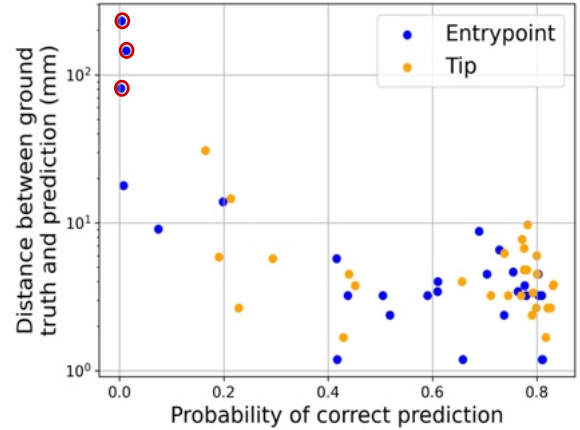


Fig 1: Prediction error as a function of likelihood: distance between ground truth and predicted coordinates in millimeters as function of the probability of correct prediction.

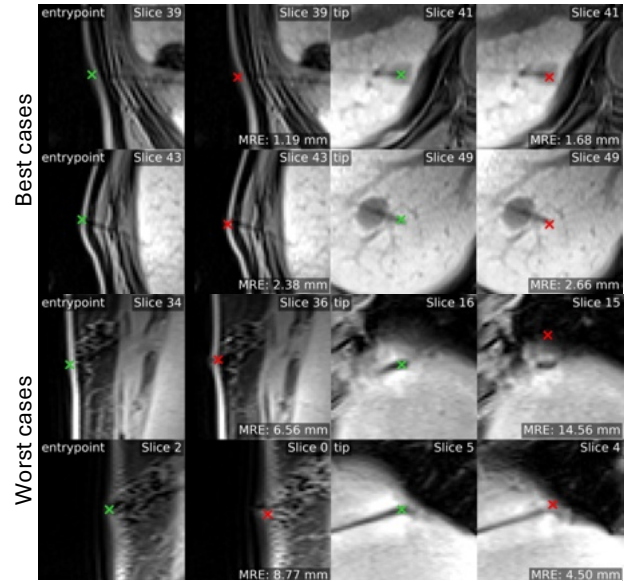


Fig 2: Best and worst cases on the upper & lower sections. Entry point and the tip are indicated: * target, * prediction.

MRI-Guided Microwave Ablation: System Comparison of Shaft Cooling and Thermocouple Regulation on Thermal Dose Distributions and Artifacts

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Purpose

MRI-guided microwave ablation (MWA) provides high rates of local tumor control [1,2,3]. Real-time proton resonance frequency shift (PRFS) thermometry allows for the prediction and modeling of necrosis extent. [1,2,3,4]. However, variations in MWA generator designs, e.g., active shaft cooling vs. uncooled antennas or thermocouple-based temperature control vs. uncontrolled power output, inherently affect tissue heating, delivered power, and MR image quality. This study evaluates how these systematically different MWA system designs impact delivered thermal dose, ablation zone geometry, and MRI thermometry monitoring capability.

Material and Methods

Fifty-four microwave ablations (MWAs) were performed *ex vivo* in bovine livers in a 1.5-T MRI system using real-time volumetric MRI thermometry. Experiments were conducted twice across all combinations of parameters for two MR-compatible systems: 1) an active-cooling system (Saberwave Eco-200G, Eco Medical, Nanjing, China) using four power settings (60, 80, 100, 150 W) and three ablation times (3, 5, 8 min) and 2) a temperature-controlled system (Avecure, MedWaves, San Diego, CA, USA) using four target temperatures (60, 80, 100, 120 °C) and four ablation times (5, 7.5, 10, 15 min). Custom Python scripts were used to process magnitude data with respect to signal loss (due to tissue boiling and susceptibility), to convert phase data to temperature and CEM43 thermal dose maps. Ablation zone dimensions and geometries were evaluated by fitting an ellipsoid (long, mid, and short axes) to CEM43 ablation volumes. Susceptibility artifacts were quantified by the ratio of short-to-mid axes (l_s/l_m), where a ratio of 1 indicates an unaffected dataset. Ablation roundness was assessed via the ratio $l_l/(0.5 \times (l_m + l_s))$.

Results

In the signal loss analysis (Fig.1), signal loss in the active-cooling system strongly correlated with ablation time rather than power settings, conversely, signal loss in the thermocouple-based system correlated more heavily with target temperature than duration. While ablation size was predominantly driven by application time across both systems (Fig.1), geometry varied significantly. Both systems exhibited comparable susceptibility artifact indicators ($l_s/l_m=0.724 \pm 0.017$ for active cooling and 0.736 ± 0.023 for thermocouple-based antennas), Fig.2. However, the active-cooling system yielded significantly rounder ablation zones (1.497 ± 0.012) compared to the more elongated zones of the thermocouple-based system (1.967 ± 0.026), Fig.3.

Conclusion

The different system features impact MR thermometry magnitude signal loss and final ablation geometry without altering the severity of susceptibility artifacts. Active shaft cooling produces significantly more spherical ablation zones. Understanding the generator-specific behaviors is critical for optimizing real-time PRFS monitoring accuracy and tailoring system selection to specific clinical tumor geometries.

Acknowledgements

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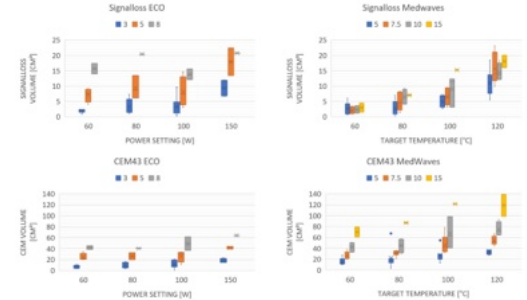


Fig 1: Top: Signal loss, down: CEM43 analysis. Left: Active shaft cooling, right: Thermocouple-based temperature control

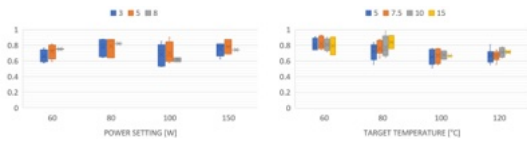


Fig 2: Susceptibility analysis for the dataset l_s/l_m . Left: Active shaft cooling. Right: Thermocouple-based temperature control

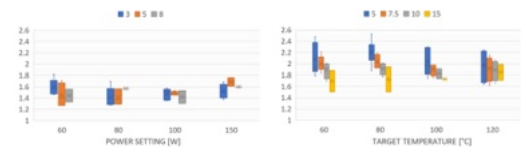


Fig 3: Roundness analysis for the dataset $l_l/(0.5 \times (l_s + l_m))$. Left: Active shaft cooling. Right: Thermocouple-based temperature control

Investigation of water-cooled MRI-guided microwave ablation system on electromagnetic interference and artifacts

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Purpose

MR-guided microwave ablation (MRgMWA) is a promising minimally invasive approach for treating localized tumors (1-3). However, its wider clinical adoption remains limited due to the availability of high-performance MRI compatible MWA systems. This work evaluates the performance, MRI compatibility and clinical integration of an improved and commercially available high-power water-cooled MWA system for MRI-guided interventions.

Material and Methods

Experiments were performed on a 1.5T MRI system using a newly improved Mima-Pro MWA platform capable of delivering up to 100W output power. System modifications included continuous electromagnetic shielding along the entire cable assembly extending to the antenna. Additional RF chokes were incorporated to suppress RF-induced currents and artifacts (4).

Phantom and ex vivo studies were conducted to evaluate image quality, signal-to-noise ratio (SNR), and applicator artifacts. SNR measurements were performed using the same small phantom setup from our prior study to enable direct comparison between configurations: baseline, MW off without RF choke, MW standby without RF choke, ablating without RF choke, MW off with RF choke, MW standby with RF choke, and ablating with RF choke. The relative SNRs were calculated by normalizing the measured SNR to the baseline without the presence of the MWA system.

Antenna artifacts were evaluated using T2-weighted turbo spin echo (TSE) and GRE-EPI thermometry sequences while varying cable routing, RF choke configurations. Clinical configurations with the generator positioned both inside MRI Zone 4 and outside Zone 4 were investigated. Real-time MR thermometry was evaluated during ex vivo porcine tissue ablation.

Results

Enhanced cable shielding and optimized RF choke implementation substantially reduced RF-induced image artifacts. Minimal antenna artifact was observed during GRE-EPI thermometry imaging, while TSE image degradation was significantly reduced.

SNR measurements demonstrated negligible SNR penalty under optimized operating conditions. When the generator was positioned inside MRI Zone 4, RF chokes effectively eliminated the remaining RF-induced noise and artifact issues during standby and active ablation. For configurations with the generator positioned outside Zone 4, careful grounding was necessary to minimize noise. Increasing the number of RF chokes further reduced RF-induced artifacts and noise.

No significant EMI-related MRI image corruption or functional issue of the microwave system was observed during simultaneous MRI and ablation operation. Ex vivo ablation experiments demonstrated stable high-power ablation performance with reliable MR thermometry monitoring throughout treatment and a desirable nearly spherical treatment zone. RF heating assessments showed minimal temperature elevation along the antenna shaft.

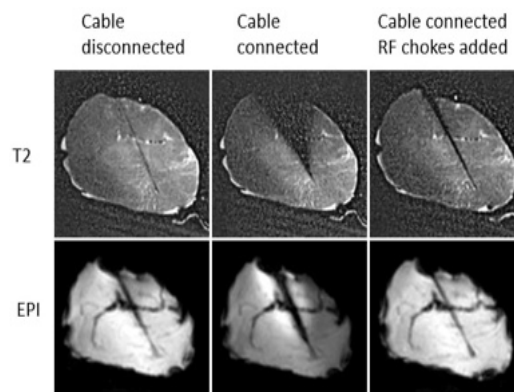
Conclusion

The modified microwave ablation system demonstrated substantially improved MRI compatibility. Practical workflow strategies enabled operation with the generator positioned either inside or outside MRI Zone 4 while maintaining minimal imaging artifacts and negligible SNR impact. These results support the feasibility of integrating high-performance MWA into clinical MRI-guided ablation workflows.

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Fig 1: Impact of pulse sequences, MW cable and RF choke configurations on Antenna artifacts.



Projection-based Receiver Interference Separation Method for Electromagnetic Interference Suppression during MR-Guide Microwave Ablation

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Purpose

Proton Resonance Frequency (PRF) shift MR thermometry enables real-time temperature mapping during microwave ablation (MWA). However, electromagnetic interference (EMI) from the MWA system may degrade image quality and temperature accuracy. Existing EMI-mitigation strategies may require MWA hardware shielding¹, lack real-time processing², or depend on auxiliary EMI-sensing coils³. We introduce the Projection-based Receiver Interference Separation Method (PRISM), a self-calibrating, software-only approach that suppresses EMI artifacts in real-time, demonstrated in phantom and in vivo studies.

Material & Methods

PRISM operates directly on multi-channel k-space data from a standard receiver coil array, requiring no calibration scans, training data, or additional EMI-sensing hardware. The algorithm first constructs two covariance matrices from central and peripheral k-space regions, then performs generalized eigenvalue decomposition on them. The eigenvalues represent the variance ratio of the central and peripheral k-space regions, intrinsically quantifying the signal-to-noise ratio of channel mixture defined by the corresponding eigenvector. Unsupervised thresholding⁴ of log-transformed eigenvalues then classifies components as either signal-bearing or EMI/noise-dominated, and only signal-bearing components are retained for image reconstruction. PRISM was implemented in the Siemens Image Calculation Environment (ICE). Porcine muscle tissue MWA experiments were performed on a 0.55T MRI system (MAGNETOM Free.XL, 510K pending, Siemens Healthineers, Forchheim, Germany). A commercial, non-MR-compatible MWA system (IntelliBlate, Varian, a Siemens Healthineers company, Palo Alto, CA, USA) was positioned inside the MRI scanner room just outside the 5-G line, with a modified prototypical water-cooled MWA probe. MR thermometry was performed using a 2D free-induction-decay echo-planar imaging sequence (EPI-FID) with rate-2 GRAPPA acceleration. MR thermometry was also acquired in one healthy volunteer, using the same sequence with respiratory triggering at end-expiration; here the MWA device was operated in standby without ablation, with the probe placed adjacent to the volunteer. Data was reconstructed without and with PRISM to compare SNR and temperature estimate stability.

Results

Figure 1 shows magnitude and phase from MR thermometry during porcine muscle tissue MWA. PRISM significantly increased image SNR during ablation (uncorrected: 9.17 ± 0.87 ; PRISM: 39.67 ± 2.41 , $p < 0.0001$). PRISM also reduced variations in temperature estimates at both unheated (pt1) and heated (pt2) regions. Figure 2 shows magnitude and phase images acquired in a healthy volunteer with the MWA device operating in standby mode. PRISM visibly improved image SNR by suppressing EMI-induced artifacts.

Conclusion

We developed and evaluated PRISM as a real-time, software-based method for suppressing EMI artifacts during MR-guided MWA. By improving image quality and stabilizing temperature measurements, PRISM may facilitate more reliable MR thermometry during MR-guided MWA. It may also generalize to other interventional settings where EMI from non-MR-compatible devices degrades image quality.

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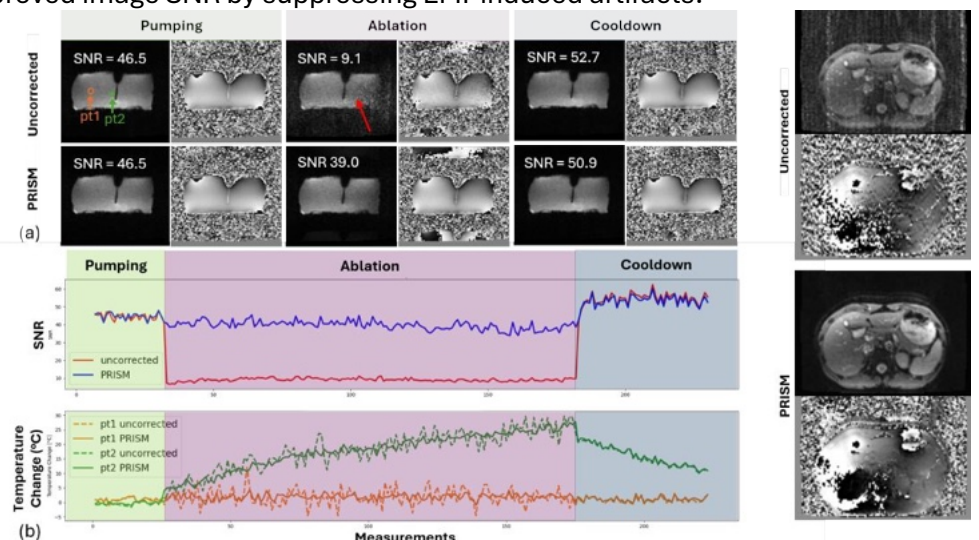


Figure 1. (a) Uncorrected and PRISM-corrected magnitude and phase images during MWA system in pumping, ablation and cool down modes. (b) SNR comparison. (c) Comparison of measured temperature change at unheated (pt1) and heated (pt2) regions.

Figure 2. Volunteer magnitude and phase images without and with PRISM during MWA device in pumping mode.

Technical Validation of Transurethral Ultrasound Ablation (TULSA) on Siemens 0.55T Free.Max MRI

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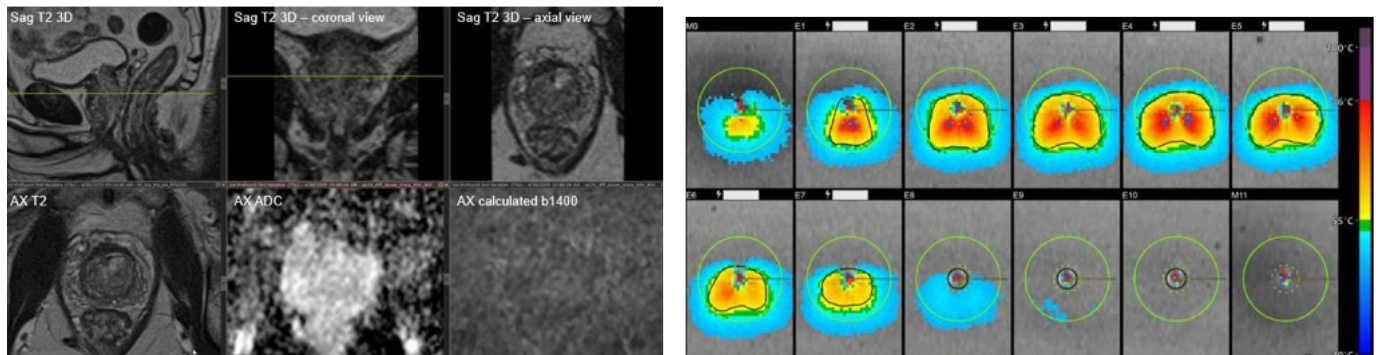
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Purpose: MRI-guided transurethral ultrasound ablation (TULSA-PRO, Profound Medical) is FDA-cleared for 1.5T and 3T magnets. Real-time thermometry guides conformal heating to a physician-defined ablation zone. Higher field strengths improve SNR but worsen susceptibility artefacts from device materials. We explored technical validation of TULSA on the Siemens 0.55T Free.Max, a lower-cost system with an 80 cm bore designed for interventional access.

Methods: Since thermometry accuracy and precision is crucial to the success of TULSA, to account for the expected loss of SNR at 0.55T, 3D segmented EPI sequences were compared to traditional 2D multi-slice (MS) approaches in phantom and volunteer studies. In addition, image quality of biparametric prostate MRI protocols (used intraoperatively during TULSA) were evaluated on volunteers using similar criteria to the PI-RADS standard for visualizing prostate cancer. Siemens Deep Resolve AI was used in image reconstruction to overcome the losses in SNR and to reduce acquisition time. Finally, as a proof-of-concept, a simulated TULSA treatment was performed in a phantom.

Results: The 3D protocol improved thermometry precision by 40% over the equivalent 2D sequence in phantoms. Prostate and adjacent structures were clearly visualized on T2-weighted and diffusion-weighted imaging in volunteers. Thermometry precision at 0.55T approached 1.5T 2D MS performance, with baseline temperature uncertainty of $\pm 1.1^\circ\text{C}$ versus $\pm 1.2^\circ\text{C}$ at 1.5T. Simulated phantom treatments achieved targeting accuracy and precision of 1.0 ± 0.9 mm.

Conclusions: TULSA thermometry and anatomic targeting were technically validated on the Siemens 0.55T Free.Max using 3D EPI sequences with AI reconstruction. The platform offers lower cost, wider bore accessibility, and reduced susceptibility artefacts from hip prostheses, supporting its potential as an interventional MRI platform for TULSA.



Single-center experience with MRI-guided transurethral ultrasound ablation (TULSA): factors affecting treatment duration and outcome

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Purpose

MRI-guided transurethral ultrasound ablation (TULSA) has primarily been evaluated in regulatory studies of whole-gland ablation for favorable-risk localized prostate cancer, with few reports including follow-up beyond one year. As the first US institution to incorporate TULSA into routine clinical use, here we report our single-center experience on focal and whole-gland TULSA with up to 4.5 years of follow-up.

Material and Methods

Among 103 patients treated with TULSA who are enrolled in our ethics-approved institutional database, those with at least 12 months of follow-up after first-line TULSA for localized prostate cancer were included for analysis. Data collection at 3-month intervals included PSA, IIEF, EPIC-26, and adverse events. MRI was collected at baseline and periodically after treatment, along with for-cause biopsy. Baseline and treatment characteristics were correlated to treatment time, as well as treatment failure defined as progression to salvage treatment.

Results

Across 58 patients included for analysis, median follow-up duration was 25 (IQR 18 to 36) months, PSA 6.4 (4.7 to 8.7) ng/ml, histologic Grade Group 1 / 2 / 3 prostate cancer was present in 7% / 81% / 12%, and clinical stage was T1c / T2 in 86% / 14%. On diagnostic prostate MRI, prostate volume was 39 cc (IQR 31 to 47, range 17 to 101 cc), PI-RADS 3/4/5 lesions were present in 19% / 53% / 28%, with capsular abutment in 12%.

Median treatment time of 135 min (IQR 120 to 168, range 89 to 269 min) was longer when a second treatment pass was used (31% of cases, median 139 vs 132 min), and increased with ablation pattern (FTS Classification, examples in Fig 1A): 185 min for whole gland (15.5% of cases), 164 min for subtotal (10%), 129 min for hemi-ablation (65.5%), and 134 min for quadrant or anterior (9%, half had second pass).

Catheter duration was 7 days with rare exceptions, maximum 12 days. Adverse events included GU infections (8.6%), retention (3.4%), and hematuria (3.4%), with no strictures or fistulae. Erections sufficient for penetration were maintained beyond 12 months by 87.5%. 100% of patients had pad-free continence, 94% leak-free.

PSA nadir was 0.9 (IQR 0.5 – 1.8) ng/ml. Secondary prostate cancer treatment was required in 12% of patients at a median of 30 (18 – 31) months after TULSA (n=4 repeat TULSA, 2 EBRT, 1 prostatectomy). Logistic regression found weak associations of treatment failure with higher PSA (adjusted odds ratio 1.19 [1.00-1.42] per ng/ml, p=0.045) and smaller gland volume (aOR 0.90 [0.81-0.99] per cc, p=0.035), but no correlation with grade group, MRI lesion PI-RADS score or location, the ablation pattern, or use of a second sweep (p>0.05).

Conclusion

This single-center experience demonstrates that focal to whole-gland ablation with TULSA is safe, preserves urinary and sexual function, and achieves promising rates of oncologic control at up to 4.5 years of follow-up. Treatment time varies between 2 and 3 hours for focal vs. whole-gland ablation.

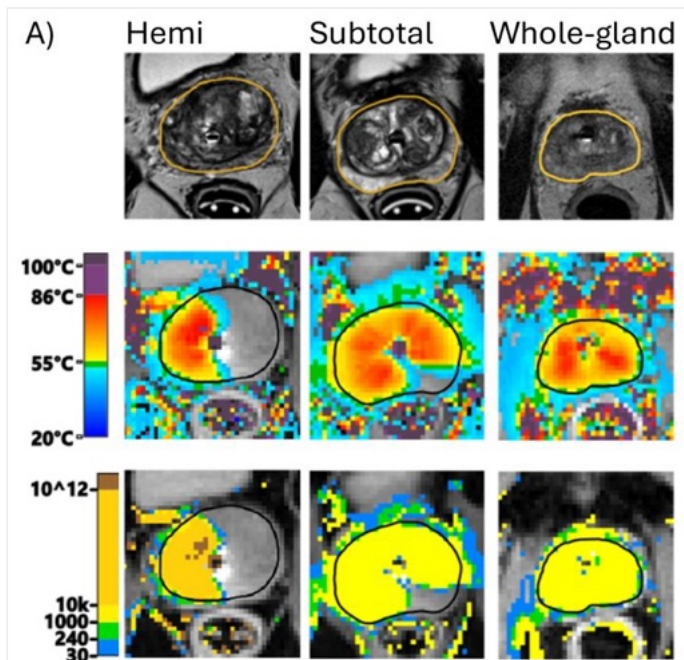


Fig 1: A) Examples of treatment plan (top), maximum temperature image (middle), and thermal dose (bottom) for hemi-ablation (left), subtotal ablation (middle), and whole-gland ablation (right) with MRI-guided transurethral ultrasound ablation (TULSA).

Title: MRI-Guided Transurethral Ultrasound Ablation (TULSA) for Treating Native Prostate Cancer, a prospective single center study

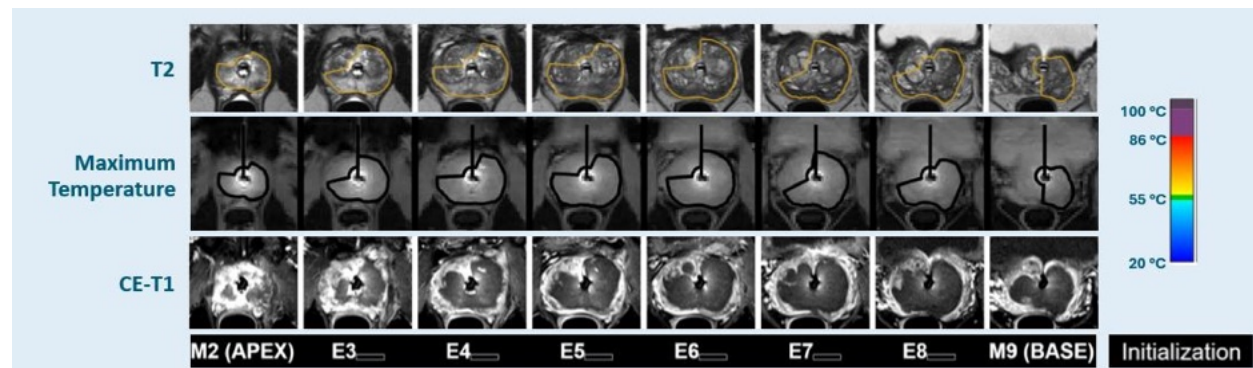
Setayesh Sotoudehnia Korani MD, Ben Leung, Ola A.E. Mohamed MD, Scott M. Thompson MD PhD, Aiming Lu PhD, Daniel A. Adamo MD, David A. Woodrum MD PhD

Purpose: This study evaluates the safety and efficacy of MRI-guided transurethral ultrasound ablation (TULSA) in patients with native prostate cancer, with a focus on PSA response, procedural complications, and oncologic recurrence.

Methods: A prospective analysis was conducted on 41 patients treated with the TULSA procedure (Profound Medical, Mississauga, ON) between March 2023 and early 2025 at a single center. Mean follow-up was 23.7 months (13.6-38.9). All patients had biopsy-proven native prostate cancer with Gleason scores of 6 or 7 (Grade Groups 1 to 3). The procedure utilized a transurethral rigid ultrasound applicator delivering targeted thermal energy under continuous real-time MR thermometry. Pre- and post-procedural data collected included PSA levels at 1 to 3, 6 to 9, and 9 to 12 months; PIRADS scores; Gleason score and grade group; prostate volume; BMI; complication rates; and biopsy-proven recurrence rates. The mean BMI was 28.4 (range 22.2 to 37.6) and the mean prostate volume was 55.1 cc (median 47 cc, range 21 to 111 cc). PIRADS scores were 3 in 4 patients (10%), 4 in 26 patients (63%), and 5 in 8 patients (20%) with available scoring. Statistical comparisons were performed using paired t-tests.

Results: The mean pre-ablation PSA was 7.70 ± 3.82 ng/mL. At 1 to 3 months post-ablation, mean PSA decreased to 2.59 ± 2.09 ng/mL (mean reduction 62.0%, $p < 0.0001$, $n=38$). At 6 to 9 months, mean PSA was 2.32 ± 1.74 ng/mL (mean reduction 65.7% from baseline, $p < 0.0001$, $n=36$). At 9 to 12 months, mean PSA was 2.52 ± 2.47 ng/mL (mean reduction 63.5% from baseline, $p < 0.0001$, $n=32$). Complications occurred in 8 of 41 patients (19.5%) and included worsened erectile dysfunction in 3 patients (7.3%), hematuria in 1 patient (2.4%), new-onset urinary retention with recurrent urinary tract infections in 1 patient (2.4%), new dry ejaculation in 1 patient (2.4%), urinary urgency in 1 patient (2.4%), and one case of initial erectile dysfunction that subsequently improved. Nine patients (22.0%) developed biopsy-proven oncologic recurrence during follow-up, with recurrences classified as in-field ($n=5$, 12%) or out-of-field ($n=4$, 10%) based on post-treatment imaging correlation.

Conclusion: MRI-guided TULSA demonstrates statistically significant and durable PSA reductions sustained through 12 months following focal ablation of native prostate cancer, with a minor complication profile. Larger prospective studies with extended follow-up are warranted to fully define the long-term oncologic efficacy of this technique.



A Simulation-to-Robot Workflow for Directional Needle-Based Therapeutic Ultrasound Ablation

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Purpose

Needle-based hyperthermia therapy uses heat to eradicate tumors through a minimally invasive approach [1]. Conventional 360° diffusing laser probes create uniform, symmetric lesions, which often do not conform to irregular tumors. Directional laser probes and needle-based therapeutic ultrasound (NBTU) applicators can generate arbitrary lesion patterns, but require case-specific tuning of transducer position, rotation speed, and ablation power to deposit heat at selected locations [2]. Designing a cost-efficient trajectory that covers the full target while minimizing healthy-tissue damage remains challenging. One solution is pre-surgical planning in simulation followed by assistive robotic execution to stably reproduce the desired trajectory. Prior work has developed a thermal-ablation simulator and an MRI-compatible neurosurgical robot [3,4]. Here, we investigate the feasibility of a simulation-guided robotic neuro-ablation pipeline and evaluate the simulation-to-real gap between the planned simulation outcome and in-vivo operation.

Material and Methods

This feasibility study used pre-surgical simulation to generate a simple two-step trajectory, then manually controlled the robot to execute it in an in-vivo swine study. The simulator was an interactive finite-element-method (FEM) planning toolkit for estimating tissue temperature distribution during ultrasound thermal ablation [3]. The trajectory delivered power at 0° for 7 min, rotated 135° clockwise, and held for another 7 min to create a “butterfly”-shaped lesion. The robotic assisted system used in this study is WPI AiM Lab MRI-compatible neurosurgical robot [4] that drove a 90° NBTU TheraVision system (Acoustic MedSystems, Illinois) at 7 W. The animal was a male swine weighing 35.7 kg with an ideal body condition score of 3.0/5.0. Temperature and Cumulative Equivalent Minutes at 43°C (CEM43) were recorded using continuous magnetic resonance thermal imaging, including a 5-min pre-ablation baseline and a 10-min post-ablation cool-down. The calculated CEM43 map was compared with the simulation result to evaluate the sim-to-real gap.

Results

Using the robotic-assisted ablation pipeline, we successfully generated a lesion in the in-vivo swine study. As shown in Fig. 1, the lesion had a similar arbitrary shape to the simulation design. The simulated-to-real ablation-area ratio was approximately 2.47:1, mainly because the simulator lacks a perfusion model; power loss and reflection during ultrasound energy transmission also likely reduced actual ablation power below the desired input.

Conclusion

Simulation provided acceptable lesion-shape guidance with a conservative setup, and the robotic system generated the arbitrary shape planned in simulation. This simulation-trajectory-robot pipeline may reduce surgeon workload, improve ablation shape/dose control, increase hyperthermia therapy success, and reduce healthy-tissue over-ablation.

Acknowledgements

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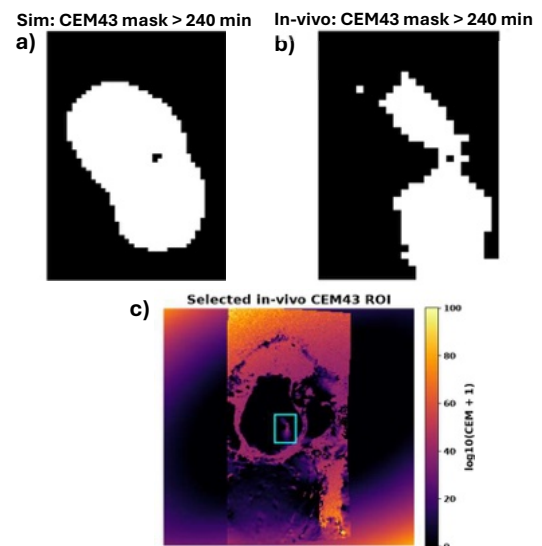


Fig. 1: CEM43 comparison. (a) Simulation: 773 pixels, 0.572700 mm²/pixel, total area 442.697100 mm². (b) In vivo: 243 pixels, 0.738568 mm²/pixel, total area 179.472111 mm². (c) Original MRI CEM43 map with lg(CEM43+1); blue box marks the ablated area for plot (a).

Characterization of MRgFUS Ablation Effects through Radiology-Pathology Data Integration

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Purpose

MRI guided focused ultrasound (MRgFUS) is an incisionless technique enabling precise thermal ablation of oncologic and other deep-seated tissue targets. Noninvasive methods for assessing post-treatment tissue viability are essential; proposed biomarkers such as non-perfused volume and thermal dose attempt to address this need^{1,2}. However, a deeper understanding of MRI sensitivity to the biological effects of FUS, such as necrosis and immune cell infiltration, would inform treatment efficacy and the use of MRgFUS as a standalone or combination therapy. This work aims to identify imaging biomarkers indicative of MRgFUS treatment response from integrated multiparametric MRI (mpMRI) and histology data.

Material and Methods

Rabbits (N=18) were injected with VX2 tumor cells in their quadriceps, and following a period of tumor growth underwent a MRgFUS thermal ablation procedure, with mpMRI collected before and after ultrasound ablation of the tumor. Three days later the rabbits underwent follow-up mpMRI before being euthanized. In vivo mpMRI and H&E data were co-registered using a previously developed multi-stage pipeline^{2,3} (Fig.1) incorporating ex vivo imaging at each tissue processing stage to link in vivo and histology data. Cell detection was performed on the H&E images in QuPath using the built-in watershed-based cell detection algorithm to quantify cellular morphology and staining

characteristics. The nucleus hematoxylin optical density mean (NHM) was selected as an indicator of immune cell presence; NHM values were averaged over all the cells within a coregistered MR voxel. These NHM values served as ground truth labels for a random forest regression model, trained using mpMRI as input features. This model directly interrogates whether and to what extent MRI is sensitive to immune cell infiltration following MRgFUS thermal ablation.

Results

Initial results (N=7 rabbits) indicate a relationship between MRI and histology data. The regression model was able to recover structural patterns in the data (Fig. 2), though notable error remains, likely reflecting a combination of registration imprecision and mpMRI predictive model limitations. Future work will aim to disentangle these error sources and refine the predictive models.

Conclusion

Overall, these techniques show promise for deepening our understanding of cellular-level MRgFUS induced tissue changes detectable by MRI.

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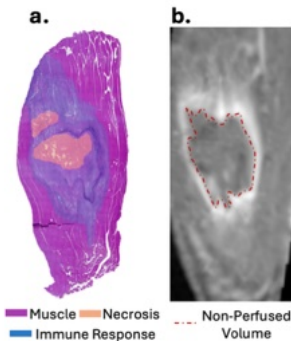


Fig 1: Post registration images of: **a.** H&E slide with muscle, necrosis and immune cell infiltration regions labeled. **b.** T1 weighted contrast enhanced image with annotated non-perfused volume

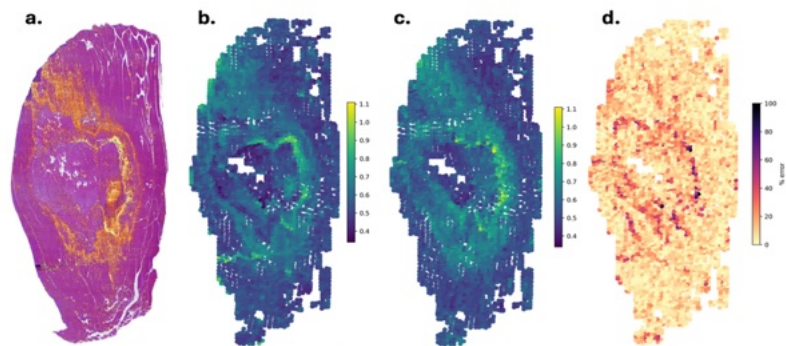


Fig 2: **a.** H&E slide with cells labeled by nucleus hematoxylin optical density mean (NHM) with yellowness indicating optical density. **b.** Registered H&E with NHM binned to MR voxels through averaging **c.** Predicted NHM from MR intensity value **d.** Percent error between **b.** and **c.**

SPION-polymer balloon catheters in 3T MRI: a phantom study

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Purpose

Magnetic resonance imaging (MRI) represents a radiation-free alternative to conventional X-ray fluoroscopy for guiding endovascular procedures. However, one fundamental limitation hindering broad clinical implementation is the sparsity of MRI-dedicated instruments, especially regarding device visualization. The purpose of this study was to evaluate the imaging properties of a newly developed balloon catheter with polymer-embedded superparamagnetic iron oxide nanoparticles (SPIONs). It was hypothesized that these SPIONs enable stable MR visualization by producing a balanced amount of susceptibility artifacts.

Material and Methods

A phantom study was conducted in a clinical 3T MRI environment (Magnetom Vida, Siemens) using a balanced steady-state free precession (bSSFP) sequence with parameters suitable for real-time imaging. The balloon dimensions were 6 mm in diameter and 20 mm in length. Three different balloon catheters were evaluated featuring SPION concentrations of 5, 20 and 30 weight percent (wt%) within the balloon's polymer (1). The experiments were conducted in a custom-made, angle-adjustable phantom setup. The chamber containing the balloon was filled with a gadolinium-doped aqueous solution (1:200).

The SPION concentration as well as the following parameters were systematically evaluated for their influence on susceptibility artifact dimensions: The balloon's spatial orientation in relation to the main magnetic field B_0 at angles of 0° (parallel to B_0), 30°, 60° and 90° (perpendicular to B_0) in two planes. The inflation state of the balloon, inflated (with saline solution) vs. deflated. The influence of the flip angle (FA) (10° to 60° in 10° increments). The influence of phase-encoding direction across different spatial orientations. Artifact dimensions were segmented independently by two blinded, board-certified radiologists.

Results

The embedded SPIONs consistently induced susceptibility artifacts that allowed for a clear identification of the balloon structure in 3T bSSFP imaging.

All three versions of the balloon were visualized sufficiently. Notably, there was no linear relationship between SPION concentration and resulting artifact dimensions, with the intermediate SPION concentration of 20 wt% leading to the smallest artifact dimensions. The spatial orientation of the balloon relative to B_0 had a major impact on visualization characteristic with higher angles generally leading to increased artifact dimensions. FA did not appreciably influence artifact dimensions; however higher flip angles led to increased SNRs. The phase encoding direction had minor influence, with one exception.

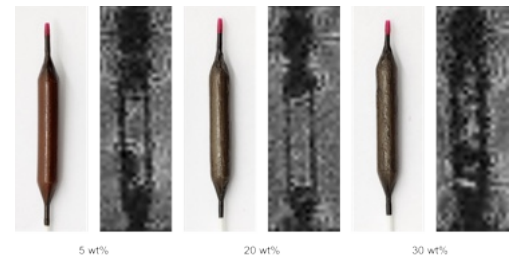


Fig 1: Balloon catheters with different SPION weight-concentrations (5wt%, 20wt% and 30wt%) with corresponding MR-images (central slice, longitudinal orientation). The balloon's brown to greyish color is due to the iron oxide particles.

Conclusion

The SPION-balloons were effectively visualized across different spatial orientations in both inflation states in this phantom proof-of-concept study. The concept of SPION-integration into polymers may be a promising approach for instrument visualization in general, enhancing the prospects of MRI-guided interventions but warrants further investigations under real-world physiological conditions.

Acknowledgements

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Evaluation of an Interventional MRI User Interface Extension for Real-Time Vascular and Cardiac Navigation

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Purpose

While MRI offers unique visualization capabilities, procedural adoption remains limited by the complexity of image navigation and workflow management during interventions. In this work, an enhanced fully integrated research graphical user interface (UI) extension for interventional MRI is evaluated, that expands upon prior work to support cardiovascular navigation using interactive live imaging.

Materials and Methods

Building upon prior development efforts, the research interface (iMRI UI Extension WIP, Siemens Healthineers, Erlangen, Germany) enhances user interaction by enabling real-time targeting and slice positioning directly within the console environment. A configurable orientation management system preserves user display preferences throughout the procedure.

The interface was evaluated during development of MRI-guided vascular and cardiac navigation workflows involving guidewire and catheter visualization, where continuous adjustment of imaging planes and target locations is frequently required. For vascular and cardiovascular navigations, the planning and stamp selection panes allow rapid selection and re-orientation to predetermined slice locations. To emphasize the utility of the tool, we demonstrated and timed navigation from a femoral vessel to the pulmonary arteries.

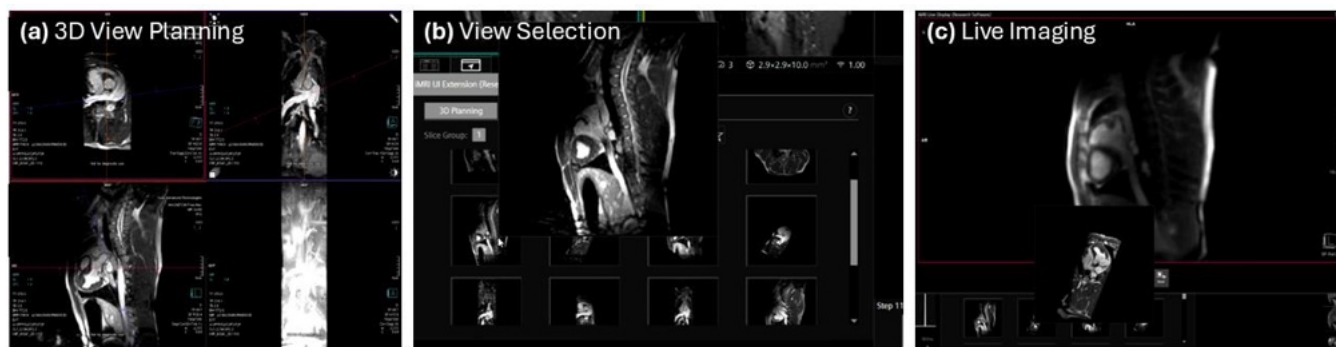


Fig 1: Cardiovascular workflow example. (a) Pre-planning of views on a 3D dataset (b) Each view is represented by a directly selectable thumbnail. Resorting and labeling feasible. (c) One-click access to views during live imaging.

Results

Following localizing images from the pelvis to the chest, 3D images (VIBE and whole heart tfi_navigated) were acquired. The 3D planning tool allowed for orientation of steps to be saved as thumbnails and loaded into the UI extension for easy and quick access during the procedure (cf. Fig. 1).

Once live imaging is initiated, the user can select the stamps to rapidly select the next step in the navigation process, or return to a prior step. Dry runs (without devices inserted or tracked) are typically less than 30 mins, with one case specific time of 26 mins and 30 secs from planning to pulmonary arteries.

Conclusions

The research software is fully integrated into the scanner software and enables easy access to predefined views and improved image visualization. Real-time vascular and cardiac navigation is facilitated while maintaining a simple and robust workflow. Interfaces such as this may play an important role in reducing workflow complexity and accelerating broader clinical adoption of iMRI.

Acknowledgements

We would like to acknowledge Siemens Healthineers, Cook Advanced Technology, and the iMRI Division of Cook Medical for our continued partnership and mutual support in advancing interventional MRI.

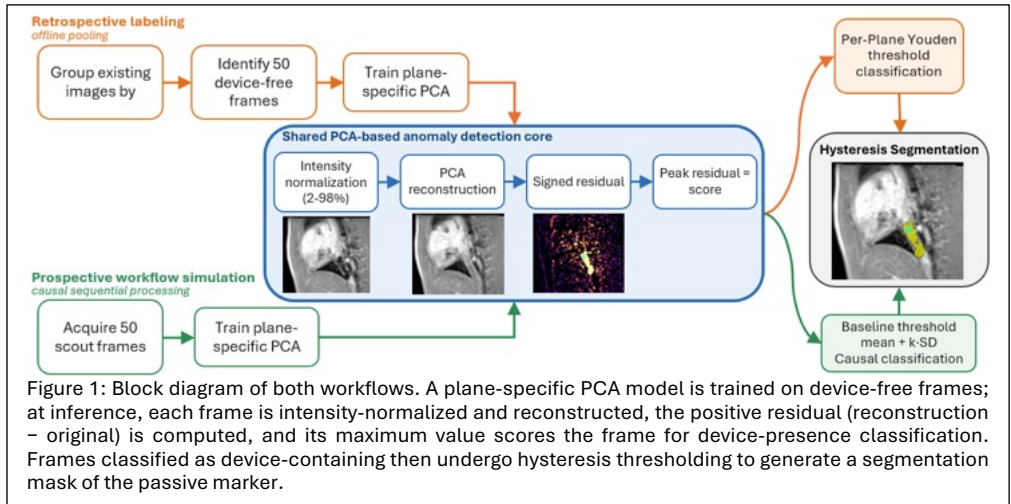
Plane-Specific Anomaly Detection for Passive Marker Identification and Localization in iCMR

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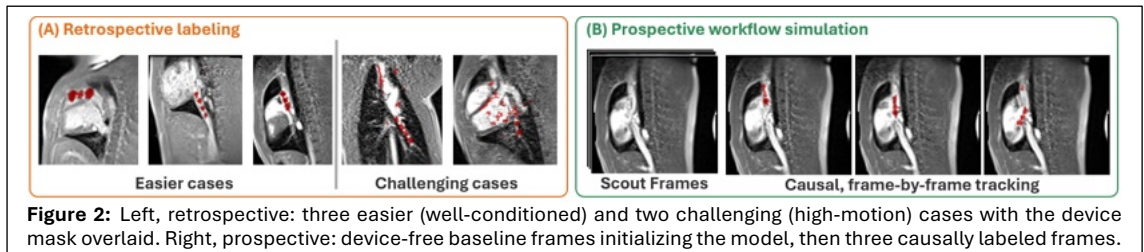
Purpose: Reliable real-time tracking of passive markers remains a challenge in interventional cardiovascular magnetic resonance (iCMR).¹⁻³ A generalized, learning-based tracker would address it, but training such a model requires large labeled real-time datasets that are impractical to annotate by hand. We propose a fast, plane-specific anomaly-detection workflow requiring only few device-free baseline frames and serving two roles toward this goal. **Retrospectively**, it automatically flags and segments passive markers in archived frames, producing labeled data to train future generalized models. **Prospectively**, the same workflow runs causally to provide device tracking within a procedure; because it is plane-specific and baseline-dependent, it is a practical foundation rather than the generalized solution itself.

Methods: The retrospective and prospective workflows share a principal component analysis (PCA)-based anomaly-detection core. A PCA model⁴ learns the device-free appearance of a given imaging plane; when frames from the same plane are passed through it, the proposed workflow classifies device presence and generates a segmentation mask of the passive marker. Detailed workflow shown in Fig. 1.



Real-time spoiled gradient echo images from 4 MR-guided pulmonary stenting studies (0.55 T)², manually labeled for device presence, were used. **For retrospective evaluation**, 2,044 frames from 11 plane groups (1-4 per study) were randomly selected, each group supplemented with 50 device-free training frames; 60 device-containing frames also received manual segmentation masks. **For simulated prospective evaluation**, one continuous series was used: its first 50 device-free frames trained the PCA model, and the remaining 150 frames were evaluated causally. Training and inference latency were recorded.

Results: **Retrospectively**, across 2,044 frames, frame-level classification reached ROC-AUC 0.967. Automatic segmentation agreed well with manual annotations in well-conditioned planes (Dice 0.70 ± 0.14 ; precision/recall 0.75/0.69) but degraded in high-motion planes (Dice 0.49 ± 0.20 ; precision/recall 0.46/0.58; Fig. 2, left). In the simulated prospective workflow, the PCA model trained in 27 ms and then ran causally on each incoming frame in 0.50 ms, segmenting the passive marker (Fig. 2, right), with 0.88 classification accuracy.



Conclusion: A

plane-specific anomaly-detection workflow accurately classified passive device-containing frames in real-time iCMR with minimal manual labeling and ran causally with rapid initialization and sub-millisecond per-frame processing. Segmentation was reliable in well-conditioned planes and coarser under bulk motion — acceptable as weak labels for training, though motivating motion-aware refinement. Because the approach is plane-specific and requires a device-free baseline for each view, it is a practical foundation rather than the generalized solution itself: it both supplies the labeled data needed to train a future view-agnostic tracking model and provides immediate device tracking, pending online scanner integration.

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Dexterous Intramyocardial Needle Ablation (d-INA): Initial Design, Fabrication, and In-Vivo Validation

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Purpose

Intraprocedural MRI guidance is of interest for improving VT ablation because of the ability of MRI to visualize ablation targets throughout the ventricular wall and visualize ablation lesion effects to confirm that the intended ablation has been achieved [1]. However, current MRI-conditional ablation devices are limited in treating ablation targets deeper within the myocardial wall. Intramyocardial needle ablation (INA) addresses this limitation using a conductive needle that penetrates the wall and delivers radiofrequency (RF) current heating from within [2]. However, existing INA catheters are not MRI-conditional, and the steerable sheaths used with them have suboptimal dexterity for navigating within the highly trabeculated LV chamber and steering around obstacles (papillary muscles), complicating ablation in areas where these circuits occur [3]. Recently, first-in-man MRI-guided VT ablation using MR-Tracking and RF ablation was demonstrated [4]. This work presents a novel MRI-conditional dexterous INA (d-INA) device to enable effective manipulation and creation of deep ablation lesions.

Material and Methods

This initial prototype consists of an outer sheath and an inner catheter, using a multi-segment design that makes them independently bidirectionally steerable, along with an integrated ablation needle assembly. Benchtop tests demonstrated that the sheath and catheter reached bending curvatures of 0.088 mm^{-1} and 0.114 mm^{-1} , respectively, and achieved stable planar “C” and “S”, and non-planar shaped configurations. Low-magnetic-susceptibility Nitinol was used for tendons, the inner catheter, and the needle assembly. Tinned copper braiding was used to improve the torsional rigidity of the sheath. The depth of the lesions created was evaluated using TWILITE MRI imaging [5], a sensitive method for assessing acute thermal lesions.

Results

Ex-vivo studies validated the system’s stiffness modulation and lesion creation capabilities. In-vivo experiments in two swine demonstrated the device’s ability to reach hard-to-access locations such as the LV summit region and achieved a 107% increase in ablation depth compared with a standard ablation catheter.

Conclusion

These results illustrate the proposed d-INA design as a promising platform for achieving deep ablation with enhanced dexterity and manipulability, leveraging intra-procedural MRI for ablation targeting and lesion assessment. The device can advance VT treatment by enabling deeper, more precise ablation in hard-to-reach regions. An MRI-conditional device, including MRI-conditional location sensors, is being constructed, which will employ an MRI workflow that detects VT circuits in scar with LGE [1], terminates the lesions with d-INA, and evaluates success with TWILITE MRI imaging.

Acknowledgements

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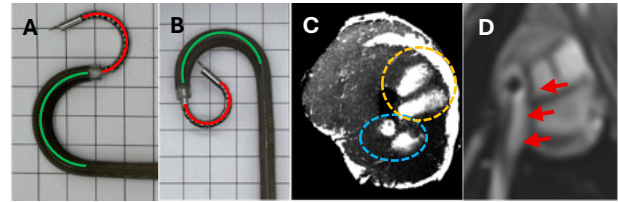


Fig 1: The proposed d-INA device and ablation results under MRI. (A) The “S” shape bending and (B) the “C” shape bending, with the fitted centerlines of the sheath (green) and the fitted centerlines of the catheter (red). (C) TWILITE MRI of lesions created by this d-INA prototype (yellow) vs. a conventional RFA catheter (blue). (D) Real-time MRI frame illustrating passive visualization of the d-INA catheter (red arrows) with acceptable magnetic susceptibility artifact as it is advanced into the superior vena cava.

MR-guided cryoablation of gluteal metastases

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Purpose:

To assess the feasibility and safety of MR-guided cryoablation of gluteal muscle metastases.

Materials and methods:

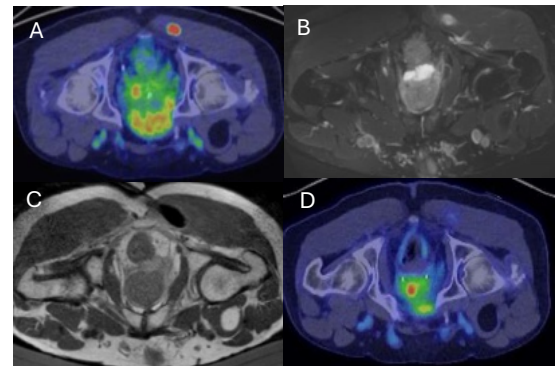
Two male patients, both 69-year-old, with metastatic malignancies underwent cryoablation for painful gluteal soft tissue metastases. One patient with renal cell carcinoma developed a biopsy-proven right gluteal metastatic nodule two years after cytoreductive nephrectomy. The second patient with prostate cancer developed rising PSA levels despite radiation therapy and androgen deprivation therapy, with one of the metastatic sites involving the left gluteal region adjacent to the sacrum. Both lesions were treated using three cryoprobes and 3-4 freeze-thaw cycles.

Results:

At 3-month follow-up, there was complete resolution of both gluteal lesions (100%). No major complications occurred.

Conclusion:

MR-guided cryoablation of gluteal muscle metastases is a feasible and safe minimally invasive modality for locoregional tumor control.



A 69-year-old male with renal cell carcinoma developed gluteal metastases. (a) PET-CT demonstrating PSMA-avid uptake in the gluteal region. (b) Axial T2-weighted MRI showing a hyperintense gluteal lesion. (c) Intraprocedural MRI demonstrating ice ball formation during cryoablation. (d) Complete resolution of the lesion at 3-month follow-up.

MR-guided laser ablation of a painful glomus tumor of the left dorsal forefoot and right knee in a 14-year-old male patient after failed surgical resection – a case report

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Purpose To assess the feasibility of MR-guided laser ablation of a recurrent glomus tumor in the left dorsal forefoot and right knee for pain relief in a pediatric patient.

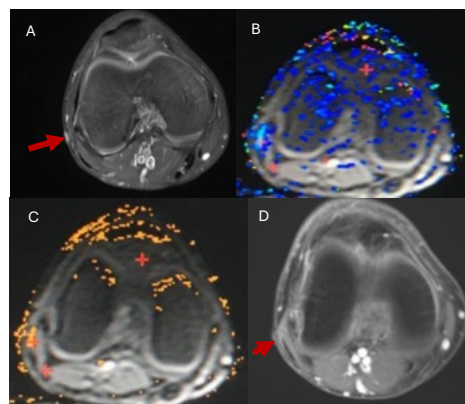
Materials and methods A 14-year-old male with recurrent glomus tumors of the knee and foot, previously treated with surgical resections performed three years apart, underwent MR-guided laser interstitial thermal therapy (LITT). Glomus tumors are rare benign neoplasms arising from the glomus body, often associated with significant localized pain. Procedures were performed in the supine position under general anesthesia. Initial placement of a bare-fiber Visualase 1.5-cm diffuser tip laser fiber (Medtronic, Minneapolis, MN, USA) was performed under ultrasound guidance and subsequently confirmed with MRI. Laser ablation was performed at 2-4 W for 1-2 minutes at each of three treatment positions, including two within the knee lesion and one within the foot lesion. Real-time MR thermometry was used to monitor ablation zones throughout treatment.

Results

At 12-month follow-up there was complete pain resolution in the knee, but there remained to be intermittent, short-lived pain in the foot despite complete resolution of both lesions on follow-up MRI. The left foot was retreated with a second ablation and pain resolved temporarily, with recurrence two years later. A third laser ablation was performed with significant improvement in the pain sustained for 1.5 years to date. No major complications occurred except for self-limited loss of sensation of the second toe at 3-month post first ablation and a minor skin burn that healed with conservative management after referral to the wound care clinic.

Conclusion

MR-guided laser ablation can be utilized for superficial glomus tumors of the leg and foot for symptomatic pain relief.



Laser ablation of a right lateral knee subcutaneous glomus tumor in a 14-year-old male. (A) Pre-ablation T2 hyperintense lesion in the lateral right knee subcutaneous tissues. (B) MR thermometry map demonstrating the ablation zone. (C) Corresponding damage map. (D) Two-year post-ablation MRI demonstrating decreased enhancement at the treatment site.

MR guided bone biopsy

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Purpose

To assess the efficacy and safety of MR-guided bone biopsy.

Materials and Methods

We retrospectively reviewed three cases (2 females, 1 male; mean age, 64 years) who underwent MR-guided bone biopsy for suspected osseous metastases. Primary cancers of both female patients were breast cancer and lesions suspected were in the sacral ala in one and iliac bone in the other patient. The primary cancer of the male patient was prostate cancer with a suspected lesion in the ischium. In all cases, CT imaging failed to identify the suspicious bone lesions, while MRI demonstrated altered bone marrow signal or contrast enhancement and so MR-guided biopsy was favored. Under MRI guidance, an MR-compatible biopsy device was inserted into the target site, and needle positioning was confirmed. After drilling into the bone, a core biopsy sample was obtained and imaging verified that the specimen was acquired from the center of the T2 signal abnormality. Samples were sent for histopathological evaluation.

Results

MR-guided bone biopsy was technically successful in all three cases (100% success rate) and no procedure-related complications were reported.

Pathological analysis confirmed metastatic lobular carcinoma to the sacral ala in one case, granulomatous disease of the ilium in another, and no evidence of malignancy in the ischium in the third case.

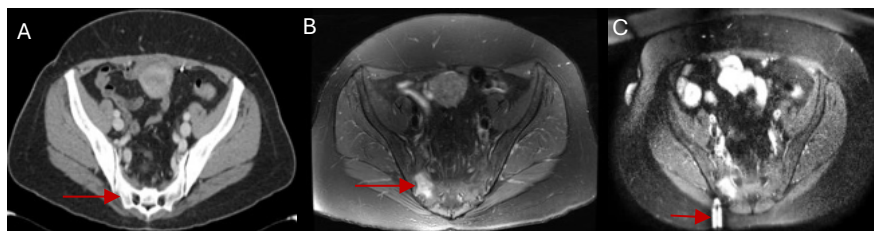


Figure 1. Computed tomography scan demonstrating an unremarkable sacrum with no evidence of sclerotic osseous metastases (a). A 3T MRI of the pelvis and hips performed 10 days later revealed a destructive lesion in the right sacral ala (b). Localizing MR image obtained during the procedure demonstrating placement of a needle marker and advancement of a 14-gauge bone biopsy introducer (c).

Conclusion

This case series demonstrates that MRI can be used to both identify radiographically occult osseous lesions and guide safe and accurate biopsy-taking.

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Combined CT- and MR- Guided Ablation of a Sacral Tumor: A Case Report

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Purpose

A 63-year-old male presents with metastatic prostate cancer. A PET scan revealed single metastasis of mid sacrum. To target lesion using CT and MR guided cryoablation for left sacral met.

Material and Methods

A 63-year-old male with metastatic prostate cancer underwent combined CT- and MRI-guided percutaneous cryoablation of a metastatic lesion involving the left sacral ala. Due to the osseous location of the lesion, initial access was obtained under CT guidance. A transosseous approach was performed by drilling through the sacral bone to facilitate placement of the cryoablation probes. Following successful probe placement, the patient was transferred from the CT suite to the MRI suite for real-time monitoring of ice-ball formation and assessment of the ablation zone. Continuous neuro monitoring was utilized throughout the procedure to evaluate nerve function and minimize the risk of neurologic injury. Cryoablation of the left sacral metastasis was performed using one IceRod and one IceSeed CryoNeedle. The procedure was completed under combined CT and MRI guidance, with CT used for osseous access and probe placement and MRI used for intraprocedural monitoring of cryoablation.

Results

Cryoablation achieved complete technical success with no immediate complications. Combined CT- and MRI-guided percutaneous cryoablation of the left sacral ala metastasis was successfully performed. CT guidance facilitated accurate transosseous access and cryoprobe placement, while MRI provided real-time visualization of ice-ball formation and monitoring of the ablation zone throughout treatment. Neurophysiologic monitoring remained stable during the procedure without evidence of nerve injury or neurologic compromise. The targeted lesion was encompassed by the planned ablation zone, and the procedure was completed without immediate complications. The patient tolerated the procedure well and was transferred for routine post-procedural recovery.

Conclusion

Combined CT- and MRI-guided percutaneous cryoablation is a feasible and effective technique for the treatment of sacral metastatic lesions. CT guidance enables accurate transosseous access and cryoprobe placement within complex bony anatomy, while MRI provides superior visualization of ice-ball formation and adjacent neural structures during ablation. The addition of neurophysiologic monitoring may further enhance procedural safety when treating lesions near sacral nerve roots. This multimodality approach allowed successful treatment of left sacral metastasis without immediate complications and may represent a valuable option for selected patients with painful or difficult-to-access sacral metastatic disease.

MSK MRI-Based Personalized Osteochondral Arthroplasty with Tissue-Specific Robotic Resection: A Feasibility Framework

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Purpose

Current robotic and navigation-assisted arthroplasty workflows are predominantly based on CT-derived bone morphology and do not explicitly account for articular cartilage, which defines true joint contact geometry and load distribution. Joint mechanics and contact forces are highly sensitive to surface conformity, alignment, and soft tissue conditions [1]. The omission of cartilage may therefore introduce systematic inaccuracies in implant positioning and joint surface restoration. This study presents and evaluates an MRI-based framework for patient-specific osteochondral arthroplasty that incorporates cartilage morphology into planning and resection to improve geometric fidelity and anatomical relevance.

Material and Methods

A framework was developed integrating high-resolution musculoskeletal (MSK) MRI for segmentation of cartilage and subchondral bone, followed by patient-specific planning of osteochondral resection. MRI enables direct visualization of articular cartilage, unlike CT-based approaches that rely solely on bone representation [2,3]. The planned geometry was used to generate robot-compatible resection trajectories with tissue-specific control parameters, including tool path, speed, and feed rate, along with provisions for real-time temperature monitoring and reaction-force feedback to account for differences in mechanical and thermal response between cartilage and bone. The workflow included: (1) MRI-based segmentation and generation of patient-specific osteochondral resection plans; (2) comparative simulation of MRI-based versus CT-based planning for representation of articular surfaces and joint conformity; and (3) preliminary benchtop studies to assess technical feasibility of integrating tissue-aware control, thermal monitoring, and force-based feedback within a robotic resection framework. Geometric conformity and surface agreement were evaluated using surface-based comparison metrics in simulation.

Results

All quantitative results are based on completed simulation studies. CT-based planning, which omits cartilage, resulted in systematic geometric deviations from cartilage-defined joint surfaces, with discrepancies on the order of approximately 1–2 mm in implant positioning and surface conformity. In contrast, MRI-based planning preserved cartilage-informed surface geometry and demonstrated improved agreement with intended osteochondral resection boundaries. Surface-based analyses showed enhanced geometric conformity using MRI-derived planning, particularly in regions of high curvature and spatially varying cartilage thickness. Preliminary benchtop studies demonstrated the technical feasibility of integrating tissue-aware tool speed and feed-rate control, real-time temperature monitoring, and reaction-force feedback within a robotic resection workflow. These studies confirmed system-level feasibility for coordinating trajectory generation, sensing, and control architecture. Quantitative validation of resection accuracy and control performance is ongoing and will be presented in future work.

Conclusion

This study demonstrates the feasibility of an MRI-based, tissue-specific framework for personalized osteochondral arthroplasty, representing a shift from bone-centric to osteochondral-informed planning and resection. By incorporating cartilage geometry, the approach addresses a limitation of CT-based workflows and enables anatomically representative joint surface reconstruction. The framework is compatible with robotic and navigation-assisted systems, supporting validation and translation. Beyond conventional total or unicompartmental arthroplasty, this framework introduces patient-specific osteochondral reconstruction using non-standardized implant geometries or grafts that match native cartilage morphology.

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Hybrid CT/MRI-Guided Cryoablation of Osteoblastic Prostate Cancer Bone Metastases

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Purpose

To evaluate the technical feasibility and locoregional tumor control of hybrid CT/MRI-guided cryoablation for osseous prostate cancer metastases.

Materials and Methods

This retrospective case series included 4 procedures in 3 male patients with metastatic prostate cancer to bone (mean age 82 years; range: 79–83). Treated sites included the 4th rib, sacrum, pubic ramus, and ischium; one patient underwent 2 procedures. Prior therapies included radiation therapy, CT-guided ablation, androgen deprivation therapy (ADT), and chemotherapy. A hybrid CT/MRI-guided cryoablation workflow was used to overcome challenges related to osteoblastic lesions and the lack of MRI-compatible bone drills. CT facilitated bone access with a bone drill and cryoneedle guide catheter placement while MRI enabled real-time ice-ball visualization to determine lesion coverage and prevent ice-ball encroachment on other structures. Procedures were performed under general anesthesia. MRI localization imaging was followed by transfer to CT for CT-guided osseous access using an 11-gauge bone drill (Arrow OnControl Powered Bone Access System, Teleflex, Wayne, Pennsylvania, USA) which were then exchanged with 13 Ga Angiocath needles (BD Angiocath™ Special IV Catheter, Becton, Dickinson and Company, Franklin Lakes, NJ, USA) placed for transfer back to MRI where stylets were then exchanged with cryoprobes. Additionally, neuromonitoring was used in 2 procedures. Cryoablation consisted of a mean 3 freeze-thaw cycles (range: 3–4). Outcomes included technical success and locoregional tumor control assessed with post-procedural gadolinium-enhanced MRI and 3-month PSMA PET or choline PET imaging.

Results

Technical success was achieved in 100% (4/4) of procedures. At 3-month follow-up, PET imaging demonstrated locoregional resolution of PET-PSMA activity on 3/4 treated lesions, for a mean follow-up of 1 year. One of 4 patients with residual pubic bone PSMA activity also had lymph node activity and was treated with 6 cycles of lutetium-177. One of four patients reported persistent pain despite imaging resolution of a right sacral metastasis and was referred for orthopedic evaluation.

Conclusion

Hybrid CT/MRI-guided cryoablation is a technically feasible approach for treating osteoblastic prostate cancer bone metastases promising locoregional tumor control.

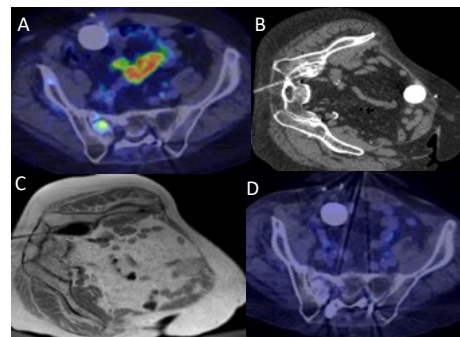


Figure 1. Seventy-nine-year-old male with prostate cancer metastatic to the right sacral ala refractory to ADT. (A) Choline PET-CT demonstrating uptake within the right sacral ala metastasis. (B) Axial CT image obtained in the left lateral decubitus position demonstrating advancement of the puncture drill into the lesion. (C) Intraprocedural T2-weighted MRI demonstrating cryoablation ice-ball formation. (D) Choline PET obtained 3 months post-procedure demonstrating complete metabolic resolution.

MRI-Guided Percutaneous Cryoablation for Pediatric Vascular Anomalies: Safety and Technical Outcomes

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PURPOSE

Pediatric vascular anomalies encompass a spectrum of venous malformations (VM), arteriovenous malformations (AVM), and lymphatic malformations (LM) that frequently cause debilitating pain, functional impairment, and progressive deformity. While sclerotherapy remains first-line for many lesions, anatomically challenging or high-flow lesions may not respond adequately or may pose significant risk to adjacent neurovascular structures. MRI-guided cryoablation allows precise iceball visualization under continuous MRI monitoring, enabling controlled ablation with reduced thermal spread and the ability to protect adjacent critical structures, including nerves, with real-time neural monitoring. This study evaluated the technical feasibility, safety, and outcomes of in-bore MRI-guided percutaneous cryoablation in a pediatric vascular anomaly population.

MATERIALS AND METHODS

This retrospective single-institution study included all pediatric patients (age 0 to 17 years) who underwent in-bore MRI-guided percutaneous cryoablation for vascular anomalies between February 2014 and May 2026. Procedures were performed under general anesthesia or CRNA-administered sedation using combined ultrasound and MRI guidance. Cryoablation was performed using IceSeed and IceRod cryoneedles (Boston Scientific, Marlborough, MA) with a mean of 3.5 freeze-thaw cycles per procedure (range 2 to 6). For foot and ankle lesions with high vascular flow, a tourniquet technique using a blood pressure cuff inflated to 120 mmHg was employed to impede flow and facilitate ice formation. Neural monitoring was performed in procedures adjacent to peripheral nerves. Technical success was defined as ablation zone encompassing the target lesion on post-procedure MRI. Procedural complications, repeat session rates, and patient demographics were recorded.

RESULTS

A total of 24 cryoablation procedures were performed in 18 unique pediatric patients (mean age 11.5 years, SD 3.9, range 0 to 17 years). Lesion types included VM in 13 procedures (54.2%), AVM in 8 procedures (33.3%), and LM in 2 procedures (8.3%). Anatomic locations included foot and ankle ($n = 10$, 41.7%), thigh and femur ($n = 8$, 33.3%), abdominal wall ($n = 2$, 8.3%), forearm ($n = 1$, 4.2%), chest wall ($n = 1$, 4.2%), and lower leg ($n = 1$, 4.2%). Notably, 3 procedures targeted vascular malformations with osseous involvement of the femur, requiring bone introducer needles placed within the femoral cortex. Technical success was documented in 17 of 17 cases with recorded post-procedure imaging evaluation (100%): complete ablation zone coverage in 12 procedures and 90% or greater coverage in 5 procedures. The mean number of procedures per patient was 1.3 (range 1 to 3). No procedural complications were recorded in any of the 24 cases (0% complication rate). General anesthesia was used in 21 procedures (87.5%) and CRNA-administered sedation in 3 procedures (12.5%).

CONCLUSION

MRI-guided percutaneous cryoablation is technically feasible and demonstrates an excellent safety profile for the treatment of pediatric vascular anomalies, with a 100% technical success rate and zero procedural complications in this series. The real-time MRI visualization of iceball growth, combined with the option for intraoperative neural monitoring, enables treatment of anatomically challenging lesions including periosseous, perineural, and plantar foot malformations. The tourniquet technique for high-flow foot and ankle lesions is an effective adjunct to facilitate adequate ice formation. These data support MRI-guided cryoablation as a safe and effective minimally invasive option across diverse pediatric vascular anomaly subtypes and anatomic locations.

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MR-Guided Cryoablation for Desmoid Fibromatosis: Technical Feasibility, Safety, and Symptomatic Outcomes

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PURPOSE

Desmoid fibromatosis is a rare, locally aggressive monoclonal fibroblastic neoplasm with high recurrence after surgery and limited systemic options. MR-guided cryoablation allows real-time intraprocedural iceball visualization, a critical advantage for tumors with complex neurovascular relationships. Evidence for this approach remains limited. We report our institutional experience with MR-guided cryoablation for refractory desmoid fibromatosis, focusing on technical feasibility, safety, and imaging and symptomatic outcomes.

MATERIALS AND METHODS

Retrospective single-center case series of patients with biopsy-proven desmoid fibromatosis who underwent MR-guided percutaneous cryoablation at Mayo Clinic (2018–2025) with ≥ 3 months imaging follow-up. Procedures used combined MRI/ultrasound guidance under general anesthesia with Galil or Boston Scientific IceRod/IceSeed cryoprobes (2–14 per session, 2–5 freeze-thaw cycles). Intraoperative neuromonitoring and saline hydrodissection were applied per anatomy. **Imaging response** was assessed by viable enhancing tumor volume (ellipsoid formula $\pi/6 \times L \times W \times H$) on contrast-enhanced MRI and classified by modified RECIST (CR/PR/SD/PD). Symptomatic response was assessed by patient-reported NRS change. Complications were graded per SIR/CIRSE classification. Wilcoxon signed-rank test compared pre- and post-ablation volumes (non-parametric given $n=10$).

RESULTS

Six patients (all female; median age 35 years, range 14–61) underwent 11 cryoablation sessions across right hand ($n=4$), left lower extremity ($n=3$), left deltoid ($n=2$), left flank ($n=1$), right paravaginal pelvis ($n=1$), and right calf ($n=1$). All had prior treatment failure: 5/6 had undergone surgical resection and 2/6 had prior embolization. Technical success ($\geq 90\%$ iceball target coverage) was achieved in 10/11 sessions (91%). No major complications (SIR $\geq$ Grade C) occurred. Minor adverse events included post-ablation edema ($n=3$), transient paresthesias ($n=2$), and neuropathic pain requiring gabapentin ($n=1$).

Volumetric analysis was performed for 11 measured session targets across all 6 patients. Mean pre-ablation viable tumor volume was $278 \pm 735 \text{ cm}^3$; post-ablation $70 \pm 201 \text{ cm}^3$. Mean volume reduction was 71.1% (SD 24.8%; median 72.0%; IQR 48.7–94.7%; range 31–100%). Volume reduction was statistically significant (Wilcoxon signed-rank, $W=0$, $p=0.002$). mRECIST response in targeted zones: CR 2/10 (20%), PR 4/10 (40%), SD 4/10 (40%), PD 0. Overall disease control rate 100%; objective response rate 60%. All SD lesions were Patient 1 hand targets managed by intentional partial ablation to preserve extensor tendons. NRS pain improvement was achieved in 5/6 patients. Median follow-up: 18 months (range 3–60).

CONCLUSION

MR-guided cryoablation is feasible and safe for desmoid fibromatosis across diverse anatomic locations. Mean volume reduction of 71.1% and 91% technical success are consistent with published cryoablation series. A 100% disease control rate in targeted zones and symptomatic improvement in 5/6 patients support both curative and palliative roles. Prospective studies with BPI/QoL endpoints are needed.

ACKNOWLEDGEMENTS

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MR-Guided Ablation in Kaposiform Hemangioendothelioma

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Purpose

Kaposiform Hemangioendothelioma (KHE) is an aggressive vascular neoplasm affecting young children. Lesions present as invasive and erythematous soft tissue masses in the craniocervical and retroperitoneal regions and are often accompanied by the Kasabach-Merritt Phenomenon (KMP), a severe consumptive coagulopathy marked by thrombocytopenia and hypofibrinogenemia secondary to vascular trapping by KHE. Our purpose is to highlight the successful treatment of two patients with refractory kaposiform hemangioendothelioma utilizing MR-guided ablation and raise awareness of MR ablation as an alternative to existing therapies.

Material and Methods

We highlight a case series of two patients with KHE aged 5-years-old and 11-years-old who were treated with MR-guided laser ablation and cryoablation, respectively. MR-guided laser ablation is performed utilizing a laser fiber applicator placed percutaneously under MR-guidance and forms an ablation zone that overlaps the lesion. MR-guided cryoablation is performed by placing cryoprobes percutaneously under MR-guidance that deliver extreme negative temperatures in freeze-thaw cycles for cell destruction. MR-guidance allows for real-time temperature monitoring via proton resonance frequency thermometry, which provides accurate measurement of temperature and rapid cycling for real-time feedback.^{1,2} Response was then measured using the residual T2 signal within the lesion.

Results

The first case was a 5-year-old male who presented with KHE of the right lower flank and retroperitoneum complicated by KMP at one year of age. Initial treatment of sirolimus for one year resulted in resolution until recurrence of KHE without KMP at age four. The patient was treated with MR-guided laser ablation of the right paraspinal and iliopsoas muscles and right retroperitoneal fascia. Evaluation three months post-ablation demonstrated non-masslike enhancement and no mass effect. The patient had excellent pain control and stable disease. The second case was an 11-year-old male who presented with KHE complicated by KMP of the right chest and abdomen at birth. Oral corticosteroids, vincristine, and sirolimus were attempted with improvement of KMP but continued growth of KHE. At two years of age, he was treated with MR-guided cryoablation which demonstrated radiologic and symptomatic improvement of KHE. Four additional treatments of MR-guided cryoablation were performed between the ages of 3 to 6 due to intermittent recurrences of KHE. Since the last treatment at 6-years-old, he has remained stable and asymptomatic, with no new radiologic findings.

Conclusion

Treatment of KHE typically consists of medical management and surgical resection. Complete surgical resection can be difficult in challenging lesion sites, while some reports suggest partial resection can trigger KMP. Medical management includes oral corticosteroids, vincristine, and sirolimus. These medications, however, carry concern for potential cumulative toxicity. MR-guided ablation in our two cases demonstrated successful treatment of KHE and identifies a potential alternative for pediatric dermatologists in the treatment of refractory KHE.

Acknowledgements

Conflict of Interest Statement: Dr. Megha Tollefson acting as a consultant for Revelare and consultant and investigator for Palvella, Dr. Scott Thompson acting as a research consultant and medical advisory board member at Boston Scientific, and Dr. David Woodrum acting as a consultant on interventional MRI for Boston Scientific, Siemens, and Cook Medical.

Previous Presentations: The cases have been presented as a poster at the International Society for the Study of Vascular Anomalies (ISSVA) meeting; May 2024; Madrid, Spain.

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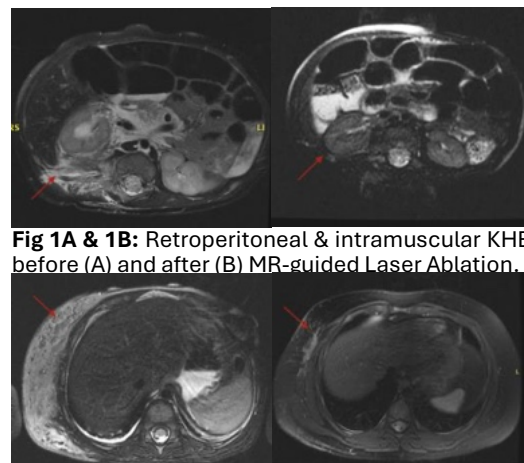


Fig 1A & 1B: Retroperitoneal & intramuscular KHE: before (A) and after (B) MR-guided Laser Ablation.

Figure 2A & 2B: Abdominal Wall KHE before (A) and after (B) multiple MR-guided Cryotherapy (4-year post-procedure follow up)

MR guided cryoablation for treatment of painful large to medium slow-flow venous malformations in the plantar foot – a retrospective study

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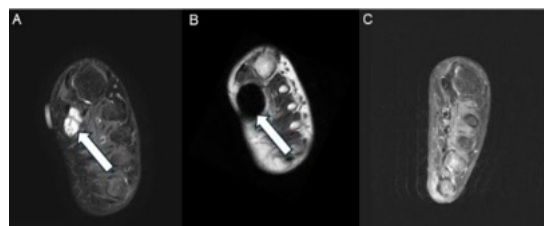
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Purpose To evaluate the safety and efficacy of MR-guided cryoablation for painful plantar foot venous malformations (VMs) in pediatric and adult patients.

Material and methods An institutional-review board-approved retrospective study of patients who underwent MR-guided cryoablation for symptomatic plantar foot VMs between 2015 and 2025. US-assisted MRI targeting and MRI monitoring were performed. Lesion dimensions, ablation sessions, freeze-thaw cycles, and recurrence rates were recorded. Pain was assessed using a visual analog scale (VAS, 0–10) or self-reported symptomatic improvement categorized as: (a) complete resolution; (b) partial improvement; (c) no improvement.

Results Eighteen patients (14 females, 4 males; mean age 31.5 years; range 10–85 years) underwent 23 cryoablation procedures with a 1-year mean follow-up (range: 1–5 years). Eight patients (44.4%) had failed prior interventions (i.e. sclerotherapy, embolization, and/or surgery). Mean freeze-thaw cycles per procedure was three. Mean lesion diameter was 3.7 cm (maximum 8.3 cm), and all lesions were T2-hyperintense on pre-treatment MRI. Clinical follow-up was available for 15 patients. At 6–12 months post-ablation, 14 (93.3%) of 15 patients achieved complete or partial improvement. At 12–24 months post-ablation, three (20%) of the 15 patients had pain recurrence with repeat treatment offered and went on to a second cryoablation. Following the second ablation at 6-month follow-up, all the re-treated patients achieved complete pain relief, however one (33.3%) of three patients had pain recurrence 3 years later treated with a third cryoablation with subsequent complete relief.



A 10-year-old female with a plantar foot venous malformation located beneath the first and second metatarsals measuring 2.3 × 1.5 × 1.0 cm. (a) Pre-ablation axial T2-weighted MR image demonstrates the lesion (white arrow). (b) Intraprocedural axial T2-weighted MR image shows adequate ice-ball coverage during a single cryoablation session (white arrow). (c) Axial T2-weighted MR image at 6-month follow-up demonstrates no residual venous anomaly. Clinically, the patient reported complete pain resolution at 6 months.

Conclusion MR-guided cryoablation appears to be a safe with short-to-mid-term effectiveness - for pain control for plantar venous malformations.

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Sequential MR-guided laser and cryoablation of a recurrent painful slow-flow venous malformation in a 6-year-old

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Purpose

To evaluate the utilization of sequential MR guided cryo-laser ablation for a refractory painful slow-flow vascular malformation in the plantar aspect of the left foot in a 6-year-old male patient.

Materials and methods

MR guided laser ablation was performed followed by MR guided cryoablation 5 years later. Procedures were performed under general anesthesia using a 1.5-T Philips MRI system with real-time MRI guidance and monitoring. Pain outcomes were assessed using patient-reported improvement. Lesion diameter was 3.7 cm in the longest axis.

Results

The patient was able to achieve pain relief for five years after the initial laser ablation then MR guided cryoablation which has given him 6 years of relief so far. No major complications occurred.

Conclusion

Sequential use of MR-guided laser ablation followed by cryo-ablation is a safe and effective strategy for pain control in plantar foot venous slow-flow malformations.

Acknowledgements

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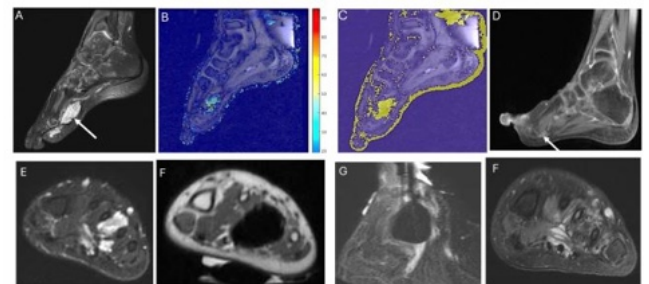


Fig 1: Magnetic resonance images of the laser ablation and subsequent cryoablation retreatment sessions for treatment of the plantar foot venous malformation. MRI sagittal T2 image of the venous malformation in the left plantar foot at 6 years old pre-laser ablation (white arrow) (a). Real-time MR thermometry demonstrates a well-defined circular region with elevated temperature during laser ablation (b). Corresponding thermal damage zone (c). MRI sagittal T2 gadolinium-enhanced image 12 months post-laser ablation showing residual venous malformation (white arrow) (d). MRI axial T2 Fat-Saturated image at 11 years old pre-cryoablation retreatment (e). MRI axial and sagittal T2 interprocedural images showing real-time ice-ball formation (f, g). MRI axial T2 image 6 months post-cryoablation showing post-ablation changes with small residual venous malformation (white arrow) (h). Images showing real-time ice-ball formation (f, g). MRI axial T2 image 6 months post-cryoablation showing post-ablation changes with small residual venous malformation (white arrow) (h).

MRI-Guided Cryoablation for Recurrent Pediatric Hand Desmoid Tumor: A Longitudinal Case Report

Laurel Woodrum, Setayesh Sotoudehnia Korani, Scott Thompson, Aiming Lu, David Woodrum

Purpose

To evaluate the feasibility, safety, and clinical outcomes of MRI-guided cryoablation for recurrent desmoid fibromatosis of the hand in a pediatric patient.

Materials and Methods

A 14-year-old female with recurrent multifocal desmoid tumor of the right dorsal hand following prior surgical resection presented with progressive pain, hyperextension deformity, and functional limitation. CT-guided cryoablation was not feasible due to inadequate soft tissue visualization. The patient underwent four MRI-guided cryoablation procedures over a 5-year period.

Procedures were performed under general anesthesia using MRI-compatible cryoprobes. Lesions were targeted using T2-weighted and contrast-enhanced imaging. Real-time MRI (balanced SSFSE) monitored ice ball formation at approximately 20-second intervals. Hydrodissection was utilized for skin protection. Ablation endpoints included tumor margin coverage or limitation by proximity to critical structures.

Results

MRI guidance enabled accurate probe placement and real-time monitoring in all sessions without major complications. Imaging follow-up demonstrated decreased enhancement and interval reduction of treated lesions. New or progressive foci were successfully managed with repeat ablation.

Clinically, the patient experienced significant pain reduction and gradual functional improvement, including decreased hyperextension and improved digit mobility. Hospital stays decreased over time, transitioning from overnight observation to same-day discharge.

Conclusion

MRI-guided cryoablation is a safe and effective minimally invasive option for management of recurrent desmoid tumors of the hand, particularly in pediatric patients where radiation avoidance and functional preservation are paramount. This case highlights the advantages of MRI for precise treatment planning, real-time monitoring, and iterative management of multifocal disease.

Keywords

MRI-guided intervention; cryoablation; desmoid fibromatosis; pediatric; soft tissue tumor

MRI-Guided Cryoablation of a Bilateral Plantar Fibromatosis

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Purpose

Plantar fibromatosis (Ledderhose disease) is a benign but often debilitating fibroproliferative disorder of the plantar fascia for which no definitive curative therapy exists. Conservative measures including orthotics, topical verapamil, extracorporeal shockwave therapy, and radiation therapy provide incomplete or transient relief in a subset of patients. We report the first case of sequential bilateral MRI-guided percutaneous cryoablation for plantar fibromatosis refractory to multiple prior therapies.

Materials and Methods

A 47-year-old male presented with bilateral plantar fibromatosis causing daily significant pain limiting ambulation to 3 to 4 hours. He had failed sequential conservative management including custom orthotics, topical verapamil, six sessions of extracorporeal shockwave therapy, and bilateral radiation therapy. He was referred to interventional radiology for ablative therapy. Two staged procedures were performed under general anesthesia using combined ultrasound and intra-bore MRI guidance. The left foot fibroma was treated first using 2 IceSeed and 1 IceRod cryoneedles; the right foot fibroma was subsequently treated using 1 IceSeed and 1 IceRod cryoneedle. In both procedures, Abbocath guides were placed under direct ultrasound guidance with cryoneedle position confirmed by MRI prior to freezing. Three freeze-thaw cycles were performed in each session with continuous MRI monitoring of iceball propagation. Dynamic contrast-enhanced imaging was obtained at the conclusion of each procedure to confirm fibroma coverage. Gelfoam was administered at needle removal sites.

Results

Both procedures were technically successful with no immediate or delayed complications. Dynamic contrast-enhanced MRI at the conclusion of each session confirmed adequate iceball coverage of the target fibromas. On limited interval imaging obtained at the time of the right foot procedure, the previously treated left foot fibroma demonstrated interval size reduction from 9 mm to 6 mm in width. On clinical follow-up, the patient reported significant symptomatic improvement with restoration of full ambulatory function.

Conclusion

Sequential bilateral MRI-guided percutaneous cryoablation is technically feasible and clinically effective for plantar fibromatosis refractory to standard therapies. Real-time intra-bore MRI guidance enables precise iceball monitoring in this anatomically compact target, supporting its role in benign musculoskeletal pain management. Larger prospective series are warranted to define optimal treatment protocols and durability of response.

MRI-Guided Microwave Ablation of Hepatic Metastases Using a High-Energy MR-Compatible Microwave System: Initial Clinical Series

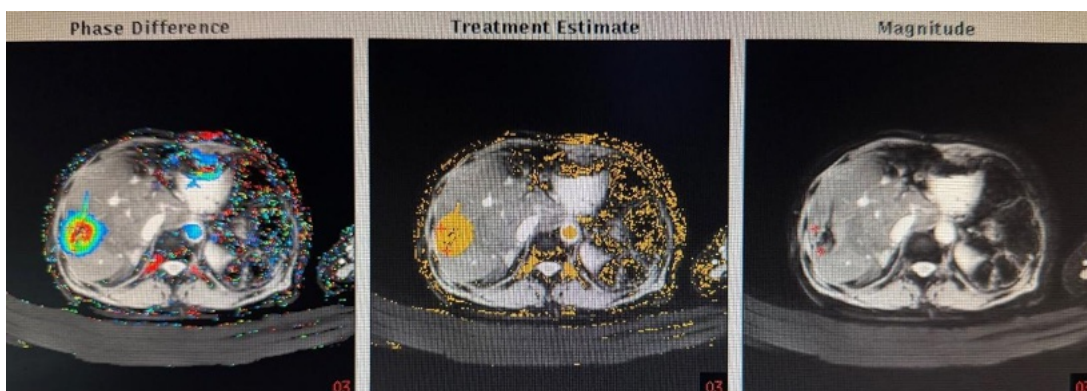
Aiming Lu PhD, Setayesh Sotoudehnia Korani MD, Ola A.E. Mohamed, Scott M. Thompson MD PhD, Daniel A. Adamo MD, David A. Woodrum MD PhD

Purpose: MRI guidance for hepatic microwave ablation (MWA) offers superior lesion conspicuity, real-time temperature mapping, and the ability to target lesions not visible on CT or ultrasound. However, widespread adoption has been limited by the scarcity of MR-compatible high-energy microwave systems. This study reports our initial clinical experience with a commercially available, MR-compatible high-energy microwave system SurBlate™, (Vison Medical) used under continuous intraprocedural MR thermometry guidance for hepatic metastases.

Materials and Methods: We performed a retrospective review of 11 consecutive patients who underwent MRI-guided percutaneous MWA of hepatic metastases using the SurBlate™ MR-compatible microwave system (Vison Medical, Santa Clara, CA) with 15 cm or 20 cm antennae at a single academic institution. All procedures were performed on a 1.5 Tesla MRI scanner (Ingenia, Philips Healthcare) under general anesthesia. Antenna placement was performed with combined ultrasound and MRI guidance. Ablation was performed with continuous proton resonance frequency (PRF)-based MR thermometry synchronized to the respiratory cycle. Ablation parameters, number of lesions treated, primary malignancy, and procedural safety were recorded. Where tumor size was not documented in the procedural record, the cohort mean was applied.

Results: Eleven patients (23 target lesions in total) were treated across the study period. Primary malignancies included neuroendocrine tumor (n=4), prostate cancer with liver metastases (n=3), pancreatic adenocarcinoma (n=1), and renal cell carcinoma (n=1); two patients had the primary cancer type not specified in the procedural record. Lesion sizes ranged from 1.5 to 4.3 cm (mean 2.9 cm) where documented. A total of 31 discrete ablation activations were delivered across all procedures. Power settings ranged from 35 to 80 W (mean 60.6 W), and individual activation duration ranged from 1 to 7 minutes (mean 3.4 min per activation). Mean total MWA time per patient was 10.1 minutes (range 1 to 22 minutes). MR thermometry demonstrated real-time thermal zone monitoring in all cases which was usable for ablation monitoring without significant electromagnetic interference artifact. Dynamic contrast-enhanced MRI performed immediately following ablation confirmed satisfactory lesion coverage in all cases. No immediate intraprocedural complications were recorded.

Conclusion: This initial clinical series demonstrates the feasibility and technical safety of MRI-guided percutaneous microwave ablation of hepatic metastases using a high-energy MR-compatible microwave system with continuous intraprocedural MR thermometry. The system enabled effective MR targeting and real-time thermal monitoring across a range of primary malignancies and lesion locations, with no immediate procedural complications. Larger prospective studies with systematic follow-up imaging are warranted to evaluate local tumor control and long-term outcomes.



MR-guided cryoablation for local control of adrenal gland metastases – a case series

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Purpose

To assess the feasibility, technical success and local control rate of MR-guided cryo-ablation for adrenal metastases.

Methods and materials

Three MR-guided percutaneous adrenal cryoablation procedures were performed in three patients (mean age, 63 years; 2 male, 1 female) with metastases from hepatocellular carcinoma (n = 1) and renal cell carcinoma (n = 2). Procedures were performed under general anesthesia in the supine feet-first position (n = 2) and in the left lateral decubitus position (n = 1), with continuous blood pressure monitoring via an arterial line in the wrist and pretreatment with labetalol. Guidance needles and IceRod/IceSeed probes (average, 2; range, 1–4) were inserted under ultrasound guidance with MRI confirmation. A total of 2–3 freeze–thaw cycles were performed. Gelfoam tract embolization was used in one patient. Postprocedural contrast-enhanced MRI was performed to assess technical success.

Results

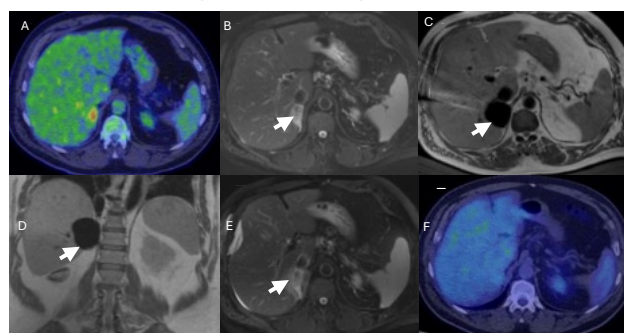
Technical success, defined as complete lesion coverage on immediate post-ablation imaging, was achieved in all cases (100%). At 3-month follow-up, all lesions demonstrated complete response (100%). Local control was maintained in three of three patients (100%) of cases at up to two years. No major complications occurred. One patient developed a small, self-limited hematoma lateral to the liver.

Conclusion

MR-guided cryo-ablation for adrenal metastases is a feasible and technically safe modality for local control of metastases to adrenal glands.

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63-year-old male with right adrenal metastasis from renal cell carcinoma. (a) Preablation FDG PET demonstrating a right-sided FDG-avid lesion. (b) Preablation T2-weighted MRI showing a hyperintense right adrenal lesion. (c, d) Axial and sagittal intraprocedural images demonstrating ice ball formation fully encompassing the tumor. (e) Follow-up PET/CT at 3 months showing decreased FDG uptake in the previously biopsy-proven right adrenal metastasis, which is no longer anatomically distinct.

MRI-guided Microwave Ablation for Primary and Metastatic Liver Tumors: a Comprehensive Single-center Experience Including Dual-antenna and Combined Cryoablation Techniques

Setaeysh Sotoudehnia Korani MD, Ola A.E. Mohamed MD, Scott M. Thompson MD, PhD, Aiming Lu PhD, Daniel A. Adamo MD, David A. Woodrum MD, PhD

Purpose: To report our center's primary experience with the safety and efficacy of MRI-guided microwave ablation (MWA), detailing dual-antenna techniques, and to describe our approach using combined-modality sessions where cryoablation targets separate, anatomically challenging lesions from MWA within the same procedure.

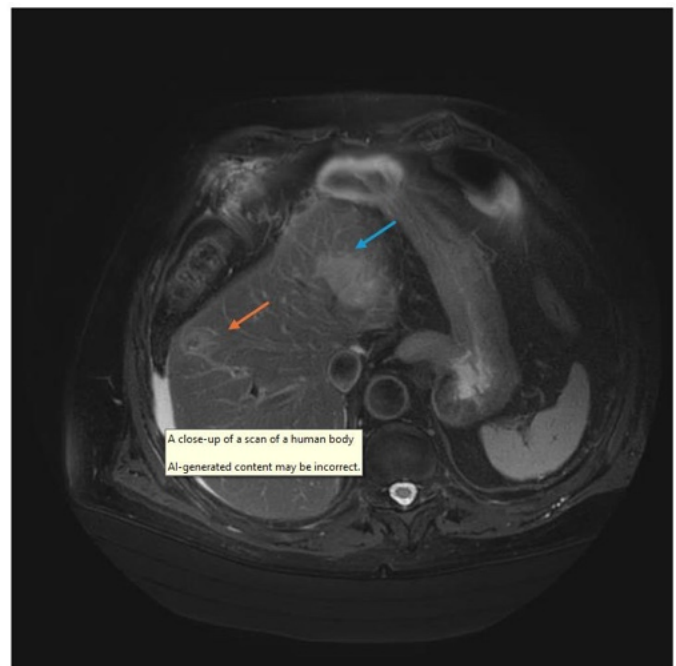
Materials and Methods: After institutional review board approval, we retrospectively reviewed two distinct patient groups treated from January 2019 to February 2025. The primary

MWA cohort included 26 patients (mean age 65.1 years) treated for primary (n=4) and metastatic (n=22) lesions; 12 of these were treated with a dual-antenna MWA technique to target larger or multiple lesions. A separate cohort of **6 patients** (mean age 68.2 years) underwent combined-modality sessions, where MWA and cryoablation targeted different lesions in the same procedure, often based on lesion proximity to sensitive structures. All procedures were performed under general anesthesia using combined MR and ultrasound guidance. MWA was monitored with real-time MR-thermometry, while cryoablation was monitored by MR-based ice-ball visualization.

Results: MWA cohort (N=26): Technical success was 100%. One-year locoregional control was 88.5% (23/26). No major complications occurred. The dual-antenna subgroup (n=12) was technically feasible and had no complications. **Combined-modality cohort (N=6):** Technical success was 100% in 7 procedures. This complex-case cohort had one minor (SIR Grade B) complication (14%, 1/7 procedures): a small renal hematoma managed conservatively. Early follow-up showed outcomes dependent on palliative vs. therapeutic intent.

Conclusion: MRI-MWA is a safe and effective primary treatment for liver tumors, and its application can be extended with dual-antenna techniques. For patients with multiple complex lesions, a combined-modality session using MWA and cryoablation on separate lesions is a feasible and safe problem-solving strategy.

Clinical Relevance: Mastery of MRI-guided MWA, from standard to dual-antenna methods, allows for effective treatment of a wide range of liver tumors. A patient-tailored approach, including the intra-procedural use of cryoablation for challenging lesions, demonstrates a sophisticated strategy to maximize minimally invasive treatment options.



Post ablation t2 images. Blue arrow (post cryo), orange arrow (post microwave)

Title: MR-Guided Renal Cryoablation: Local Control and Technical Parameters in a Single-Center Cohort

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Purpose: To evaluate local tumor control and technical parameters in patients undergoing MR-guided renal cryoablation.

Materials and Methods: Retrospective single-center study of 14 patients who underwent 17 MR-guided renal cryoablation procedures between 2010 and 2025. Right kidney: 8 (57%), left: 5 (36%), bilateral: 1 (7%). Histology: primary RCC in 11 (79%), RCC metastasis (1), HCC metastasis (1), atypical cystic lesion (1). A median of 2 cryoprobes (IQR 2 to 3) and 3 freeze-thaw cycles (range 2 to 4) were used per procedure. Gelfoam tract hemostasis and hydrodissection were employed selectively. Technical success, local recurrence, and complications were recorded [1,2].

Results: Technical success was 100%. Primary local control was achieved in 12 of 14 patients (85.7%) after the index ablation; two failures occurred (one in-field recurrence at 3 years; one immediate residual tumor treated in a planned second session). Secondary local control after salvage ablation was 100% (14/14). No out-of-field recurrences were observed. No immediate procedural complications were recorded. Median follow-up was 51 months (IQR 38 to 78; range 6 to 178 months).

Conclusion: MR-guided renal cryoablation achieved high primary local control with all failures confined to the ablation zone and effectively salvaged by repeat ablation, yielding 100% secondary control [1,2]. Real-time MR ice-ball visualization enables precise margin assessment while protecting adjacent structures, supporting its role as a durable, repeatable minimally invasive therapy for RCC and select non-RCC renal lesions.

Acknowledgements:

The authors thank the interventional radiology nursing and anesthesia teams at Mayo Clinic for their contributions to procedural care.

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Title: MRI-Guided Percutaneous Liver Biopsy: 15-Year Single-Center Experience at a Tertiary Referral Institution

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Purpose: To evaluate the diagnostic yield, safety, and clinical characteristics of MRI-guided percutaneous liver biopsy, with emphasis on lesions occult or nondiagnostic on ultrasound.

Materials and Methods: Retrospective review of all MRI-guided percutaneous liver biopsies performed at a single institution from January 2010 through July 2025. Fifty-five patients (mean age 62 years, range 18 to 83; 52% male) underwent biopsy under general anesthesia on a 1.5 T MRI system using an 18-gauge spring-loaded core biopsy device via a coaxial introducer. Ultrasound guided introducer placement in most cases; intermittent MRI confirmed needle position. Eighty percent had a history of malignancy; 70% had a prior nondiagnostic ultrasound-guided biopsy. Diagnostic success was defined as histopathology concordant with imaging and clinical follow-up [1,2].

Results: Diagnostic concordance was achieved in 51 of 55 cases (92%). Pathology revealed malignancy in 21 cases (38%) and benign findings in 34 (62%). Lesion size ranged from 4 mm to 38 mm; several lesions were occult on ultrasound but clearly visualized on MRI. No major complications occurred. Three patients (5%) required overnight admission for pain control; no other acute complications were recorded.

Conclusion: MRI-guided percutaneous liver biopsy demonstrates high diagnostic yield (92%) and an excellent safety profile over a 15-year experience [1,2]. It is particularly valuable for lesions occult on ultrasound or following prior nondiagnostic biopsy, enabling diagnosis and guiding management in a challenging oncologic population despite requirements for general anesthesia and specialized resources.

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The authors thank the interventional radiology nursing and anesthesia teams at Mayo Clinic for their contributions to procedural care.

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An End-to-End Framework for Real-Time 2D Multislice MRI-Guided Liver Interventions

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Purpose

Real-time MRI-guided percutaneous liver interventions require high temporal resolution and reliable needle tracking to support radiologist decision-making and closed-loop robotic control [1]. We developed an end-to-end framework by integrating open-source Pulseseq [2] sequence design, online reconstruction, automated deep learning-based needle localization/tracking, and real-time visualization for 2D multislice MRI-guided liver interventions. We performed reconstruction parameter selection to balance temporal resolution, reconstruction latency, and needle tracking performance to support real-time applications.

Material and Methods

Figure 1 shows the structure of the proposed framework. An interleaved 2D 3-parallel-slice golden-angle (GA) radial sequence (TE/TR=1.72/3.8ms; flip angle=10°; slice thickness=5mm; slice gap=0mm; resolution = 1.56 × 1.56mm²; field-of-view=300×300mm²) was implemented using Pulseseq [2] and acquired on a 3T scanner (MAGNETOM Vida, Siemens). The Framework for Image Reconstruction Environments (FIRE; Siemens) [3] was used for online image reconstruction. Incoming radial spokes were streamed to the FIRE pipeline, where sliding-window compressed sensing reconstruction [4] was performed in real time. Reconstruction parameters were selected to minimize reconstruction latency while maintaining image quality sufficient for downstream multislice needle tracking using our keypoint detection network [5]. Reconstructed images and tracking results were streamed to 3D Slicer [6] via OpenIGTLink [7] for real-time rendering/display. The framework was evaluated online during real-time needle insertion and angulation in an abdominal phantom.

Results

The proposed framework achieved an end-to-end latency of 340.8 ms/frame, including 279.2 ms/frame for sliding-window reconstruction (93.0 ms/slice), 36.7 ms/frame for needle tracking, and 24.9 ms/frame for OpenIGTLink transfer and 3D-rendered visualization. The overall latency remained below the temporal resolution of 444.6 ms/frame (39 spokes/slice, 3 slices/frame, 117 spokes/frame), enabling continuous real-time image guidance without accumulating processing delay.

Conclusion

We developed a real-time end-to-end framework that achieved a temporal resolution of 444.6 ms/frame (3 slices) with the overall latency below the image frame interval. The proposed framework has the potential to facilitate real-time MRI-guided liver interventions by providing real-time imaging, needle tracking, and display.

Acknowledgements

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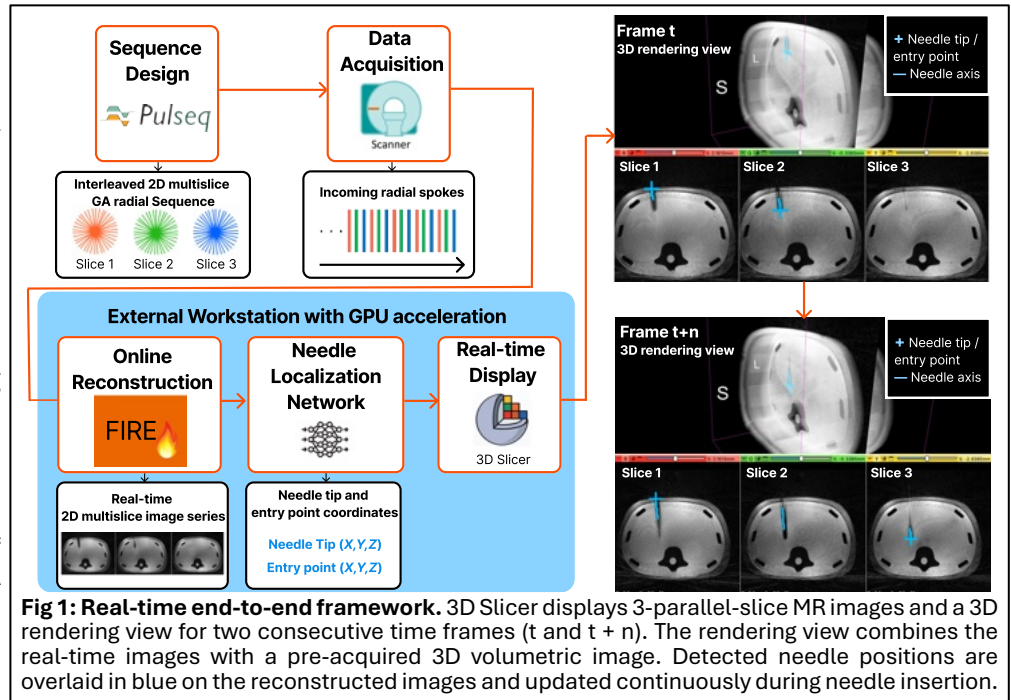


Fig 1: Real-time end-to-end framework. 3D Slicer displays 3-parallel-slice MR images and a 3D rendering view for two consecutive time frames (t and t + n). The rendering view combines the real-time images with a pre-acquired 3D volumetric image. Detected needle positions are overlaid in blue on the reconstructed images and updated continuously during needle insertion.

MR-Guided Microwave Ablation of Liver Tumors: Outcomes in Local Tumor Control and Determinants of Treatment Success

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Purpose

To evaluate local tumor control and determinants of treatment success following magnetic resonance (MR)-guided microwave ablation (MWA) of liver malignancies.

Material and Methods

This single-center retrospective study evaluated prospectively included patients undergoing MR-guided MWA with real-time thermometry between September 2021 and February 2025. Procedures were performed under general anesthesia using a 1.5-Tesla MR scanner and two MR-compatible MWA systems (output power: system 1, 40 W; system 2, 100 W). Complications were assessed according to CIRSE. Technical success was defined as total lesion coverage on MR imaging one day post-ablation. Local recurrence and hepatic progression were analyzed using follow-up liver MR imaging every three months. Determinants of treatment success, including lesion size, tumor entity, ablation duration, type of microwave system used, lesion and antenna conspicuity (6-point Likert scale), adjacent vascular structures (yes/no), liver cirrhosis (yes/no) as well as antenna position (central/non-central), were analyzed.

Results

Forty-eight liver lesions in 40 patients (median age 62.5 years, 70.0% male) were treated. MWA system 1 and system 2 () were used in 33/48 (68.8%) and 15/48 lesions (31.3%), respectively. Technical success was achieved in 46/48 lesions (95.8%), with one peri-procedural complication (CIRSE grade 3, 2.5%). Mean follow-up was 21.0±10.9 months with one patient lost to follow-up. Local recurrence occurred in 4/47 lesions (8.5%), all treated with system 1, while no recurrence was observed with system 2 (p=0.206). Adjacent vascular structures were associated with higher local recurrence rates (p=0.019). No local recurrence occurred in lesions <20 mm.

Conclusion

MR-guided MWA of liver tumors is safe, achieving high technical success and local tumor control. Adjacent vascular structures increase the risk of local recurrence and may warrant more aggressive treatment. Local recurrence occurred in lesions ≥20 mm and with low-power MWA systems, highlighting the influence of technical factors and the importance of selecting lesions according to device-specific ablation.

Acknowledgements

Siemens Healthineers (Erlangen, Germany) provided access to the real-time sequences and associated software.

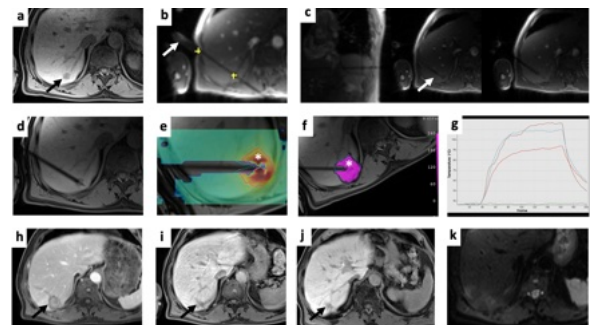


Fig 1: Eighty-year-old male with metastatic colorectal carcinoma undergoing magnetic resonance (MR)-guided microwave ablation (MWA) using the ECO-200G system.

(a) Axial T1-weighted (T1w) MRI after injection of gadoteric acid shows a liver lesion (arrow) in segment 7. (b) Defining and digital marking of entry point and target lesion (yellow tags) for procedure planning in use of real-time T1w MR sequences and finger-pointing (arrow). (c) Real-time T1w MR-sequences with 3 orthogonal slices confirmed guidance and angulation of MWA antenna (arrow) while allowing interactive graphical modification of the slice geometry. (d) Periprocedural axial T1w MRI confirmed optimal antenna placement. (e) MRI thermometry image shows temperature distribution within the liver during the ablation (asterisk). (f) Thermal dose image shows the predicted volume exposed to lethal dose of thermal energy (asterisk). (g) Graphical depiction of procedural temperature curves at varying distances from the antenna tip (see also panel e). (h) Portal-phase MRI on the first postprocedural day shows the extent of the ablation zone (arrow), complete coverage of the lesion was achieved. (i, j) Axial T1w MR images in the hepatobiliary phase after three (i) and six (j) months, as well as diffusion-weighted imaging (DWI) after six months (k) reveal decrease in lesion size accompanied by corresponding post-therapeutic changes consistent with complete response (arrows).

An Adaptable, Patient-Conforming Robotic Template for MRI-Guided Abdominal Needle Interventions

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Purpose

Percutaneous needle placement for abdominal targets such as the liver requires high accuracy under image guidance, yet closed-bore CT and MRI restrict patient access and often require iterative advance-and-check workflows. Existing rigid body-mounted needle-guidance robots only partially address this, remaining constrained by limited reachable workspace, profile height, strict mounting requirements, and poor adaptation to patient-specific geometry and oblique trajectories [1,2]. This work develops an Adaptable Robotic Template (ART), a patient-conforming platform that expands the insertion workspace at reduced height, with a surface-aware kinematic framework for consistent needle positioning and orientation control across curved, patient-specific anatomies in CT and MRI.

Material and Methods

ART comprises a flexible planar stage that conforms to the abdominal surface and a central aiming frame for needle orientation control (Fig. 1). Four independently actuated tendons route from the template corners to the aiming frame to control its position, with needle entry location and insertion orientation treated as decoupled objectives defined by a user-selected entry point and target point. Three variants share this architecture: a manual MR-safe template, a CT-compatible robotic version, and an MRI-compatible robotic version.

Because the template deforms to the patient surface, its deployed geometry and pose cannot be assumed from a fixed rigid-body model. The abdominal surface was therefore reconstructed from 16 sparse fiducial points. Candidate estimators (global polynomial fits, local regression, radial basis functions, and piecewise interpolants) were benchmarked in simulation across planar, spherical, and paraboloidal geometries under Gaussian noise (0–1.5 mm) and assessed by global root-mean-square error, noise robustness, and cross-sectional consistency. Tendon lengths over the reconstructed surface were computed as surface-constrained geodesics, from which entry-point feasibility, aiming-frame position, and needle orientation were derived.

Results

The moving least-squares quadratic (MLS-quad) estimator provided consistent surface reconstruction, achieving a noise-free mean RMSE of 0.33 ± 0.87 mm across 17 simulated surfaces and increasing to 0.56 ± 0.80 mm under 0.6 mm noise. MLS-quad also maintained strong cross-sectional consistency, with weighted height RMSEs of 0.057 mm and 0.111 mm along the x- and y-sections, respectively, and lower slope error than interpolating radial-basis and scattered-linear methods. The decoupled, surface-constrained formulation extended cable-driven planar-robot control to variable curved surfaces and expanded the achievable insertion-angle range relative to fixed-frame body-mounted systems. Benchtop testing is ongoing for manual and robotic template versions, with 400 points to be evaluated across five physical surfaces before in-scanner testing.

Conclusion

An adaptable, patient-conforming robotic template was developed as a compact, surface-aware alternative to rigid body-mounted needle guidance, compatible with both CT and MRI. A sparse-point surface-modeling and geodesic kinematic framework enables consistent control across patient-specific curved anatomies, and initial fabrication and simulation results support feasibility. Benchtop characterization is underway, with phantom validation and subsequent MRI cadaver studies at Mayo Clinic planned to establish targeting accuracy and procedural integration in the interventional MRI suite.

Acknowledgements

This work was supported by the NSF GRFP and Mayo Clinic.

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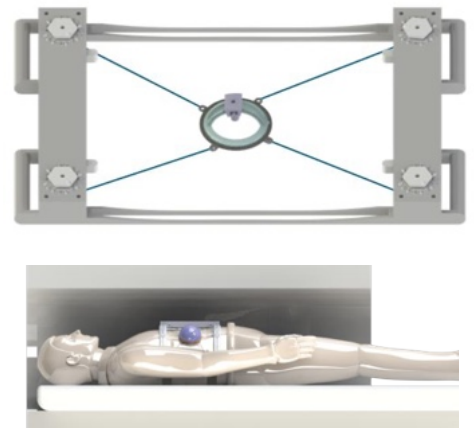


Fig 1: (Top) Top-down view of the manual MR-safe Adaptable Template. Four independently controlled tendons (dark lines) route from the corners to the central aiming disk to position the needle entry point on the conforming template. The four manual knobs are replaced with piezoelectric motors for MR-compatible robotic version. (Bottom) Robot mounted to patient in MR-scanner bore.

MRI-Guided Percutaneous Cryoablation of Hepatic Tumors in Pericardiac Locations: A Case Series

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Purpose

To report the feasibility, cardiac safety, and local tumor control of MRI-guided percutaneous cryoablation for hepatic tumors in pericardiac locations — a subset where heat-based ablation carries risk of myocardial thermal injury and CT/ultrasound guidance offers limited iceball margin visualization.

Material and Methods

IRB-approved retrospective single-center case series. From a prospectively maintained database of MRI-guided hepatic cryoablation procedures (January 2019–January 2025), five patients with hepatic lesions at pericardiac or subdiaphragmatic locations (lesion-to-cardiac structure distance ≤ 10 mm on pre-procedural MRI) and ≥ 12 months follow-up were identified. Tumor types: cholangiocarcinoma (n=1), prostate metastases (n=2), colorectal/rectal metastases (n=2). Mean lesion size 15.4 ± 4.8 mm (range 10–20 mm); mean lesion-to-cardiac distance 5.0 ± 1.7 mm (range 3.2–7.5 mm). All procedures performed in a 1.5T wide-bore MRI suite (Philips Ingenio) under general anesthesia; 17-gauge IceRod cryoneedles (Boston Scientific VisualICE) deployed via US-guided Abbocath introducer; 3–4 freeze–thaw cycles with continuous T2-weighted iceball monitoring. Primary endpoints: technical success and absence of cardiac complications. Secondary endpoint: in-field local tumor control.

Results

All five procedures were technically successful (100%). Post-ablation MRI confirmed iceball coverage with preserved T2 signal interface between the ablation margin and cardiac silhouette in all cases. No cardiac arrhythmia, pericardial effusion, phrenic nerve palsy, or other cardiac complication occurred in any patient during or after the procedure. At median follow-up of 27 months (range 19–44 months), no patient developed in-field local tumor progression. Technique efficacy was confirmed at 3-month imaging in all cases. One patient (prostate metastasis) developed out-of-field hepatic progression at 27 months, representing systemic disease rather than local failure. The closest cardiac proximity in the series was 3.2 mm (Patient 5, colorectal metastasis, anterior hepatic dome), with the longest recurrence-free follow-up of 44 months. Patient characteristics and outcomes are summarized in Table 1 and representative imaging in Fig 1.

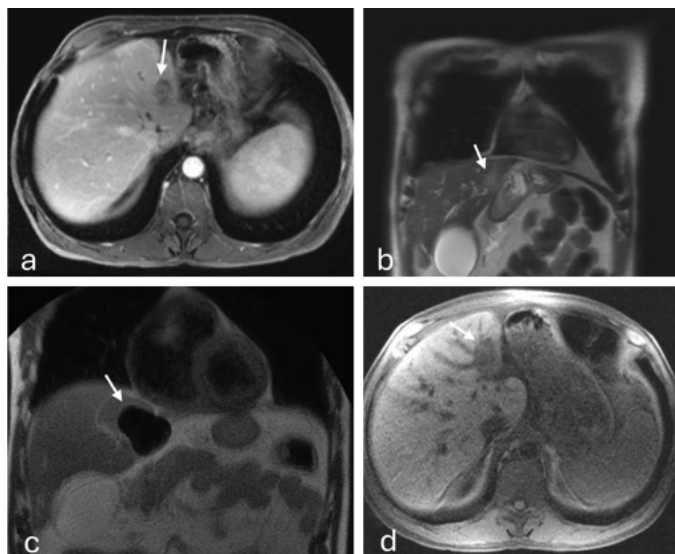


Fig1

Figure 1. 59-year-old male with cholangiocarcinoma metastasis in hepatic segment II, 7.5 mm from the cardiac silhouette. (a) Pre-procedural axial contrast-enhanced T1-weighted MRI demonstrating the index lesion (arrow) in the left hepatic lobe adjacent to the diaphragm. (b) Coronal T2-weighted MRI showing the lesion (arrow) in relation to the overlying cardiac border. (c) Intraprocedural coronal T2-weighted MRI demonstrating the hypointense iceball (arrow) with preserved T2 signal between the ablation zone and the myocardium. (d) Follow-up axial contrast-enhanced T1-weighted MRI showing a non-enhancing ablation zone (arrow) without local tumor progression.

Conclusion

MRI-guided percutaneous cryoablation of pericardiac hepatic tumors is feasible and safe, achieving 100% technical success and zero cardiac complications at iceball-to-cardiac distances as close as 3.2 mm. Real-time

T2-weighted iceball monitoring enables precise margin control without myocardial contact, a capability not reliably achievable with CT or ultrasound guidance. All five treated lesions remained free of in-field recurrence at median follow-up of 27 months (range 19–44 months), supporting MRI guidance as the preferred modality for cryoablation of hepatic lesions in pericardiac and subdiaphragmatic locations.

Optimization of Flip Angle for Dynamic Contrast-Enhanced MR Lymphangiography with Gadopiclenol

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Purpose

Dynamic contrast-enhanced MR lymphangiography (DCE-MRL) is an important diagnostic tool for assessing lymphatic anatomy and flow over a wide range of lymphatic disorders. Gadopiclenol, a new paramagnetic macrocyclic non-ionic gadolinium-based contrast agent (GBCA), has higher relaxivity than other GBCAs and necessitates additional optimization of MR sequence parameters such as flip angle and echo time. The purpose of this work was to optimize the flip angle in sequences used for MR lymphangiography.

Material and Methods

The MR signal in spoiled gradient-echo pulse sequences, such as the post-contrast sequences for MR Lymphangiography, can be maximized for a given tissue using the so-called “Ernst angle.”(1) Due to the higher relaxivity of gadopiclenol compared with other GBCAs, the T1 of contrast-enhanced (CE) lymph fluid is shorter, with a larger Ernst angle for a given TR. The Ernst angle was approximated by assuming a gadopiclenol concentration of 2.5 mM, modeling lymph fluid as blood plasma, and using T1 values from the literature (2-3). Lymphatic volume will vary with patient anatomy, but these assumptions provide a reasonable estimate until more sophisticated estimates can be made.

While Ernst angle maximizes signal for a specific tissue, it may not maximize contrast relative to surrounding tissues. By incorporating a second tissue, as shown in Figure 1, the flip angle to optimize image contrast between contrast-enhanced lymph and liver was calculated using the signal equation for spoiled gradient-echo pulse sequence.(3)

Results

Signal contrast between CE lymph fluid and liver tissue increased from flip angle = 20° to 35° and decreased at flip angles greater than 35°. Under the assumptions used here, the flip angle that maximized image contrast between contrast-enhanced lymph and liver tissue was approximately 35 degrees. Representative comparisons are shown in Figures 1 and 2.

Conclusion

Due to the higher relaxivity and greater T1 and T2 shortening effects of gadopiclenol, MR sequence parameters may require adjustment to optimize image quality. Specifically, flip angles may need to be increased to operate near peak MR signal or image contrast, and echo times may need to be reduced to better capture the MR signal. Future work will continue optimization through preclinical and phantom studies.

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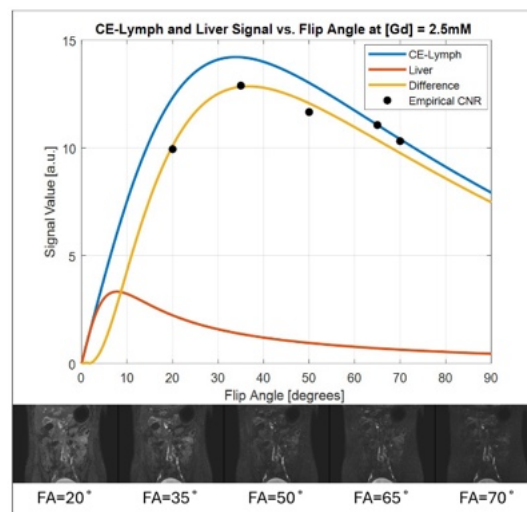


Fig 1: DCE-MRL lymphatic and liver signal intensity versus flip angle. Solid lines were computed from the spoiled gradient-echo signal equation and empirical CNR points were measured in the images shown. The flip angle providing the highest CE-lymph signal is lower than the flip angle providing the highest contrast between CE-lymph and liver.

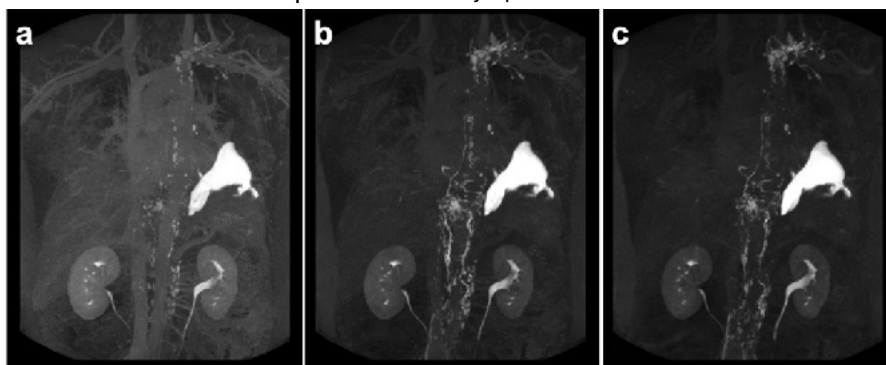


Fig 2: Post-gadopiclenol enhanced T1-weighted coronal MR images of the chest, abdomen and upper pelvis at flip angles of a) 20 degrees, b) 35 degrees and c) 54 degrees.

First-in-Human Safety and Feasibility of Nodal MR Lymphangiography with Gadopiclenol

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Purpose

To evaluate the preliminary technical development, safety, and feasibility of dynamic contrast-enhanced MR lymphangiography (DCE-MRL) performed with nodal injection of gadopiclenol.

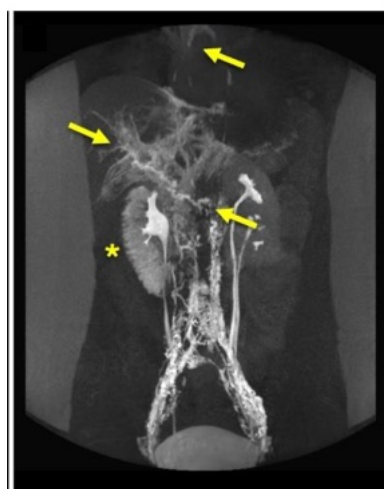
Material and Methods

This institutional review board-approved retrospective study included consecutive patients referred to the complex lymphatic disorders clinic who underwent DCE-MRL at 1.5 T with gadopiclenol administered via inguinal nodal injection. Technical performance, safety, and feasibility for imaging central conducting lymphatic anatomy, flow, and drainage were assessed.

Results

A total of 23 patients (12 female) underwent DCE-MRL (1 with gadopiclenol (2). Contrast was administered via inguinal lymph nodes in all patients (23/23, 100%). No serious or nonserious adverse events were observed (0/23). Evaluation of central conducting lymphatic anatomy, flow, and drainage into the central venous system was feasible in all patients (23/23) for both non-traumatic (Figure 1) and traumatic (Figure 2) lymphatic flow disorders.

Fig 1: Non-Traumatic Lymphatic Flow Disorder. A 21-year-old female with history of protein-losing enteropathy and sequela of chronic lymphatic fluid losses including hypoalbuminemia, hypoproteinemia, hypogammaglobulinemia and leukopenia. Vascular ultrasound and CT imaging had demonstrated left venous thoracic outlet compression (not shown). Post-gadopiclenol enhanced T1-weighted coronal MIP of lower chest, abdomen and pelvis shows markedly abnormal central conducting lymphatic flow (yellow arrows) with findings consistent with central conducting lymphatic hypertension due to impaired drainage of the thoracic duct into left venous angle and retrograde flow within the pancreaticoduodenal and hepatic lymphatics with macroscopic lymphatic leak into the duodenum (yellow asterisk).



Conclusion

Dynamic gadopiclenol-enhanced nodal magnetic resonance lymphangiography was safe and feasible for imaging central conducting lymphatics.

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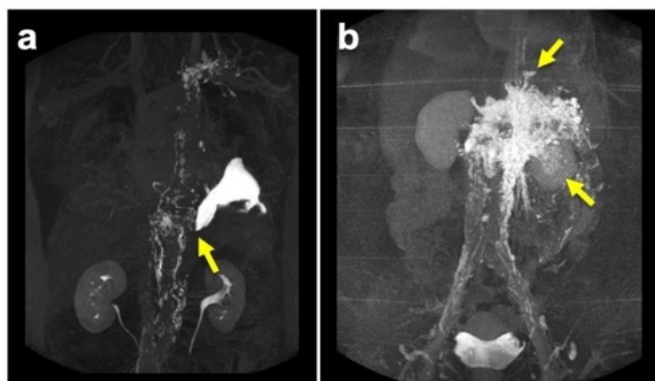


Fig 2: Post-gadopoclenol T1-weighted coronal maximum intensity projections (MIPs) of Traumatic Lymphatic Leaks and/or Flow Disorders. a) 59-year-old male post bilateral lung transplant with left chylothorax. Abnormal lymphatic flow is shown from left retroperitoneal lymphatics to left pleural space from approximately T10-T12 (yellow arrow). b) 50-year-old female with history of non-traumatic left chylothorax and chyluria worsened following thoracic duct embolization at outside hospital. Abnormal central conducting lymphatic flow is shown with markedly dilated abdominal lymphatics and reflux into left pleural, left renal, mesenteric, pancreatic, splenic and hepatic lymphatics with no in-line drainage into central venous system (yellow arrows).

Feasibility of MR guided lung biopsy at 0.55T using 3D stack-of-spirals UTE

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Purpose

Image-guided lung biopsy is critical for diagnosing pulmonary lesions, with CT being the most widely used imaging modality¹. While MRI potentially offers advantages of radiation-free guidance and superior soft-tissue contrast, imaging of lung with MRI remains challenging due to low proton density, short T2*, and susceptibility gradients at tissue-air interfaces. Consequently, MRI-guided lung biopsy has been explored only in limited studies to date². Modern mid-field MRI systems are attractive for lung imaging due to prolonged T2* and reduced susceptibility. High-resolution structural imaging has been achieved at 0.55T using ultrashort echo time (UTE), T2 weighted BLADE, and diffusion³. Furthermore, modern mid-field MRI shows significant potential guiding interventions with reduced device heating and improved bore accessibility⁴. In this study, we evaluated the feasibility of using UTE imaging at 0.55T MRI to visualize needles inserted into lung parenchyma in a porcine model.

Material and Methods

Simulated lung biopsies were performed in three porcine models. Data were acquired on a 0.55T scanner (MAGNETOM Free.Max, Siemens Healthineers AG, Forchheim, Germany). A 19cm iLoop Interventional coil (Noras MRI Products GmbH, Höchberg, Germany) was placed around the point of needle insertion, and a 12-channel surface coil was placed underneath the pig during imaging. A prototype coaxial needle with proprietary MR-visible markers (Cook Medical, Bloomington, IN, USA) was inserted into the lung parenchyma percutaneously. Needle placement was assessed with immediate post-procedural imaging using a Spiral VIBE Ultra Short Echo (UTE) research application sequence with stack-of-spirals acquisition⁴. The imaging volume was reformatted into three 2D views: two along the needle insertion axis, and one perpendicular to it. The contrast-to-noise ratio (CNR) of needle artifact was calculated as the absolute difference between the mean signal intensity of the needle region-of-interest (ROI) and the lung parenchyma ROI, divided by the standard deviation (SD) of a noise-only region (e.g. airway).

Results

Figure 1a shows the representative multiplanar views of the UTE MRI images of a porcine lung with prototype needle inserted. The UTE images provided high-resolution visualization of the lung parenchyma at 0.55T, enabling clear delineation of both vessels and airways. Needle artifacts are distinctly visible against the background parenchyma, and both the markers and the tip of the needle can be identified (red and green arrows). Figure 1b summarizes the calculated CNR values of needle artifact obtained from three porcine models.

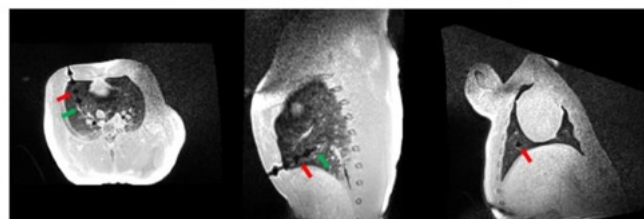
Conclusion

In this work, our preclinical study demonstrated that UTE imaging at 0.55T allowed needle visualization in the lung. The mid-field MRI, combined with UTE acquisition, offers substantial lung parenchyma signal and distinct contrast of the needle. These findings support further development of mid-field MRI-guided pulmonary interventions, which may become a radiation-free, contrast-rich alternative to CT-guided procedures.

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a) 3D UTE MRI of Porcine Lung with Needle Inserted (Multiplanar Views)



b) CNR of the Needle Artifacts

Porcine Lung #	1	2	3	Mean $\pm$ STD
CNR	7.1	7.4	9.8	8.1 $\pm$ 1.5

Fig 1: a) Multiplanar views of the UTE MRI images of a porcine lung with needle inserted. Red arrows indicate the needle markers, while green arrows indicate the needle tip. b) Table lists the calculated CNR values of needle artifacts obtained from three porcine models. CNR is calculated as the absolute difference between the mean signal intensity of the needle ROI and the lung parenchyma ROI, divided by SD of a noise-only region (e.g. airway).

Combined MRI-Guided Microwave Ablation and Cryoablation for Hepatic Tumors in a Single Session: A Single-Center Feasibility and Safety Report

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Purpose

To evaluate the feasibility, safety, and mid- to long-term local control outcomes of combined MRI-guided microwave ablation (MWA) and cryoablation for hepatic tumors performed in a single session, leveraging the complementary strengths of each modality under real-time MRI guidance.

Material and Methods

This IRB-approved retrospective study included 6 patients (5M/1F; mean age 68.2 years) who underwent 7 combined MRI-guided MWA and cryoablation procedures for 17 hepatic lesions between December 2022 and February 2025. Primary malignancies: prostate cancer (n=4), neuroendocrine tumor (n=1), renal cell carcinoma (n=1). Mean lesion size 2.2 cm (range 1.0–3.5 cm). All procedures performed under general anesthesia with combined ultrasound/MRI guidance in a 1.5T wide-bore suite (Philips Ingenio). Cryoablation (2–3 cryoneedles, 3 freeze–thaw cycles; Boston Scientific VisualiCE) was preferentially applied to lesions adjacent to bile ducts, vessels, or the diaphragm/heart; MWA (1–2 antennae, 65–75 W for 5–10 minutes) was used for deeper or larger lesions. One case employed simultaneous dual-antenna MWA for a large capsular lesion. MRI provided real-time iceball monitoring and soft-tissue margin assessment throughout. Technical success, complications (SIR classification), and local control were assessed through September 2025.

Results

Technical success was achieved in 100% of procedures (17/17 lesions; 9 MWA, 8 cryoablation). One minor complication occurred (SIR Grade B: small renal hematoma, 14%), managed conservatively. At a median follow-up of 22 months (range 7–33 months), durable local control was achieved in 16 of 17 lesions (94.1%). The single locally uncontrolled lesion was subsequently retreated with MRI-guided cryoablation with success. Three of 6 patients (50%) remained completely disease-free (no local or systemic disease) at last follow-up; 2 patients developed out-of-field systemic progression unrelated to the treated ablation zones; 1 patient died of systemic disease progression. No local recurrence was identified in any ablation zone at last follow-up imaging.

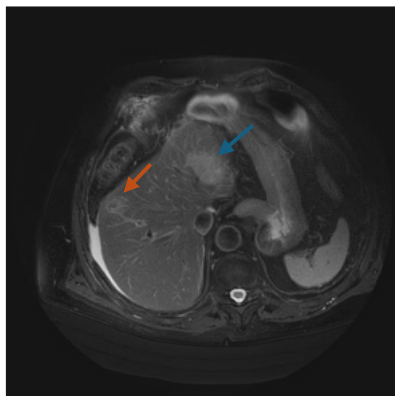


Figure 1: Post ablation T2 images. Blue arrow (post cryoablation), orange arrow (post microwave).

Conclusion

Combined MRI-guided MWA and cryoablation in a single session is technically feasible, safe, and achieves high local tumor control (94%) for complex hepatic tumors. Selecting ablation modality by lesion location — cryoablation for critical-structure proximity, MWA for deep or larger lesions — and using MRI for real-time margin assessment maximizes both safety and efficacy. The ability to deploy dual modalities, including simultaneous dual-antenna MWA, within one session expands therapeutic versatility for challenging hepatic disease. These

results support further evaluation in larger prospective cohorts.

Acknowledgements

No external funding was received. The authors declare no conflict of interest. IRB-approved retrospective study; informed consent waived.

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MRI-Guided Percutaneous Cryoablation of Anatomically Challenging Primary and Metastatic Liver Tumors: A Five-Year Single-Center Experience

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Purpose

To evaluate the technical success, locoregional tumor control, and safety of MRI-guided percutaneous cryoablation for primary and metastatic liver tumors at anatomically challenging locations, where real-time iceball visualization confers a critical advantage over CT or ultrasound guidance.

Material and Methods

This IRB-approved retrospective single-center study included 20 consecutive patients (10M/10F; mean age 61.1±13.8 years) who underwent 24 MRI-guided percutaneous cryoablation sessions treating 34 hepatic lesions between January 2019 and February 2025. All lesions were at anatomically challenging locations: hepatic dome/subdiaphragmatic (n=6 sessions), pericardiac ≤10 mm from heart (n=6), perigallbladder (n=3), IVC-adjacent (n=3), periductal (n=1), and subcapsular/liver edge (n=6). Tumor types included prostate cancer metastases (n=7), colorectal metastases (n=5), neuroendocrine tumor metastases (n=3), HCC (n=3), and cholangiocarcinoma metastases (n=2). Mean maximum lesion diameter was 12.1±7.3 mm (range 3–33 mm). All procedures used a 1.5T wide-bore MRI suite with IceRod/IceSeed cryoneedles (Boston Scientific VisualICE); iceball growth was monitored with T2-weighted sequences every ~20 seconds. Gelfoam was injected through the Abbocath introducer upon needle removal in all cases. Technical success, local tumor control, and complications (SIR Adverse Events Classification) were assessed.

Results

Technical success was 100% (24/24 sessions). Primary technique efficacy was achieved in 29/34 lesions (85.3%); residual disease was identified in 5 lesions (14.7%), concentrated at perigallbladder and IVC-adjacent sites. Local tumor progression (LTP) = 0% among the 29 completely ablated lesions (median follow-up 31.5 months). Two residual lesions were successfully re-treated with repeat cryoablation; 3 underwent surgical resection. Outfield progression occurred in 4 patients (20%), reflecting systemic disease behavior. By tumor type: prostate metastases achieved 100% primary technique efficacy (10/10 lesions); colorectal 80% (8/10); NET 85.7% (6/7); HCC 66.7% (2/3); cholangiocarcinoma 75% (3/4). One minor complication occurred (SIR Grade A: superficial skin burn from arterial line cable contact). No hemorrhage, biliary injury, bowel perforation, or cryoshock was recorded. Six patients died from systemic disease progression unrelated to the procedure.

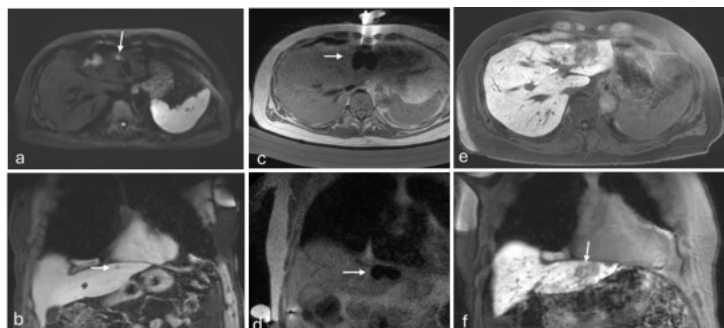


Figure 1. Representative case of complete MRI-guided percutaneous cryoablation: hepatic dome prostate cancer metastasis. (a) Axial DWI pre-ablation: restricted diffusion at hepatic dome (arrow). (b) Coronal LAVA pre-ablation: lesion abutting diaphragm (arrow). (c) Axial intraprocedural T2-weighted MRI: signal-void iceball encompassing target with cryoneedle in situ (arrow). (d) Coronal intraprocedural T2-weighted MRI: iceball extent confirmed in coronal plane (arrow). (e) Axial LAVA post-contrast at 1 month: no residual tumor enhancement (arrow). (f) Coronal LAVA post-contrast at 1 month: complete ablation zone, no marginal enhancement (arrow).

Conclusion

MRI-guided cryoablation achieved perfect technical success and strong local tumor control at difficult liver locations, with zero recurrence in completely treated lesions. The key advantage is real-time iceball visibility on T2 MRI, eliminating radiation exposure while allowing precise, safe treatment where CT and ultrasound fall short.

Acknowledgements

No external funding was received. IRB-approved retrospective study; informed consent waived.

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MRI-Guided Cryoablation for Residual Limb Neuroma Pain After Transtibial Amputation and Targeted Muscle Reinnervation: A Case Report

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Purpose

To describe the management of persistent residual limb neuroma pain following transtibial amputation and targeted muscle reinnervation (TMR), highlighting the role of MRI/ultrasound-guided cryoablation in refractory neuropathic pain control.

Material and Methods

A 45-year-old male firefighter with a history of Lisfranc fracture, multiple foot/ankle surgeries, complex regional pain syndrome (CRPS), and right transtibial amputation developed chronic residual limb pain refractory to conservative measures. Despite TMR, he experienced focal neuropathic pain at the common peroneal nerve (CPN) remnant and tibial/medial sural region. Diagnostic nerve blocks and CT-guided nerve injections confirmed symptomatic involvement of the CPN and medial sural cutaneous nerve as the pain generators. Subsequently, percutaneous MRI/ultrasound-guided cryoablation was performed under general anesthesia using one IceSeed and one IceRod cryoprobe (Boston Scientific VisualICE) with three freeze–thaw cycles. MRI guidance enabled real-time iceball visualization around the target nerves to ensure complete coverage and minimize collateral injury.

Results

Cryoablation achieved complete technical success with no immediate complications. Follow-up MRI demonstrated expected post-ablation signal changes with no recurrent neuroma formation at the treated sites. Clinically, the patient reported resolution of his most intense stump pain adjacent to the fibular remnant, improved prosthesis tolerance, and greater functional activity. Pain scores decreased significantly from a baseline of 3–7/10 to minimal daily pain. The patient required prosthesis refitting due to secondary muscle atrophy from expected denervation but remained satisfied with his overall pain control and functional status.

Conclusion

In patients with focal neuropathic residual limb pain after transtibial amputation and TMR, MRI-guided cryoablation of symptomatic neuromas provides a safe, effective, and minimally invasive treatment option. Real-time MRI iceball visualization ensures precise coverage of target nerves while minimizing collateral injury. Diagnostic nerve blocks are a valuable tool to identify pain generators and guide patient selection. This case demonstrates durable pain relief and improved functional outcomes following cryoablation, supporting its role in the multidisciplinary management of refractory residual limb pain.

Acknowledgements

No external funding was received. The authors declare no conflict of interest. Informed consent was obtained from the patient for publication of this case report.

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MR-guided cryoneurolysis for painful neuropathy – a case series

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Purpose

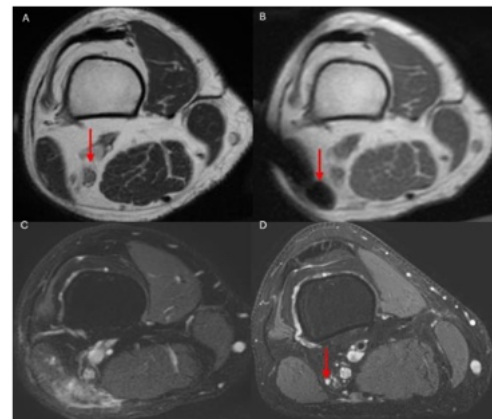
To assess the feasibility and effectiveness of MR-guided cryoneurolysis for control of neuropathic pain.

Materials and methods

This retrospective case series included eight patients (3 female, 5 male; mean age, 53 years [range, 38–69]) with functionally debilitating pain. Nine procedures were performed, including repeat treatment in one patient. Treated nerves included the posterior femoral cutaneous (n = 5), posterior antebrachial cutaneous (n = 1), posterior antecubital brachial cutaneous (n = 1), S1 nerve root from metastatic prostate cancer involvement (n = 1), and the right common peroneal and medial sural cutaneous nerves in one patient (n = 1). Four patients underwent CT-guided lidocaine injection before treatment for target confirmation. Needles and cryoprobes were placed under combined ultrasound and MRI guidance. Procedures used a mean of three freeze–thaw cycles and two cryoprobes (range, 1–5). Adjunctive techniques included Gelfoam injection of the needle tract (n = 1), neuromonitoring (n = 4), and saline hydrodissection to protect the forearm skin (n = 1) and to move the rectum adjacent to the S1 nerve (n = 1).

Results

Follow-up was available for seven of eight patients. At 3 months, six of seven patients reported pain relief (86%), while one continued to have pain despite treatment of the S1 metastatic disease from prostate cancer. At 6 months, pain recurred in two of six patients (33%), while four (67%) maintained pain relief for up to 2 years. One patient with recurrent pain underwent repeat cryoablation with sustained relief for an additional 6 months and the other was lost to follow-up. No immediate post-ablation complications occurred aside from one self-limited hematoma.



Right peroneal nerve cryoablation for treatment of residual limb pain following transtibial amputation. (a) Pre-ablation T2 showing the common peroneal nerve (red arrow). (b) Intraprocedural T2-weighted image demonstrating ice-ball formation (red arrow). (c) Immediate post-ablation T2 fat-suppressed image showing edema and post-ablation changes. (d) Follow-up image demonstrating mildly decreased T2 signal and slight enlargement of the common peroneal nerve with mild surrounding soft-tissue edema and enhancement, compatible with interval cryoablation.

Conclusion

MR-guided cryoneurolysis is a safe and effective minimally invasive modality to control neuropathic pain.

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Methods for simulating advanced MR guided cortical brain gene therapy

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Purpose: MR-guided intraparenchymal brain delivery of AAV-based gene therapies has made huge strides over the last two years when localized genetic correction is needed. Convection-enhanced delivery (CED) to the gray matter regions in the striatum using minimally invasive catheters has resulted in FDA-approval of Kebilidi (PTC Therapeutics) and the promising AMT-130 trials for Huntington's Disease (uniQure). Successful treatment of the vast majority of monogenic neurodegenerative diseases, like lysosomal storage diseases, will depend on whether low resistance, penetrating sulci deleteriously shunt treatment away from cortical folds.

Non-animal models are needed to speed advances in brain coverage with CED. We present a gel model of the cortex which will ultimately allow us to 1) characterize our ability to image and predict unwanted shunting of gene therapies to penetrating sulci, and 2) determine if cortical treatment in infancy when the brain surface is much smoother will simplify the crucial, complete coverage of genetic therapies at the cortical surface.

Methods:

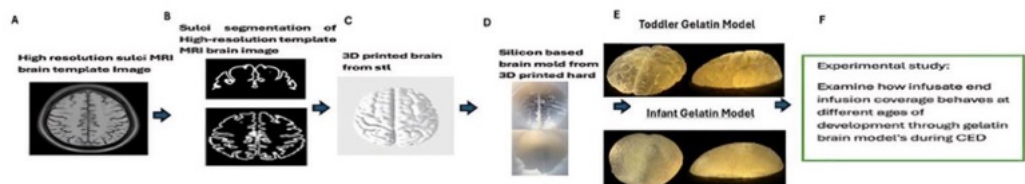


Figure 1: Preliminary use of (A-B) 3D T1-W brain scans to build (C) 3D printed cortical surfaces that are filled with (E) gel to create models of highly folded 5 year old cortex and the much smoother infant cortex.

Gel phantoms often simulate CED in neurological research and training. We undertook morphological image processing of 3D T1-weighted GRE MRI to generated STL files of cortical surfaces of the typical 2 month and 5-year age mark, as shown in Fig. 1 A-C. We next fabricated silicone-based brain molds from these models (Fig. 1D). We used the molds to generate gelatin brain models, doped to reflect gray matter properties, also shown in Fig. 1E. The brain model was inserted in a bath doped to simulate the surrounding CSF.

We first hypothesized whether rapid 3D positive contrast imaging of sulci using bSSFP could capture all the sulci folds from the original T1-weighted template. Sulci maps of these cortical models were derived after

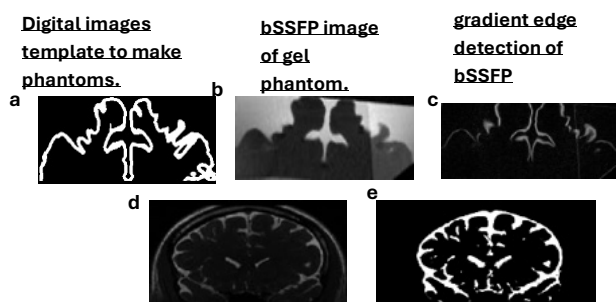


Figure 2 Illustration shows that high definition b) bSSFP images can be used to obtain c) gold standard sulci mapping from gel phantoms generated from a) high-definition structural images of both 5 year old and 2 month old brain. Example images d) bSSFP images from adult brain and e) sulci map.

obtaining a 3D bSSFP scan with isotropic 0.5 mm resolution of the gel structure. We used the structural term within the Structural Similarity Index Metric (SSIM) to calculate the similarity between the high resolution bSSFP of the phantom brain and the original, high resolution T1-W GRE used to build the models.

Results: The 3D T1-W scan used to generate the corresponding gel brain phantom is highlighted in Fig 2 a. The acquired bSSFP images of the gel

phantom and the corresponding gradient sulci maps are shown in Fig 2b and Fig 2c, respectively. The SSIM value between the original template Fig 2a to create the gel phantom and the gradient of bSSFP of the gel phantom was 0.915. Fig 2d and Fig 2e shown an example of bSSFP for 5-year-old subject.

Conclusion: We were able to design subject-specific brain cortical phantoms that can serve as gold models of sulci mapping. This brain phantoms will be utilized in further study for convection enhance delivery studies to compare the percentage of cortical volume that is likely treatable for various catheter types and trajectories. The models will allow testing of the hypothesis that the infant brain will suffer less unwanted shunting to penetrating low resistance sulci.

Brain Laser Interstitial Thermal Therapy (LITT) in Drug-Resistant Epilepsy: Case Examples

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Purpose

Laser interstitial thermal therapy (LITT) is a minimally invasive, MR-guided, heat-based ablation technique that is increasingly applied to intracranial pathologies. One major application of LITT is drug-resistant epilepsy, where it can be an alternative to open surgery. Using stereotactic navigation, laser fibers are precisely sited in focal lesions known to trigger seizure activity (seizure onset zones – SOZ), taking care to follow a trajectory through brain parenchyma that respects important white matter tracts. Laser activation deposits thermal energy at the tip of the fiber, creating an area of coagulative necrosis. The temperature change is monitored using real-time MR thermography techniques, allowing the confident assessment of an ablation zone that ideally encompasses the target completely without involving adjacent critical structures such as eloquent regions. LITT can also be used in non-lesional epilepsy where there is no clear SOZ. In this context, the white matter tracts that permit the spread of seizure activity (e.g. the corpus callosum) are ablated. We illustrate the use of LITT in a selection of epilepsy cases, focusing on important technical and anatomical factors that ensure safety and efficacy.

Material and Methods

All cases are discussed in a multidisciplinary meeting with input from neurology, neurosurgery and neuroradiology before proceeding to LITT. The procedures are performed by a neurosurgeon in collaboration with a neuroradiologist. A stereotactic frame (Leksell Stereotactic System; Elekta AB, Stockholm, Sweden) and neuronavigation (Stealth; Medtronic, Minneapolis, MN, USA) are used to place skull anchoring bolts and plan fiber trajectories. We use the Visualase MR-guided laser ablation system (Medtronic, Minneapolis, MN, USA). Intraprocedural MRI, specifically MR thermometry (based on the proton resonance frequency shift), is used to monitor the ablation. This provides near real-time feedback of temperature within the brain parenchyma, and temperature limit-markers are used to set upper and lower bounds. Ablation zones are derived from this data; these “thermal damage thresholds” are a function of temperature and time (according to the Arrhenius equation).

Lesional cases involve treatment of a defined SOZ. In terms of etiology, these can be divided into lesions involving the temporal lobe (i.e. mesial temporal sclerosis), tumor (hypothalamic hamartoma; dysembryoplastic neuroepithelial tumor [DNET]; developmental lesions (malformations of cortical development e.g. focal cortical dysplasia; encephaloceles); vascular lesions (cavernoma), and other miscellaneous causes.

For non-lesional cases, we focus on treatment of drop attacks in epilepsy syndromes such as Lennox-Gastaut syndrome. LITT is used to ablate the corpus callosum, effectively performing a minimally invasive “disconnection” procedure. Ablation may be total or partial, and partial cases may be extended to complete disconnection if required. We describe a standardized approach with trajectories targeting the genu, posterior body, splenium and anterior body. Structures at risk are identified and avoided as much as possible.

Results

Our selected cases span the spectrum of causes of drug-resistant epilepsy, broadly categorized as lesional and non-lesional. Common to all cases is the careful planning of fiber trajectories, avoiding blood vessels and minimizing traversal of sulci and ventricles. This involves integration of standard structural imaging and advanced techniques such as diffusion tensor imaging (DTI); functional and physiological imaging (e.g. functional MRI, positron emission tomography PET); as well as results from stereotactic electroencephalography (SEEG). A thorough knowledge of white matter tracts and functional anatomy (i.e. eloquent regions) is required to minimize postoperative neurological deficits.

Conclusion

LITT has an increasingly important role in the treatment of selected cases of drug-resistant epilepsy. A multidisciplinary approach is essential to ensure proper patient selection, treatment planning, procedural success and long-term follow up.

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Microwave ablation of liver tumor assisted by magnetic resonance thermometry based on proton resonance frequency displacement method

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Purpose: The purpose of this study is to explore the application value of magnetic resonance temperature measurement technology based on proton resonance frequency displacement method in microwave ablation of liver tumors. **Methods:** From June 2023 to December 2025, 40 patients (33 cases of primary liver cancer and 7 cases of liver metastases) were included in this study. 3.0T MRI(Ingenia, Philips, Netherlands) was used to guide MWA operation and MRT monitoring, and the scanning sequence is shown in Table 1. T1W-mDixon-WipBH (breath holding) and T2WI-SPIR-RT sequence (respiratory gated trigger scanning) were used to guide puncture. After the puncture is in place, start the MRT temperature measurement software and set the isotherms of 50°C and 60°C. TAMP sequence was used to monitor the temperature change during ablation, and a pseudo-color temperature distribution map was generated, and 60°C isothermal surface was sketched as the predicted ablation area. Postoperative scanning was used to evaluate the efficacy and complications of ablation. One month after operation, the activity and technical effectiveness of ablation lesions were evaluated by contrast-enhanced MRI. **Results:** MWA was completed in 42 lesions (33 hepatocellular carcinoma lesions and 9 hepatic metastases), and there were no serious complications and tumor residue or recurrence. Ablation focus showed typical "target sign" on T1WI, low signal focus was completely covered by high signal ablation focus, and showed low signal on T2WI. The safe ablation edge was set to 5-10mm outside the focus. MRT workstation records the dynamic evolution of thermal field during ablation. The ablation center changes from low temperature color to red color representing high temperature. With the increase of temperature, the center of the lesion changes into a completely black area, and a red isotherm of 60°C can be seen at the edge of the black area, a yellow isotherm of 50°C can be seen within a few millimeters outside the red isotherm, and a blue isotherm below 50°C can be seen at the outer edge. One month after operation, the ablation focus showed slightly high signal on T1WI, low signal on T2WI, and no enhancement was found. **Conclusion:** Magnetic resonance thermometry assisted microwave ablation of liver tumor can directly and quantitatively monitor the distribution of thermal field, improve the safety of operation and reduce complications.

Keywords: Thermometry, Microwave ablation, Magnetic resonance imaging.

Table 1 Sequence-related acquisition parameters.

Sequence	TE/ms	TR/ms	FA/°	GAP/mm	ST/mm	FOV/mm	NEX	T (s)
T1W-mDixon-WipBH	1.5	4.2	10	2	3	380×289	1	11
T1W-mDixon-WipFB	1.5	4.2	10	2	3	380×289	1	11
TMAP	8.0	16	30	3	3	280×280	1	3
T2WI-SPIR-RT	70	1100	90	3	5	380×289	1	32

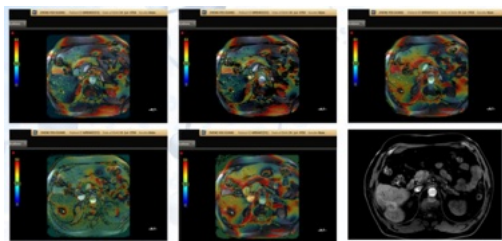


Fig.1 With the increase of temperature in the process of microwave ablation of liver cancer patients, MR Temperature Measurement Software can display the pseudo-color in the ablation area gradually changing from low temperature to red high temperature color in real time.

Respiratory-Triggered Real-Time PRF Thermometry in the Liver at 0.55 T

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Purpose

Respiratory-triggered real-time PRF thermometry is used for ablation monitoring in the liver at 1.5 T [1]. For MR-guided interventions, low-field MRI offers reduced device heating and susceptibility artifacts [2]. So far, liver thermometry at 0.55 T was investigated using segmented echo-planar imaging with retrospective gating, as well as prospective gating (10 breathing cycles per volume) [3]. Here, we evaluated respiratory-triggered single-shot 2D EPI thermometry in the liver at 0.55 T (one volume per breathing cycle) in volunteers. In a phantom experiment, a microwave ablation was performed with the same protocol to demonstrate thermometry during heating.

Material and Methods

Acquisition: Experiments were performed at 0.55 T (MAGNETOM Free.XL (510k pending), Siemens Healthineers AG, Forchheim, Germany). In two volunteers, respiratory-triggered (exhale position) single-shot EPI series were acquired with a pilot-tone navigation system (2 series per volunteer, each series consisting of at least 30 dynamic repetitions). For the heating experiment, a commercial, non-MR-compatible microwave ablation system (IntelliBlate, Varian Medical Systems Inc. - A Siemens Healthineers Company, Palo Alto, CA, USA) was used in combination with a prototypical probe inserted into a porcine tissue phantom. EPI scans were triggered every 4 s. Acquisition parameters: FOV = 284 mm x 350 mm; slice thickness = 5 mm; 7 slices; matrix = 128x104; TE = 34 ms; acq. duration = 694 ms; BW = 1002 Hz/px; flip angle = 90°. **Postprocessing:** PRF temperature mapping was done with the averaged, first 5 measurements as baseline. Phase drift was assessed in a background ROI. **Evaluation:** Temperature standard deviation was evaluated in an ROI, centrally placed in the liver of the volunteers. For the phantom experiment, temperature was evaluated over time in two pixels in proximity of the probe.

Results

A median temporal standard deviation within the selected ROIs between 1.4 °C and 1.9 °C was found for the four volunteer measurements (Fig. 1). For the heating experiment, electromagnetic interference from the non-MR-compatible ablation system is impacting SNR most visibly during the heating phase (Fig. 2). Still, temperature-time curves clearly depict the different periods (pre-heating, heating, post-heating).

Conclusion

Our data indicates that respiratory-triggered single-shot 2D multi-slice EPI PRF thermometry in the liver for monitoring of thermal ablations seems feasible. More accurate triggering in ventilated patients could potentially further improve temperature accuracy.

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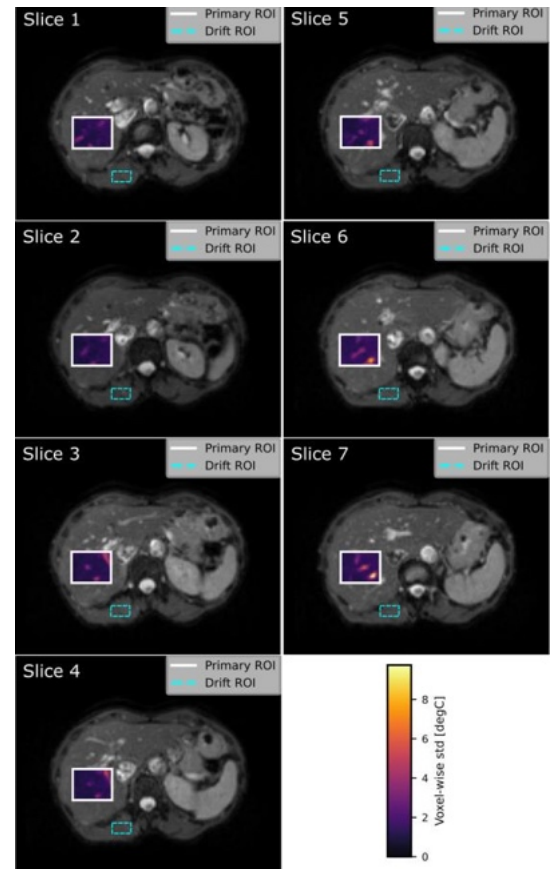


Fig 1: Voxel-wise temperature uncertainty across a multi-slice ROI, calculated from the PRF-derived temperatures in the volume acquired with the 2D echo-planar imaging sequence for one example measurement series.

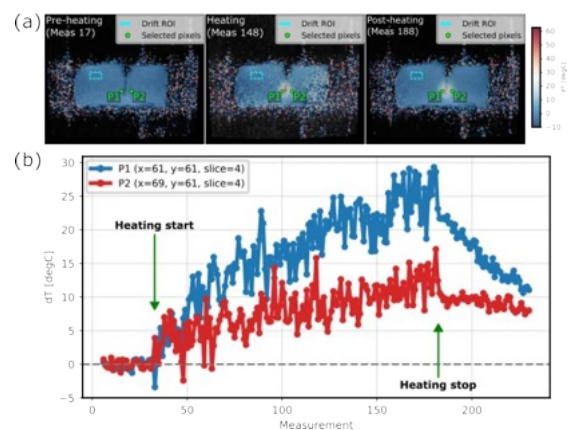


Fig 2: MW-induced heating in a porcine phantom. (a) Temperature map showing central imaging slice during pre-heating, heating and cooldown phase. (b) Temperature over time for two selected pixels.

Volumetric 3D Stack-of-Spirals PRFS Thermometry at 0.55 T for MR-Guided Hepatic Microwave Ablation: Initial Feasibility

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Purpose: Volumetric proton resonance frequency shift (PRFS) thermometry enables monitoring of magnetic resonance (MR)-guided hepatic microwave ablation (MWA) by resolving temperature changes across the intended ablation zone and safety margin¹. Contiguous 3D Stack-of-Spirals PRFS thermometry has recently been developed and clinically evaluated at 1.5 T², but its feasibility at 0.55 T remains unknown³. Wide-bore 0.55 T MRI is attractive for intervention because it improves procedural access and mitigates susceptibility and off-resonance constraints in spiral readouts⁴. However, lower field strength reduces both PRFS phase sensitivity and signal-to-noise ratio (SNR), making temperature precision the key feasibility constraint at 0.55 T. This feasibility experiment evaluated 3D Stack-of-Spirals PRFS thermometry at 0.55 T in a phantom ablation and in one free-breathing volunteer.

Material and Methods: Thermometry was performed on a 0.55 T wide-bore interventional MR system (MAGNETOM Free.XL*, Siemens Healthineers, Erlangen, Germany) using a dual-echo 3D Stack-of-Spirals spiral-in/in sequence (TE=19.2/37.0 ms; TR=51.9 ms with fat saturation every third TR; 400×400×60 mm³; 2.5 mm isotropic; per-TR motion navigator). One healthy volunteer was scanned without heating during a 16-min free-breathing dynamic acquisition. A static fat-containing pork phantom was scanned with the same sequence and duration during 3 min baseline, 8 min MWA at 30 W, and 5 min cooling, with the probe approximately orthogonal to B₀. Experiments were performed using a commercially available, non-MR-compatible microwave ablation system (IntelliBlate, Varian Medical Systems, Inc. - A Siemens Healthineers Company, Palo Alto, CA, USA) in combination with a modified, prototypical probe. Post-ablation T2-weighted HASTE was resampled to the thermometry grid for qualitative comparison with the CEM43-derived ablation zone⁵. Reconstruction, B₀ drift correction, and temperature calculation followed the recently published respiratory-resolved framework², retaining approximately 33% end-expiratory data and reconstructing one temperature volume per respiratory cycle using a five-breath sliding window; phantom data were reconstructed at the same nominal temporal resolution. Temperature precision was the time-averaged spatial standard deviation within a 147-voxel cylindrical ROI in unheated tissue (Fig. 1a, f).

Results: Phantom MWA produced focal heating near the probe with progressive temperature increase and visible frame-to-frame variability in local temperature curves (Fig. 1). A spatially confined CEM43-derived ablation zone was obtained; post-ablation T2-weighted HASTE served for qualitative, not quantitative comparison. Temperature precision was 3.19 ± 1.44 °C in the phantom and 2.33 ± 0.27 °C in the free-breathing volunteer.

Conclusion: Volumetric 3D Stack-of-Spirals PRFS thermometry at 0.55 T was feasible in phantom MWA and free-breathing abdominal imaging. Despite the absence of motion, the phantom showed higher variability than the volunteer, indicating that technical noise sources in the interventional setup remain relevant. Sequence optimization and denoising may further improve thermometry performance. Larger cohorts are required to assess robustness and quantitative ablation-zone accuracy for MR-guided hepatic MWA.

Acknowledgements:

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*MAGNETOM Free.XL is still under development and not yet commercially available. Its future availability cannot be guaranteed.

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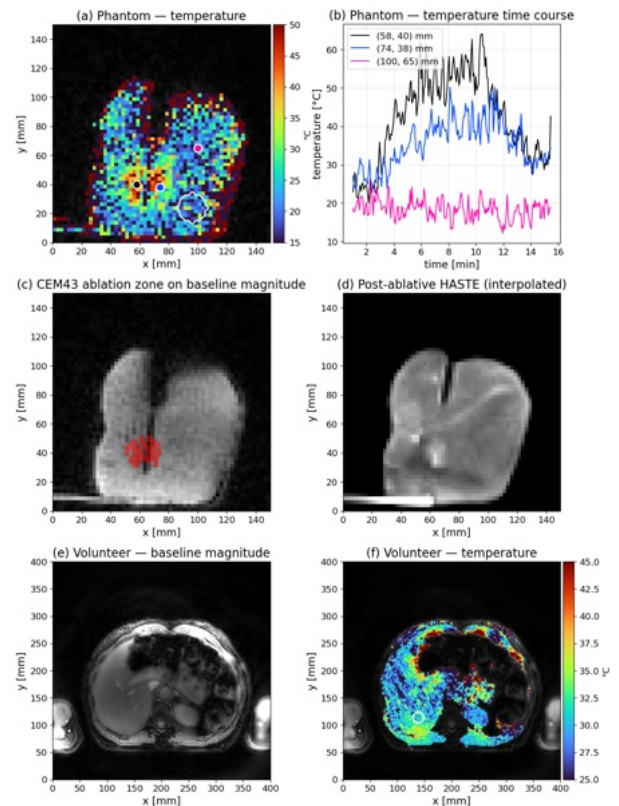


Fig. 1 0.55 T 3D Stack-of-Spirals PRFS thermometry in a phantom and volunteer. (a,b) Phantom temperature map and local temperature time courses during MWA. (c,d) Baseline overlay with CEM43-derived ablation zone and post-ablation reference imaging. (e,f) Volunteer baseline magnitude image and free-breathing temperature map.

A subvoxel correction of boiling-induced susceptibility artifacts in MR thermometry and dosimetry for monitoring MW thermoablation

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Purpose: Real-time monitoring of microwave liver ablation (MWA) using MRI thermometry can be hindered by susceptibility artefacts [1] caused by boiling [2]. The latter provoke unphysical temperature deviations that prevent lesion size prediction. This study proposes a correction based on removing the contribution of the susceptibility artifact using subvoxel sources of susceptibility.

Material and Methods: The protocol was approved by the ethics committee for animal experimentation. MWA (N=23) were performed using an AveCure microwave system (14-gauge large antenna, MedWave, USA, 7'30, target temperature of 80°C) on seven pig livers. Acquisitions were performed under respiratory gating using multi-shot gradient-echo EPI sequences on 1.5T and 3T Siemens. The voxel size was 2.3 x 2.3 x 3.0 mm³, slice gap 1.5 mm with 7 slices. 3D post contrast T1w images were acquired at the end of the experiment, to visualize the non-perfused volumes at a resolution of 1.2x1.2x1.2 mm³. The proposed correction of susceptibility artifacts was then applied after temperature calculation. The cumulative TD based on the Sapareto equation (CEM43) was then computed as the lesion volume ('TD ablation zone') Susceptibility correction methodology consists in generating a susceptibility distribution that reproduces the main stable component of the artifact through time. The susceptibility values were determined at a finer resolution (subvoxel) and using adjacent slices contribution by solving a linear inverse problem which was set to nullify the artifact. The data analysis was performed retrospectively, aimed at assessing the ability of the method to reproduce the artifacts and correct the temperature deviations. Comparisons were performed with the postprocedural 'T1w ablation zone' against two 'TD ablation zones': one given by the uncorrected data and one given by the proposed correction.

Results : Out of seven pigs with 23 MWA, 3 (13%) MR temperature acquisitions were excluded due to low signal-to-noise ratio (SNR). Susceptibility artifacts caused by boiling were observed in 11 (48%) acquisitions. Figure 1 presents a selection of corrections over six MWA with varying degrees of: artifact intensity, shapes of the artifact, orientations of the ablation probe in respect to the slice and B0 magnitudes (1.5T and 3T). The geometric and overlap metrics indicate an increase of agreement between corrected-and-T1w measures versus uncorrected-and-T1w: (i) overestimation of lesion volume (T1w-TD) was reduced in average from -1.02 to -0.59 cm³ (42% reduction), most noticeable in Y (26% reduction in average from -3.46 to -2.55 mm) and Z axis (17% reduction in average from -3.20 to -2.67 mm); (ii) median volumetric Dice increased by 4.8% (0.62 to 0.65).

Conclusion : The use of subvoxels was motivated to diminish the main pitfall in the dipole model: for positions close to the point of non-zero susceptibility. The prior selection is a potential limitation. A perfect correction would require a time dependent simulation. However, the artifact is relatively stable through time. Comparison with T1w imaging showed improvements in prediction of lesion volume. The proposed algorithm takes into account partial volume effects and the contribution of adjacent slices. It possesses the necessary adaptability for simulating a diverse range of deformed dipole-like artifacts seen in experimental data.

Acknowledgements : This work received financial support from the CNRS through the MITI interdisciplinary programs (CHAUPATAT) through its exploratory research program and from the French National Investments for the research programs ANR-DFG (SMART-HEAT)

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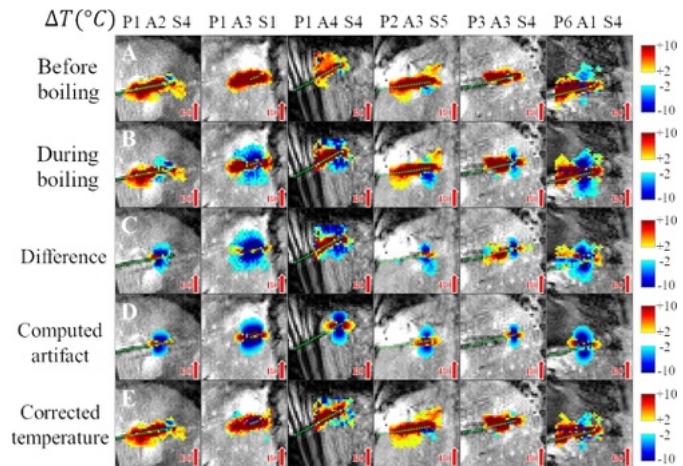


Figure 1 : Spatial distribution of temperature in six different in-vivo liver ablation experiments before and after correction. Temperature maps before (A) and during (B) susceptibility artifacts. (C) A priori of corrupted temperature distribution related to the artifact appearance. (D) Estimated artifact. (E) Corrected temperature maps.

MR-Compatible Testbed for MR Thermometry in Liver Microwave Ablation with Heat-Sink Effect

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Purpose: Liver microwave ablation (MWA) is a minimally invasive treatment for unresectable primary and metastatic liver tumors [1]. One uncertainty arising from MWA is that the heat sink effect of blood flow can distort the expected temperature distribution and reduce ablation efficacy [2]. Vendor-provided ablation charts may not fully capture patient-specific thermal behavior, particularly in the presence of perfusion heat-sink effects, and conductive heat transfer. This study aimed to develop an MR-compatible experimental testbed for controlled investigation of perfusion-induced heat-sink effects during MR-guided MWA.

Material and Methods: We developed an MR-compatible testbed to reproduce flow-induced heat-sink effects during MR-guided MWA under controlled experimental conditions. The testbed comprised a 2% agar-based tissue-mimicking phantom with an embedded silicone vessel channel to simulate blood perfusion. Water flow through the vessel was controlled by a peristaltic pump to reproduce cooling effects during ablation. A clinical MR-compatible microwave ablation probe (MedWaves AveCure, 16G, 15 cm length) was inserted through a custom 3D-printed lid that also secured four fiber-optic temperature sensors for ground-truth temperature measurements. The phantom container was placed at the scanner's isocenter, and a VIBE image sequence

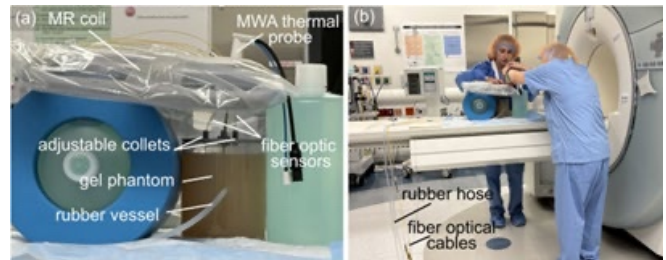


Fig 1: MR-compatible phantom setup inside the 3T MRI scanner room, highlighting the inserted MWA probe, embedded perfusion-mimicking vessel, and aligned fiber-optic temperature sensors. Water circulation is done with a rubber hose and connected to the reservoir outside the scanner room.

was acquired for spatial localization of all components. The MWA system was set to operate at 60°C for 12 minutes. Before ablation started, a 3D gradient-echo (GRE) sequence was used to acquire the reference phase for Proton Resonance Frequency Shift (PRFS) MR thermometry. During ablation, fiber-optic temperature readings were recorded every minute, and repeated GRE phase images were acquired for PRFS temperature mapping. MRI data were processed using a 3D Slicer [3] module developed in-house to perform PRFS-based temperature estimation [4] for each time point. Numerical simulations were performed in MATLAB using a finite-difference method to solve the transient heat conduction equation [5].

Results: The MR-compatible phantom testbed successfully reproduced perfusion-induced heat-sink effects during MWA. PRFS-based temperature maps showed asymmetric heating patterns, with the thermal distribution displaced away from the perfused vessel channel, indicating convective cooling due to simulated blood flow. Fiber-optic ground-truth measurements confirmed progressive, spatially variable heating, while PRFS-based estimates captured the relative heating pattern, but underestimated temperature rise compared with sensor measurements, consistent with voxel averaging near steep thermal gradients and PRFS sensitivity to baseline mismatch caused by susceptibility changes during heating. The heating zone reached approximately 75% of its final size within the first 5 minutes and then approached a plateau. Numerical simulations qualitatively reproduced the observed asymmetric heating.

Conclusion: In this study, we developed an MR-compatible phantom testbed designed to simulate MWA performance under heat-sink conditions. The platform reproduced flow-induced asymmetric heating and provided fiber-optic ground-truth measurements for comparison with PRFS-based temperature estimates. This controlled testbed can support systematic validation and optimization of MR thermometry methods for MR-guided microwave ablation.

Acknowledgements

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Using Fiber Bragg Grating Sensor Arrays to validate MR Thermometry

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Purpose

Single-point fiber-optic temperature sensors are commonly used as a reference for validating MR thermometry algorithms. However, as they measure temperature only at their distal tip, multiple fibers are required to sample different locations, increasing experimental complexity [1]. Fiber Bragg Grating (FBG) sensor arrays overcome this limitation by integrating multiple optical gratings along a single fiber, enabling distributed temperature measurements with only one insertion [2]. Due to their inherent MR compatibility, FBG sensor arrays are a promising reference standard for MR thermometry validation. This study investigates their feasibility for monitoring MR-guided microwave ablation procedures in an MRI environment.

Material and Methods

Experiments were performed using a microwave ablation generator (ECO Medical, China) in an ex-vivo chicken tissue phantom. Two FBG sensor arrays (FiSense, Germany) were deployed: one was attached longitudinally along the microwave applicator shaft, while the second was inserted perpendicular to the applicator. The experimental setup is illustrated in Figure 1.

MR thermometry based on the Proton Resonance Frequency Shift (PRFS) method was performed on a 3T MRI system using a gradient-echo (GRE) sequence (FoV 200x141 mm, TR = 50 ms, TE = 15 ms, flip angle = 40°, matrix = 96 × 68, slice thickness = 4 mm, bandwidth = 260 Hz/pixel). Temperature maps were continuously acquired throughout a 5-minute ablation at 20 W.

Results

Figure 2 presents the dynamic temperature profiles measured by the FBG sensor arrays together with the corresponding temperatures extracted from selected locations in the MR thermometry maps. Both measurement techniques showed excellent agreement, with a root-mean-square error (RMSE) below 1.8°C across most sensor locations. A notable deviation was observed for sensor 27, located closest to the ablation center, where the corresponding MR thermometry measurement (MR Zone 3) increasingly overestimated the temperature after approximately 350 s. This discrepancy is likely caused by excessive tissue heating and dehydration, resulting in a breakdown of the linear relationship between PRFS and temperature.

Conclusion

Compared with conventional single-point fiber-optic sensors, FBG sensor arrays enable the acquisition of multiple independent ground-truth temperature measurements (up to 30 sensing points per fiber) using only a single fiber insertion. This substantially improves the spatial sampling of temperature distributions while simplifying the experimental setup. Consequently, FBG sensor arrays provide a robust reference for the validation of advanced MR thermometry algorithms, including methods for motion compensation and susceptibility correction, allowing a more comprehensive assessment of these algorithms.

Acknowledgements

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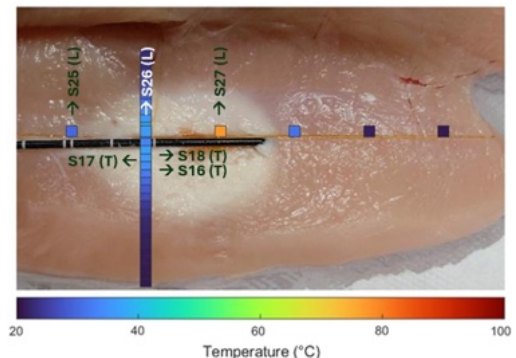


Fig 1: FBG sensor arrangement and corresponding FBG thermal map overlay.

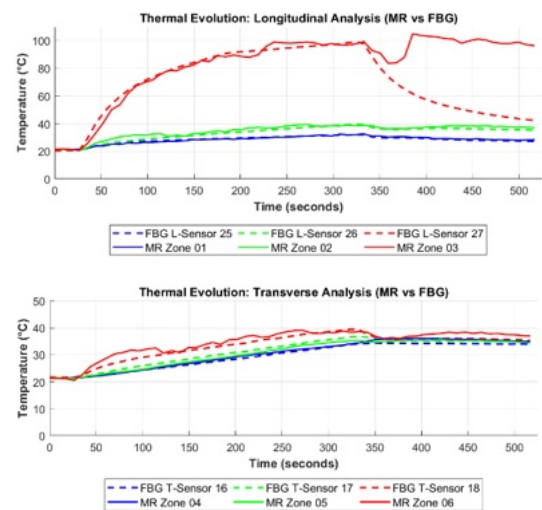


Fig 2: Thermal profiles comparing FBG arrays (dashed) and MR (solid) for longitudinal (top) and transverse (bottom) configurations over a 525-second timeline.

Title: Safety and Efficacy of Live Intraoperative Neuromonitoring During Interventional MRI Guided Ablation: A Retrospective Analysis

Authors: Setayesh Sotoudehnia Korani; Angela Bathke, APRN; Matt Hoffman, D.O./Ph.D, Tatsuya Oishi M.D., Ola Mohamed M.D., Scott M. Thompson, M.D, Daniel Adamo, M.D.; Ph.D.; Aiming Lu, Ph.D., David A. Woodrum, M.D., Ph.D.

Purpose This retrospective analysis aims to assess the safety and efficacy of intraoperative neuromonitoring with motor evoked potentials (MEP) during MRI guided ablation procedures across a broad range of lesions located near named peripheral nerves or neural plexuses.

Materials and Methods Data were collected from a single institution between April 2022 and April 2026. Fifty one MRI guided ablation procedures were included in this study, performed in patients with lesions in proximity to named peripheral nerves or neural plexuses. Indications included vascular malformations, desmoid tumors, oncologic lesions including nodal and osseous metastases, and other soft tissue or osseous lesions adjacent to nerves including the sciatic, femoral, tibial, and posterior femoral cutaneous nerves, the brachial plexus, and the lumbosacral plexus. All procedures were performed under MRI guidance with cryoablation as the primary ablative modality. Subdermal needle electrodes were placed by trained neuromonitoring technologists under neurology direction, with electrode positioning dictated by the anatomic location of the nerve at risk and the treatment target.

Results Of the 51 procedures, vascular malformations accounted for 13 cases, oncologic lesions including nodal and osseous metastases accounted for 5 cases, desmoid tumors accounted for 3 cases, and the remaining 30 procedures addressed other nerve adjacent soft tissue or osseous lesions. Postprocedural outcomes were assessed at short term, 6 month, and 12 month follow up. In general, immediate postprocedural deficits have been infrequent, including isolated cases of transient motor deficit and needle insertion site burns, with the large majority of patients reporting no adverse effects at all follow up intervals. Detailed outcomes for the full cohort are being finalized and will be presented in full.

Conclusion Live intraoperative MEP monitoring during MRI guided ablation appears feasible and generally safe across a wide range of lesion types and anatomic locations. Larger cohort experience continues to refine understanding of electrode placement, safety precautions, and the anatomic boundaries within which this technique can be safely applied.

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Technical Performance and Quality Assessment of Clinical Interventional MR Microwave Applicators for Microwave Ablation Procedures

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Purpose

MRI-guided microwave ablation (MWA) of tumor lesions is a powerful technique achieving high rates of complete ablation and local tumor control [1,2,3]. The proton resonance frequency shift (PRFS) method enables real-time temperature monitoring using MR thermometry [1,2,3], allowing prediction and modeling of necrosis extent [4].

For reliable clinical application, accurate estimation of the energy deposited into tissue is essential to ensure effective treatment planning, adequate ablation zones, and optimal clinical outcomes [5]. In particular, MWA performance is strongly influenced by cable length. However, in MR-guided MWA, long cables are often required to ensure shielding and MR safety. This study aims to demonstrate that different MR-compatible MWA generators and cable lengths significantly affect the delivered power and thermal dose in tissue. To this end, an analysis pipeline was developed to provide a robust evaluation of the effective thermal performance of these systems under varying conditions.

Material and Methods

A total of 21 experiments were conducted using two MWA systems (Saberwave Eco-200G, Eco Medical, Nanjing, China, and MedWaves Avecure, MedWaves, San Diego, CA, USA). The Eco system was tested with four cable lengths (2, 4, 5, and 7 m) and compared to the MedWaves system (single cable, 7.44 m) (Fig. 1). Temperatures were measured in 300 ml of 0.9% saline solution with a fiber-optical thermometer (Neoptix, Reflex, Canada), and lesion shapes were evaluated in ex vivo tissue samples. The Eco generator output was set to 100 W (600 s) for water experiments and 80 W (600 s) for tissue experiments. The MedWaves system was operated with an unattainable target temperature to capture its maximum heating performance.

Results

All experiments demonstrated a strong correlation between cable length and heating performance (Fig. 2). At a generator power setting of 100 W, the delivered power ranged from 29.4 W (7 m) to 65.2 W (2 m). Fig. 3 shows the temperature-time curves of the water experiments and the corresponding calculated power output over time (with and without heat loss correction). For a valid system comparison, device-specific differences must be considered. While the Eco system incorporates active cooling, the MedWaves system does not, which affects both power delivery and ablation characteristics.

Conclusion

System-dependent parameters have a relevant impact on power delivery and should be considered in the planning and evaluation of MWA procedures.

Acknowledgements

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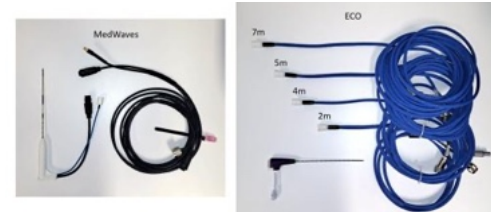


Fig 1: MWA antennas and cable lengths compared in this study.

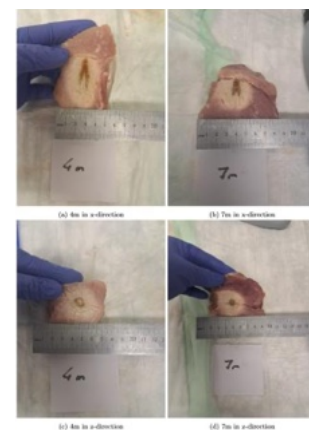


Fig 2: Tissue mimicking experiment with ablation zones; 4m and 7m

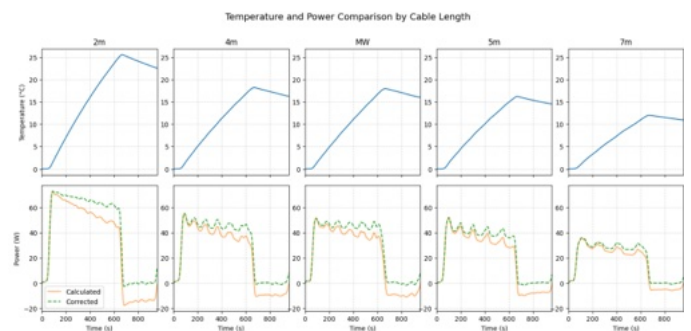


Fig 3: Temperature-time curves (blue) and power-time curves (orange: uncorrected, green: corrected for heat loss) of the heating procedure.

Effective Wire Heating Risk Mitigation for Intraoperative Neurophysiological Monitoring during MRI-Guided Ablations

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Purpose: Intraoperative neurophysiological monitoring (IONM) is a critical technique to monitor neural injury when ablating tumors near nerves. However, IONM during MRI-guided ablations is hindered by electrode wire radiofrequency (RF)-heating and skin-burn risk. Wire routing and low specific absorption rate (SAR) sequences have been proposed to mitigate this risk and successfully integrated into clinical practice. However, low-SAR sequences can limit MRI monitoring performance and complicate the workflow, motivating a passive, hardware-based mitigation strategy. This work evaluated RF chokes as an approach to reduce wire RF heating risk during MRI-guided ablations.

Materials and Methods: *Ex vivo* experiments in a 1.5 T interventional MRI suite were performed using a porcine tissue phantom placed near the scanner bore to simulate an MRI-guided ablation with high RF-heating risk. Four pairs of IONM electrodes were inserted into the phantom at locations representing different levels of RF-heating risk (Figure 1a). MRI was performed using a T2-weighted (T2W) fast-spin-echo (FSE) sequence at a SAR of 1.8 W/kg, first without and then with RF chokes added to the electrode wires. The number of RF chokes and their placement along the IONM wires were systematically varied. Temperatures at each electrode insertion site were recorded continuously using fiber-optic sensors.

Results: Without RF chokes, significant temperature elevations ($>20^{\circ}\text{C}$) were recorded at electrode insertion sites during MRI (Figure 1b). When RF chokes were added to wires #1–3, their temperature elevations were reduced to negligible levels during the execution of the same sequence; however, a $\sim 20^{\circ}\text{C}$ rise persisted at wire #4. Adding RF chokes to all four wires reduced heating to negligible levels across all sites, demonstrating the effectiveness of the approach. The heating reduction effect was observed whether RF chokes were applied to individual wires or to wire bundles.

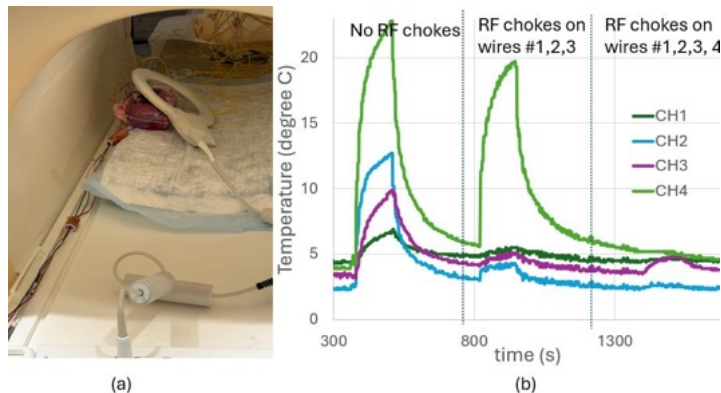


Figure1: (a) Porcine tissue phantom with IONM wires, temperature sensors and RF chokes. (b) Temperature changes due to wire RF heating during MRI.

Conclusion: With IONM electrodes in the scanner bore, RF heating risk is substantially reduced by adding RF chokes to the IONM wires. RF chokes represent a practical, passive mitigation strategy that complements wire routing and enables the use of higher-SAR sequences for improved MRI monitoring quality.

Reliability of vendor provided gradient data for interventional off-center MR Imaging

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Purpose

Limited physician workspace remains a major challenge in interventional MRI. Off-center patient positioning can provide improved procedural access but introduces image distortions that complicate accurate slice selection. We previously proposed a separate slice-selection correction method [1] to enhance workflow and image quality. However, its performance is dependent of the vendor-provided gradient information, which becomes insufficiently accurate at increasing distances from isocenter. Therefore, this study aimed to evaluate the validity of the vendor-provided gradient data across the imaging volume and to identify the off-center distance at which relevant deviations from the actual gradient fields emerge.

Material and Methods

A structured phantom [2] was used to assess the accuracy of vendor-provided gradient field data on a 3-T MAGNETOM Skyra system (Siemens Healthineers) equipped with a Skyra GC99 gradient system. The phantom was embedded in a water-filled cube and imaged using a three-dimensional gradient-echo sequence. Marker structures spaced at 2-cm intervals enabled slice-wise evaluation throughout the imaging volume. Expected gradient field geometries were calculated from the vendor-provided gradient data and overlaid onto the acquired images. Spatial deviations between predicted and measured marker positions were quantified to evaluate the accuracy of the provided gradient information.

Results

Table 1: Example of maximum z deviations of phantom lines to the vendor provided gradient distortion to the real image in mm;

Z in mm	0	20	40	60	80	100	120	140	160	180	200
Max $ \Delta Z $ in mm	1±1	2±1	3±1	3±1	4±1	4±1	4±1	4±1	5±1	8±1	13±1

The analysis demonstrated that the vendor-provided gradient field data accurately represented the actual gradient behavior within an approximately 16-cm radius around scanner isocenter. In this region, the calculated field geometries showed good agreement with the measured phantom data, resulting in only minor spatial deviations. Beyond this distance, agreement progressively deteriorated. Increasing discrepancies between simulated field geometries and measured phantom positions were observed, indicating that the vendor-provided gradient information does not fully capture gradient field characteristics in peripheral regions of the imaging volume. The magnitude of these deviations increased with distance from isocenter. Representative deviations measured at the left phantom boundary are summarized in Table 1.

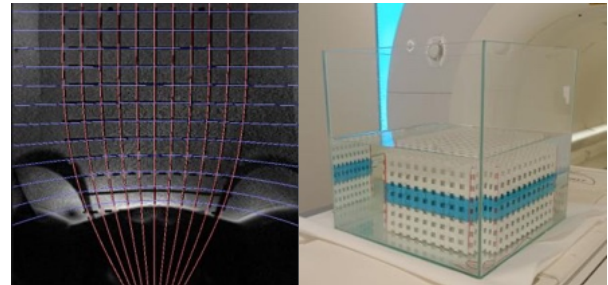


Fig 1: Photo of the phantom (right) and an example slice image of the phantom overlaid with the expected grid from the vendor provided data (left).

Conclusion

Vendor-provided gradient field data were accurate within approximately 16 cm of scanner isocenter but showed increasing deviations at larger off-center distances. In agreement with previous studies on gradient nonlinearity [3,4], these findings define the practical validity range of vendor-provided gradient information for gradient-based slice-selection correction in interventional MRI.

Acknowledgements

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Mitigation of RF Heating and Needle Artifacts in MRI-Guided Interventions Using RF Chokes

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Purpose: Metallic cryoneedles and conductive components used in MRI-guided interventions can induce radiofrequency (RF) heating and image artifacts that compromise patient safety and procedural accuracy. Previous studies have demonstrated that cryoneedle-induced artifacts are predominantly RF-related and correlate with RF heating. Although workflow adjustments, hardware configuration optimization, and sequence selection can mitigate these issues, such measures may be impractical or insufficient in certain clinical settings. This study systematically investigates the use of RF chokes combined with imaging sequence optimization to reduce RF-induced heating and artifacts under realistic operating conditions.

Material and Methods: Experiments were performed on a 1.5T MRI scanner (Ingenia, Philips Healthcare) using ex vivo porcine tissue phantoms. The Visual-ICE cryoablation system (Boston Scientific) was used with one or two MRI-conditional cryoneedles inserted at ~45° to the patient table. The phantom was positioned laterally to simulate elevated RF heating conditions. Cryoneedle temperatures were continuously monitored with fiber-optic sensors. The high-pressure gas supply line (HPGSL) was alternately connected to or disconnected from the mobile control panel (MCP) to vary RF coupling. Custom non-ferrous RF chokes placed on the HPGSL suppressed common-mode currents. Three sequences were evaluated: T2-weighted FSE (TE=60 ms, SAR=2.1 W/kg), proton density-weighted FSE (TE=7 ms, SAR=1.9 W/kg), and gradient echo (GRE; TE=1.6 ms, SAR<0.1 W/kg).

Results: As shown in the images in Figure 1, with the HPGSL disconnected, only mild temperature rises

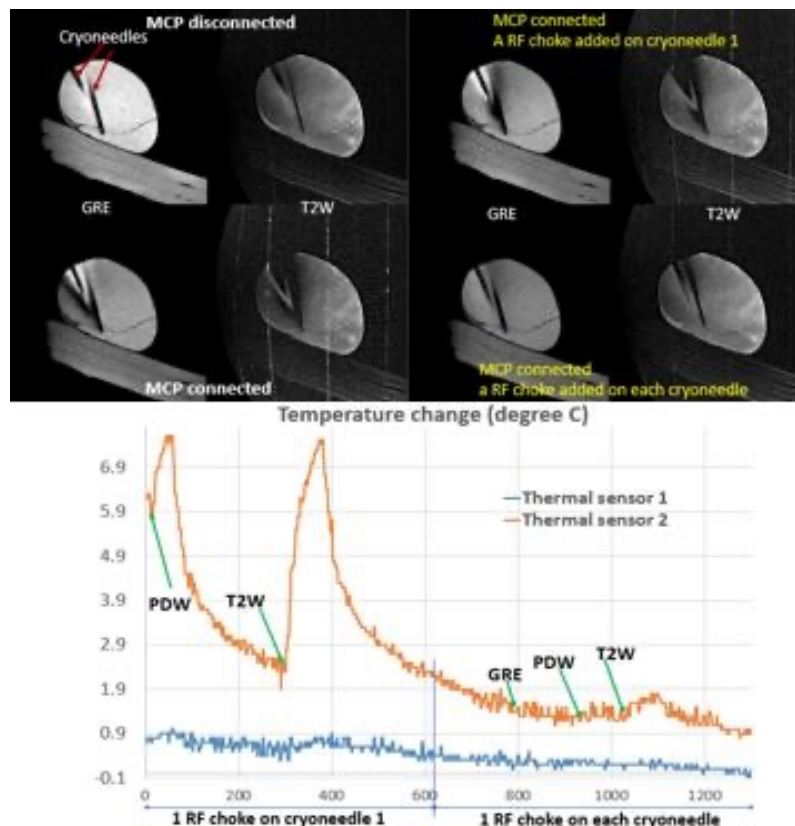


Figure 1. Plots of temperature changes at cryoneedle insertion sites and representative axial MRI images demonstrating RF heating and artifact reduction with RF chokes and low-SAR GRE sequences under varying hardware configurations.

occurred during FSE acquisitions and cryoneedles were clearly visualized. Connecting the HPGSL to the MCP produced significant RF heating and pronounced artifacts in T2W and PDW images; GRE showed less artifacts but compromised needle visibility. Adding RF chokes substantially reduced both heating and artifacts: a single choke selectively suppressed artifacts near the treated needle, while chokes on both needles restored clear visualization of the needles across all sequences. GRE consistently exhibited the lowest artifact susceptibility with no measurable RF heating. Artifact and heating reduction efficacy depended on both the number and placement of RF chokes, with dual chokes providing marginally greater improvement.

Conclusion: RF-induced heating and image artifacts in MRI-guided cryoablation are directly correlated and effectively mitigated by RF chokes. Combining RF choke hardware, wire-path optimization, and low-SAR sequence selection (GRE) offers a practical, broadly applicable approach to improve both safety and image quality in MRI-guided procedures involving

conductive instruments.

Quantitative Investigation of Off-Centre Positioning Effects on MR Image Quality in an Interventional Setting based on Phantom study

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Purpose

In interventional magnetic resonance (iMRI) or intraoperative MR (IoMRI) practice, patient anatomy and procedural access often require imaging with off-centre positioning. This deviation may degrade image quality due to increased susceptibility to system nonlinearities, main field (B_0) inhomogeneity, and gradient imperfections. A comprehensive, quantitative assessment of this effect is essential for establishing quality assurance protocols and ensuring diagnostic confidence during procedures.

To systematically evaluate the degradation of key image quality parameters as a function of off-centre displacement in a clinical iMR environment using the Siemens homogenous MRI phantom based on American College of Radiology (ACR) MRI accreditation method.

Material and Methods

The large vendor provided homogenous MRI phantom was imaged on a clinical Siemens Sola 1.5T MRI system (Siemens, Germany). The phantom was sequentially positioned at isocentre and at systematic table offsets (80mm, 100mm, 110mm, 120mm, 130mm, 140mm, 150mm, 160mm, 170mm, 180mm) in the superior-inferior (S/I) directions. Standard ACR T1 spin echo sequences were performed at each position. Quantitative metrics of the homogeneity slice per ACR guidelines were measured: the mean intensity (SNR based on the method provided by National Manufacturers Electrical Association (NEMA))¹, geometry distortion, the percent signal ghosting and image intensity uniformity².

Results

Measurable degradation in image quality parameters was observed with increasing off-centre distance. The most pronounced effects were a significant decrease in mean intensity which decrease <5% at offset ≤ 100 mm, <10% at offset ≤ 130 mm, <15% at offset ≤ 150 mm and >20% at offset larger than 150mm respectively shown as Figure 1. The image uniformity decreased >91% at offset ≤ 100 mm, >85% at offset ≤ 140 mm, and <85% at offset larger than 150mm respectively. Geometric distortion also increased beyond ACR tolerance limits at offsets (180cm). Ghosting artifacts showed minimal change within 180mm but deteriorated at the largest displacement.

Conclusion

Off-centre positioning in iMRI or IoMRI, while often necessary, leads to quantifiable degradation in specific image quality metrics, primarily imaging intensity and uniformity. The homogeneity phantom provides a robust framework for this assessment. For high-quality iMRI or IoMRI guidance, protocols should minimize off-centre distances where possible, and site-specific tolerances (e.g., ≤ 150 cm offset) should be established if intensity degrading 15% is acceptable through routine QA. Otherwise 100mm is recommended for 5% intensity degrading. This study provides a baseline for protocol optimization and quality control in interventional MRI practice.

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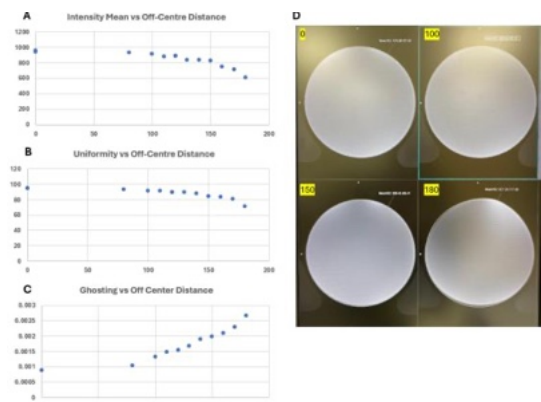


Fig 1: Measurable degradation in image quality parameters was observed with increasing off-center distance. (A) mean Intensity, (B) uniformity, (C) ghosting and (D) geometry distortion vs off-center distance respectively.

Large Language Model-Supported Multi-Agent Pipeline for Speech-Based Workflow Analysis in Interventional Radiology: A Proof-of-Concept Study

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Purpose

Automated workflow analysis could support documentation, training, and intraprocedural assistance in interventional radiology (IR) [1,2], but prevailing methods rely on supervised learning over annotated procedural video [1], a paradigm poorly suited to IR. On the data side, IR-specific resources are scarce, as the only IR dataset to date remains unreleased [3]. On the modality side, neither video source fits well: Room-mounted cameras capture little workflow-relevant content, since IR is minimally invasive and key actions are small and often occluded. Imaging-derived video is tied to a single guidance modality, whereas IR spans fluoroscopy, ultrasound, CT, and MRI, and models trained on one modality rarely transfer to the others. Intraprocedural speech and large language models (LLMs) jointly address these gaps: speech is modality-independent, conveys the operator's intent and reasoning, and is easily anonymized, while LLMs show strong medical performance without fine-tuning [4], enabling annotation-free inference.

Material and Methods

In this experimental proof-of-concept study, the pipeline derives a three-level phase-step-action hierarchy from intraprocedural speech alone. Audio is transcribed into timestamped text and chunked, after which specialized LLM agents extract atomic actions, aggregate them bottom-up into phases, and refine these top-down into ordered steps with tool linkage (Fig. 1). Transcription and the analysis agents were implemented via the OpenAI API, using GPT-4o-audio-preview and GPT-4o, respectively (OpenAI, San Francisco, CA, USA). The pipeline was evaluated zero-shot on three IR procedure videos recorded at Hannover Medical School in an instructional setting, with the interventionalist instructed to verbalize all actions and explanations. Five independent interventional radiologists (1–15 years' experience) rated 15 automatically generated phases on a five-point Likert scale, yielding workflow segmentation quality and system usability. One-tailed one-sample t-tests against the neutral value 3 were used.

Results

Workflow segmentation quality averaged 4.11 ± 0.93 (neutral value 3; $p = 0.028$), with phase and step naming and content rated higher than temporal accuracy at both levels. System usability averaged 3.90 ± 0.99 ($p = 0.056$); presentation clarity (4.60 ± 0.89) and clinical/educational relevance (4.40 ± 0.89) were rated highest, while effort for clinical use and willingness to use were rated lowest. Experts noted occasional over-segmentation and rare hallucinated steps.

Conclusion

An LLM-supported multi-agent paradigm can generate structured IR workflow representations directly from explicit intraprocedural speech, without annotated training data, removing the dependence on modality-specific video and manual annotation. As a small-panel feasibility demonstration under controlled conditions, the findings warrant larger-scale validation before clinical use.

Acknowledgements

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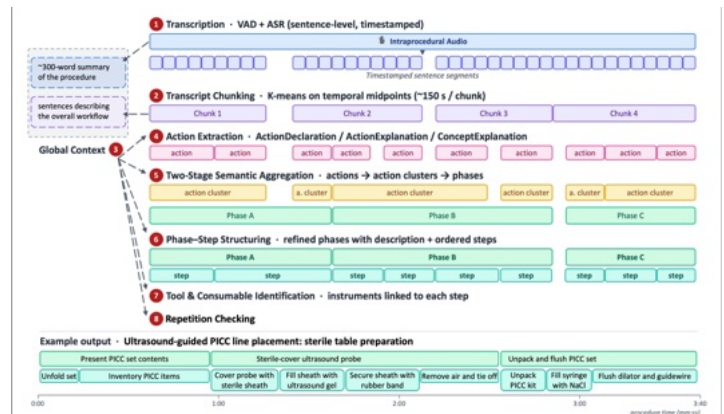


Fig 1: Overview of the LLM-supported multi-agent pipeline: speech is transcribed (1), chunked (2) and summarized into a global context (3); actions are extracted (4), aggregated bottom-up into phases (5), refined top-down into ordered steps (6), linked to tools/consumables (7) and checked for repeated structure (8). Bottom: example output (sterile table preparation before PICC placement).

High impedance dielectric cable combination to reduce burn risk from Intraoperative Neuromonitoring during MRI.

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Purpose

MRI-guided cryoablation procedures can benefit from intraoperative neuromonitoring (IONM) to monitor neural function during treatment. However, when IONM cable lengths approach a substantial fraction of the RF wavelength, the cable can behave as a monopole antenna, resulting in RF-induced currents and localized heating [1, 2]. This study proposes a high-impedance dielectric cable (HIDC) to suppress RF-induced currents at 1.5T.

Material and Methods

The HIDC was designed by winding the IONM cable around a dielectric medium to form an artificial inductance-capacitance (LC) resonator. The cable windings provide inductance, whereas the dielectric material provides capacitance that generates a high impedance at 64 MHz. The design was evaluated using electromagnetic (EM) simulations and bench-top measurements (Fig. 1).

RF-induced heating was assessed in a virtual human model under three conditions: no IONM, conventional IONM, and IONM with the HIDC. The HIDC consists of an eight-turn rectangular solenoid formed by the IONM cable ($20 \times 20 \text{ mm}^2$, 1.5 mm spacing) embedded within a dielectric material (relative permittivity = 4; $30 \times 30 \times 20 \text{ mm}^3$) (Fig. 1A). The number of turns and the geometric parameters were optimized through EM simulations (Fig. 1B) and bench-top measurements to maximize impedance at 64 MHz.

Results

In human model simulations performed at a high specific absorption rate for a duration of 5 minutes (Fig. 1C), the maximum temperature changes (ΔT) were $+72^\circ\text{C}$ and $+6^\circ\text{C}$ for the conventional IONM and HIDC configurations, respectively, compared with $+2^\circ\text{C}$ in the absence of the IONM system. Network analyzer measurements (Fig. 1D) demonstrated that the HIDC substantially increased cable impedance at 64 MHz, reducing the reflection coefficient (S11) from -10 dB to -0.5 dB at 64 MHz, indicating a significant reduction in RF current flow.

Conclusion

The proposed HIDC substantially increases cable impedance at 64 MHz while preserving normal IONM operation, demonstrating its potential to reduce RF-induced heating during MRI-guided cryoablation.

Acknowledgements

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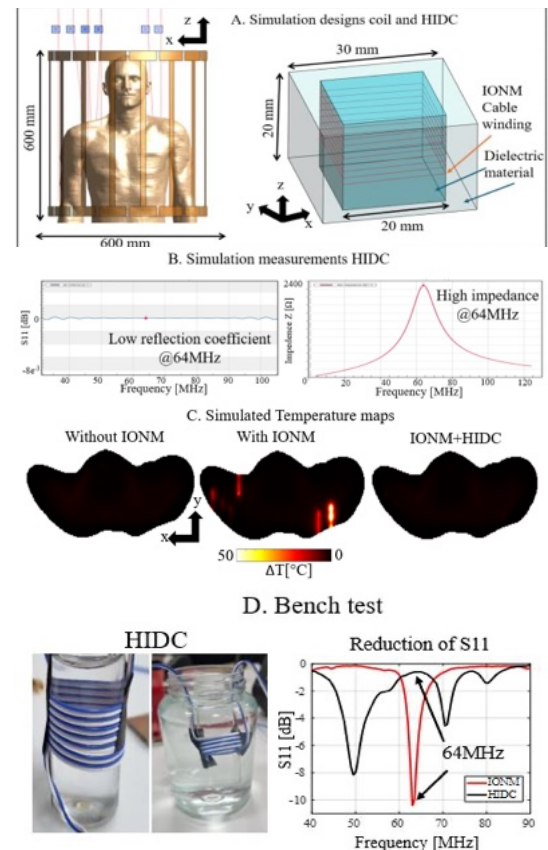


Fig 1: Simulation and experimental evaluation of the proposed HIDC. (A) Simulation setups showing the whole-body coil, IONM cable configuration, and HIDC design. (B) Impedance response of the HIDC in the simulation. (C) Computed temperature maps of the shoulder region obtained from human model simulations. (D) Bench-top experimental setup, in which the IONM cable was wound around a cylindrical water phantom and submerged into another water phantom, along with the measured reflection coefficient (S11) before and after implementation of the HIDC.

Electromagnetic-Thermal Analysis of RF-Induced IONM Lead Heating at 1.5T MRI: Quantitative Effects of Geometric Parameters

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Purpose

MR-guided ablation relies on MRI for accurate lesion visualization and treatment monitoring, but MRI provides limited information regarding nerve function during the procedure. Consequently, intraoperative neurophysiological monitoring (IONM) can be used to monitor nerve function and reduce the risk of nerve injury. However, conductive IONM components, including electrodes and long connecting cables, can couple with MRI radiofrequency (RF) fields and cause localized tissue heating. This study quantitatively evaluated the effects of cable length and needle position on RF-induced heating in a 1.5T MRI environment using electromagnetic (EM) and thermal simulations, providing a quantitative basis for minimizing thermal risks.

Material and Methods

EM and thermal simulations were performed in Sim4Life using a 64-MHz quadrature birdcage coil and a torso-equivalent cylindrical phantom. The IONM needle and cable were modeled as ideal conductors (Fig. 1A). RF-induced heating was calculated using finite-difference time-domain (FDTD) EM simulations with the Pennes bioheat equation. RF power was normalized to a whole-body specific absorption rate (SAR) of 2 W/kg. The effects of cable length (0 to 3000 mm) and needle position along the x- and z-axes were systematically evaluated. Resulting temperature elevations were analyzed to determine the relative contributions of these geometric parameters.

Results

RF-induced heating was highly localized at the needle tip (Fig. 1B). As cable length increased, temperature rise exhibited a non-linear dependence, with pronounced peaks occurring between 1000 and 1500 mm and a minimum observed at approximately 2000 mm (Fig. 1C). Temperature rise also increased with needle position along the z-axis as a larger portion of the needle-cable assembly became exposed to the RF field, reaching a maximum at $z = 350$ mm (Fig. 1D), consistent with position-dependent heating effects previously observed in an experimental study [1]. This trend was consistently observed across different needle positions, although the magnitude of the maximum temperature rise varied with needle position. In contrast, variations in x-axis position produced minimal changes in heating, likely due to the geometric symmetry of the phantom and RF coil configuration.

Conclusion

RF-induced heating in IONM systems is strongly dependent on cable length and needle position within the MRI RF field. These findings provide a quantitative basis for understanding RF-induced heating and improving procedural safety during MR-guided interventions.

Acknowledgements

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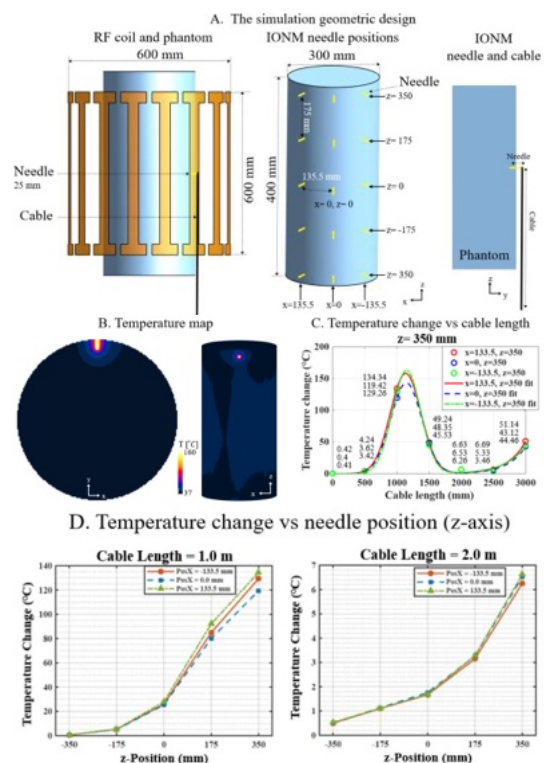


Fig 1: Simulation setup and results. (A) 3D model of the RF coil, phantom, and IONM system, with the relevant geometric parameters. (B) Computed temperature distribution within the phantom. (C) Temperature rise as a function of cable length. (D) Temperature rise as a function of the IONM needle position along the z-axis.

Title: Safety in Zone 4 of the Interventional MRI Suite during MR Guided Cryoablation – From an MR Technologist Perspective

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Purpose:

To present, from a technologist's perspective, the techniques and workflows that promote safety awareness, foster a secure working environment, and enhance competency within the interventional MRI (iMRI) suite, thereby reducing the risk of injury or harm to patients and staff.

Methods:

MR technologists play a vital role in maintaining MRI safety through personnel screening, multidisciplinary collaboration, and implementation of safe procedural workflows. This presentation highlights key workflow strategies from the MR technologist perspective, focusing on (i) anesthesia considerations, (ii) patient positioning and setup, and (iii) cryoablation-specific safety measures.

Results:

i) Anesthesia: Patients undergo MRI safety screening and receive hearing protection before general anesthesia. An MR-safe tube management system and MR-conditional IV pole are used to organize and secure anesthesia and monitoring lines during table movement.

(ii) Patient Positioning: Patients are positioned supine with gel padding applied to pressure points and skin-to-skin contact areas to improve comfort and reduce the risk of pressure injuries, RF heating, and burns. MR-safe arm boards help prevent bore contact and line compression.

(iii) Cryoablation-Specific Considerations: Metallic cryoablation needles and gas lines present RF heating risks. Gas lines are carefully routed away from the patient and aligned near the scanner's central axis using a custom guide. Low-SAR imaging protocols are utilized when gas lines remain connected during MRI to minimize RF-induced heating.

Conclusion:

Strategies addressing anesthesia management, patient positioning, and cryoablation-specific safety considerations were implemented to promote a safe environment for both patients and staff within the interventional MRI (iMRI) suite. These workflows have effectively reduced potential risks and enhanced procedural safety. Their success is largely due to the expertise, vigilance, and strong safety culture maintained by MR technologists and the multidisciplinary team involved in iMRI procedures

Ergonomic Comparison of CT- and MR-Guided Needle Interventions Using Digital Human Model Simulation

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Purpose

MRI guidance eliminates radiation exposure and improves tissue visualization compared to CT but may introduce ergonomic challenges due to bore geometry and procedural access constraints. Simulation-based evaluation offers a proactive approach to identifying and mitigating these risks. This study aimed to compare ergonomic outcomes across CT-guided and MR-guided needle interventions using different system geometries, with a focus on muscle fatigue and spinal loading.

Material and Methods

This simulation-based study employed Santos Pro™, a physics-based digital human modeling platform, to predict postures during needle-guided interventions (Figure 1). Simulations included a vertical (90°) and near-side 45° needle approach, positioned at CT isocenter and at 12 cm offset for MRI systems. Three anthropometric avatars (5th percentile female, 50th and 95th percentile male) were modeled following prior validation.

Systems evaluated included conventional MRI with a 70 cm bore, MAGNETOM Free.XL (100 cm flared bore), and conventional CT (Siemens Healthineers AG, Erlangen, Germany). CT simulations incorporated a 4.3 kg trunk load to account for lead vest use. Primary outcomes were muscle endurance time (indicator of fatigue) and joint torque (indicator of biomechanical loading) at the shoulder and low back. Outcomes were compared across modalities and geometries.

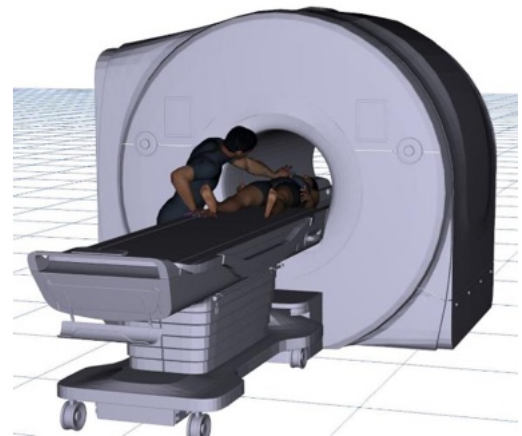


Figure 1 - Simulating a 50th percentile stature interventionalist performing an MRI-guided needle intervention.

Results

Shoulder muscle endurance times were 8–9% lower for MRI systems compared with the conventional CT. In contrast, low back endurance time was greatest for MAGNETOM Free.XL, indicating reduced fatigue relative to other systems. Low back extensor torque was highest for the conventional CT and conventional MRI with a 70cm, with values 29–64% greater than those observed for MAGNETOM Free.XL. These findings indicate reduced spine loading with the wider bore MR system.

Conclusion

MR-guided interventions using MAGNETOM Free.XL reduce ergonomic risk by improving procedural access and decreasing spinal loading compared to both CT-guided procedures and standard-bore MRI systems. The elimination of lead vest requirements further contributes to reduced biomechanical strain. These findings support the ergonomic advantage of wide-bore, radiation-free MR-guided interventions.

Acknowledgements

This study was funded by Cook Medical and Siemens Healthineers AG.

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MR-guided Ablation in Patients with Active Implanted Medical Devices

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Purpose

MR-guided interventions can present challenges such as i) limitations on imaging parameters, ii) generation of artifacts, and iii) procedural complexity. We aim to outline how treatment planning, workflow efficiency, and procedural outcomes are affected in our MR-guided interventional practice.

Material and Methods

Patients who underwent MR-guided interventions in our practice between 1/2/2024 and 6/11/2026 were reviewed for presence of AIMDs. For each patient with an AIMD present, the model and type of device was recorded, in addition to the type of MR-guided procedure that was performed. The workflow for each type of device was reviewed and summarized to identify differences between different workflows.

Results

Device counts per procedure are summarized in Table 1. In 30 months of data, procedures were performed on patients with pacemakers (n=25), infusion pumps (n=1), hypoglossal nerve stimulators (n=1), and other neuro-stimulators (n=2). All images were diagnostic, despite artifact (Figure 1). For all devices, when the patient is scheduled for an MR-guided intervention, the MR Safety Officer (MRSO) is contacted to determine whether the device is classified as “high risk” requiring physicist support and will verify patient eligibility and safety staff availability. The MRSO also ensures that pre-procedure appointments are scheduled and provides MRI technologists with device specific checklists applicable to each patient. During the MR-guided procedure, MR technologists and/or MR physicists will ensure that i) the device checklist is followed, ii) that all conditions from the device manufacturer are met, and iii) that artifacts from the device are managed as best as possible through MR pulse sequence parameter changes. Device specific workflows are also used:

Pacemakers: Pre-procedure, Heart Rhythm Services will program the pacemaker for MRI. During the procedure, anesthesia will monitor the patient and Normal RF Operating Mode is used to restrict whole-body SAR ≤ 2 W/kg. Post-procedure, Heart Rhythm Services will program the pacemaker back to normal mode.

Infusion pumps: Pre-procedure, the patient is scheduled in the outpatient Pain Clinic for pump interrogation and a detailed note is added to the electronic medical record stating pump status (e.g. “in working order”) and type of pain medication. During the procedure, Normal RF Operating Mode is used. Post-procedure, the patient will return to the Pain Clinic to verify a working pump and no evidence of adverse effects from procedure.

Hypoglossal Nerve Stimulators: Pre-procedure, the patient must bring device remote control to enable verification that stimulation is off. During the procedure, a Medical Physicist will closely monitor the exam to ensure that MR conditions are met (e.g. $B1^+_{rms} \leq 3.2$ or $1.9 \mu T$ for scan regions 4 and 5, respectively¹).

Other Neuro-Stimulators: Pre-procedure, the patient must bring device remote control to verify that the stimulator is full-body MR eligible and to start MRI mode. During the procedure, manufacturer provided MR conditions are met. Post-procedure, the patient uses the device remote control to turn the stimulator on.

Conclusion

AIMDs can create additional challenges for MRI-guided procedures including artifacts and limitations on scan parameters that can affect image quality. However, all of these challenges can be overcome with careful planning, workflows, and procedures, making it possible to provide patients with the care they need.

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	Prostate Cryoablation	Other Cryoablation	Biopsies	MR Lymph- angiogram	TULSA
Pacemaker	12	2	2	8	1
Infusion Pump	-	1	-	-	-
Hypoglossal Nerve Stimulator	-	-	-	1	-
Other Neuro- stimulator	-	1	-	1	-

Table 1: Number and Types of AIMDs and MR-guided Procedures



Fig 1: MR-susceptibility “blooming” artifact from an implanted cardiac pacemaker. The artifact didn’t confound the MR lymphangiogram.

Title: Safety in Zone 4 of the Interventional MRI Suite during MR Guided Cryoablation – A Nursing Perspective

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Purpose: To share, from a nursing point-of-view, techniques and workflows to create high awareness, a safe work environment, and competence in the interventional MRI (iMRI) suite and prevent injury or harm to patients and employees.

Methods: Nurses play a vital role in carrying out established safety measures in routine clinical practice. Roles for nurses include safety screening of patients entering Zone 4, assisting and working with physicians and physicists during the procedure, and optimizing and carrying out workflows to ensure MR safety. Here, we share key parts of our optimized workflows from a nursing perspective sorted into three categories relating to (i) anesthesia, (ii) patient positioning for prevention of skin injury (iii) cryoablation-specific considerations.

Results: (i) Anesthesia: All patients undergoing a cryoablation procedure are screened and given ear protection prior to general anesthesia. To ensure that airway and other vital anesthesia lines are not pinched in the scanner while moving the table in and out, an MR-safe anesthesia tube tree was acquired and placed near the head for line routing and an MR conditional IV pole (checked with a hand magnet each time prior to entering Zone 4) is used to hold anesthesia fluids and arterial lines. (ii) Patient Positioning: The majority of cryoablation cases are in the supine position with proper padding placed before starting the scan. Gel pads are placed on the table and under points of prolonged table contact (e.g. heels, elbows, knees etc.) to ensure patient comfort and to prevent pressure sores. We have added using Mepilex Borders at the knees if the patient's knees are close to the bore to prevent skin tears with the movement in and out of the scanner for cryo needle placement. A Mepilex could be placed at the coccyx as well to prevent pressure injury, especially for longer cases. Additional padding is used to prevent heating and RF burns at skin-to-skin contact areas such as between hands and thighs, the axillary, or any other areas where skin might fold over upon itself. Finally, MR safe arm boards with gel padding are placed on the lateral sides of the hands and arms to improve patient comfort, prevent patient contact with the edge of the bore, and prevent IV lines from pinching along the bore. Nursing assessment of the skin should be done both prior to cryoablation and after the cryoablation and any changes should be documented appropriately.

(iii) Cryoablation-Specific Considerations: Cryoablation gas lines and needles are metallic and subject to RF heating risks. As such, the gas lines must be positioned and carefully monitored during table movement to ensure that they do not come into direct contact with the patient's skin and cause heating or burns during the MR exam. To reduce potential heating of the gas lines, a number of strategies were developed: (1) the gas lines are routed straight out of the bore along a line close to the central Z-axis when possible using a custom developed gas line guide (2) When MR imaging is performed while the cryo gas lines are connected to the mobile gas connection panel, low SAR sequences should be used to prevent additional RF heating along the gas lines or at the insertion site of the cryo needles.

Conclusion: Strategies addressing anesthesia, patient positioning, and cryoablation-specific concerns were developed to enhance patient and employee safety in the iMRI suite. These have minimized potential harm to the patient and have been successful due to the experience and high awareness of the MR nurses, technologists and other staff working in the iMRI practice.

Title: CPT Code Based Reimbursement Disparity Between Prostate Cryoablation (55873) and Radical Prostatectomy (55866): Implications for Interventional Radiology Practice and Value Fee Schedule

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Purpose: Prostate cryoablation (CPT 55873) is an established, image guided minimally invasive treatment for localized prostate cancer increasingly performed under MRI guidance. Despite comparable single session resource utilization and growing evidence for favorable functional outcomes, the current Medicare reimbursement structure assigns substantially lower relative value units (RVUs) to cryoablation than to radical prostatectomy. The purpose of this study is to compare the CPT based reimbursement, total episode of care costs, and clinical outcomes between MRI/ultrasound guided prostate cryoablation (CPT 55873) and robotic assisted laparoscopic prostatectomy (RALP; CPT 55866), and to evaluate whether the current reimbursement differential reflects actual value delivery.

Materials and Methods: A retrospective analysis was performed using institutional and Medicare claims data (2018 to 2024). Patients with clinically localized, low to intermediate risk prostate cancer (NCCN criteria) who underwent primary treatment with CPT 55873 (cryoablation) or CPT 55866 (RALP) were identified. Physician reimbursement, facility charges, total 30 day episode costs (including anesthesia, imaging, and post procedure care), length of hospital stay, complication rates (Clavien Dindo), 90 day readmission rates, and functional outcomes (continence, erectile function via validated instruments) were compared. RVU assignments were cross referenced with the CY 2026 CMS Physician Fee Schedule. Cost per quality adjusted life year (QALY) was estimated using published utility values.

Results: Under the CY 2026 Medicare Physician Fee Schedule (conversion factor \$33.40), CPT 55873 carries a physician reimbursement of approximately \$777 (22.28 total RVUs) compared to approximately \$1,464 for CPT 55866 (41.95 total RVUs), a 47% lower reimbursement for cryoablation. Despite this differential, cryoablation was associated with significantly shorter hospitalization (median 1 day versus 2 to 3 days), lower 30 day total episode costs, no transfusion requirement, and substantially reduced rates of urinary incontinence and erectile dysfunction. When analyzed on a value per episode basis, prostate cryoablation demonstrated superior or equivalent cost effectiveness across all NCCN low and favorable intermediate risk patient cohorts. Biochemical recurrence free survival at 5 years was comparable between modalities in propensity matched cohorts.

Conclusion: The current CPT reimbursement framework significantly undervalues prostate cryoablation relative to radical prostatectomy when total episode costs, recovery burden, and functional outcomes are considered. Notably, CPT 55873 does not distinguish between CT guided, ultrasound guided, and MRI guided approaches, so the added technical complexity and resource intensity of in bore MRI guidance are not separately recognized under the current code. This disparity may discourage adoption of MRI guided cryoablation by interventional radiologists and urologists despite its favorable value profile. Advocacy for RVU rebalancing and a distinct CPT descriptor for MRI guided prostate cryoablation is warranted to support equitable, value based prostate cancer care.

Cost-Effectiveness of MR-Guided versus CT-Guided Biopsy of Small Focal Liver Lesions: A Decision-Analytic Analysis

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Purpose

To determine the cost-effectiveness of MR-guided versus CT-guided biopsy of small focal liver lesions (<20 mm) from the perspective of US Medicare, and to test the robustness of this finding across alternative care settings and a second health system.

Material and Methods

A decision-tree cost-effectiveness analysis was built in TreeAge Pro. Effectiveness was defined as clinical success (diagnostic adequacy of the biopsy). Clinical inputs were derived from a propensity-score-matched single-center cohort (1): clinical success 97% (MR) versus 79% (CT), periprocedural complications 0% versus 13%. In the base case, costs reflected CPT-based Medicare reimbursement for a standard hospital outpatient biopsy. The model captured procedure costs, minor complications, and follow-up including re-biopsy after non-diagnostic sampling; major complications were excluded. Zero-event complication rates were handled with the Haldane correction. Uncertainty was assessed with one-way deterministic sensitivity analyses and a probabilistic sensitivity analysis (30,000 Monte Carlo iterations). As additional analyses, two alternative US care settings (ambulatory and inpatient) and a second health system (German statutory health insurance, GKV) were evaluated.

Results

In the base case, MR guidance was the dominant strategy, yielding higher clinical success at lower cost than CT, with a per-patient saving of \$539. In probabilistic sensitivity analysis, MR was the dominant strategy (more effective and less costly) in 74% of the 30,000 iterations, and the result was robust in one-way sensitivity analyses. The economic advantage was driven primarily by the higher diagnostic yield of MR — fewer non-diagnostic biopsies and downstream re-procedures — rather than by differences in procedural cost. Additional analyses were consistent: MR remained less costly in the alternative US settings (−\$107 fully ambulatory, −\$656 inpatient) and in the German GKV perspective (MR €1,363.55 vs CT €1,799.45; Δ −€435.90, CT absolutely dominated).

Conclusion

Under US Medicare, MR-guided biopsy of small focal liver lesions was the dominant strategy, combining higher clinical success with lower cost than CT guidance. This finding was robust across alternative care settings and a second health system.

Acknowledgements

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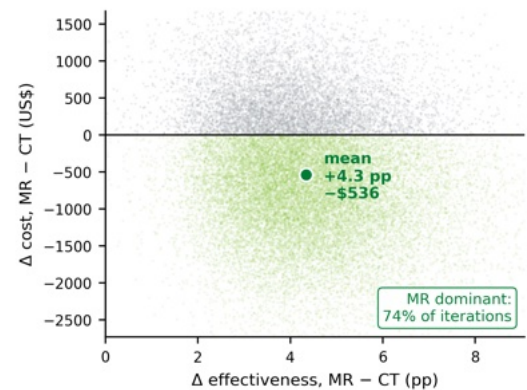


Fig 1: Cost-effectiveness plane for the US Medicare base case, comparing MR-guided with CT-guided biopsy of small focal liver lesions (<20 mm). Each point represents one of 30,000 probabilistic Monte Carlo iterations; the x-axis shows the incremental effectiveness (difference in clinical success, MR minus CT, in percentage points) and the y-axis the incremental cost in US dollars (MR minus CT). The green marker is the mean estimate (−\$536); the deterministic base case was −\$539 per patient. Green points denote MR dominance (higher clinical success at lower cost); grey points denote iterations in which MR was more effective but more costly. MR was dominant in 74%.

Reimbursement Disparity Between CT Guided and MR Guided Percutaneous Ablation Procedures Under the CY2026 Medicare Physician Fee Schedule

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Purpose

Magnetic resonance imaging offers superior soft tissue contrast and real time thermal monitoring compared with computed tomography for percutaneous tumor ablation, yet these technical advantages are not reflected in current Medicare reimbursement policy. CPT 77013 (CT guidance for parenchymal tissue ablation) and CPT 77022 (MR guidance for parenchymal tissue ablation) are valued identically despite the added complexity of MR guided procedures. The purpose of this study is to compare Calendar Year 2026 Medicare Physician Fee Schedule valuations for CT guided and MR guided percutaneous ablation, and to characterize how recent CMS policy changes affect this reimbursement disparity.

Material and Methods

This is a publicly sourced, retrospective health policy analysis. Work relative value units, total relative value units, and national Medicare payment estimates for CPT 77013 and CPT 77022 were obtained from the Calendar Year 2026 CMS Physician Fee Schedule final rule, effective January 1, 2026, and cross referenced against American Urological Association coding summaries and an industry interventional oncology coding guide. Hospital Outpatient Prospective Payment System status indicators and ambulatory payment classification assignments were reviewed for both guidance codes. The effect of two Calendar Year 2026 policy changes, a 2.5 percent efficiency adjustment to work relative value units and a 50 percent reduction in facility based indirect practice expense allocation, was assessed for the guidance codes and for representative ablation procedure codes (CPT 55873, prostate cryoablation, and CPT 50593, renal cryoablation), with surgical alternatives (CPT 55866 and related laparoscopic and open prostatectomy codes) used as comparators.

Results

CPT 77013 and CPT 77022 carry equal work relative value units under the Calendar Year 2026 fee schedule despite the added sequence planning and multiplanar thermometry interpretation required for MR guidance, and CPT 77022 retains Hospital Outpatient Prospective Payment System status indicator N, meaning payment is packaged into the primary procedure with no separate facility payment, identical to CT guidance. CPT 55873 (prostate cryoablation) carries a work relative value unit of 13.26 under the Calendar Year 2026 schedule, a reduction from 13.60 in 2025 attributable to the new efficiency adjustment, while the surgical comparator CPT 55866 (robotic assisted radical prostatectomy) remains valued at approximately 21 to 22 work relative value units, roughly 1.6 times higher despite shorter inpatient recovery and lower complication rates reported for image guided ablation in the published literature. The new Calendar Year 2026 site of service policy further reduces total relative value units for facility based ablation procedures by approximately 10 percent, disproportionately affecting MR guided cases, which are concentrated in hospital based settings due to magnet access requirements.

Conclusion

Despite the technical and interpretive advantages of MR guidance for percutaneous ablation, the Calendar Year 2026 Medicare Physician Fee Schedule continues to value MR and CT guidance identically and packages both without separate facility payment. New Calendar Year 2026 policies, including the efficiency adjustment and the site of service indirect practice expense reduction, compound this disparity for MR guided procedures, which are concentrated in facility settings. The continued absence of a guidance code that reflects the added complexity of MR guided ablation represents an unresolved policy gap with direct implications for the sustainability of in-bore MRI guided interventional programs.

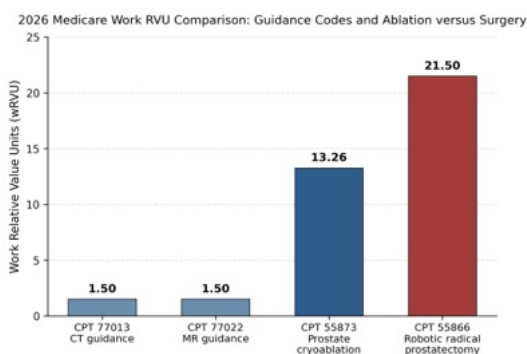


Fig 1: 2026 Medicare work RVUs for CT guidance (77013), MR guidance (77022), prostate cryoablation (55873), and robotic prostatectomy (55866). Guidance codes are valued identically by modality, while the surgical comparator is valued far higher despite shorter recovery.

Author Index

Abramowitz, Matthew	69
Acevedo Bado, Andrea	43
Acevedo, Andrea	88
Adamo, Daniel	5, 30, 47, 54, 55, 56, 57 60, 62, 63, 64, 65, 81, 85 89, 96, 97, 98, 101, 102, 103 105, 106, 108, 109, 110, 111, 112 113, 117, 121, 122, 123, 124, 133 135, 137, 146, 148
Adams, Samuel	44, 91
Aghazadeh Mohandesi, Nessa	104
Ahmed, Azam S	38, 71
Ahmed, Jalil	21
Ajala, Afis	70, 82
Alexander, Andrew L	38, 71
Ali, Ridaa	19
Ambruster, Gunter	78
Anderson, Katelyn	104
Angela, Bathke	61
Antaki, Majd	32
Arbabzadah, Sahar	7
Armbrüster, Günter	93
Armstrong, Aimee	18, 94
Arora, Sandeep	42
Arts, Erik	16
Augustine, Matthew	113
Avetikovi, Nino	6, 52, 83
Aydin, Ahmed	19
Babu, Silpa	12
Baert, Gregory	37
Bagci, Ulas	100
Bales, Charles	36, 76, 77
Barkhausen, Jörg	22
Bathke, Ang	142, 145
Bathke, Angela	54, 56, 57, 64, 96, 97, 101 105, 106
Battaglia, Jean-Luc	6, 37, 52, 83
Bauer, Laura Maria	84, 134
Begovic, Jovan	42, 43
Belker, Othmar	7, 15, 72, 129, 132
Bendel, Emily	97, 105, 106, 119
Bermudez, Dalton	125
Bernardes, Mariana	131

Bernardes, Mariana C.	45, 53
Bezerra, Hiram	19
Bhagavatula, Sharath	59
Bhandari, Ajay	19
Bhurwani, Mohammad Mahdi	42, 43
Bhushan, Chitresh	70, 82
Bitton, Rachelle	29
Bjarnason, Haraldur	97, 105, 106
Block, Walter	125
Block, Walter F	38, 71
Bock, Michael	20, 23, 46
Bogner, Balazs	46
Bolster, Bradley	120
Bolster, Bradley D.	128
Bomers, J.G.R.	28
Bonaventura, Justina	91
Borm, Paul	22
Bosboom, D.G.H.	28
Bosman, Witek	16
Bour, Pierre	52, 130
Brahm, Gary	42
Brandner, Eric	143
Brandt, Jan	143
Breto, Adrian	69
Brink, Wyger	16, 75
Browne, Jacinta	63
Bruesch, Inga	7
Brunsing, Ryan	29
Burgers, Ida	6, 52
Burich, Aidan	125
Burke, Courtney	142, 145
Busch, Joseph	42, 87
Butt, Ateeb	94
Buzug, Thorsten M.	22
Buzurovic, Ivan	33
Caoili, Elaine	25
Cavusoglu, Murat	19
Chaffin, Julia	82
Chambers, Sean	143
Chan, Heather	82
Chan, O L	138
Chen, Baoying	9
Chen, Hui	139
Chen, Jin	127
Chen, Qiaozhi	72
Chen, Yen-Chun	19
Chen, Yue	95, 116

Chiang, Jason	114
Choi, Suin	140, 141
Clarke, Gina	42, 43
Cleary, Kevin	13, 39, 48
Cohen, Martin	42, 43
Collins, Jeremy	118, 119
Corea, Joseph	14
Cormack, Robert A	32
Corradini, Stefanie	31
Cosgrove, Garth	36
Costa, Daniel	42, 43
Courtney, Burke	61
Cowles, Tobias	19
Cox, Jayden	14
Cruz, Moises	146, 148
Dai, Qing	114
Damgrave, Roy	75
Dang, Yanli	9
Daniel, Bruce L	32
Dantas, Eber	130
Daudelin, Andrew	90
David, Mills	70
Davidson, Justin	143
De Melo Antunes, Mariana	6, 52
Delgado-Esbensshade, Timoteo	114
Delhem, Nadira	37
Dell, Annika C.	22
Demir, Alican	5
Demirel, Omer Burak	63
Denecke, Timm	10
Deng, Sichen	68
Deniz, Sinan	4, 115, 147
Derre, Benoit	37
Desclides, Manon	6, 37, 52, 83
Dewalle, Anne-Sophie	37
Dietrich, Olaf	4, 6, 31, 52, 66, 67, 83 84, 115, 134, 147
Donato Jr, Francisco	12, 17
Dong, Fang	93
Dora, Chandler	43, 88
Duan, Isabella	73
Dumoulin, Charles	39
Dumoulin, Charles L	32
Dupuis, Andrew	19
Düring, Klaus	20
Düx, Daniel	7, 72, 129

Eisenmann, Marcel	7
Ellermann, Jutta	100
Fabritius, Matthias P	84, 134
Fabritius, Matthias Philipp	4, 31, 66, 67, 115, 147
Fallahi, Hojjat	86, 128, 129
Faller, Thibaut	52, 130
Fan, Richard	29
Fan, Yuqing	131
Fasbender, Sarah	61, 99, 142, 145
Faust, Jonas	78, 129
Faust, Jonas Frederik	86, 128
Favazza, Christopher	81, 135, 137, 142, 145
Favazza, Christopher P.	140, 141
Felgendreff, Philipp	7
Fernandes, Christi	81
Fischer, Gregory	36, 76
Fischer, Gregory S.	90
Fischer, Steven	143
Fiveland, Eric	70
Fleury Rosa, Suelia	131
Foley, Lori	53
Fordham, Nicholas	58
Foroughi, Pezhman	5
Fowlkes, J. Brian	82
Francoeur, Jacynthe	68
Franson, Dominique	78
Frey, Gregory	42, 43, 88
Friedrich, Thomas	22
Fritsch, Christoph	7
Fritsch, Elena	46
Fuchs, Frederik	67
Fütterer, J.J.	28
Fütterer, Jurgen	24
Garcia, Ana Sofia	132
Garrow, Sarah	79
Gaston, Sandra	69
Gehling, Katie	142, 145
Gerlach, Thomas	7, 15, 72, 80, 132, 136
Gesché, Valentine	22
Ghanouni, Pejman	14, 29, 42
Glandorf, Julian	7
Gleichner, Connor	61
Gouges, Bastien	37
Grissom, William	79
Griswold, Mark	19
Guan, Xingmin	87

Guenette, Jeffrey P.	45
Gulani, Vikas	25
Guo, Rui	127
Gutberlet, Marcel	7, 15, 72, 74, 129, 132, 136 139
Gutt, Moritz	129
H. Holmes, James	82
Haghpanah, Alireza	31, 67
Harlan, Kenny	5
Hartleben, Björn	7
Hata, Nobuhiko	40, 59, 76, 77
Heimer, Heather	61, 99, 142, 144
Hensen, Bennet	7, 15, 74, 129, 132, 139
Hermann, Paul E	92
Hernandez, Daniel	140, 141
Hernandez, Renato	14
Herzka, Daniel	19
Hoffman, Matt	133, 135
Holmes, James	70
Holmes, James H	12, 17
Hong, Charles	95
Hoogenboom, M.	28
Horstmann, Dominik	129
Hu, Jintao	9
Huang, Bingyu	127
Hubmann, Max Joris	7
Hussain, Hero	25
Ilia, Evangelia	16, 75
In, Myung-Ho	140, 141
In, Myunggho	61, 81, 135, 137, 142, 145
Iordachita, Iulian	68
Iranpour, Negaur	29
Jaitli, Nirvaan N	38, 71
Jarrard, David	26
Jiang, Yiwei	76, 90
Jiang, Yun	25
Jin, Ning	86
Johnson, Sara	44
Joshi, Sarang	44, 91
Jung, Matthias	46
Kalayeh, Kourosh	82
Kallumpurath, Zainudeen	63
Kamp, Benedikt	21
Katemann, Christoph	27

Katie, Gehling	61
Kaufmann, Timothy	126
Kaur, Tejinder	25
Kavran, Michael	19
Kaza, Evangelia	33
Kella, Naveen	43
Kerr, Christina	50
Kim, Gayoung	32
Kim, Jin Seob	68
Kim, Kyoung-Nam	140, 141
Kim, Tae Wan	77
Kinney, Ramsey	100
Kloekner, Roman	22
Knavel Koepsel, Erica	3
Knavel Koespel, Erica	47, 102
Knavel-Koepsel, Erica	26
Knudsen, Beatrice	91
Knull, Lucas	80
Kolandaivelu, Aravindan	95
Krafft, Axel	11, 87
Krafft, Axel J.	86, 128
Krieger, Axel	1, 53
Krieger, Joshua	120, 143, 146, 148
Krüselmann, Paula	21
Kuhl, Christiane K.	27
Kuroda, Kagayaki	35
Kurup, Anil	124
Kwok, K Y	138
L. Brace, Christopher	82
LaViolette, Peter	34
Lafont, Rebecca	130
Lam, Van Khanh	48
Lee, Eunwoo	140, 141
Lee, Junghoon	32
Lee, Philip	138
Lehman, Vance	126
Leiner, Tim	63
Leonard, Simon	53
Leung, Ben	87, 89
Levesque, Vincent M	45
Lewey, Lauren	146, 148
Li, Allen	138
Li, Gang	39
Li, Zhiqiang	63
Lilieholm, Thomas	38, 71
Lilieholm, Tom	26
Lin, Zhengyu	127

Linden, Philipp	115
Liu, Feng	73
Liu, Yixuan	94
Loew, Wolfgang	32, 39
Lomas, Derek	57, 60
Lu, Aiming	5, 8, 30, 47, 54, 55, 56 57, 61, 62, 63, 64, 65, 81 85, 89, 96, 97, 98, 101, 102 103, 105, 106, 107, 108, 109, 110 111, 112, 113, 116, 117, 121, 122 123, 124, 133, 135, 137, 140, 141 142, 145
Lu, David	114
Luttmann, Areli	20
Ma, Mijia	58
Ma, Yinsong	68
Macura, Katarzyna	42
Maier, Alexander	20, 23
Maier, Florian	4, 11, 78, 86, 93, 115, 120 128, 129
Makela, Pietari	42
Markley, Michael	19
Marlatt, Lauren	26
Mason, Andrew	113
McKenzie, Carly	61, 99
McKernan, Michael	60
McDonald, Robert	119
Mei, Chang-Sheng	51
Meng, Xiaosong	43
Mengwasser, Jörg	7
Mesropyan, Narine	27
Meyer, Hans Jonas	10
Meyer, Marius R.	22
Miller, Kai	126
Mills, David	82
Minko, Peter	21
Mitra, Jhimli	70, 82
Mohamed, Ola	5, 30, 47, 54, 55, 56, 57 60, 62, 64, 65, 89, 96, 97 98, 99, 101, 102, 103, 105, 106 108, 109, 110, 111, 112, 113, 117 121, 122, 123, 124, 133
Moin, Adnaan	43
Monfaredi, Reza	39
Moreira, Pedro	36, 53, 58, 76
Morgan, Brendan	19
Mori, Satoshi	76

Mueller, Samuel	21
Murakami, Ryo	76, 77
Musa, Juna	105
Myenderse, Lance	56
Mynderse, Lance	30, 42, 57, 62, 87
Männel, Georg	22
Müller, Samuel	22
Nam, Taewoo	140, 141
Nankani, Ayush	39
Nardone, Luigi	52, 84, 115, 134
Niatsetski, Yury	32
Nie, Pin	9
Nuernberger, Darius	21
Oakes, Terrence R	38, 71
Obereisenbuchner, Florian	4
Odeen, Henrik	44
Oishi, Tatsuya	133, 135
Oluigbo, Chima	39
Ooms, Nathan	11, 93, 120
Orlande, Helcio	130
Ozenne, Valéry	6, 37, 52, 83, 130
Page, Ariana	44
Pallares-Lupon, Nestor	52
Pang, Jianing	86, 120
Pannicke, Enrico	139
Paralikar, Atharva	39
Patel, Chirag	42
Pathak, Ram	43, 88
Pawlowski, Gary	19
Payne, Allison	44, 91
Petrisor, Doru	48
Piktel, Joseph	19
Piwowarczyk, Jakub	48
Pollack, Alan	69
Pooler, Angela	61, 99, 144
Princenthal, Robert	42, 43
Pugazandi, Anbarasi	132
Puhr-Westerheide, Daniel	4, 31, 66, 67, 115, 147
Purysko, Andrei	42
Quesson, Bruno	52, 130
Rajan, Purnima	5
Raman, Steven	42
Regler, Lisa	20

Reiss, Simon	20, 23, 46
Requardt, Martin	87
Reyns, Nicolas	37
Ricke, Jens	4, 31, 66, 67, 84, 115, 134 147
Risk, Michael	26
Robb, Fraser	17
Roll, Jesse	11, 93, 120
Rose, Georg	7, 72, 74, 132
Rossman, Phillip	137
Roumeliotis, Michael	32
Runkel, Alexander	46
Sagdic, Serdar	25
Sakhardande, Jayant D	12, 17
Salles, Hippolyte	83
Samsonov, Alexey A	12
Schlünz, Anton	72, 74
Schmidt, Ehud	95
Schmidt, Ehud J	32
Schmidt, Vanessa F	84, 134
Schmidt, Vanessa F.	4, 31, 66, 67, 115, 147
Schnitzer, Moritz	4
Schnitzer, Moritz Ludwig	147
Schrauben, Eric M	12
Schröer, Simon	7
Scott, Jason	11, 93
Seethamraju, Ravi	32, 53
Seibt, Janis	136
Seidensticker, Franziska	22
Seidensticker, Max	4, 6, 31, 52, 66, 67, 83 84, 115, 134, 147
Seidensticker, Ricarda	4, 115
Sella, David	42, 43, 88
Sewell, Sharmila	29
Shah, Nadeem	80
Sharma, Karun	48
Shea, Jill	44
Sheikh, Khadija	32
Shen, Guofeng	51
Sheth, Vipul	29
Shi, Shu-Fu	114
Shinyanbola, Deborah	59
Sieren, Malte M.	22
Simonetti, Orlando	94
Sin, Oscar	138
Singh, Amanpreet	44, 91
Singh, Mrinal	60, 126

Siu, Jimmy	138
Song, Daniel	32
Song, Sangeun	100
Sonn, Geoff	29
Sotoudehnia Korani, Setayesh	5, 30, 47, 54, 55, 56, 57 60, 62, 63, 64, 65, 87, 89 96, 97, 98, 101, 102, 103, 105 106, 107, 108, 109, 110, 111, 112 113, 117, 121, 122, 123, 124, 133 146, 148
Sotoudehnia, Setayesh	145
Sotoudenhnia Korani, Setayesh	99
Speck, Oliver	15, 72, 74, 80, 132, 136
Spieler, Benjamin	69
Spreter, Felix	23
Staruch, Robert	42, 88
Steiner, Edward	43
Stenzel, Wayne	61, 99, 119, 145
Stevenson, William	95
Stinson, Eric	81, 118, 119, 135, 137, 142, 144 145
Stinson, Eric G.	140, 141
Stoianovici, Dan	48
Stoyanova, Radka	69
Su, Pan	120
Taracila, Victor	17
Tavakkolmoghaddam, Farid	76
Tempany, Clare	53, 58, 59
Thieße, Gina	74
Thompson, Scott	5, 30, 47, 54, 55, 56, 57 60, 62, 63, 64, 65, 85, 89 96, 97, 98, 101, 102, 104, 105 106, 107, 108, 109, 110, 111, 112 117, 118, 119, 121, 122, 123, 124 133, 135, 137, 144, 146, 148
Ting, Samuel	94
Tirkes, Temel	42
Todd, Nicholas	41
Tokuda, Junichi	32, 45, 53, 58, 76, 95, 116 131
Tollefson, Megha	97, 104, 105, 106
Tse, H F	138
Tuna, Eser	19
Tuncali, Kemal	53, 58
Uller, Wibke	20
Usmani, Hasan	104

Vangel, Mark	58
Vannod Michel, Quentin	37
Velikina, Julia M	12
Verloh, Niklas	20
Vick, Ralf	15
Vincent, Jana	17
Virkar, Pooja	20
Viswanathan, Akila N	32
Vogt, Ivan	74
Volk, Martin	72
Wacker, Frank	2, 7, 15, 72, 74, 129, 132 139
Waddle, Spencer	63
Wallaengen, Veronica	69
Wang, Eric	77
Wang, Shiyue	36
Wang, Yang	36, 76, 90
Wang, Yifan	95
Weber, Oliver M.	27
Wegner, Franz	22
Wei, Jianli	94
Wei, John	25
Wei, Juan	138
Weiss, Jakob	46
Welch, Brian	54, 103, 113
Wells, Shane	25, 70, 82
Wells, Shane A	12, 17
Wendt, Dennis	22
Weng, Xin	9
Wilcox, Samuel	116
Wildgruber, Moritz	4, 115, 147
Willemen, M.	28
Wilms, Lena Marie	21
Woodrum, David	5, 30, 42, 47, 54, 55, 56 57, 60, 61, 62, 63, 64, 65 81, 85, 87, 89, 96, 97, 98 99, 101, 102, 103, 104, 105, 106 107, 108, 109, 110, 111, 112, 113 116, 117, 118, 119, 121, 122, 123 124, 133, 135, 137, 142, 144, 145 146, 148
Woodrum, David A.	140, 141
Woodrum, Laurel	107
Wu, Holden	49, 50, 114
Wu, Mingming	4, 84, 115, 134

Xiao, Li	138
Yang, Kehan	77, 90
Yarmolenko, Pavel	39, 48
Yeo, Desmond	70, 82
Yeo, Desmond T	12, 17
Yerdelen, Seben Sena	7
Yimsuk, Adisorn	35
Yu, Mandy	85
Yüksel, Can	27
Zhang, Haichong	76, 77
Zhao, Sophie	91
Zhao, Yueran	51
Zhao, Zhanyue	36
Zhou, Chang	95
Zhou, Wenqi	50, 114
Zhu, Kaiguo	9
Zimmermann, Silke	10
Zong, Shenyan	51
Öcal, Osman	4, 31, 66, 67, 115